Expansions about the short-range end of the long-range Ising model in one, two and three dimensions

The dual of the chain through eighth order, the crossover coefficients c_2 ($d = 2$) and c_1 ($d = 3$), their checks and their status

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Abstract

We collect the expansions of the thermal scaling dimension $\Delta_\mathcal{E}$ of the critical long-range Ising model about the point where it joins the short-range Ising universality class, in $d = 1, 2, 3$. In $d = 1$ the strong–weak dual of the chain of Benedetti, Lauria, Mazáč and van Vliet gives $\Delta_\mathcal{E} = 1 + \sum_n C_n^{\mathrm{SR}} \Delta_\sigma^{n/2}$; its published coefficients are $C_1^{\mathrm{SR}} = -2$ and $C_2^{\mathrm{SR}} = \frac{1}{2}$. In the matched two-coupling prescription defined in Sec. 2.1, the archived integral expressions for C_3^{SR} to C_7^{SR} reduce exactly to rational combinations of zeta values, C_5^{SR} to C_7^{SR} by exact word-algebra certificates. The step from the kink integrals to those expressions cannot be rerun, because 21 stage programs of the archive are missing; it is backed by an independent re-derivation of the inputs by Mellin–Hepp sector integration, which reproduces C_3^{SR} to C_7^{SR} within statistical errors of 3×10^{-14} to 3×10^{-5} (largest pull 0.9 over the 25 inputs of C_5^{SR} to C_7^{SR}). $C_8^{\mathrm{SR}} = 247.381204805\ldots$ is a numerical record printed to about 85 digits, of which a second evaluation of one input cross-checks about 57 decimal places; its rational part, 88, is exact, and a weight-seven zeta-value formula matches all printed digits but is not proved. The independent re-derivation gives $C_8^{\mathrm{SR}} = 242.8 \pm 2.4$ (one standard error), 1.9 to 4.0 standard errors below the record depending on the error estimator; the difference sits in one input integral. A third calculation, in a scheme defined by the magnetization at the integrable point $s = 1$, where the Bethe-ansatz solution gives the whole $s = 1$ part, shares no input with the other two and gives $C_8^{\mathrm{SR}} = 247.38118 \pm 0.13$, an empirical numerical uncertainty. It agrees with the record to 2.5×10^{-5}, reproduces C_3^{SR} to C_7^{SR} to 7×10^{-26}, and so indicates that the low independent value is the one in error; the re-derivation's own input comparison locates the difference in that integral. In $d = 2$, $c_1 = 0$, and $c_2 = -4.0662701481325\ldots$ is a closed form in Gamma values and three ${}_4F_3(1)$ constants whose identification with the physical coefficient rests on one unproved uniform-remainder step. A Monte Carlo integration of the regulated integral gives -4.075 ± 0.019 (one standard error), and fourth-order perturbation theory with the truncated cylinder Hamiltonian gives $-4.0670(24)$, where the error is a linear sum of systematic diagnostics; its relative difference from the closed form is 1.9×10^{-4}. In $d = 3$, $c_1 = 0.2732$, with the estimated range $[0.2690, 0.2760]$, conditional on assumed gaps in the omitted spectrum and on assumed ranges of the input data. Every result carries its status word and caveats. The appendices give in full the derivations, proofs, methods and checks of the supplied packages behind these results (the independent re-derivation, the Monte Carlo and the cylinder continuum analysis are described in the main text), including the formal arguments that two couplings close under renormalization of the dual and that its prescription extracts the local beta functions. Appendix H maps each number to the package and command that reproduce it, and a last appendix gives the magnetization-scheme calculation of C_8^{SR} that the paper quotes.

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1 Introduction and conventions

1.1 The short-range end

The critical long-range Ising (LRI) model in d dimensions has couplings $J(r) \propto r^{-(d+s)}$. Its spin field has the exact dimension $\Delta_\sigma = (d-s)/2$ throughout the long-range regime, and its fixed point joins the short-range Ising (SRI) universality class at

$$s_* = d - 2\Delta_\sigma^{\mathrm{SR}}. \tag{1.1}$$

Near s_* the natural small parameter is

$$\delta = \frac{s_* - s}{2} = \Delta_\sigma - \Delta_\sigma^{\mathrm{SR}}. \tag{1.2}$$

This note collects what is known about the thermal dimension $\Delta_\mathcal{E}$ near that end in $d = 1, 2, 3$. These are the results quoted in Sec. 3.2, "Near the short-range end", of the accompanying paper [1] (its table of short-range coefficients, the closed forms of the dual of the chain, the c_2 row of its table of perturbative Hamiltonian truncation, and the closed-form appendix for c_2), with the derivations, checks and status of each. The paper's resummation of the expansion about the generalized-free-field end, $s = d/2$, also imposes these values at the other end of the long-range window, $s = s_*$.

Two different expansions apply. In $d = 1$ the short-range end is $s_* = 1$, where $\Delta_\sigma^{\mathrm{SR}} = 0$ (for $s > 1$ the chain has no finite-temperature transition); the expansion is that of a strong–weak dual and proceeds in $\sqrt{\Delta_\sigma}$ (Sec. 2). In $d = 2, 3$ the fixed point near s_* is the SRI conformal field theory (CFT) coupled to a generalized free field (GFF), and the expansion proceeds in integer powers of δ (Secs. 3 and 4). Table 1 lists the conventions.

d	s_*	$\Delta_\sigma^{\mathrm{SR}}$, $\Delta_\mathcal{E}^{\mathrm{SR}}$	variable	expansion
1	1	0, 1	$x = \Delta_\sigma = \frac{1-s}{2}$	$\Delta_\mathcal{E} = 1 + \sum_{n\geq 1} C_n^{\mathrm{SR}}\, x^{n/2}$
2	$\frac{7}{4}$	$\frac{1}{8}$, 1	$\delta = \frac{7/4-s}{2} = \Delta_\sigma - \frac{1}{8}$	$\Delta_\mathcal{E} = 1 + c_1\delta + c_2\delta^2 + O(\delta^3)$
3	1.963702388(48)	0.518148806(24), 1.41262528(29)	$\delta = \frac{s_*-s}{2}$	$\Delta_\mathcal{E} = \Delta_\mathcal{E}^{\mathrm{SR}} + c_1\delta + O(\delta^2)$

Table 1: Conventions. The $d = 3$ data are those of Ref. [15]. In $d = 1$ the variable δ of Eq. (1.2) is x; the dual of Sec. 2 uses $\bar\delta = 1 - s = 2x$, which its computational sources write δ (as in the input name N_δ). A coefficient of $(s_* - s)^n$ is $c_n/2^n$: the coefficient of $(7/4 - s)^2$ in $d = 2$ is $c_2/4$, and that of $s_* - s$ in $d = 3$ is $c_1/2$. The paper writes $C_n^{\mathrm{SR}}(d)$ for all three: $C_n^{\mathrm{SR}}(1) = C_n^{\mathrm{SR}}$ multiplies $\Delta_\sigma^{n/2}$, while $C_n^{\mathrm{SR}}(2) = c_n$ and $C_n^{\mathrm{SR}}(3) = c_n$ multiply δ^n (so $c_2 = C_2^{\mathrm{SR}}(2)$ and $c_1 = C_1^{\mathrm{SR}}(3)$).

The symbol b_0 follows the sources and has two unrelated meanings: in $d = 1$ it is the charge of the compact field of the dual (Sec. 2.1); in $d = 2$ and $d = 3$ it is the one-loop coefficient of the beta function of the crossover coupling (Secs. 3.2 and 4). A few other letters are reused, always with a local definition: h is the current coupling of the dual in Sec. 2 and the g^4 coefficient of $\gamma_\mathcal{E}$ or of the cylinder observable in Sec. 3 (h_0, $h(\lambda)$); W is an input of C_8^{SR}, the Gaussian source factor of Sec. 3 and, as W_k, the cylinder counterterms; κ is the reference inverse length of the dual (Appendix A), the finite redefinition parameter of Sec. 3.2 and, as κ_r, a counterterm coefficient of Appendix F; and χ is the generalized free field in $d = 2, 3$, while the bare current of the dual is χ_B and its renormalized current primary (Appendices A and C) is written χ there.

1.2 Sources and who produced them

The note rests on seven packages of the release. Four were written by another AI system, OpenAI GPT-6 astra (via ChatGPT), and supplied by the author: a research compendium for the $d = 1$ dual (its derivations, code, exact expressions, certificates and numerical records of C_3^{SR} to C_8^{SR}; "the compendium"; its original archive, of which the release ships a pruned copy, is "the archive"), a closed-form study of c_2 in $d = 2$ ("the study", also called the closed-form or final-method study), a base study of c_2 with the truncated cylinder Hamiltonian, and a package for c_1 in $d = 3$ ("the c_1 package"). Claude Code (Anthropic) sessions reviewed each of them ("the review" of each), re-derived every input of C_5^{SR} to C_8^{SR} independently, ran the Monte Carlo of c_2, and wrote the cluster layer and the continuum analysis of the cylinder calculation, and computed C_8^{SR} in the magnetization scheme (Sec. 2.7); OpenAI Codex reviewed the plans and the code of those two calculations. Two formal arguments about the dual, that two couplings close under its renormalization and that its prescription extracts the local beta functions, are also by GPT-6 astra.

The main text states each result with its basis and caveats; the appendices give the derivations behind them, edited to the notation of this note: the compendium in Appendices A (the model, the regulator and the finite-response renormalization), B (the third to sixth orders) and C (the exact certificates and the seventh and eighth orders); the two arguments about the dual in Appendix D; the study in Appendix E; the method of the cylinder calculation in Appendix F; and the c_1 package in Appendix G. Section 5 gives the authorship, and Appendix H the folder of each package. A third calculation of C_8^{SR}, in a scheme defined by the magnetization at the integrable point $s = 1$, was made for the project after the other packages: Sec. 2.7 summarizes it, Appendix I gives it in full, and its code and records are the package `integ_d1`.

1.3 Status words

Each quoted result carries one of the following status words.

literature a published value, cited where it is used and quoted as published; where this note re-derives it as a check, Table 2 says so;

proved closed form within the prescription ($d = 1$) an exact expression into which the archived integral expressions of the matched two-coupling prescription (Sec. 2.1) reduce exactly, with exact certificates where the reduction is computer-assisted; it covers neither the step from the kink integrals to those expressions nor the prescription itself;

closed form conditional on one unproved step (c_2) an exact expression whose evaluation is proved, but whose identification with the physical coefficient needs a step that is not proved;

numerical a number obtained by numerical integration, extrapolation or sampling, with the kind of its error stated: a statistical standard error, a linear sum of systematic diagnostics that is not a standard deviation, or, for a record without an error estimate, the precision to which it is cross-checked;

conjectured an exact expression supported numerically but without a proof;

conditional a statement or range that holds only under explicitly stated, unproved assumptions.

The file `results/sr_values.json` uses the short forms literature, closed form, numerical, conjectured and conditional, with the qualification in a note or caveat field; it lists c_1 in $d = 3$ as a numerical retained value and an estimated range with the status conditional. Errors are always labelled with their kind. A pull is (our value − claimed value)/(quoted standard error).

1.4 Results and their status

Table 2 gives every result of this note that the paper quotes, and three that it does not print (the largest pull of the recomputed inputs, the independent value of C_8^{SR} and the agreement of the two independent codes for c_1), with its status word, its basis and its caveats. The summaries and checks follow in Secs. 2 to 4, the derivations in Appendices A to G and I, and the authorship in Sec. 5.

Table 2: The results near the short-range end, with the status words of Sec. 1.3. In $d = 1$ "the prescription" is the matched two-coupling prescription of Sec. 2.1. The paper prints six significant digits; the longer strings are evaluations of the closed forms and of the C_8^{SR} record, rounded to 15 significant digits (all digits are in `results/sr_values.json`).

result	value	status	basis and caveats
$d = 1$			
C_1^{SR}, C_2^{SR}	-2, $\frac{1}{2}$	literature	Refs. [3, 4]; re-derived from the leading flow (Sec. 2.2).
C_3^{SR} to C_7^{SR}	Eq. (2.12):	proved closed form within the prescription	The archived integral expressions of the beta inputs reduce exactly to zeta values: C_3^{SR}, C_4^{SR} by analytic period evaluations (with four archived mixed response coefficients), C_5^{SR} to C_7^{SR} by exact word-algebra certificates (Sec. 2.4). The step from the kink integrals to these expressions cannot be rerun (Appendix H.5); the independent re-derivation reproduces C_3^{SR} to C_7^{SR} numerically (Sec. 2.6). Prescription caveat: Secs. 2.1 and 2.9, Appendix D.
C_3^{SR}	$5.72736813369645\ldots$		
C_4^{SR}	$-12.1961914926697\ldots$		
C_5^{SR}	$29.2378115809900\ldots$		
C_6^{SR}	$-58.6656974742253\ldots$		
C_7^{SR}	$51.0689886129281\ldots$		
$\pi_0 C_8^{\mathrm{SR}}$ (rational part)	88	proved closed form within the prescription	Exact rational projection, from two separate exact calculations (Sec. 2.5); the same two qualifications as C_3^{SR} to C_7^{SR}.
C_8^{SR}	$247.381204804994\ldots$	numerical	Record printed to about 85 digits (printing floor 2×10^{-85}); a second evaluation of its input P_6 cross-checks about 57 decimal places; six inputs are single numerical records; no error bound (Sec. 2.5).
Eq. (2.15)	weight-seven formula for C_8^{SR}	conjectured	Matches the record to 3.1×10^{-86}; no certificate; the record is its only high-precision support (the magnetization-scheme value agrees to 2.5×10^{-5}).
inputs of C_5^{SR} to C_7^{SR}, independent	largest pull 0.9	numerical	One standard error (the largest of three statistical estimators); 25 inputs, largest pull 0.915, rms 0.34; the coefficients within 0.78 (C_6^{SR}); the paper says only that the recomputed inputs match.

result	value	status	basis and caveats
C_8^{SR}, independent	242.8 ± 2.4	numerical	One standard error (largest of three estimators); 1.9 to 4.0 standard errors below the record, depending on the estimator; the difference sits in one input integral, $A_9[\varepsilon^0 \ell^1]$ (Sec. 2.6); the magnetization-scheme value (next row) indicates that this value is the one in error; not printed by the paper.
C_8^{SR}, magnetization scheme	247.38118 ± 0.13	numerical	Empirical numerical uncertainty, a linear sum of the differences between three runs and a validated roundoff estimate, not a standard deviation or a bound; no input shared with the two hard-core calculations, no nine-kink integral; reproduces C_3^{SR} to C_7^{SR} to 7×10^{-26}; 2.5×10^{-5} from the record; two check criteria amended after their data were seen; the paper quotes it as 247.38 ± 0.13 (its Sec. 3.2); package `integ_d1` (Sec. 2.7, Appendix I).
$d = 2$			
c_1	0	literature	BRRZ [12]; re-derived ($I_2(0) = 0$, Sec. 3).
c_2	$-4.06627014813252\ldots$	closed form conditional on one unproved step	The evaluation of $\mathcal{I}$ is proved; the identification with the physical coefficient needs the uniform-remainder step (Sec. 3.6); a formal result in critically tuned analytic minimal subtraction.
c_2, Monte Carlo	-4.075 ± 0.019	numerical	One standard error (statistical); 0.5 standard errors from the closed form (Sec. 3.7).
c_2, cylinder	$-4.0670(24)$	numerical	Linear sum of systematic diagnostics (64% regulator model), not a standard deviation; 1.9×10^{-4} relative from the closed form, which was known but is not an input; error rule revised after the first result; 11 of the 25 points are archived records of the base study (Sec. 3.8).
$d = 3$			
c_1	0.2732; estimated range [0.2690, 0.2760]	conditional	Retained 60-primary value $0.27315242\ldots$ (Sec. 4.2). The range holds if no omitted primary lies below the gaps 8 ($\mathbb{Z}_2$-even) and 7 ($\mathbb{Z}_2$-odd), the inputs lie within twice their quoted errors, a published constraint on $\lambda_{\epsilon\epsilon\epsilon'}$ applies and the chosen margins and allowances suffice; it is neither a bound nor a confidence interval, and mild relaxations extend it (Secs. 4.3 and 4.4).
c_1, two independent codes	agree to 10^{-12}	numerical	Absolute difference in c_1 of the review's two re-implementations from the package (Sec. 4.2); not printed by the paper.
c_1, BRRZ	0.27(4)	literature	Ref. [12].

2 One dimension: the dual of the chain

2.1 The dual, its two couplings and the prescription

Near $s = 1$ the domain walls of the critical chain form a dilute gas of alternating kinks and antikinks with a logarithmic interaction [6, 10]. Benedetti, Lauria, Mazáč and van Vliet recast this gas as a two-state system (a qubit) coupled to a compact Gaussian field $\phi \sim \phi + 2\pi/b_0$ of negative dimension [3, 4]. With $\bar\delta = 1 - s$ and $f_{\bar\delta}(r) = (r^{\bar\delta} - 1)/\bar\delta$, the field has covariance $-2f_{\bar\delta}(|X - Y|)$ on neutral test functions, and the path-ordered evolution

$$\mathrm{P}\exp\int \mathrm{d}X\,[g\,\mathcal{O}_g + h\,\mathcal{O}_h], \qquad \mathcal{O}_g = \hat\sigma_+ {:}e^{\mathrm{i}b_0\phi}{:} + \hat\sigma_- {:}e^{-\mathrm{i}b_0\phi}{:}, \qquad \mathcal{O}_h = \hat\sigma_3\chi_B, \qquad \chi_B = \tfrac{\mathrm{i}}{\sqrt{2}}\partial\phi, \tag{2.1}$$

reproduces the alternating gas because $\hat\sigma_\pm^2 = 0$ (Appendices A.2 and A.3 derive the gas and this representation). The bare current χ_B is a different field from the generalized free field χ of Secs. 3 and 4 (and from the renormalized current primary that Appendix A.6 writes χ); the symbol table of the dual is Table 4. The critical chain is the sector invariant under a $U(1)$ that shifts ϕ and rotates the qubit. The fugacity g and the current coupling h flow; the current can be resummed into the effective charge $b = b_0 - \sqrt{2}h$.

The matched two-coupling prescription. The coefficients C_3^{SR} to C_8^{SR} are computed in one fixed prescription, taken from the compendium and derived in Appendices A.4 to A.9:

(i) *Regulator.* Kinks at ordered positions $0 < X_1 < \cdots < X_n < 1$ on an interval of unit length, with a hard core $X_{i+1} - X_i \geq \varepsilon$ on consecutive kinks only (none between a kink and an endpoint), and $\ell = \log\varepsilon$ (Eq. (A.14)).

(ii) *Exact current resummation.* Before the hard core is imposed, a field-dependent change of frame removes the bulk current interaction. It shifts the wall charge to $b = b_0 - \sqrt{2}h$ and leaves charges $\pm h/\sqrt{2}$ at the two endpoints of the interval, whose contraction with the walls is an *endpoint source* $\eta = \sqrt{2}\,h\,b$ (Appendix A.4).

(iii) *Matching at the cutoff.* Normal ordering at the cutoff defines the charge $B = b\,\varepsilon^{\bar\delta/2}$ and the fugacity g, keeping the $\bar\delta$-dependent normal-ordering factor of the wall interaction exactly (finite-tail matching); the essential couplings are

$$u = g^2, \qquad v = B^2 = 1 + z, \qquad \beta_u = 2uF(u, z, \bar\delta), \qquad \beta_z = (1 + z)H(u, z, \bar\delta), \tag{2.2}$$

in units with $b_0 = 1$, so that $z = 0$ is the marginal point (Appendix A.5).

(iv) *Vacuum normalization.* The ordered moments of the regulated gas (Sec. 2.3) are divided by the vacuum moments, as complete series in ε and ℓ, before any coefficient is extracted (Appendix A.7).

(v) *Two projected responses.* The beta functions of u and z are solved, order by order, from two responses only: the odd (charged) response and the current response, each projected on its coefficient $[\varepsilon^0\ell^1]$, through a finite-response Jacobian that carries their finite parts $[\varepsilon^0\ell^0]$ (Appendix A.9).

"Matched" refers to (iii), and "two-coupling" to the assumption that (u, z) close under renormalization: no further coupling enters their flow at the orders computed. At leading order (Appendix A.6)

$$F = z + u + \cdots, \qquad H = -\bar\delta + 4u + \cdots, \qquad u_* = \frac{\bar\delta}{4} + O(\bar\delta^2), \quad z_* = -\frac{\bar\delta}{4} + O(\bar\delta^2), \tag{2.3}$$

so the fugacity at the fixed point is of order $\sqrt{\bar\delta}$, which is why the series runs in $\sqrt{\Delta_\sigma}$.

What "proved within the prescription" covers. In this prescription the beta inputs are fixed regulated integrals. A *proved closed form within the prescription* means that the archived exact expressions of these integrals reduce exactly to rational combinations of zeta values (Sec. 2.4). Two steps lie outside it. First, the step from the kink integrals to the archived expressions cannot be rerun, because 21 stage programs of the archive are missing (Appendix H.5); it is backed numerically by the independent re-derivation of Sec. 2.6. Second, whether the prescription computes the coefficients of the dual (that the dual describes the chain is the published duality proposal), which rests on the two-coupling closure and on the prescription extracting the local beta functions; two unrefereed arguments support both formally (Appendix D: the closure statement in Appendix D.2, the prescription statement in Appendix D.3). Their status is stated once, in the caveat "Two-coupling closure" of Sec. 2.9, together with the evidence that bears on it.

2.2 Stability invariants and the thermal series

At the fixed point $F = H = 0$, so the trace and determinant of the stability matrix in (u, z) are

$$T = 2u_*F_u + (1 + z_*)H_z, \qquad D = 2u_*(1 + z_*)(F_uH_z - F_zH_u), \tag{2.4}$$

with the derivatives taken before the fixed point is substituted. They are invariant under regular changes of couplings, although individual beta coefficients are not (Appendix A.10). With

$$T(\bar\delta) = \frac{\bar\delta}{2} + \sum_{m\geq 2} T_m\bar\delta^m, \qquad D(\bar\delta) = -2\bar\delta + \sum_{m\geq 2} D_m\bar\delta^m, \tag{2.5}$$

the two dimensions are

$$\Delta_\pm = 1 + \frac{T}{2} \pm \sqrt{\frac{T^2}{4} - D}, \qquad \Delta_\mathcal{E} = \Delta_-, \tag{2.6}$$

assuming, as in Ref. [1], that the relevant operator is not crossed by another level. The square root is $\sqrt{2\bar\delta}$ times a power series in $\bar\delta$, so the odd powers of $\sqrt{\Delta_\sigma}$ come from the eigenvalue splitting and the even powers from $T/2$. At leading order $\Delta_\pm = 1 \pm \sqrt{2\bar\delta} + \bar\delta/4 + O(\bar\delta^{3/2})$ [4], which is

$$C_1^{\mathrm{SR}} = -2, \qquad C_2^{\mathrm{SR}} = \frac{1}{2}. \tag{2.7}$$

Expanding Eq. (2.6) gives the exact conversions

$$\begin{array}{lll} C_3^{\mathrm{SR}} = D_2 - \frac{1}{16}, & C_4^{\mathrm{SR}} = 2T_2, & C_5^{\mathrm{SR}} = 2D_3 - \frac{T_2}{2} + \frac{1}{4}\left(D_2 - \frac{1}{16}\right)^2, \\ C_6^{\mathrm{SR}} = 4T_3, & C_7^{\mathrm{SR}} = 4D_4 - T_3 - T_2^2 + \frac{1}{2}C_3^{\mathrm{SR}}C_5^{\mathrm{SR}}, & C_8^{\mathrm{SR}} = 8T_4, \end{array} \tag{2.8}$$

and in general $C_{2m}^{\mathrm{SR}} = 2^{m-1}T_m$. These are algebraic identities; they do not determine an uncomputed invariant (C_9^{SR} needs D_5).

2.3 Beta inputs and finite-response renormalization

The invariants are polynomials in Taylor coefficients of F and H at the marginal point $v = 1$, $\bar\delta = 0$, which the computation calls beta inputs. They come from ordered moments of the regulated gas, odd moments A_{2m+1}, vacuum moments Z_{2m} and current moments Q_{2m}, with up to nine kinks and their derivatives in the charge v, the tail parameter $\bar\delta$ and the endpoint source η. After the vacuum normalization (iv), the coefficients $[\varepsilon^0\ell^0]$ and $[\varepsilon^0\ell^1]$ are taken, and the finite-response Jacobian (v) turns them into beta coefficients (Appendices A.7 to A.9; the inputs of each order, evaluated or given by their archived exact expressions, are in Appendices B and C). Five inputs (a, p, b, c, f)

fix C_3^{SR} and C_4^{SR}, 25 fix C_7^{SR}, and ten more $(W, S, U, J_9, L_3, M_{21}, M_{12}, N_2, N_\delta, K_v)$ enter C_8^{SR}. For example (the input $b = b_5$ is not the charge b)

$$\begin{aligned} &a = -4\zeta(2), \qquad c = -16, \qquad p = 8 - 2\zeta(2) - 7\zeta(3) - 8\log 2, \\ &b = -18 - 6\zeta(2) + \tfrac{28}{3}\zeta(3) + 32\log 2 - 12\log 3 - 4\log^2 2 - 8\,\mathrm{Li}_2(-\tfrac{1}{2}), \\ &f = 32 - \tfrac{136}{3}\zeta(3) - 64\log 2 + 48\log 3 + 16\log^2 2 + 32\,\mathrm{Li}_2(-\tfrac{1}{2}), \end{aligned} \tag{2.9}$$

and $D_2 = \frac{1}{2} - \frac{a}{2} - \frac{c}{8}$, $T_2 = -\frac{a}{8} + \frac{b}{4} - \frac{c}{32} + \frac{f}{16} + \frac{p}{2}$. The regulator constants $\log 2$, $\log 3$ and $\mathrm{Li}_2(-\frac{1}{2})$ are genuine parts of the individual inputs and cancel from the invariants. The eighth order is affine in the ten new inputs,

$$C_8^{\mathrm{SR}} = B_{\mathrm{old}} + 4W + 2S + \tfrac{3}{4}U + \tfrac{1}{4}J_9 + \tfrac{3}{32}L_3 - \tfrac{1}{4}M_{21} + \tfrac{1}{2}M_{12} - \tfrac{1}{16}N_2 + \tfrac{1}{8}N_\delta + \tfrac{1}{32}K_v, \tag{2.10}$$

where the polynomial B_{old} in the 25 earlier inputs (printed in Appendix C.12) has the value $-861.66356723547977920\ldots$.

2.4 Closed forms through the seventh order

The invariants reduce to

$$\begin{aligned} &D_2 = \tfrac{5}{2} + 2\zeta(2), \qquad T_2 = 2 - 2\zeta(2) - 4\zeta(3), \qquad D_3 = \tfrac{21}{4} - 13\zeta(2) + 11\zeta(3) + 11\zeta(4), \\ &T_3 = \tfrac{13}{2} - \tfrac{19}{2}\zeta(2) + 26\zeta(3) - 20\zeta(4) - 68\zeta(5) + 28\zeta(2)\zeta(3), \\ &D_4 = 13 - \tfrac{179}{8}\zeta(2) + \tfrac{47}{2}\zeta(3) - \tfrac{493}{8}\zeta(4) + 289\zeta(5) - 130\zeta(2)\zeta(3) + \tfrac{499}{4}\zeta(6) - 76\zeta(3)^2, \end{aligned} \tag{2.11}$$

and Eq. (2.8) gives

$$\begin{aligned} C_3^{\mathrm{SR}} &= \frac{39}{16} + 2\zeta(2), \qquad C_4^{\mathrm{SR}} = 4 - 4\zeta(2) - 8\zeta(3), \\ C_5^{\mathrm{SR}} &= \frac{11249}{1024} - \frac{361}{16}\zeta(2) + 24\zeta(3) + \frac{49}{2}\zeta(4), \\ C_6^{\mathrm{SR}} &= 26 - 38\zeta(2) + 104\zeta(3) - 80\zeta(4) - 272\zeta(5) + 112\zeta(2)\zeta(3), \\ C_7^{\mathrm{SR}} &= \frac{1798583}{32768} - \frac{90637}{1024}\zeta(2) + \frac{453}{4}\zeta(3) - \frac{16835}{64}\zeta(4) + 1224\zeta(5) \\ &\quad - 540\zeta(2)\zeta(3) + \frac{4335}{8}\zeta(6) - 320\zeta(3)^2. \end{aligned} \tag{2.12}$$

Their values are in Table 2. Each C_n^{SR} with $n \le 7$ lies in the rational span of multiple zeta values of weight at most $n - 1$.

How the closed forms are established. C_3^{SR} and C_4^{SR} follow from analytic evaluations of the four- and five-insertion periods, closed by a finite-part trilogarithm identity proved with explicit primitives and a real-branch Spence–Kummer identity; the four mixed response coefficients of Eq. (B.102) that attach the periods to the inputs are archived values, not derived here (Appendix B, which also assembles C_5^{SR} and C_6^{SR} from their inputs and evaluates the analytic inputs $A = \frac{1}{2}b_{3,vv}$, $E = b_{3,v\bar{\delta}}$ (Appendix B.1) and the two endpoint constants of C_5^{SR}; the other inputs of these orders are archived exact expressions). From C_5^{SR} on, the exact archived expressions of the beta inputs, rational functions of the cutoff multiplying iterated integrals, are lifted to a free rational vector space of words of regularized hyperlogarithm constants. The difference between the archived expression and the proposed closed form is then written as an explicit rational combination of shuffle products of proved zero relations (shuffle, stuffle, duality, a convergent confluence lemma and a beta-integral identity), and a standalone verifier checks it word by word in exact rational arithmetic, with no numerical period and no assumption of numerical independence (Appendix C.1):

target	weights	nonzero words	certificate rows
D_3 (hence C_5^{SR})	2–4	170	476
C_6^{SR}	2–5	1 945	56 381
D_4, C_7^{SR} (with all product terms)	2–6	25 212	588 878

A separate standard-library checker parses the archived input expressions, lifts them into the word algebra and verifies the trace assembly and the target coefficients, so the certificates are tied to the inputs. The certificates prove the reduction of the archived regulated-integral expressions to zeta values. They do not reintegrate the original kink kernels, and they do not test the prescription itself (Sec. 2.1).

2.5 The eighth order

Rational part. The formal weight-zero projection π_0, which removes positive-weight constant periods after integration, is computed exactly (Appendix C.4; the trace reduction of the eighth order is in Appendix C.3). A decomposition into four raw residues (of the six-kink and eight-kink current moments, the seven-kink tail and the nine-kink fugacity) gives, after the first two are combined with the known terms,

$$\pi_0 C_8^{\mathrm{SR}} = -\frac{111389}{24} - \frac{910891}{8} + \frac{1423087}{12} = 88, \tag{2.13}$$

and a separate literal full-Jacobian calculation gives $\pi_0 T(\bar\delta) = \frac{\bar\delta}{2} + 2\bar\delta^2 + \frac{13}{2}\bar\delta^3 + 11\bar\delta^4 + \cdots$, hence $\pi_0 C_8^{\mathrm{SR}} = 8 \times 11 = 88$. None of the raw integrals was inferred from the answer or from an earlier conjectured value, 160, which this result excludes.

Full value. Four of the ten new inputs were computed in four separate stages: the six-kink mixed logarithm $P_6 = 1261.944897843140491372852392951531767542124984174\ldots$ (an exact period expression with 66 404 terms, evaluated at 70 and 100 working digits), from which $N_\delta = -4P_6 + 11218.914021881803142\ldots$; the seven-kink tail coefficient $U = -1296.687960291526491656\ldots$; the eight-kink current derivative $K_v = -1486.962109278823406128\ldots$; and the ninth-order fugacity $J_9 = -273.724515999324646514\ldots$, a cancellation between pieces of order 6×10^5. The other six $(W, S, L_3, M_{21}, M_{12}, N_2)$ enter as *inherited numerical records*: decimal values from one recorded 100-digit evaluation, with no surviving exact expression and no second evaluation (Appendix C.5 describes all ten). The assembly reduces to $C_8^{\mathrm{SR}} = C_{\mathrm{fixed}} - \frac{1}{2}P_6$ with $C_{\mathrm{fixed}} = 878.353653726563794093781271\ldots$, and gives

$$C_8^{\mathrm{SR}} = 247.381204804993548407355074766099998652159151220 9\ldots. \tag{2.14}$$

How many digits. Four facts bear on the digits of Eq. (2.14); none is an interval bound.

- *Cross-check.* The two evaluations of P_6 give values of C_8^{SR} that differ by 2.2×10^{-57}, the precision of the 70-digit run. This cross-checks about 57 decimal places, and only through P_6.
- *Printing floor.* The record is assembled exactly from 36 numbers (the 35 inputs, with N_δ given as P_6 and its remainder), all decimal strings of 85 to 96 decimals except the exact $c = -16$. Changing each by half a unit in its last printed digit changes C_8^{SR} by about 2×10^{-85} in all, about 87 significant digits, quoted as about 85. This is the precision to which the inputs are written, not a demonstrated accuracy: the six inherited inputs have neither exact expressions nor second evaluations.
- *The formula.* The 100-digit value agrees with the conjectured Eq. (2.15) below to 3.1×10^{-86}. This is consistent with both, but the formula has no high-precision support other than this record, so the agreement is not independent evidence for the record's digits.

- *A third value.* The magnetization-scheme value of Sec. 2.7, 247.38118 ± 0.13, agrees with the record to 2.5×10^{-5}; it bears on the first few digits only.

The weight-seven candidate. An integer-relation search (Appendix C.6) on the 70-digit value, in the basis $\{\zeta(2), \zeta(3), \zeta(4), \zeta(5), \zeta(2)\zeta(3), \zeta(6), \zeta(3)^2, \zeta(7), \zeta(2)\zeta(5), \zeta(3)\zeta(4)\}$ with the rational part 88 subtracted, gives

$$C_8^{\mathrm{SR}} \stackrel{?}{=} 88 - 178\zeta(2) + 110\zeta(3) + 113\zeta(4) + 4892\zeta(5) - 2376\zeta(2)\zeta(3) - \frac{8630}{3}\zeta(6) + 1872\zeta(3)^2 - 12168\zeta(7) + 6048\zeta(2)\zeta(5) + 1536\zeta(3)\zeta(4). \tag{2.15}$$

The 100-digit evaluation differs from Eq. (2.15) by 3.1×10^{-86}, and a search with the rational part left free recovers 88. In the review of the compendium (Sec. 2.8) the relation stayed the only one when 26 further constants were added to the basis. A looser earlier search had produced a different relation that failed at higher precision. Equation (2.15) is therefore conjectured: the residual is strong numerical evidence, not an interval enclosure or an exact certificate. A proof would assemble every input of C_8^{SR} as an exact expression in one regularized-period convention (which requires recovering or regenerating the six inherited inputs), subtract Eq. (2.15) and certify the difference at every weight through seven, products included, as was done for C_7^{SR}.

2.6 Independent re-derivation of the inputs

The compendium's raw integral expressions had been replayed only with its own code. Every input of C_5^{SR} to C_8^{SR} was therefore recomputed from scratch.

- *Definitions.* The 35 inputs were written as Taylor coefficients of the auxiliary beta functions from the compendium's text alone (the definitions now in Appendices A.9 and C.3).
- *Renormalization map.* The response-to-beta map, the vacuum normalization, the fixed point and the invariants were derived again from first principles (symbolically); the map agrees with the compendium's (Eq. (A.47)). All required raw coefficients must be supplied; a missing one is an error, never a zero.
- *Engine.* A new numerical engine computes the ε expansion of the hard-core moments: a Mellin transform in the cutoff, Hepp sectors (total orderings of the gaps) with the divergent variables subtracted analytically, and rank-one lattice quasi-Monte Carlo with six independent random shifts per job.
- *Direct check.* The raw moments were also integrated directly at finite cutoffs (nested Gauss–Legendre, relative accuracy about 10^{-14}) and compared with the engine's series.
- *Independence.* No integration code or raw data of the compendium was used. Its numbers were read by a script only in a comparison step that refused to run before the independent results had been written and hashed. Some compendium numbers were visible earlier in task descriptions and reports; the result records disclose each case, and none entered a computation.

Three error estimators were computed: linear propagation with the replicate covariance, the standard error of six replicate assemblies (one per random shift), and a sector-pooled estimate from the scatter of the shifts within each integration sector, summed over sectors. The protocol, fixed before the comparison, quotes the largest of the three; all three are statistical estimates, not bounds.

Through the seventh order (production about 3.3×10^3 core-hours on the symmetry cluster). All 25 inputs agree with the compendium: the rms pull is 0.34, and the largest, 0.915, is that of

the input the compendium calls Ω (the coefficient of $u^3 z$ in F, Eq. (C.18)). The coefficients agree within 0.78 standard errors:

	independent	closed form	pull
C_5^{SR}	$29.237811563 \pm 1.35 \times 10^{-7}$	$29.2378115809900\ldots$	-0.13
C_6^{SR}	$-58.66569861 \pm 1.45 \times 10^{-6}$	$-58.6656974742253\ldots$	-0.78
D_4	$-2.53463048 \pm 7.85 \times 10^{-6}$	$-2.53462979953825\ldots$	-0.09
C_7^{SR}	$51.0689860 \pm 3.14 \times 10^{-5}$	$51.0689886129281\ldots$	-0.08

The errors are one standard error. Five checks use no compendium number. They are the proved low-order closed forms (17 quantities, among them C_3^{SR} and C_4^{SR} with errors 3.3×10^{-14} and 9.8×10^{-10}; largest pull 0.52), the cancellation of 702 negative powers of ε in the normalized responses (largest pull 2.37), 39 higher-logarithm predictions (largest pull 0.96), a renormalization-group second-logarithm test derived for this work (six responses, largest pull 1.09), and the direct finite-cutoff comparison. In the last, a moment passes when its two leading coefficients agree with the direct fit, that is, no pull reaches 3 and their joint two-degree-of-freedom χ^2 stays below 11.83; 44 of the 47 moments pass. The other three are flagged: two depend on the fit window, and one comes from a single production coefficient that is 4 replicate standard errors low and changes C_7^{SR} by 7×10^{-10}. C_7^{SR} and D_4 are effectively one test, dominated by the eight-kink current moment.

The eighth order (about 1.9×10^4 core-hours, on the symmetry cluster and a personal desktop). All ten new inputs and all 201 raw scalars are the project's own. Table 3 compares them with the compendium.

input	coefficient	compendium	independent	pull
W	4	-37.70811151495	$-37.70811151495 \pm 1.1 \times 10^{-10}$	-0.02
S	2	878.1837337477	$878.1837337543 \pm 3.1 \times 10^{-9}$	$+2.13$
U	$\frac{3}{4}$	-1296.687960	-1296.687551 ± 0.0015	$+0.27$
J_9	$\frac{1}{4}$	-273.72452	-291.94 ± 9.48	-1.92
L_3	$\frac{3}{32}$	-0.6386928106	$-0.6386928106 \pm 9 \times 10^{-11}$	-0.02
M_{21}	$-\frac{1}{4}$	-171.2355765656	$-171.2355765656 \pm 1.2 \times 10^{-9}$	-0.01
M_{12}	$\frac{1}{2}$	-422.7150579699	$-422.7150579699 \pm 6.1 \times 10^{-9}$	0.00
N_2	$-\frac{1}{16}$	189.742012494	$189.742009475 \pm 8.5 \times 10^{-6}$	-0.35
N_δ	$\frac{1}{8}$	6171.134430509	$6171.134427764 \pm 9.1 \times 10^{-6}$	-0.30
K_v	$\frac{1}{32}$	-1486.962109	-1487.004626 ± 0.158	-0.27
$C_8^{\mathrm{SR}} - \frac{1}{4} J_9$		315.812334	315.811301 ± 0.0051	-0.20
C_8^{SR}		247.381204805	242.825 ± 2.371	-1.92

Table 3: The new inputs of C_8^{SR}, with their coefficients in Eq. (2.10): compendium values and the independent re-derivation (one standard error, the largest of three estimators). The $+2.13$ of S is 7.5×10^{-12} relative and moves C_8^{SR} by 1.3×10^{-8}.

The result is

$$C_8^{\mathrm{SR}} = 242.83 \pm 2.37 \qquad \text{(independent; one standard error)}, \tag{2.16}$$

1.92 standard errors below the record with the quoted error (rounded, 242.8 ± 2.4; for the other estimators see the next paragraph). Everything except the J_9 term is tested sharply: $C_8^{\mathrm{SR}} - \frac{1}{4} J_9$ agrees at -0.20σ, which checks B_{old}, the seven inputs of the re-derivation's first tier (the six inherited ones and N_δ), U and K_v together to 2×10^{-5} of C_8^{SR}. The whole difference sits in J_9, and within it in one raw coefficient, $A_9[\varepsilon^0 \ell^1]$ of the nine-kink odd moment: independent minus compendium is -20.4 ± 9.3 (-2.19σ, 3.3×10^{-5} relative), while the other four pieces of J_9 agree.

How large the tension is. The size of the pull depends on the error estimator, because $\sigma(C_8^{\mathrm{SR}})$ comes almost entirely (93%) from $A_9[\varepsilon^0\ell^1]$. With the same central value 242.825:

estimator	$\sigma(C_8^{\mathrm{SR}})$	pull
sector-pooled (quoted, the largest of the three)	2.37	−1.92
per job the larger of the replicate and pooled projections	2.42	−1.89
chunk-pooled (the post-assembly statistics check)		−2.09
linear propagation	1.37	−3.32
replicate spread (five degrees of freedom)	1.15	−3.98

The independent value therefore lies 1.9 to 4.0 standard errors below the record, depending on the estimator. Through the seventh order the same rule, the largest of the three, gives an rms pull of 0.34 over the 25 inputs, so there it is conservative. For A_9 the replicate error is 0.52 of the sector-pooled one, and an archive-free comparison of production with an independent probe on a subsample of its sectors differs by +2.16 standard errors, so $\sigma(A_9)$, and with it $\sigma(C_8^{\mathrm{SR}})$, may be underestimated. Which value is right is indicated by the third calculation of Sec. 2.7, which agrees with the record. A second independent evaluation of $A_9[\varepsilon^0\ell^1]$ with fresh random shifts (about 9×10^3 core-hours) would show whether the low value is a fluctuation or a systematic error. A 1% measurement of C_8^{SR} cannot test the eleven rational coefficients of Eq. (2.15) in any case; that would need A_9 to about 10^{-9} relative, beyond the present method.

2.7 The eighth order from the integrable point

A third calculation of C_8^{SR} uses neither the matched two-coupling prescription nor any input of the two calculations above. At $s = 1$ the wall gas is the Coulomb gas of the anisotropic Kondo model [5, 6], equivalently of the ohmic spin–boson model [7], which is solvable by Bethe ansatz [8, 9]. The calculation (Appendix I) has three parts.

- *Scheme.* The gas, without a hard core, is put in a uniform magnetic field, and its couplings $\hat{u}$ and $\hat{z}$ are defined from the magnetization (the magnetization scheme). The flow of $\hat{z}$ is then exact, $\hat{H} = 4\hat{u} - \bar{\delta}$, and only one function $\hat{F}$ has to be computed; T and D follow as in Sec. 2.2, with a factor $1 - \bar{\delta}/2$ that converts the field scale to a length scale.
- *The exact part at $s = 1$.* The whole $\bar{\delta} = 0$ part of $\hat{F}$, to all orders in the fugacity, follows from the magnetization at $s = 1$, derived from the Bethe-ansatz equations by a Wiener–Hopf solution as a Mellin–Barnes integral (Eq. (I.4)). It contains the counterpart of J_9 exactly, so no nine-kink integral is needed.
- *Kink clusters.* The $\bar{\delta}$-dependent coefficients come from connected clusters of two to eight kinks with logarithm insertions, continued analytically in the charge v to $v = 1$ by sector decomposition. Their poles at $v = 1$ must cancel in $\hat{F}$. These cancellations, the exact clusters at $s = 1$ and the closed forms of C_3^{SR} to C_7^{SR} are the checks.

The same assembly reproduces the closed forms of C_3^{SR} to C_7^{SR} to at most 7×10^{-26} relative. C_8^{SR} needs two new coefficients, a six-kink one, $f_{202} = 172415.83018563 \pm 1.7 \times 10^{-7}$, and an eight-kink one, $f_{301} = -514275.75 \pm 0.17$, which enter as $2f_{202} + \frac{3}{4}f_{301}$. The result is

$$C_8^{\mathrm{SR}} = 247.38118 \pm 0.13 \qquad \text{(magnetization scheme; empirical numerical uncertainty).} \tag{2.17}$$

The central value is given to five decimals because it is compared with the record below. The uncertainty is $\frac{3}{4}$ of that of f_{301}, a linear sum of the differences between three complete runs (two quadrature grids and two contour radii) and of a roundoff estimate validated by recomputing 56 sectors in double-double arithmetic. It is not a standard deviation or a bound, and it cannot exclude an error common to all three runs. The central value was frozen before any comparison; two criteria of the checks were amended after their data had been seen (Appendix I.7). The code

and records of this calculation are the package `code/integ_d1/`, whose replay rebuilds every record from the shipped engine outputs (Appendix H).

Comparison. Equation (2.17) agrees with the record (2.14) to 2.5×10^{-5} (10^{-7} relative, 0.0002 of its uncertainty) and lies 4.56 above the independent value, 1.9 of that value's standard error. It shares no input with either hard-core calculation and never uses A_9. Since $C_8^{\mathrm{SR}} - \frac{1}{4}J_9$ of the independent re-derivation agrees with the compendium (Sec. 2.6), the low independent value lies in its J_9, and within J_9 in $A_9[\varepsilon^0\ell^1]$, the one piece that deviates. Whether that deviation is a fluctuation with an underestimated error or a systematic error is not known. At 5×10^{-4} relative, Eq. (2.17) supports the leading digits of the record but can neither identify nor certify the rational coefficients of Eq. (2.15).

What the integrable point tests directly. Of the $s = 1$ coefficients of the flow up to the order that C_8^{SR} needs, only two combinations are invariant under changes of coupling, and both agree with the compendium; one of them, $b + \frac{f}{4} - \frac{a}{2} - \frac{c}{8} - A$, vanishes identically with the closed forms of Sec. 2.3 (Appendix I.3). All the others, J_9 among them, are coordinate choices of the hard-core scheme, so the $s = 1$ solution alone cannot test A_9.

2.8 Review of the compendium

The compendium was reviewed on 2 October 2026 by a multi-agent workflow (ten review dimensions and adversarial verification). It found no error in any coefficient or proof claim: the conversions and invariants of C_3^{SR} to C_7^{SR} were recomputed from their stated formulas, the C_5^{SR} to C_7^{SR} certificates were found sound (mutation tests: altered rows are rejected), and every task of the archive's reproduction script passed on Linux, including a fresh evaluation of the exact P_6 expression that reproduces it byte for byte. These tasks replay the archived expressions, certificates and records; none regenerates the raw kink integrals, which is impossible because 21 ordered-integration stage programs are missing from the archive (Appendix H.5). The other defects concern the archive as well: two files whose names differ only by case (they collide on case-insensitive file systems), a reconstruction step that does not recreate two historical paths, and bibliographic errors. None affects a quoted number.

2.9 Caveats

- *Two-coupling closure.* The expansion assumes that the renormalization of the dual involves only the couplings g and b, and that the prescription of Sec. 2.1 extracts their β functions. Two unrefereed arguments, given in full in Appendix D, support both formally, coefficientwise in g, h and $\bar{\delta}$ and so at every order. The first (Appendix D.2) argues that, for a symmetry-preserving local subtraction, every bulk ultraviolet counterterm lies in $\mathrm{span}\{\mathbf{1}, \mathcal{O}_g, \mathcal{O}_h\}$, so no third bulk coupling is needed (h and b are two coordinates on one direction). A literal two-operator truncation would nevertheless be wrong: finite irrelevant terms, such as $:\chi_B^2:$ from $\mathcal{O}_g\mathcal{O}_g$, feed the marginal β functions, and the full multi-kink integrals of the archive include them (Appendices D.6 and D.9). The second (Appendix D.3) argues that the prescription extracts the local β functions. A divergence at a fixed endpoint of the interval can only multiply the identity, so it cancels when the responses are divided by the vacuum moments (Appendix D.10), while the finite endpoint-source terms remain and are those of the archive's recurrence (Eq. (A.47), Appendix D.13); the two-response system then has the local β functions as its solution, up to a finite change of (g, b), which leaves T and D, and so the series, unchanged. Neither argument checks the historical high-order integrations, which are tested only numerically, by the independent re-derivation and by the magnetization scheme of Sec. 2.7 (the certificates concern the later reduction to zeta values), nor the identification

of the dual with the chain (Appendix D.14). The magnetization scheme, with a different regulator and different responses but the same two-coupling description, reproduces C_3^{SR} to C_7^{SR} to 7×10^{-26} and the record of C_8^{SR} within its uncertainty: numerical support for the prescription, not a test of the closure. Independently, the NRG (next item) confirms C_3^{SR} to C_5^{SR} without using the dual.

- *The NRG test in numbers.* The numerical renormalization group (NRG) of the accompanying paper [1, 2] does not use the dual. At $\Delta_\sigma = 0.01$, the residual of its $\Delta_{\mathcal{E}}$ after each added order of the series is 4.7×10^{-3}, 9.8×10^{-4}, 2.4×10^{-4} and 5.3×10^{-5} through C_2^{SR} to C_5^{SR}; the NRG error there is 4.7×10^{-5} (a linear sum of chain-depth, truncation and discretization parts, not a standard deviation), so the last residual is 1.1 times that error. The test is of the partial sums, and with them of C_3^{SR} to C_5^{SR}; the term of C_6^{SR} (5.9×10^{-5}) is of the size of the NRG error, and the higher terms are below it.
- *Level crossing.* $\Delta_{\mathcal{E}}$ is identified with Δ_-, assuming that no other level crosses it.
- C_8^{SR} *is not proved.* Its record is printed to about 85 digits and cross-checked to about 57 decimal places (Sec. 2.5), and the zeta-value form is conjectured. The magnetization-scheme value agrees with the record to 2.5×10^{-5}, within its empirical uncertainty of 0.13 (Sec. 2.7); the independent value lies 1.9 to 4.0 standard errors below, with the difference traced to one input integral.
- *Numerical errors are not bounds.* The compendium's high-precision evaluations are checked by repeated runs with different settings (Appendix C.7), not by interval arithmetic; the independent errors are statistical, and the magnetization-scheme uncertainty is an empirical numerical estimate.
- *Side results.* Appendices C.8 to C.11 also record general-formula tests and results derived from published data of the dual (the shadow relation, the logarithmic endpoint). They hold under the regularity assumptions stated there, are not quoted by the paper and are not results of Table 2.

3 Two dimensions: c_2

3.1 The crossover theory

In $d = 2$ the long-range fixed point meets the short-range Ising CFT at $s_* = 2 - 2\Delta_\sigma^{\mathrm{SR}} = 7/4$. Near s_*, Behan, Rastelli, Rychkov and Zan (BRRZ) describe it as the Ising CFT coupled to a generalized free scalar χ of dimension $\Delta_\chi = 15/8 - \delta$ through $g \int \sigma\chi$ [11, 12]; the relevant thermal coupling is tuned to the critical surface. With unit-normalized fields, $\Delta_\sigma^{\mathrm{SR}} = 1/8$, $\Delta_{\mathcal{E}}^{\mathrm{SR}} = 1$, $c = 1/2$, $C_{\sigma\sigma\epsilon} = 1/2$ and $C_{\epsilon\epsilon\epsilon} = 0$. The exact Ising correlators $\langle \sigma_1 \cdots \sigma_{2n} \rangle$ and $\langle \epsilon\epsilon\sigma_1 \cdots \sigma_{2n} \rangle$ needed here follow from a neutral-charge (bosonization) formula [13]. Appendix E derives every step of this section in full: the conventions in Appendix E.1 and the correlators in Appendix E.3.

First order. With AC the analytic continuation in the exponents (defined, with the circular finite part FP used below, in Appendix E.1), the first-order energy shift is proportional to

$$\begin{aligned} I_2(\delta) &= \mathrm{AC} \int_{\mathbb{C}} |1-z|^{-4+2\delta} \Big(\frac{|1+z|^2}{4|z|} - 1 \Big) \, \mathrm{d}^2 z \\ &= \frac{\pi\delta^2}{4(1-\delta)^2} \frac{\Gamma(\delta)\Gamma(\frac{1}{2}-\delta)}{\Gamma(1-\delta)\Gamma(\frac{1}{2}+\delta)} = \frac{\pi}{4} \big[\delta + (2 + 4\log 2)\delta^2 + \cdots \big], \end{aligned} \tag{3.1}$$

which vanishes at $\delta = 0$ (Appendix E.4). This reproduces the known cancellation $c_1 = 0$ [12], and the expansion starts at $c_2\delta^2$.

3.2 Renormalization-group reduction

In pole-only analytic minimal subtraction (Appendix E.2), $\beta(g) = -\delta g + b_0 g^3 + O(g^5)$ and $\gamma_{\mathcal{E}} = h_{\rm MS} g^4 + O(g^6)$, so $g_*^2 = \delta/b_0$ and

$$c_2 = \frac{h_{\rm MS}}{b_0^2} = -\frac{4 z_{4,1}}{b_0^2}, \tag{3.2}$$

where $z_{4,1}$ is the coefficient of g^4/δ in $\log Z_{\mathcal{E}}$. The one-loop coefficient is exact:

$$b_0 = \frac{9\pi^2}{8}\Big[\frac{\Gamma(3/4)}{\Gamma(1/4)}\Big]^2 = 1.26840430893914254379\ldots, \tag{3.3}$$

obtained from the four-spin correlator by a quadratic change of variables and the complex beta integral (Appendix E.4); it agrees with the BRRZ value 1.268404. The split between a g^2 and a g^4 term is prescription dependent: if $\gamma_{\mathcal{E}} = a_1\delta g^2 + h g^4$, then $c_2 = a_1/b_0 + h/b_0^2$ is invariant under $\mathcal{E}_R \to e^{\kappa g^2}\mathcal{E}_R$, which shifts a_1 by 2κ and h by $-2\kappa b_0$. Minimal subtraction has $a_1 = 0$ and $h = h_{\rm MS} = c_2 b_0^2 \approx -6.54$.

3.3 Localized source, elliptic cover and the finite contact

The fourth-order logarithm is extracted with the coupling smeared by a localized source $f(w) = e^{-|w|^2}$ about the origin and with the two energy operators at 0 and ∞. Write S_4 for the four-spin correlator $\langle\sigma\sigma\sigma\sigma\rangle$ and H_4 for the same correlator with the two energies inserted, and S_{ij}, H_{ij} for the corresponding two-spin functions of spins i and j. The fourth-order term of the logarithm of the energy two-point function has the connected kernel $\mathcal{F} = H_4 - S_4 - H_{12}H_{34} + S_{12}S_{34}$ for the GFF pairing (12)(34), and the three equal pairings give $3/4! = 1/8$; its integral over the four spin positions is $A_4(\delta)$. The local coupling counterterm $b_0 g^3 f^3/(2\delta)$ contributes at the same order through the second-order integral $A_{a,b}(\delta)$, the two-spin integral with Gaussian weights $e^{-a|w_1|^2 - b|w_2|^2}$: at fourth order it gives $b_0 A_{3,1}(\delta)/\delta$ (Appendix E.5), so that

$$z_{4,1} = \operatorname*{Res}_{\delta=0} A_4 + b_0 B_{31}, \qquad B_{31} = A_{3,1}(0) = \frac{\pi^2}{96}(12 - \pi\sqrt{3} - 9\log 3) = -0.34223970692408023655\ldots. \tag{3.4}$$

A Möbius map returns the energies to 0 and ∞ (Appendix E.6); the integral over the overall scale gives $A_4(\delta) = \frac{\pi}{8}\Gamma(2\delta)J(\delta)$, where $J(\delta)$ is the remaining integral over the four-spin cross ratio z and the two energy positions, weighted by $W^{2\delta}$, with W the Gaussian source factor carried through the map. The energy positions are uniformized by the elliptic curve E_z, the double cover of the t-plane branched at 0, z, 1 and ∞, on which the Ising kernel becomes an algebraic expression in Weierstrass functions, and the torus integration defines the elliptic finite part

$$\mathcal{I} = \mathrm{FP}\int_{\mathbb{C}} \mathrm{d}^2 z\, |z|^{-15/4}\frac{\mathcal{A}(z)^2}{4}\Big[w_+|c_+ - 2q|^2 + w_-|c_- - 2q|^2 - w_0|c_0 - 2q|^2\Big], \tag{3.5}$$

with the torus area $\mathcal{A} = 16\,\mathrm{Re}[\overline{K(z)}K(1-z)]$, the circular mean $q = E(z)/(4K(z)) - (2-z)/12 - \pi\overline{K(z)}/(K(z)\mathcal{A})$, the Ising block weights $w_\pm = |1 \pm \sqrt{1-z}|/(2|z|^{1/4}|1-z|^{1/4})$, $w_0 = |z|^{-1/4}$, and $c_\pm = (z - 2 \pm 3\sqrt{1-z})/6$, $c_0 = (2-z)/12$. In this notation the four-spin function at cross ratio z is $S_4(z) = w_+ + w_-$.

The flat torus finite part is not yet the limit of the source-weighted integral $J(\delta)$. When both energies approach the same branch point, the first derivative of the source factor $W^{2\delta}$ survives against a vanishing zeroth-order angular residue and leaves a finite contact $A_{\rm br}$ (Appendix E.7). It is related exactly to the lower Gaussian integral,

$$A_{\rm br} = 4B_{31} - \frac{\pi^2}{4}, \tag{3.6}$$

and at fixed modulus z the regulated energy integral equals the torus integrand plus $4A_{\mathrm{br}}[S_4(z) - |z|^{-1/4}]$. The energy-primary term of the shrinking spin pair cancels exactly in the connected logarithm, and no scalar with a logarithmic threshold remains at the modulus boundaries (Appendix E.8).

3.4 Assembly and the matching term

Integrating the fixed-modulus comparison over z, with $\mathrm{AC}\int |z|^{-15/4}[S_4 - |z|^{-1/4}]\,\mathrm{d}^2z = -b_0/\pi$, gives $J(0) = \mathcal{I} - 16b_0B_{31}/\pi + \pi b_0$ (Appendix E.9), so B_{31} cancels and

$$z_{4,1} = \frac{\pi\mathcal{I} + \pi^2 b_0}{16}, \qquad \boxed{c_2 = -\frac{\pi\mathcal{I}}{4b_0^2} - \frac{\pi^2}{4b_0}.} \tag{3.7}$$

The finite matching term $-\pi^2/(4b_0) = -1.94527965797909034563\ldots$ is comparable to the period term $-\pi\mathcal{I}/(4b_0^2) = -2.12099049015343120711\ldots$. The study's own source-contact checks (Appendix E.15) and its review showed that it does not depend on the shape of the source (analytically, and numerically for two other shapes), and that it equals a_1/b_0, where $a_1 = -\pi^2/4$ is the $O(\delta)$ slope of the leading-order formula of BRRZ with the δ-deformed χ propagator; the finite redefinition with $\kappa = -\pi^2/8$ maps minimal subtraction to that scheme. The cylinder calculation of Sec. 3.8 finds the same $a_1 = -\pi^2/4$ independently (Appendix F.2). Without the matching term c_2 would be about -2.12, which the Monte Carlo of Sec. 3.7 excludes by about 100 standard errors.

3.5 Closed form of the elliptic finite part

The integral $\mathcal{I}$ is reduced by a Landen substitution and a single-valued period pairing to periods of $K(t)^2$, $K(t)K(1-t)$ and $K(1-t)^2$ over $0 < t < 1$, then by contour reflection to symmetric elliptic moments, and finally, through summand-level rational telescoping certificates with their nonzero endpoint terms, to three convergent hypergeometric values (Appendices E.10 to E.14). With $R_\Gamma = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$, so that $b_0 = 9\pi^2/(8R_\Gamma^2)$, and

$$\mathsf{P} = {}_4F_3\left(\begin{matrix}-\frac{1}{2}, -\frac{1}{2}, \frac{3}{2}, \frac{3}{4} \\ 1, 2, \frac{1}{4}\end{matrix}; 1\right), \quad \mathsf{Q} = {}_4F_3\left(\begin{matrix}\frac{1}{2}, \frac{3}{2}, \frac{3}{2}, \frac{9}{4} \\ 3, 3, \frac{7}{4}\end{matrix}; 1\right), \quad \mathsf{T} = {}_4F_3\left(\begin{matrix}\frac{3}{2}, -\frac{1}{4}, \frac{3}{4}, \frac{3}{4} \\ \frac{1}{4}, \frac{9}{4}, \frac{9}{4}\end{matrix}; 1\right), \tag{3.8}$$

the coefficient is

$$c_2 = \frac{16}{R_\Gamma^2} - \frac{R_\Gamma^2}{9} - \frac{\pi^2 R_\Gamma^2}{16}\Big(\mathsf{P} - \frac{5R_\Gamma^2}{384}\mathsf{Q}\Big)\Big(\mathsf{P} - \frac{5R_\Gamma^2}{128}\mathsf{Q} - \frac{512}{25\pi R_\Gamma^2}\mathsf{T}\Big), \tag{3.9}$$

which evaluates to

$$c_2 = -4.06627014813252155274002760857287648752990114118 66\ldots. \tag{3.10}$$

Each series has $\sum b_j - \sum a_i = 2$, so its terms decay as n^{-3}, it converges absolutely and its unaccelerated remainder is $O(N^{-2})$. The three constants need no analytic continuation, and the expression contains no unevaluated integral, parameter derivative or unresolved subtraction constant. No reduction to Gamma values alone has been established (an integer-relation search found none). The constants are

R_Γ	2.95867511918863889231	b_0	1.26840430893914254379
P	1.62562945952539344424	B_{31}	−0.34223970692408023655
Q	1.28812240964200676062	$\mathcal{I}$	4.34474465231888563075
T	0.76385352476101335168	$z_{4,1}$	1.63550416445718870488

Checks of the evaluation (Appendix E.15). The six-function and three-function expressions and a half-interval Frobenius series of the moments agree to 2.53×10^{-80} (the 80-digit arithmetic floor). An endpoint-subtracted quadrature with Carlson's integrals, which uses no Frobenius coefficients, agrees to 1.40×10^{-61}, a difference set by the finite upper limit of its last integral (doubling that limit lowers it to 7.4×10^{-77}). Exact tail inequalities for the three series give $-4.066270148133018 < c_2 < -4.066270148131205$ (floating-point propagation, not a certified rounding). The base study of the cylinder calculation (Sec. 3.8) contains an integer-interval evaluation that certifies the value of the closed expression to 51 decimal places (Appendix F.14; its code is not in the release, Appendix H.5). In the review, $\mathcal{I}$ was also evaluated directly from its defining integral, without the reduction, to 51 digits in one implementation and to 5×10^{-15} by z-plane integration in another, and every intermediate identity of the chain was checked at 30 to 80 digits.

3.6 The step that is not proved

Equation (3.7) needs one step that is not proved: exchanging the limit $\delta \to 0$ with the nested subtracted integrals of the global comparison. In each collision sector the comparison subtracts the local OPE and Taylor terms, restores powers by their Mellin continuation and spin terms by their circular finite parts, and then takes the regulator limit of the subtracted remainder. That exchange needs an integrable bound on the remainder, uniform near $\delta = 0$, including the degenerating shapes where sectors intersect. The study classifies the channels and gives a general subtraction argument (Appendix E.8), but this step is incomplete in two respects: not every sectorwise subtraction is written out, and the uniform bound is not proved. We count both as the one unproved step of the abstract and Table 2. Its weakest point is where a same-branch contact meets the corner $z \to 0$.

The review of the study (summarized in `docs/c2_closed_d2/REVIEWS.md`) looked for each mechanism by which an $O(1)$ term could arise there, and found none. There is no scalar with a logarithmic threshold at dimension two (the energy primary cancels), no dependence of the leading regulator on $\arg z$ at the corners, and no term above threshold from subleading regulator terms combined with spin-one terms. Two numerical tests bear on the step: the Monte Carlo of Sec. 3.7 integrates the full regulated integral, and the cylinder calculation of Sec. 3.8 uses a different regularization. The identification of Eq. (3.9) with the physical coefficient is therefore a formal perturbative result in the critically tuned analytic minimal-subtraction prescription, conditional on this step; the evaluation of $\mathcal{I}$ itself is proved.

3.7 Independent Monte Carlo of the regulated integral

The review integrated the complete regulated eight-dimensional integral over the four insertion positions by Monte Carlo, with the exact Ising correlators built by bosonization and no Möbius map, elliptic cover, contact derivation or sector subtraction. Because it integrates the whole regulated integral, it tests the global exchange step numerically. Three configurations with different channel weights were run, two on the symmetry cluster and one on a personal desktop, with 4.26×10^{10} samples in all (142 chunks, about 135 core-hours); all three enter the result. Their inverse-variance combination gives

quantity	Monte Carlo	closed-form route
$J(0)$	10.559 ± 0.039	10.5404
$z_{4,1}$	1.6391 ± 0.0076	1.635504
c_2	-4.075 ± 0.019	-4.066270

The errors are statistical (one standard error). The recombination gives $c_2 = -4.0752 \pm 0.0189$, quoted as -4.075 ± 0.019; it differs from the closed form by -0.009, 0.5 standard errors. The two configurations run on the symmetry cluster alone give -4.0724 ± 0.0197. The three configurations

agree on the integral itself ($\chi^2 = 1.29$ for two degrees of freedom). A source-shape control, the difference between a quartic and a Gaussian source, reproduces its prediction when the three are combined (16.185 ± 0.053 against 16.150), but the three configurations disagree among themselves on it ($\chi^2 = 7.6$ for two degrees of freedom). The control does not enter c_2; the disagreement is unexplained and left open. The same code reproduces b_0, the vanishing first-order shift, B_{31}, and $z_{2,1} = -b_0/2$, which gives $\Delta_\sigma = \frac{1}{8} + \delta$.

3.8 Fourth-order perturbation theory on the cylinder

Setup. A separate calculation (its method in full in Appendix F) does Rayleigh–Schrödinger perturbation theory for the Ising CFT times the GFF on the radial cylinder, with $H_0 = D_{\text{Ising}} + D_\chi$ and $V = \int_0^{2\pi} \mathrm{d}\theta\, \sigma\chi$. It shares with the closed-form route only the crossover description, the exact Ising data and b_0; it uses none of $\mathcal{I}$, the hypergeometric form, the branch contact or the proposed value. The observable is $X = 2E_{\mathcal{E}} - E_d - E_0$, with $|d\rangle = L_{-1}|\mathcal{E}\rangle$, which equals $\Delta_{\mathcal{E}} - 1$ at a conformal fixed point and removes the direct fourth-order local terms (Appendix F.7). In the scheme defined by the cylinder Hamiltonian with its locally derived counterterms, $H = H_0 + gV + \sum_{k\geq 2} g^k W_k$, write $X = a_1\delta g^2 + hg^4 + \cdots$. With h_0 the value of h once the regulator is removed, the invariant combination of Sec. 3.2 gives

$$c_2 = \frac{a_1}{b_0} + \frac{h_0}{b_0^2}; \tag{3.11}$$

here $h_0 \approx -3.41$, while minimal subtraction has $a_1 = 0$ and $h_{\text{MS}} \approx -6.54$. The second-order coefficient is an exact meromorphic continuation (Appendix F.2), $a(\delta) = \pi^2\Gamma(1 - \frac{\delta}{2})^2\Gamma(\delta - 1)/[\Gamma(\frac{\delta}{2})^2\Gamma(2-\delta)] = -\frac{\pi^2}{4}\delta + O(\delta^2)$, so $a_1 = -\pi^2/4$ and $a_1/b_0 = -1.94527965797909\ldots$. The fourth-order coefficient h is computed at $\delta = 0$ with a family of smooth positive oscillator regulators of order $r = 1, 2, 3$ and length λ, locally derived thermal and spin-two counterterms, and exact chiral conformal blocks as spectral generating functions (Appendices F.6, F.7 and F.8). The spectral sums run over intermediate states labelled by three level indices and are truncated at a cube $0 \leq n, m, \ell < N$, with $N \leq 1024$; they are extrapolated first in N at fixed (r, λ) and then in $\lambda \to 0$. No fit has the physical δ as its variable. As a lower-order check, the renormalized magnetic gap gives $[g^2](E_\sigma - E_0) = 1.268404004672631$ at $r = 3$, $\lambda = 1/32$, which differs from b_0 by -3.04×10^{-7}; it was not used to tune the counterterms (Appendix F.12).

Continuum analysis. The base study, with $N \leq 384$, gave $c_2 \simeq -4.07$ (representative extrapolations -4.0669, -4.0680 and -4.0713 for $r = 1, 2, 3$). The continuum analysis uses 25 fixed-regulator points (r, λ), for each the record with the largest N. Fourteen are new production records with $N = 768$ or 1024, computed on SDSC Expanse and the symmetry cluster ($\lambda = 0.05$ to 0.175). The other eleven are archived records of the base study, computed by GPT-6 astra (Sec. 5): $N = 256$ at $\lambda = 0.125$ to 0.25 for $r = 1$ and 0.175 to 0.35 for $r = 2$, and $N = 384$, 256, 256 and 128 at $\lambda = 0.2$, 0.25, 0.35 and 0.5 for $r = 3$. Their spectral sums are taken as archived and extrapolated under the same rules; their auxiliary sums were recomputed on the symmetry cluster at the common cutoffs of all points (below). Nine of them are among the 23 points of the joint fit. The rules were all fixed in code before the final numbers were computed and are copied into every output:

- *Large-N limit at fixed (r, λ).* The leading large-N correction, a power $N^{-7/4}$ from the faces of the truncation cube with an analytically derived coefficient (Appendix F.18), is subtracted; the remaining tail is fitted with sets of powers and exponentials fixed in advance, on windows that are fixed fractions of N, and the median of the selected fits is $h(\lambda)$.
- *Checks of that limit.* The same extrapolation is repeated on the sums truncated at $N' = N/2, \ldots, N$, and records with different N are compared. Points with λN below a minimum (50, 60 and 75 for $r = 1, 2, 3$) are flagged, since the cutoff must resolve the regulator; the five flagged points, all archived, stay in the fits, and only records above the minimum enter

the cross-N comparison. The spectral coefficients, computed by discrete Cauchy integrals on a grid of points on a torus (Appendix F.8), are extracted again on a second grid with a different radius, for every record, and roundoff is estimated. Three auxiliary truncations (of the mixed thermal sums, of the collapsed sums and of the two-index leading-tail calculation; Appendices F.9 and F.18) are set to common values for all points (16384, 800000 and 8192), and their shifts under a change (to 400000 and 2048 for the last two) are propagated.

- *Regulator limit* (Appendix F.10). Per order, $h(\lambda) = h_0 + k_1\lambda^2 + k_2\lambda^2 \log\lambda + k_3\lambda^3$, the preferred model fixed in advance, on $\lambda \le 0.25$ for $r = 1, 2$ and $\lambda \le 0.35$ for $r = 3$ (the smallest λ is 0.05, 0.0625 and 0.1; the points $(2, 0.35)$ and $(3, 0.5)$ lie outside these windows). Alternative models and windows are admitted by fixed rules, and the regulator-model error is the largest deviation of an admissible fit.
- *Choice of estimator.* A rule fixed in advance chooses between the per-order results and a joint fit. Here the three orders are consistent (each pairwise difference of h_0, at most 1.9×10^{-3}, is below the sum of the two errors), and the joint fit with a shared h_0 is accepted (23 of the 25 fixed-regulator points enter it).

The closed form was never an input, a prior or a selection criterion. The per-order and joint results are

	points	h_0	c_2	error in c_2
$r = 1$	9	-3.414150	-4.06739	0.01006
$r = 2$	7	-3.412774	-4.06653	0.00417
$r = 3$	7	-3.412269	-4.06622	0.00346
joint	23	-3.413609 ± 0.003794	-4.0670498	0.0023580

so that

$$c_2 = -4.0670498 \pm 0.0023580, \qquad \text{quoted } -4.0670(24). \tag{3.12}$$

The error is a linear sum of systematic numerical diagnostics, not a standard deviation and not a rigorous bound. Its components in c_2 are: the choice of regulator model 1.52×10^{-3} (64%); the large-N extrapolation of the points, propagated through the fit, 4.04×10^{-4} (17%); the cutoff and model spread of the mixed thermal sums 2.14×10^{-4} (9%); roundoff 8.7×10^{-5} (4%); coefficient extraction, from the comparison of the two Cauchy grids, 7.5×10^{-5} (3%); the cutoff of the collapsed sums 6.2×10^{-5} (3%); and the cutoff of the leading-tail calculation 2.5×10^{-9}. The difference from the closed form is -7.8×10^{-4}, or $|-4.0670498 + 4.0662701|/4.06627 = 1.9 \times 10^{-4}$ relative, 0.33 of the quoted error.

Caveats, as the paper states them.

- The closed form was known before this analysis, but it enters the analysis in no way.
- The error rule was revised after the first result had been seen. The first complete analysis gave the same central value with $E(c_2) = \pm 2.21$ (absolute, in c_2). Of that, 99.9% was the coefficient-extraction term, bounded by a worst case over sign patterns: the weights $\sum_n |v_n|$ of the large-N fits on the raw sums, up to about 2.4×10^5, times differences between two coefficient grids of 5×10^{-7} to 7×10^{-5}, measured at six of the 25 fit points and transported to the others. The revised rule, written after that result and approved in the Codex review, measures the grid difference directly at all 25 points (20 new extraction-check records), carries its smooth part with its measured sign and keeps only the rough residual at worst case; it gives 7.5×10^{-5} in c_2. Every other component is identical in both analyses (2.28×10^{-3} in all). The quoted error therefore fell from ± 2.21 to ± 0.00236, about 940-fold, entirely through the replaced extraction bound, and the central value is unchanged by construction. The superseded result is archived.

- The two $N = 1024$ coefficient grids were rebuilt on the symmetry cluster, and the last nine extraction-check computations ran there, because SDSC Expanse's file system stopped serving two small files of the grids built there (an author-approved change of the plan). The grids are a deterministic function of their parameters; a pilot at $N = 512$ built on both machines agreed to 6.9×10^{-11} in the raw sums.
- A difference between two Cauchy grids measures extraction sensitivity; it does not bound an error that both share. The regulator extrapolation dominates the error and is not rigorously bounded.

3.9 Summary for $d = 2$

The closed form is $c_2 = -4.06627014813\ldots$, conditional on one unproved step. The Monte Carlo of the regulated integral gives -4.075 ± 0.019, one standard error (0.5% relative), and differs from it by 0.5 standard errors. The cylinder Hamiltonian gives $-4.0670(24)$, where the error is a linear sum of diagnostics with no probability meaning, and differs from it by 1.9×10^{-4} relative, 0.33 of that error. The Monte Carlo integrates the full regulated integral without the exchange step, and the cylinder calculation uses a different regularization; both test the unproved step numerically. No published $O(\delta^2)$ value exists in two dimensions to compare with.

4 Three dimensions: c_1

4.1 The first-order coefficient as a ratio of integrated correlators

In $d = 3$ the same crossover theory applies with $\Delta_\chi = 3 - \Delta_\sigma^{\mathrm{SR}} - \delta$. With $\beta(g) = -\delta g + b_0 g^3$ (b_0 again the one-loop coefficient, now in $d = 3$) and $\gamma_{\mathcal{E}} = a_0 g^2$,

$$c_1 = \frac{a_0}{b_0}, \qquad a_0 = -2\pi I_\epsilon, \quad b_0 = -2\pi I_\sigma, \tag{4.1}$$

where, with one external state at infinity,

$$\begin{aligned} I_O &= \mathrm{AC} \int_{\mathbb{R}^3} \mathrm{d}^3 y \, |e - y|^{-2(3-\Delta_\sigma^{\mathrm{SR}})} \big[H_O(y) - |e - y|^{-2\Delta_\sigma^{\mathrm{SR}}} \big], \\ H_O(y) &= \lim_{R \to \infty} R^{2\Delta_O} \langle O(0)\sigma(y)\sigma(e)O(Re)\rangle, \qquad O \in \{\sigma, \epsilon\}, \end{aligned} \tag{4.2}$$

with e a unit vector and AC the analytic continuation through the power-divergent short-distance terms. The denominator comes from the protected spin dimension, $a_\sigma = b_0$ (Appendices G.2 and G.3). Both integrals are evaluated on a folded region with conformal-block sums of the 3d Ising CFT: 60 primaries from the spectrum of Simmons-Duffin [14] (the stable entries with $\Delta < 24$), with $\Delta_\sigma^{\mathrm{SR}}$, $\Delta_{\mathcal{E}}^{\mathrm{SR}}$ and the stress-tensor couplings from Ref. [15] and the Ward identity, radial and small-z block recurrences, and analytic continuation of the power-divergent terms. Appendix G derives the folding (Appendix G.4), the blocks and the integration (Appendices G.5 and G.15) and lists the retained spectrum (Table 6).

4.2 Retained-spectrum value

With the retained 60 primaries,

$$a_0 = 3.3474594916917857, \qquad b_0 = 12.254914105481044, \qquad c_1 = 0.27315242382601645, \tag{4.3}$$

quoted as 0.2732. The long strings are numerical regression targets, not physical digits: the uncertainty of c_1 is set by the omitted spectrum, through the estimated range of Sec. 4.3 (the three

meanings of precision are separated in Appendix G.1). The numbers agree with BRRZ [11, 12] component by component: $I_\sigma = -b_0/(2\pi) = -1.950430$ against $-1.950(5)$, $b_0 = 12.2549$ against $12.26(3)$, $a_0 = 3.3475$ against $3.3(5)$, and c_1 against their estimate $0.27(4)$; later bootstrap studies found 0.269 [21] and 0.27 [22].

Implementations. The package's full workflow, rerun from scratch, reproduces every shipped number bit for bit in the review's run and on the Intel nodes of the symmetry cluster, and within the replay's tolerances on its AMD nodes (5×10^{-10} for a_0, b_0 and c_1). The review of the package wrote two independent implementations, A and B, each with its own conformal blocks, quadrature and analytic continuation, neither importing the package; they agree on c_1 to 10^{-12}, including all 60 per-family contributions. Both are included in `review_indep/` of the package and rerun by its replay on the retained spectrum: on the symmetry cluster (Intel and AMD nodes) A differs from the package's c_1 by at most 6×10^{-14}, B by at most 5×10^{-13}, and A from B by at most 4×10^{-13}, all absolute (the replay tolerance is 10^{-12}). For a_0 and b_0 the agreement is only at the level of 2×10^{-12} relative, so the claim of 10^{-12} is made for c_1 alone. The package's own two integrated forms (`blocks` and `kernels`) agree to about 1.5×10^{-13} but share the folded domain, so they are a consistency check, not an independent one. The blocks satisfy the Casimir equation and agree with the Zamolodchikov-type recursion of implementation B to 6×10^{-13}; the package's own block checks, against a small-z series and a closed form on the diagonal, are in Appendix G.5.

4.3 Estimated range

The omitted spectrum is controlled only under explicit hypotheses:

- every primary missing from the retained list and exchanged in $\sigma \times \sigma$, $\sigma \times \epsilon$ or $\epsilon \times \epsilon$ has dimension at least 8 if it is $\mathbb{Z}_2$-even and at least 7 if it is $\mathbb{Z}_2$-odd (published spectra motivate these gaps but do not prove them [14, 18, 19, 20]);
- each retained input lies within twice its quoted error (the input box). The errors of the higher families are spreads over spectra at fixed derivative order, not rigorous; of the five low inputs from Ref. [15], $\Delta_\sigma^{\mathrm{SR}}$ and $\Delta_{\mathcal{E}}^{\mathrm{SR}}$ have rigorous errors there, the two OPE coefficients and c do not (Appendix G.6);
- the conditional bootstrap result $\lambda_{\epsilon\epsilon\epsilon'} = 1.5362(12)$ of Ref. [17], with its sparsity assumptions, applies;
- the margins added to the linearized crossing constraints and to the integrated ratio, which cover nonlinear terms and sampled remainders, and the sampled allowances for the denominator ($|\Delta b_{\mathrm{in}}| \le 0.001428$), for the reference bounds (a factor 1.001) and for the quadrature, are large enough.

The hypotheses are discussed in Appendices G.9 (gaps and $\lambda_{\epsilon\epsilon\epsilon'}$), G.8 (the input box), and G.10 and G.14 (the margins and their sampled tests). Under them, an identical-channel positive remainder, a Cauchy–Schwarz bound on the even mixed remainder, a sign-aware integration of the odd remainder and an affine linear program over crossing constraints and inputs (Appendices G.9 to G.12) give the nominal interval [0.2697959, 0.2752135] (ten constraint updates); doubling the crossing margins gives [0.2692738, 0.2754104] (the package's recorded run, eight updates; ten give [0.26927, 0.27541]). The outward-rounded estimated range is

$$c_1 \in [0.2690,\ 0.2760] \qquad \text{(conditional)}, \tag{4.4}$$

to be read next to the retained value 0.2732, about -0.0042 and $+0.0028$ from it. It is neither a certified bound nor a confidence interval. A weaker-gap scenario (6 and 6) gives [0.26400, 0.27842] (recorded run, eight updates; ten give [0.26420, 0.27834]).

4.4 How conditional the range is

The review found, and the revised package now shows, that the range is set by the assumed input box and the gap hypotheses rather than by numerical precision.

- *Feasibility* (Appendix G.10). With all inputs at their central values, the final linear program is infeasible: 68 of its 552 constraints, all in the spin four-point channel, are violated. With the nominal program held fixed, the smallest feasible displacement is 0.574 of the box half-width, that is, 1.15 quoted errors for the input that moves most (it is neither a common displacement of all inputs nor a significance). Underestimated spectrum errors and a missing contribution below dimension 8 are natural explanations, and the chosen numerical relaxation may contribute (Appendix G.10); the data cannot tell which.
- *Box size* (Appendix G.13). Replacing each quoted error e_k by me_k, so that the inputs range over $|u_k - u_k^{(0)}| \le 2me_k$, and rebuilding the program (sensitivities, crossing secants and margins; the $\lambda_{\epsilon\epsilon\epsilon'}$ window held fixed) gives, with ten constraint updates each,

m	≤ 0.6	0.8	1 (nominal)	2
interval	infeasible	$[0.27027, 0.27492]$	$[0.26980, 0.27521]$	$[0.26708, 0.27684]$

  The infeasibility at $m = 0.6$ (± 1.2 quoted errors) does not contradict the 1.15 above: that minimization kept the nominal program fixed, whereas this scan rebuilds it.
- *Gaps* (Appendices G.9 and G.13). Mild relaxations of the gap hypotheses extend the interval outside the estimated range $[0.2690, 0.2760]$, while every interval still contains the retained value:

(even, odd) gaps	$(8, 6.5)$	$(7, 7)$	$(8, 6)$	$(6, 5.5)$
interval	$[0.26890, 0.27541]$	$[0.26891, 0.27629]$	$[0.26720, 0.27576]$	$[0.25839, 0.27948]$

  Lowering the odd gap mainly lowers the lower end, because the odd remainder enters with a definite sign; lowering the even gap, as in $(7, 7)$, widens both ends. Unlisted odd primaries are estimated at about 7.3 to 7.7, only 0.3 to 0.7 above the assumed gap: spin six at about 7.3 [18] and 7.73(5) [20], spin three at 7.53(6) [20].

The package contains its review revision of 8 October 2026, which adds these feasibility and sensitivity tests, the missing derivations (now in Appendix G) and explicit guards, with the numerical values unchanged; its `replay-full` reruns the feasibility diagnostics and the gap, margin and box scenarios. The three-dimensional Monte Carlo of the accompanying paper [1] cannot test c_1: its only accepted point, at $s = 1.6$ ($\delta = 0.18$), is too far from s_* to isolate the linear term.

5 Authorship and provenance

The author of the project chose the problems, approved plans and production runs, and decided between the options the agents presented; the author wrote none of the computations below. They were done as follows.

- C_1^{SR}, C_2^{SR}: published by Benedetti, Lauria, Mazáč and van Vliet.
- C_3^{SR} *to* C_8^{SR}: computed by another AI system, OpenAI GPT-6 astra (via ChatGPT), and supplied by the author as a research compendium with its derivations, code, exact expressions, certificates and numerical records. It was reviewed by Claude Code (Anthropic) on 2 October 2026 (Sec. 2.8). The derivations of Appendices A to C are GPT-6 astra's.

- *Independent re-derivation of the inputs of C_5^{SR} to C_8^{SR}*: Claude Code sessions, 2 to 7 October 2026, with adversarial verification by separate agents for independence, statistics and mathematics; production on the symmetry cluster, part of the eighth-order work on a personal desktop.
- *The two-coupling closure and prescription arguments* (Appendix D): written by OpenAI GPT-6 astra (via ChatGPT) and supplied by the author on 15 October 2026, the second with its checker; not refereed. Claude Code assessed both at the author's request, checked the finite algebra of the first independently, reran the checker of the second and re-derived its beta recurrence independently (Appendix D.15; `docs/closure_d1/REVIEWS.md`).
- *c_2 closed form*: derived by OpenAI GPT-6 astra (via ChatGPT), supplied by the author (the final-method study; its derivation is Appendix E). Reviewed by Claude Code on 8 October 2026; the review wrote and ran the Monte Carlo of Sec. 3.7.
- *Cylinder Hamiltonian for c_2*: the base study was written by OpenAI GPT-6 astra (via ChatGPT) and supplied by the author; its method is Appendix F. All four archives it consolidates are GPT-6 astra's: the final-method study, the independent formula, elliptic and local checks (with the integer-interval certificate of the closed expression), and the two Hamiltonian packages (the solver). Its 17 archived raw records ($N = 128$ to 384) were computed by GPT-6 astra in its own environment; their machine, dates and software versions are not recorded. Eleven of them are points of the continuum analysis (nine of the 23 in its joint fit; Sec. 3.8), with their auxiliary sums recomputed on the symmetry cluster. Claude Code sessions wrote the cluster layer for SDSC Expanse and the symmetry cluster, ran the production and wrote the arbitrary-N continuum analysis. OpenAI Codex reviewed the analysis plan in seven rounds, approving it at revision 7, and inspected the cluster layer and the analysis code twice each. The fixes made after the last inspection of each were not re-inspected (for the analysis, four hardening findings with no numerical change, by the author's choice).
- *c_1 in $d = 3$*: the package was written by OpenAI GPT-6 astra (via ChatGPT), supplied by the author and dated 7 October 2026; its derivation is Appendix G. It was reviewed by Claude Code on 7 October 2026; the two independent re-implementations of the review (`review_indep/`) were written by Claude Code review agents. The package contains its review revision of 8 October 2026 (UTC; 7 October local time; a full rerun with explicit guards, the missing derivations and the sensitivity tests; numbers unchanged), which OpenAI GPT-6 astra made after our review and the author supplied; the four earlier archives it was built from are also GPT-6 astra's. Claude Code checked the revision on 8 October 2026.
- *Magnetization-scheme value of C_8^{SR}* (Sec. 2.7, Appendix I): Claude Code sessions, 14 to 17 October 2026, on the symmetry cluster (about 2.7×10^4 core-hours, including pilots and failed jobs); OpenAI Codex reviewed the plan and inspected the code (Appendix I.7). Its code and records ship as `code/integ_d1/` (commit 8bd5a22), with `public.mk` and `public/` added by the same session on 18 October 2026.
- *This note*: written by a Claude Code session on 15 October 2026 from the sources above, and revised the same day after an independent readability and precision review. On 16 October 2026, at the author's request that each direction of the release have a single note, the derivations of the sources, which had accompanied the packages as six separate notes by GPT-6 astra, were brought into Appendices A to G by Claude Code sessions, edited to the notation and status words of this note and checked against the sources (A to C from the compendium, D from the two notes on the closure and the prescription, E from the study, F from the base study, G from the c_1 package); short explanatory steps were added in the merge and checked, and none changes a result. The separate notes are not part of the release, nor are the scripts that only build them; the few TeX blocks of the compendium that the checks parse are kept verbatim in `code/dual_d1/public/compendium_extract.json` (Appendix H). On

17 October 2026, at the author's request, the magnetization-scheme calculation was added as Sec. 2.7 and Appendix I, from its records and documentation, which are not in the release.

6 Open problems

- A proof of Eq. (2.15): exact expressions for all inputs of C_8^{SR} and a weight-seven certificate.
- The cause of the low independent value of C_8^{SR}, which Sec. 2.7 indicates is the one in error and the re-derivation's own input comparison locates in $A_9[\varepsilon^0\ell^1]$: a rerun with fresh random shifts would show whether it is a fluctuation with an underestimated error or a systematic error.
- Beyond the magnetization scheme of Sec. 2.7, which checks the prescription argument of Appendix D numerically with a second regulator and different responses but assumes the same two-coupling description: a check that does not assume it, for example a higher-order analytic bootstrap (a ring also needs the monodromy sector of the compact boson), and a source-level audit of the historical high-order integrations, whose 21 stage programs are missing from the archive.
- C_9^{SR}, which needs the next determinant coefficient D_5 (a rational-part guess for it, $D_5^{(0)} = \frac{141}{4}$, is untested; Appendix C.8).
- Interval enclosures of the compendium's high-order numerical integrals (Appendix C.7), and a proof of the weight bound that the closed forms through C_7^{SR} obey (Appendix C.8).
- In $d = 2$, the uniform remainder bound that would make Eq. (3.9) a theorem within the prescription.
- In $d = 2$, the disagreement of the three Monte Carlo configurations on the source-shape control ($\chi^2 = 7.6$ for two degrees of freedom), which does not affect c_2.
- In $d = 3$, gap hypotheses with independent support, or input errors with a calibrated meaning, so that the range of c_1 stops depending on chosen margins.

A The dual of the chain: model, regulator and finite-response renormalization

This appendix gives the derivations of the compendium (written by OpenAI GPT-6 astra; Sec. 1.2) behind Secs. 2.1 to 2.3, edited to the notation of this note. It starts from the lattice model and derives the alternating domain-wall gas and its matrix representation (Appendices A.2 and A.3). It then defines the regulated finite-interval responses of the matched two-coupling prescription (Appendices A.4 to A.8) and the finite-response Jacobian and subtraction recurrence that turn them into beta coefficients (Appendix A.9). The last part explains why the invariants T and D need the higher-insertion integrals (Appendix A.10). The beta inputs of the individual orders and their evaluation are in Appendices B and C.

Apart from the published results quoted in Appendix A.6 and the continuum derivation of the wall gas from the lattice model (Appendices A.2 and A.3), everything here either defines the prescription or derives exact relations within it. Two things are not established here: that the dual describes the critical chain, which is the published duality proposal [3, 4], and that the renormalization of the dual involves only the couplings g and b and the prescription extracts their local beta functions, which is the subject of the caveat "Two-coupling closure" in Sec. 2.9 and of Appendix D.

A.1 Notation

As in the main text, $\bar{\delta} = 1 - s$ and $x = \Delta_\sigma = \bar{\delta}/2$; the compendium writes δ for $\bar{\delta}$ (record and input names such as N_δ keep their spelling), C_n for C_n^{SR}, b_{dim} for the resummed charge b and D_{2m} for Ξ_{2m}. The hard core is ε and $\ell = \log\varepsilon$; the thermal operator is $\mathcal{E}$, with dimension $\Delta_\mathcal{E}$. Table 4 collects the symbols of the dual that this appendix and the two following ones use.

Table 4: Symbols of the dual of the chain. The calligraphic $\mathcal{C}_n$ and $\mathcal{K}_n$ are normalized responses (archived programs call them C_n and K_n); they are never the thermal coefficients C_n^{SR}.

symbol	meaning
X, X_i, q_i	coordinate along the chain; ordered kink (wall) positions; wall charges ±1, alternating, $q_i = (-1)^{i+1}$
ε, ℓ, $\Omega_n(\varepsilon)$	dimensionless hard core between consecutive kinks; $\ell = \log\varepsilon$; ordered domain of n kinks, Eq. (A.14)
κ, $f_{\bar{\delta}}$	reference inverse length (set to 1 from Appendix A.4 on); wall interaction function, Eq. (A.5)
ϕ, φ	Gaussian field of the dual; coarse-grained Ising field (different fields)
$\hat{\sigma}_3$, $\hat{\sigma}_\pm$, σ	qubit matrices; spin primary (unit normalized)
$\mathcal{O}_g$, $\mathcal{O}_h$, χ_B, χ	kink operator and current operator of Eq. (2.1); bare current $\frac{\mathrm{i}}{\sqrt{2}}\partial\phi$ (written χ in Eq. (2.1)); renormalized unit-normalized current primary
g, h, b_0, b	auxiliary wall fugacity and current coupling; reference charge; resummed charge $b = b_0 - \sqrt{2}h$
η, ∂_η	endpoint source $\sqrt{2}hb$, Eq. (A.17); derivative with respect to η at $\eta = 0$ (the compendium's script-E operator)
B, v, z; Y, u	essential charge, $v = B^2$, $z = v - 1$; essential wall fugacity (renamed g after matching), $u = g^2$
F, H	essential beta functions divided by $2u$ and by $1 + z$, Eq. (A.23)
T, D; T_m, D_m	trace and determinant of the stability matrix in (u, z); their coefficients of $\bar{\delta}^m$
W_n, Ξ_{2m}	wall weight of n ordered kinks; total length of the reversed domains, Eq. (A.31)
A_{2m+1}, Z_{2m}, Q_{2m}	odd, vacuum and current moments, Eq. (A.31)
$\mathcal{C}_{2m+1}$, $\mathcal{K}_{2m}$	vacuum-normalized odd and current responses, Eq. (A.37)
I_0, I_1	coefficients $[\varepsilon^0\ell^0]I$ and $[\varepsilon^0\ell^1]I$ of a response I; e.g. $\mathcal{C}_{3,1}$, $\mathcal{K}_{4,0}$, $Q_{2,0}$
S_j	charge-weighted sums of $\log^j$ of the kink separations, Eq. (A.41)
j_n, λ_{2m}	entries of the finite-response Jacobian, Eq. (A.42)
b_n (n odd), h_n (n even)	coefficients of the auxiliary fugacity and current beta functions, Eq. (A.43) (archived scripts write η_4, η_6 for h_4, h_6); unrelated to b_0, b, h and η
$a, p, b, c, f, \ldots$	beta inputs: Taylor coefficients of F and H at $v = 1$, $\bar{\delta} = 0$ (Sec. 2.3); the input b is b_5, not the charge

A.2 The classical model and the short-range end

The object of interest is the thermal dimension of the *classical* one-dimensional long-range Ising critical family, not the spectrum of the quantum transverse-field Ising chain. A microscopic representative is the chain of spins $m_i = \pm1$ with reduced Hamiltonian (energy divided by temperature)

$$\mathcal{H}_{\mathrm{red}} = -K_{\mathrm{nn}} \sum_i m_i m_{i+1} - K_{\mathrm{lr}} \sum_{i<j} \frac{m_i m_j}{|i-j|^{1+s}}, \qquad K_{\mathrm{lr}} > 0. \tag{A.1}$$

The nearest-neighbour coupling changes the core energies of the walls and the nonuniversal critical temperature; it need not be present separately. The long-range critical regime of interest is $\frac{1}{2} < s < 1$.

The endpoint $s = 1$ is marginal, and for $s > 1$ there is no finite-temperature transition [3, 4].

The two ends of the long-range window. The familiar coarse-grained action is

$$S[\varphi] = \frac{c_s}{4} \int \mathrm{d}X\, \mathrm{d}Y\, \frac{[\varphi(X) - \varphi(Y)]^2}{|X - Y|^{1+s}} + \int \mathrm{d}X \Big(\frac{r_\varphi}{2} \varphi^2 + \frac{\lambda_\varphi}{4} \varphi^4 \Big), \qquad c_s > 0. \tag{A.2}$$

Its nonanalytic quadratic kernel is proportional to $|p|^s$ in momentum space and gives $\Delta_\varphi = (1-s)/2$, which is Δ_σ. The quartic coupling has engineering dimension $1 - 4\Delta_\varphi = 2s - 1$. The expansion about the generalized free field (GFF) is therefore weak near $s = \frac{1}{2}$, whereas the dual used here is weak near $s = 1$. These are opposite ends of the same critical family, not the same perturbative expansion [4, 23].

What sits at the short-range end in $d = 1$. For the nearest-neighbour chain alone, the transfer matrix is

$$\mathsf{T}_{\mathrm{nn}} = \begin{pmatrix} e^{K_{\mathrm{nn}}} & e^{-K_{\mathrm{nn}}} \\ e^{-K_{\mathrm{nn}}} & e^{K_{\mathrm{nn}}} \end{pmatrix} = \mathcal{N} \exp(\gamma \hat{\sigma}_1), \qquad \tanh \gamma = e^{-2K_{\mathrm{nn}}}, \tag{A.3}$$

with a constant $\mathcal{N} > 0$. Its transfer Hamiltonian is a single two-state system, $\hat{H} = -\gamma \hat{\sigma}_1$ up to an additive constant, and the inverse correlation length is $\xi^{-1} = 2\gamma = \log \coth K_{\mathrm{nn}}$. At zero temperature $\gamma \to 0$, and the symmetric normalized trace gives $\frac{1}{2} \operatorname{tr} \hat{\sigma}_3^2 = 1$: the spin $\hat{\sigma}_3$ then has dimension zero. This endpoint spin sector plays, in $d = 1$, the role of the short-range fixed point ($\Delta_\sigma^{\mathrm{SR}} = 0$, Table 1); it is not an ordinary finite-temperature one-dimensional conformal field theory. In $d = 2, 3$ the crossover theory couples the short-range fixed point to a generalized free field through the spin (Sec. 3). In $d = 1$ the analogous diagonal coupling of the dual, $h\, \mathcal{O}_h = h \hat{\sigma}_3 \chi_B$, commutes with $\hat{\sigma}_3$ and cannot flip the spin, so operators that create kinks are essential. The higher-dimensional crossover construction therefore has a special one-dimensional realization, the dual of Sec. 2.1 [4, 11, 12].

A.3 Alternating domain walls, the Gaussian field and the matrices

Domain walls. Let $m(X) = \pm 1$ be piecewise constant along the chain, with walls at X_i of charge q_i:

$$\partial_X m(X) = 2 \sum_i q_i\, \delta(X - X_i), \qquad q_i = \pm 1, \tag{A.4}$$

where $\delta(X)$ is the Dirac distribution. Ising walls necessarily alternate in sign, and a configuration that returns to its initial spin has an even number of walls and zero total charge. With a reference inverse length κ, define

$$f_{\bar{\delta}}(r) = \frac{r^{\bar{\delta}} - \kappa^{-\bar{\delta}}}{\bar{\delta}}, \qquad f_0(r) = \log(\kappa r), \qquad f_{\bar{\delta}}''(r) = -(1 - \bar{\delta})\, r^{-2+\bar{\delta}} \quad (r > 0). \tag{A.5}$$

From Appendix A.4 on $\kappa = 1$, so that $f_{\bar{\delta}}(r) = (r^{\bar{\delta}} - 1)/\bar{\delta}$ as in Sec. 2.1. Take the continuum tail action, relative to the uniform configuration, to be

$$S_{\mathrm{tail}}[m] = \frac{\kappa_s}{4} \int \mathrm{d}X\, \mathrm{d}Y\, \frac{[m(X) - m(Y)]^2}{|X - Y|^{2-\bar{\delta}}}, \tag{A.6}$$

the long-range term of Eq. (A.1) with $1 + s = 2 - \bar{\delta}$. Since $\partial_X \partial_Y f_{\bar{\delta}}(|X - Y|) = (1 - \bar{\delta})|X - Y|^{-2+\bar{\delta}}$ away from contact points, two integrations by parts give

$$S_{\mathrm{tail}} = -\frac{\kappa_s}{2(1 - \bar{\delta})} \int \mathrm{d}X\, \mathrm{d}Y\, m'(X)\, m'(Y)\, f_{\bar{\delta}}(|X - Y|) + \text{core and boundary terms}. \tag{A.7}$$

Inserting the wall charges, removing the self-energies and absorbing the positive coefficient into $b_{\mathrm{w}}^2 = 2\kappa_s/(1-\bar{\delta})$ gives the wall action

$$S_{\mathrm{walls}} = -2b_{\mathrm{w}}^2 \sum_{i<j} q_i q_j \, f_{\bar{\delta}}(X_j - X_i) + N_{\mathrm{w}} E_{\mathrm{core}} + \text{boundary terms}, \tag{A.8}$$

with N_{w} the number of walls and E_{core} their core energy. As a check of the sign, an isolated pair of opposite charges at separation r has interaction energy $2b_{\mathrm{w}}^2 f_{\bar{\delta}}(r)$, which increases with r and is positive beyond the reference length $1/\kappa$. Short-distance details are encoded by a fugacity and by local core counterterms.

On a finite interval the regulated neutral gas is

$$Z_{\mathrm{gas}} = \sum_{m\geq 0} g^{2m} \int_{\Omega_{2m}(\varepsilon)} \prod_i \mathrm{d}X_i \, \exp\Big[2b_{\mathrm{w}}^2 \sum_{i<j} q_i q_j \, f_{\bar{\delta}}(X_j - X_i)\Big], \tag{A.9}$$

up to the overall two-state trace and reference-length normalizations, with the ordered hard-core domain $\Omega_n(\varepsilon)$ of Eq. (A.14). The positions are ordered, so there is no additional divisor $(2m)!$. The odd projected responses of Appendix A.7 carry a specified compensating external charge; they are not neutral partition functions.

The Gaussian field and its zero mode. The wall weight is reproduced by vertex operators of a Gaussian field. Choose a real Gaussian field modulo constant shifts, with covariance on neutral test functions

$$C_{\bar{\delta}}(X - Y) = -2 f_{\bar{\delta}}(|X - Y|), \tag{A.10}$$

the covariance of Sec. 2.1. With a finite-volume momentum cutoff and the zero mode removed, it is defined by the positive quadratic action

$$S_0[\phi] = \frac{\mathcal{A}_{\bar{\delta}}}{2} \int \frac{\mathrm{d}p}{2\pi} \, |p|^{1+\bar{\delta}} \, |\widetilde{\phi}(p)|^2, \qquad \mathcal{A}_{\bar{\delta}} > 0. \tag{A.11}$$

The constant in $C_{\bar{\delta}}$ is immaterial for neutral observables, and ϕ itself is not a shift-invariant local primary. Its increments have nonnegative variance. The formal scaling dimension of ϕ is $-\bar{\delta}/2$, but this does not license treating its unphysical zero mode as a negative-norm primary.

Compactifying the constant mode with period $2\pi/b_0$ implements charge neutrality for the vertices $V_\pm = \kappa \, {:}e^{\pm \mathrm{i} b_0 \phi}{:}$. For $\sum_i q_i = 0$, Wick's rule gives

$$\Big\langle \prod_i {:}e^{\mathrm{i} b_0 q_i \phi(X_i)}{:} \Big\rangle_0 = \exp\Big[-b_0^2 \sum_{i<j} q_i q_j \, C_{\bar{\delta}}(X_i - X_j)\Big] = \exp\Big[2b_0^2 \sum_{i<j} q_i q_j \, f_{\bar{\delta}}(X_j - X_i)\Big], \tag{A.12}$$

which is the nonlocal wall weight of Eq. (A.8) with $b_0^2 = b_{\mathrm{w}}^2$. At $\bar{\delta} = 0$ the two-point function of opposite charges is proportional to $r^{-2b_0^2}$.

Matrices enforce the Ising constraint. With these vertices the operators of Eq. (2.1) are $\mathcal{O}_g = \hat{\sigma}_+ V_+ + \hat{\sigma}_- V_-$ and $\mathcal{O}_h = \hat{\sigma}_3 \chi_B$, with $\chi_B = \frac{\mathrm{i}}{\sqrt{2}} \partial\phi$ the bare current (the subscript distinguishes it from the renormalized primary χ of Appendix A.6). Their path-ordered evolution from U to V is

$$\mathcal{D}(V, U) = \mathrm{P} \exp \int_U^V \mathrm{d}X \, \big[g \, \mathcal{O}_g(X) + h \, \mathcal{O}_h(X)\big]. \tag{A.13}$$

Because $\hat{\sigma}_+^2 = \hat{\sigma}_-^2 = 0$, only alternating charges survive in its expansion, so the traced matrix expansion matches the regulated Ising gas, whereas an unconstrained cosine interaction would also admit nonalternating walls. Spin correlators are obtained by inserting $\hat{\sigma}_3$ in the ordered trace. Two insertions multiply a wall history by $(-1)^{N_{\mathrm{between}}}$, with N_{between} the number of walls between them, which is exactly the product of the two Ising spins.

The auxiliary model also contains charged objects that are not physical operators of the chain. Constant shifts of ϕ compensated by rotations of $\hat{\sigma}_\pm$ define a $U(1)$ action, and the physical operator algebra is its invariant sector (Sec. 2.1). The equality of consistently regulated wall weights and matrix histories is a direct algebraic statement. The identification of the critical continuum limit with the long-range Ising CFT is the published duality proposal and its tested consequences [4]; this calculation supplies no all-orders theorem on the lattice scaling limit.

A.4 The regulator and exact current resummation

Work on an interval of length one, with $\kappa = 1$. All ordered integrations use

$$\Omega_n(\varepsilon) = \{0 < X_1 < \cdots < X_n < 1 : \ X_{i+1} - X_i \geq \varepsilon\}, \qquad \ell = \log \varepsilon. \tag{A.14}$$

Only consecutive kink separations have a hard core; there is no separate hard core between a wall and an endpoint. Current insertions are resummed *before* this regulator is imposed. An independent hard core on unresolved current insertions would change the finite parts of the responses, and could not be combined with the formulas below.

Change of frame. The resummation is a field-dependent change of frame,

$$R(X) = \exp\big[\mathrm{i}h\,\hat{\sigma}_3\,\phi(X)/\sqrt{2}\big], \qquad \mathcal{D}(V,U) = R(V)\,\widetilde{\mathcal{D}}(V,U)\,R(U)^{-1}, \tag{A.15}$$

which removes the bulk derivative interaction $h\,\mathcal{O}_h$. Conjugation by R multiplies $\hat{\sigma}_\pm$ by $e^{\mp\mathrm{i}\sqrt{2}h\phi}$ and so shifts the bulk wall charge to

$$b = b_0 - \sqrt{2}h, \tag{A.16}$$

the effective charge of Sec. 2.1 (the beta input b of Sec. 2.3 is a different quantity). The endpoint matrices $R(1)$ and $R(0)^{-1}$ remain. Equivalently, one integrates by parts on the interval I, $\int m\,\partial\phi = m\phi|_{\partial I} - \int m'\phi$. The endpoint charges have magnitude $h/\sqrt{2}$, and their contraction with a wall produces the *endpoint source*

$$\eta = \sqrt{2}\,h\,b = \sqrt{2}\,b_0 h - 2h^2. \tag{A.17}$$

The coefficient of h^2 is -2. An intermediate higher-logarithm check in the archive used -1; the beta coefficients below use only first h derivatives at $h = 0$ and are unaffected, but checks that involve second or higher source derivatives are.

Endpoint factors. With alternating charges $q_i = (-1)^{i+1}$, the endpoint factors of n kinks are

$$\begin{aligned} U_n^{\mathrm{odd}}(\eta) &= \exp\Big\{\eta \sum_i q_i \big[f_{\bar{\delta}}(X_i) + f_{\bar{\delta}}(1 - X_i)\big]\Big\} \quad (n \text{ odd}), \\ U_n^{\mathrm{even}}(\eta) &= \exp\Big\{\eta \sum_i q_i \big[f_{\bar{\delta}}(X_i) - f_{\bar{\delta}}(1 - X_i)\big]\Big\} \quad (n \text{ even}). \end{aligned} \tag{A.18}$$

We write ∂_η for the derivative with respect to η at $\eta = 0$ (the compendium writes it as a script E). By the reflection $X \to 1 - X$, a *first* source derivative may be represented after integration by $2\sum_i q_i f_{\bar{\delta}}(X_i)$; this replacement is not valid before squaring, for a second derivative. At the reference charge $v = b_0^2$ the coupling h enters the moments through the squared charge $v = b^2 = (b_0 - \sqrt{2}h)^2$ of the wall weight W_n (Eq. (A.31); it equals the essential B^2 of Appendix A.5 at $\bar{\delta} = 0$) and through η, so that

$$\partial_h\big|_{h=0} = \sqrt{2}\,b_0\,(-2\partial_v + \partial_\eta). \tag{A.19}$$

For example, for a single kink at $\bar{\delta} = 0$ and $b_0 = 1$ the odd endpoint factor is $X^\eta(1 - X)^\eta$, so the one-kink moment is Euler's beta function, $A_1(h) = \Gamma(1 + \eta)^2/\Gamma(2 + 2\eta)$, and

$$A_1(h) = 1 - 2\sqrt{2}\,h + (12 - 2\zeta(2))h^2 + O(h^3). \tag{A.20}$$

A.5 Essential couplings and finite-tail matching

Normal ordering at the cutoff and the charge shift give the essential charge B and the essential wall fugacity Y,

$$B = (b_0 - \sqrt{2}h)\,\varepsilon^{\bar\delta/2}, \qquad Y = g_{\rm aux}\,\kappa\varepsilon\,\exp\Big[\frac{b_0^2(\kappa^{-\bar\delta} - \varepsilon^{\bar\delta})}{\bar\delta}\Big], \tag{A.21}$$

where $g_{\rm aux}$ is the fugacity of the auxiliary (unmatched) theory. The h dependence cancels from Y once both normal-ordering factors are included. At unit reference length and $b_0 = 1$,

$$Y = g_{\rm aux}\exp\Big[\ell + \frac{1 - e^{\bar\delta\ell}}{\bar\delta}\Big] = g_{\rm aux}\Big[1 - \frac{\bar\delta}{2}\ell^2 + \bar\delta^2\Big(\frac{\ell^4}{8} - \frac{\ell^3}{6}\Big) + \cdots\Big]. \tag{A.22}$$

Charge derivatives require keeping b_0 variable until they are taken; setting $b_0 = 1$ prematurely loses terms. With Y renamed g, the essential flow is written as in Sec. 2.1,

$$u = g^2, \qquad v = B^2 = 1 + z, \qquad \beta_u = 2uF, \qquad \beta_z = (1+z)H, \qquad \beta = \mu\frac{\mathrm{d}}{\mathrm{d}\mu} = -\frac{\mathrm{d}}{\mathrm{d}\ell}. \tag{A.23}$$

The finite $\bar\delta$ factors of Eq. (A.22) must be matched together with the endpoint integrals. It is not consistent to compute at $\bar\delta = 0$ in one scheme and append the classical-scaling term in unrelated coordinates. The finite-tail factor is needed, for example, for the input p of Sec. 2.3 (Appendix B.9).

A.6 Leading spectrum and operator dictionary

This subsection collects the published leading-order results [3, 4] in the notation of this appendix. The published field-theory and bootstrap methods agree at their stated orders. They give

$$\Delta_\pm = 1 \pm \sqrt{2\bar\delta} + \frac{\bar\delta}{4} + O(\bar\delta^{3/2}), \qquad \Delta_\sigma = \frac{\bar\delta}{2}, \qquad \Delta_\chi = 1 - \frac{\bar\delta}{2}. \tag{A.24}$$

The coefficients $C_3^{\rm SR}$ and beyond are not quoted from those papers; they belong to the computation described in this and the following appendices and carry its conditions (Sec. 2.1).

OPE and leading flow. At $\bar\delta = 0$ and $b_0 = 1$ the ordered OPE of two kink operators is

$$\mathcal{O}_g(X)\,\mathcal{O}_g(0) = X^{-2}\,\mathbb{1} + \sqrt{2}\,X^{-1}\,\mathcal{O}_h(0) + \cdots, \qquad X > 0, \tag{A.25}$$

and the two ordered mixed OPEs, of $\mathcal{O}_g$ with $\mathcal{O}_h$ in either order, each contain $\sqrt{2}\,\mathcal{O}_g/X$. The beta functions of the dual through the published order are

$$\beta_g = -2\sqrt{2}\,gh + 2gh^2 + g^3 + \cdots, \qquad \beta_h = -\sqrt{2}\,g^2 + 2hg^2 + \cdots. \tag{A.26}$$

In the essential couplings, with $z = b^2 - 1 = -2\sqrt{2}h + 2h^2$ at $b_0 = 1$ and $\bar\delta = 0$, they become $\beta_g = gF$ and $\beta_z = 2b\,\partial_h b\,\beta_h = (1+z)H$ with $F = z + u + \cdots$ and $H = 4u + \cdots$; at $\bar\delta \neq 0$ the classical scaling of $B = b\,\varepsilon^{\bar\delta/2}$ adds $-\bar\delta$ to H, so that $H = -\bar\delta + 4u + \cdots$, whose fixed point $u_* = \bar\delta/4$, $z_* = -\bar\delta/4$ is given in Sec. 2.1. This explains the expansion in $\sqrt{\bar\delta}$: the fugacity itself is of order $\sqrt{\bar\delta}$, although u and z are of order $\bar\delta$. In the coordinates (g, v) the leading stability matrix is

$$M_* = \begin{pmatrix} \bar\delta/2 & g_* \\ 8(1 - \bar\delta/4)\,g_* & 0 \end{pmatrix} + \text{higher-order terms}. \tag{A.27}$$

A linear beta eigenvalue λ corresponds to the dimension $1 + \lambda$; with $g_*^2 = \bar\delta/4$ the eigenvalues of M_* give $\Delta_\pm$ of Eq. (A.24). The relevant branch is the thermal primary $\mathcal{E} = \mathcal{O}_-$, so $\nu^{-1} = 1 - \Delta_\mathcal{E}$, and the other eigenvalue gives the leading correction exponent $\omega = \Delta_+ - 1$. Only the orders shown in Eq. (A.24) are controlled by this truncation. Expanding the small matrix to more orders does not compute the higher coefficients (Appendix A.10).

Spin dimension. A direct wall-gas check gives, for the spin two-point function at separation r, $G_\sigma(r) - 1 = -4g^2 \log(r/\varepsilon) + \cdots$, because a kink pair changes the spin product only when one wall lies between the insertions. Thus $\gamma_\sigma = 2g^2 + \cdots$, which reproduces $\Delta_\sigma = \bar{\delta}/2$ at the fixed point.

Equation of motion and the shadow pair. The covariant derivative of an insertion is $D_X\mathcal{O} = \partial_X\mathcal{O} - [g\,\mathcal{O}_g + h\,\mathcal{O}_h, \mathcal{O}]$. In particular

$$D_X\hat{\sigma}_3 = 2g(\hat{\sigma}_+V_+ - \hat{\sigma}_-V_-), \qquad C_{\bar{\delta}}^{-1}\phi = \frac{\mathrm{i}b}{2}\,D_X\hat{\sigma}_3 \tag{A.28}$$

away from contact terms, where $C_{\bar{\delta}}^{-1}$ is the inverse of the covariance (A.10). The nonanalytic Gaussian kernel and this equation of motion give the protected shadow pair of Eq. (A.24), $\Delta_\sigma + \Delta_\chi = 1$: the antisymmetric flip operator $\hat{\sigma}_+V_+ - \hat{\sigma}_-V_-$ becomes the descendant of σ. The bare current χ_B can mix with σ through power-divergent counterterms and needs a finite normalization; all CFT data below refer to the renormalized, unit-normalized primary χ, not to an unsubtracted current [4].

Published OPE coefficients. For completeness, with $\tau = \sqrt{\bar{\delta}} = \sqrt{2\Delta_\sigma}$, with $\pm$ labelling the operators $\mathcal{O}_\pm$ of dimensions $\Delta_\pm$, and with the published operator signs [4, eqs. (5.115) and (5.120)],

$$\begin{aligned}
c_{\sigma\chi\pm} &= \frac{1}{\sqrt{2}} \mp \frac{\tau}{16} - \frac{\tau^2}{256\sqrt{2}} + O(\tau^3),\\
c_{\sigma\sigma\pm} &= \pm\frac{\tau}{\sqrt{2}} - \frac{15}{16}\tau^2 \pm \frac{1}{\sqrt{2}}\Big(\frac{543}{256} - \frac{\pi^2}{6}\Big)\tau^3 + O(\tau^4),\\
c_{\chi\chi\pm} &= -\frac{\pi^2}{2}\tau^2 \mp \frac{11\pi^2}{16\sqrt{2}}\tau^3 + O(\tau^4),\\
c_{+++} &= \frac{3}{2} + \frac{39}{16\sqrt{2}}\tau + O(\tau^2), \qquad c_{---} = \frac{3}{2} - \frac{39}{16\sqrt{2}}\tau + O(\tau^2),\\
c_{++-} &= -\frac{1}{2} + \frac{\tau}{16\sqrt{2}} + O(\tau^2), \qquad c_{+--} = -\frac{1}{2} - \frac{\tau}{16\sqrt{2}} + O(\tau^2).
\end{aligned} \tag{A.29}$$

These OPE coefficients, including their π^2 terms, are published inputs, not loop results of the compendium. The compendium takes its published inputs from Ref. [4], Secs. 2–3 and App. B, Secs. 4.1–4.4, Eqs. (5.48), (5.53)–(5.54) and (5.115)–(5.120), and App. E.

A.7 Ordered moments and vacuum normalization

Moments. At unit reference length and with $q_i = (-1)^{i+1}$, define the wall weight of n ordered kinks and, for an even number of kinks, the total length of the reversed domains,

$$W_n(v, \bar{\delta}; \boldsymbol{X}) = \exp\Big[2v \sum_{i<j} q_i q_j \,\frac{(X_j - X_i)^{\bar{\delta}} - 1}{\bar{\delta}}\Big], \qquad \Xi_{2m}(\boldsymbol{X}) = \sum_{j=1}^{m} (X_{2j} - X_{2j-1}), \tag{A.30}$$

and the odd, vacuum and current moments

$$\begin{aligned}
A_{2m+1} &= \int_{\Omega_{2m+1}(\varepsilon)} W_{2m+1} \prod_i \mathrm{d}X_i,\\
Z_{2m} = \int_{\Omega_{2m}(\varepsilon)} W_{2m} \prod_i \mathrm{d}X_i, &\qquad Q_{2m} = \int_{\Omega_{2m}(\varepsilon)} \Xi_{2m} W_{2m} \prod_i \mathrm{d}X_i.
\end{aligned} \tag{A.31}$$

When the endpoint source is switched on, the integrands also carry the endpoint factors of Eq. (A.18), on which all ∂_η derivatives act (as in the example of A_1, Eq. (A.20)). The A moments are coefficients of charged projected responses, with their compensating external operator understood; they do not contradict the zero-mode neutrality of Z. At zero endpoint source $A_1 = 1$.

Gap variables and the first kernels. Integrating the first position gives a translation factor $1-R$ in the gap variables $r_i = X_{i+1} - X_i \geq \varepsilon$, with $R = \sum_i r_i \leq 1$. At $v = 1$ and $\bar{\delta} = 0$, $W_n = \prod_{i<j}(X_j - X_i)^{2q_iq_j}$; naming the consecutive gaps r, t, w, y, the first kernels are

$$W_3 = \frac{(r+t)^2}{r^2t^2}, \tag{A.32}$$

$$W_4 = \frac{(r+t)^2(t+w)^2}{r^2t^2w^2(r+t+w)^2}, \tag{A.33}$$

$$W_5 = \frac{(r+t)^2(t+w)^2(w+y)^2(r+t+w+y)^2}{r^2t^2w^2y^2(r+t+w)^2(t+w+y)^2}. \tag{A.34}$$

(The compendium names the second gap t'; it is unrelated to the trace T.) At $v = 1$, $\bar{\delta} = 0$ the two-kink moments are elementary, $Z_2 = \int_\varepsilon^1 (1-r)r^{-2}\,\mathrm{d}r$ and $Q_2 = \int_\varepsilon^1 (1-r)r^{-1}\,\mathrm{d}r$:

$$Z_2 = \varepsilon^{-1} + \ell - 1, \qquad Q_2 = -\ell - 1 + \varepsilon. \tag{A.35}$$

These elementary expressions fix the signs of the finite parts of the responses.

Vacuum normalization before coefficient extraction. The responses are divided by the vacuum before any coefficient is extracted. One forms generating functions in the fugacity and divides the *integrated* numerator by the integrated vacuum denominator,

$$\sum_{m\geq 0} \mathcal{C}_{2m+1}\, g^{2m} = \frac{\sum_{m\geq 0} A_{2m+1} g^{2m}}{\sum_{m\geq 0} Z_{2m} g^{2m}}, \qquad \sum_{m\geq 1} \mathcal{K}_{2m}\, g^{2m} = \frac{\sum_{m\geq 1} Q_{2m} g^{2m}}{\sum_{m\geq 0} Z_{2m} g^{2m}}, \qquad Z_0 = 1, \tag{A.36}$$

so that $\mathcal{C}_1 = A_1$ and $\mathcal{K}_2 = Q_2$. At zero source ($A_1 = 1$) this gives, through the orders used,

$$\begin{aligned}
\mathcal{C}_3 &= A_3 - Z_2, \\
\mathcal{C}_5 &= A_5 - A_3 Z_2 + Z_2^2 - Z_4, \\
\mathcal{C}_7 &= A_7 - A_5 Z_2 + A_3(Z_2^2 - Z_4) - Z_2^3 + 2Z_2Z_4 - Z_6, \\
\mathcal{K}_4 &= Q_4 - Q_2 Z_2, \\
\mathcal{K}_6 &= Q_6 - Q_4 Z_2 + Q_2(Z_2^2 - Z_4), \\
\mathcal{K}_8 &= Q_8 - Q_6 Z_2 + Q_4(Z_2^2 - Z_4) + Q_2(-Z_2^3 + 2Z_2Z_4 - Z_6).
\end{aligned} \tag{A.37}$$

Source derivatives differentiate *every* factor, including A_1; for example

$$\partial_\eta \mathcal{C}_3 = \partial_\eta A_3 - (\partial_\eta A_1)\, Z_2 - \partial_\eta Z_2. \tag{A.38}$$

For a response I define

$$I_0 = [\varepsilon^0 \ell^0]\, I, \qquad I_1 = [\varepsilon^0 \ell^1]\, I, \tag{A.39}$$

so that, for instance, $\mathcal{C}_{3,1}$ is the single-logarithm coefficient of $\mathcal{C}_3$ and $Q_{2,0} = -1$ the finite part of Q_2. The products in Eq. (A.37) must be formed, as complete series in ε and ℓ, before these projections are applied: since $Z_2 \sim \varepsilon^{-1}$, a term such as $\varepsilon\ell$ in another factor contributes to I_1. Multiplying only finite parts is not a valid normalization rule. Derivatives are likewise taken before the projection: $\partial_v \mathcal{C}_{n,0}$ and $\partial_\eta \mathcal{C}_{n,0}$ mean the finite parts of $\partial_v \mathcal{C}_n$ and $\partial_\eta \mathcal{C}_n$.

A.8 Charge and tail insertions

Derivatives of the wall weight in the charge v and the tail parameter $\bar{\delta}$ insert logarithms of the kink separations. Let

$$S_j = \sum_{i<k} q_i q_k \log^j(X_k - X_i). \tag{A.40}$$

Since $[(X_k - X_i)^{\bar{\delta}} - 1]/\bar{\delta} = \log(X_k - X_i) + \frac{\bar{\delta}}{2}\log^2(X_k - X_i) + \frac{\bar{\delta}^2}{6}\log^3(X_k - X_i) + \cdots$, the derivatives at $v = 1$, $\bar{\delta} = 0$ are

$$\begin{aligned} \partial_v W_n &= 2S_1 W_n, & \partial_v^2 W_n &= 4S_1^2 W_n, \\ \partial_{\bar{\delta}} W_n &= S_2 W_n, & \partial_v \partial_{\bar{\delta}} W_n &= (S_2 + 2S_1 S_2) W_n, \\ [\bar{\delta}^2]\, W_n &= \Big(\frac{1}{3}S_3 + \frac{1}{2}S_2^2\Big) W_n. \end{aligned} \tag{A.41}$$

The second term of the mixed derivative and both terms of the second tail insertion are essential. Tail derivatives also act on the endpoint factors of Eq. (A.18), through $f_{\bar{\delta}}$. A Taylor coefficient such as $[\bar{\delta}^2]$ includes the factor $1/2!$; it is not the second derivative itself.

A.9 The finite-response Jacobian and the subtraction recurrence

The beta functions are extracted from two vacuum-normalized responses at $h = 0$: the odd response, the projection of the ordered evolution on the odd spin-flip vertex, with finite parts built from the $\mathcal{C}_{2m+1}$, and the current response, its projection on the current, built from the $\mathcal{K}_{2m}$ (with $\mathcal{K}_2 = Q_2$). The regulator convention subtracts logarithmic coupling counterterms with no additional finite coupling redefinition in these coordinates. The finite parts of the responses are nevertheless required: the single logarithm of a normalized response is not by itself a beta coefficient, because finite parts of lower orders, multiplied by lower beta coefficients, also contribute to it.

Jacobian entries. Set

$$\begin{aligned} j_1 &= -\frac{2}{1+\bar{\delta}}, \qquad j_n = -2\partial_v \mathcal{C}_{n,0} + \partial_\eta \mathcal{C}_{n,0} \quad (n = 3, 5), \\ \lambda_2 &= -2Q_{2,0} - 4v\,\partial_v Q_{2,0} + 2v\,\partial_\eta Q_{2,0}, \\ \lambda_4 &= -2\mathcal{K}_{4,0} - 4v\,\partial_v \mathcal{K}_{4,0} + 2v\,\partial_\eta \mathcal{K}_{4,0}. \end{aligned} \tag{A.42}$$

The integral defining j_1 is $\int_0^1 [f_{\bar{\delta}}(X) + f_{\bar{\delta}}(1-X)]\,\mathrm{d}X$, the source derivative $\partial_\eta A_1$ of the one-kink moment. The same definition of j_7 is used at the eighth order (Appendix C.3).

Auxiliary beta coefficients. Write the beta functions of the auxiliary couplings at $h = 0$ as

$$\beta_g = b_3 g^3 + b_5 g^5 + b_7 g^7 + \cdots, \qquad \frac{\beta_h}{\sqrt{2}\,b_0} = -g^2 + h_4 g^4 + h_6 g^6 + \cdots, \tag{A.43}$$

so that b_n (n odd) are the odd fugacity coefficients and h_n (n even, $h_2 = -1$) the normalized current coefficients. Archived scripts call h_4 and h_6 by η_4 and η_6; these are not the endpoint source η. The coefficients depend on the charge $v = b_0^2$ and on $\bar{\delta}$, and their Taylor coefficients at $v = 1$, $\bar{\delta} = 0$, after the matching factors of Appendix A.5 are restored, are the beta inputs of Sec. 2.3. For example $b_3 = 1$ and $h_2 = -1$ at that point give the leading terms u of F and $4u$ of H, while $b = b_5$, $c = -4h_4$ and $k = -4h_6$ are the coefficients of u^2 in F and of u^2 and u^3 in H; the complete dictionary is in Appendices B.1 and C.3.

The finite-response Jacobian. Through the required orders the Jacobian is

$$\mathcal{J} = \begin{pmatrix} 1 + 3\mathcal{C}_{3,0}g^2 + 5\mathcal{C}_{5,0}g^4 & \sqrt{2}\,b_0\,(j_1 g + j_3 g^3 + j_5 g^5) \\ \sqrt{2}\,b_0\,(2Q_{2,0}g + 4\mathcal{K}_{4,0}g^3) & 1 + \lambda_2 g^2 + \lambda_4 g^4 \end{pmatrix}, \tag{A.44}$$

and the vector of single-logarithm coefficients is

$$R_1 = \begin{pmatrix} \mathcal{C}_{3,1}g^3 + \mathcal{C}_{5,1}g^5 + \mathcal{C}_{7,1}g^7 \\ \sqrt{2}\,b_0\,(-g^2 + \mathcal{K}_{4,1}g^4 + \mathcal{K}_{6,1}g^6) \end{pmatrix}. \tag{A.45}$$

The entries have a direct reading. The first column holds the g derivatives of the finite parts of the odd response, $g + \mathcal{C}_{3,0}g^3 + \mathcal{C}_{5,0}g^5$, and of the current response, $h + \sqrt{2}\,b\,(Q_{2,0}g^2 + \mathcal{K}_{4,0}g^4)$. The second column holds their h derivatives at $h = 0$, taken with Eq. (A.19); the terms $-2Q_{2,0}$ and $-2\mathcal{K}_{4,0}$ of λ_2 and λ_4 are the derivative of the charge $b = b_0 - \sqrt{2}h$ that multiplies the current moments. The leading $-g^2$ in the second row of R_1 is $Q_{2,1} = -1$, from Eq. (A.35).

The subtraction recurrence. The prescription solves

$$\mathcal{J}\begin{pmatrix} \beta_g \\ \beta_h \end{pmatrix} = R_1 \tag{A.46}$$

order by order in g. With $v = b_0^2$ (the factors $2v$ are $(\sqrt{2}b_0)^2$) this gives

$$\begin{aligned} b_3 &= \mathcal{C}_{3,1} + 2vj_1, \\ h_4 &= \mathcal{K}_{4,1} - 2Q_{2,0}b_3 + \lambda_2, \\ b_5 &= \mathcal{C}_{5,1} - 3\mathcal{C}_{3,0}b_3 - 2vj_1h_4 + 2vj_3, \\ h_6 &= \mathcal{K}_{6,1} - 2Q_{2,0}b_5 - 4\mathcal{K}_{4,0}b_3 - \lambda_2 h_4 + \lambda_4, \\ b_7 &= \mathcal{C}_{7,1} - 3\mathcal{C}_{3,0}b_5 - 5\mathcal{C}_{5,0}b_3 - 2vj_1h_6 - 2vj_3h_4 + 2vj_5. \end{aligned} \tag{A.47}$$

The classical-scaling and matching contributions of Eq. (A.21) are then restored to pass from the auxiliary couplings (g_{aux}, b) to the essential couplings (u, z). This derives the recurrence; it is not an assignment of a raw single logarithm to a physical beta function. The recurrence uses only first h derivatives at $h = 0$, which is why the coefficient of h^2 in Eq. (A.17) does not enter it. Its extension through the ninth order (h_8 and b_9, with $\mathcal{C}_9$) is in Appendix C.3, Eqs. (C.29) and (C.30).

The lowest order as a check of signs. The normalized three-kink response has $\mathcal{C}_{3,1} = 5$ (its expansion is in Appendix B.2), and $j_1 = -2$ at $\bar{\delta} = 0$, so $b_3 = 5 - 4 = 1$. With $\mathcal{K}_{4,1} = -8 + 4\zeta(2)$, $Q_{2,0} = -1$, $\partial_v Q_{2,0} = -2$ and $\partial_\eta Q_{2,0} = -2\zeta(2)$, one has $\lambda_2 = 10 - 4\zeta(2)$ and $h_4 = 4$, that is $c = -4h_4 = -16$ (Sec. 2.3).

Scope. Equation (A.46) defines the beta functions of the prescription. That its solution is the local beta function of the dual, up to a finite change of (g, b) that leaves T and D unchanged, is the formal argument of Appendix D (Appendices D.3 and D.13), summarized in the caveat "Two-coupling closure" of Sec. 2.9.

A.10 Stability invariants and the limits of the leading flow

Trace and determinant. Linearizing $(\beta_u, \beta_z) = (2uF, (1+z)H)$ about a fixed point with $F = H = 0$, the terms $2F$ and H of the derivatives vanish, and the stability matrix in (u, z) is

$$\begin{pmatrix} 2u_* F_u & 2u_* F_z \\ (1+z_*)H_u & (1+z_*)H_z \end{pmatrix}, \tag{A.48}$$

with the derivatives taken before the fixed point is substituted. Its trace and determinant are the T and D of Sec. 2.2, and its eigenvalues $\lambda_\pm = T/2 \pm (T^2/4 - D)^{1/2}$ give the dimensions $\Delta_\pm = 1 + \lambda_\pm$ of Eq. (2.6). A regular change of couplings transforms the exact stability matrix by similarity, so T and D are invariant although individual beta coefficients are not. The even coefficients need only the trace, $C^{\mathrm{SR}}_{2m} = 2^{m-1} T_m$ (Eq. (2.8)): a new even coefficient needs the next trace coefficient but not the next determinant coefficient.

Why the leading flow cannot be extrapolated. If one regarded the leading $F = z + u$ and $H = -\bar\delta + 4u$ as exact, the invariants would be $T = 2u_* = \bar\delta/2$ and $D = -8u_*(1+z_*) = -2\bar\delta + \bar\delta^2/2$, so $D_2 = \frac{1}{2}$ and $T_2 = 0$, and Eq. (2.8) would give $C^{\mathrm{SR}}_3 = \frac{7}{16}$ and $C^{\mathrm{SR}}_4 = 0$ instead of the values of Eq. (2.12). Terms omitted in that truncation change both results. For instance, a term azu in F is of order $\bar\delta^2$ at the fixed point, but it adds au_*, of order $\bar\delta$, to the derivative F_z and so changes D_2 (indeed $D_2 = \frac{1}{2} - \frac{a}{2} - \frac{c}{8}$, Sec. 2.3). A term zu^2 in H has a value of order $\bar\delta^3$, but its derivative H_z is of order $\bar\delta^2$. Power counting must be applied *after* differentiation. This is the distinction between a higher-insertion calculation and a formal re-expansion of an already truncated flow, and it is why the beta inputs include derivatives in z and $\bar\delta$ of integrals with more kinks than the order would suggest. For the same reason, expanding the leading matrix M_* of Eq. (A.27) to more orders does not compute the higher coefficients.

B The dual of the chain: the third to sixth orders

This appendix gives the derivations of the compendium (written by OpenAI GPT-6 astra; Sec. 1.2) for the coefficients C^{SR}_3 to C^{SR}_6, edited to the notation of this note. It builds on the model, the regulator, the moments and the subtraction recurrence of Appendix A, and gives the calculations that Secs. 2.3 and 2.4 summarize for these orders; the inputs that rest on archived exact expressions are marked as such below.

Appendices B.2 to B.4 assemble the invariants D_2, T_2, D_3 and T_3 from the beta inputs. The remaining subsections evaluate the inputs: the Mellin representation of the regulated moments and the rules for finite parts, with the charge-derivative period $\mathcal{R}_4$ (Appendix B.5); a trilogarithm lemma and the Spence–Kummer identity on which it rests (Appendix B.6); the closed forms of the periods $\mathcal{P}_4$ and $\mathcal{S}_5$ (Appendix B.7); the counterterm calculation that attaches the periods to the inputs (Appendix B.8); the charge and tail derivatives at three insertions (Appendix B.9); and the three-kink and endpoint inputs of D_3 (Appendix B.10). What these derivations establish differs by order:

- C^{SR}_3 and C^{SR}_4. The three periods $\mathcal{P}_4$, $\mathcal{R}_4$ and $\mathcal{S}_5$ are evaluated in closed form with explicit primitives, a table of elementary finite parts and the trilogarithm lemma, without numerical recognition. Their attachment to the five inputs a, p, b, c, f goes through response coefficients and counterterms: the mixed response coefficients of Eq. (B.102) are taken from the archive without derivation, the counterterm algebra is displayed here in full, and the compendium checks the charge derivative of the fourth-order counterterm and the three-insertion derivatives with its code.
- C^{SR}_5. The inputs A and E and two endpoint constants are evaluated here. The inputs G, q, k are archived exact period expressions, and the combination of them that enters D_3 is

reduced to zeta values by the exact certificate of D_3 (Appendix C.1; Sec. 2.4).

- C_6^{SR}. The six new inputs are archived exact period expressions, and the closed form of C_6^{SR} is established by the exact certificate at weights two to five (same appendix). This appendix gives the assembly, a convergent check of one input and checks that do not use the new single logarithms.

Everything holds within the matched two-coupling prescription of Sec. 2.1, with the caveats stated there and in Sec. 2.9.

B.1 Conventions and the beta inputs

The notation is that of Table 4. In particular $\bar\delta = 1 - s$; ε is the hard core and $\ell = \log\varepsilon$; $\mathcal{C}_{2m+1}$ and $\mathcal{K}_{2m}$ are the vacuum-normalized responses of Eq. (A.37); a second subscript 0 or 1 denotes the coefficient $[\varepsilon^0\ell^0]$ or $[\varepsilon^0\ell^1]$ (Eq. (A.39)); ∂_η is the derivative with respect to the endpoint source at $\eta = 0$; and b_n (n odd) and h_n (n even) are the auxiliary beta coefficients of Eq. (A.43), determined by the recurrence (A.47). Derivatives in v and $\bar\delta$ are taken at the marginal point $v = 1$, $\bar\delta = 0$, before any fixed point is substituted. We write $\log^2 2$ for $(\log 2)^2$, and similarly for other powers.

The beta inputs of orders three to six. The beta inputs are the Taylor coefficients of F and H at the marginal point. In the compendium's dictionary, with the matching factors of Eq. (A.22) restored as described in Appendix A.9, and with a subscript v or $\bar\delta$ denoting a derivative of the auxiliary coefficient, the inputs used here are

$$\begin{gathered} a = b_{3,v}, \qquad p = b_{3,\bar\delta}, \qquad A = \tfrac{1}{2}b_{3,vv}, \qquad E = b_{3,v\bar\delta}, \qquad V = \tfrac{1}{2}b_{3,\bar\delta\bar\delta}, \\ b = b_5, \qquad G = b_{5,v}, \qquad R = b_{5,\bar\delta}, \qquad J = b_7, \\ c = -4h_4, \qquad f = -4h_{4,v}, \qquad q = -4h_{4,\bar\delta}, \qquad L = -2h_{4,vv}, \qquad M = -4h_{4,v\bar\delta}, \\ k = -4h_6, \qquad N = -4h_{6,v}. \end{gathered} \tag{B.1}$$

Taylor factors appear only for repeated derivatives; the mixed derivatives E and M carry none. The input $b = b_5$ is not the resummed charge $b_0 - \sqrt{2}h$, and the input q is not a kink charge or a Mellin variable.

Local symbols. From Appendix B.5 on, the letters e, t, w, x, y and z are integration or auxiliary variables defined where they are used; there x and z are never Δ_σ or the coupling z. Where the compendium uses one letter for two quantities, the local quantity is renamed; the renamed symbols are defined at first use.

B.2 Third and fourth orders: D_2 and T_2

Beta functions. To the first new orders the matched beta functions have the form

$$F = z + (1 + az + p\bar\delta)u + bu^2 + \cdots, \qquad H = -\bar\delta + 4u + cu^2 + fzu^2 + \cdots. \tag{B.2}$$

The one-pair calculation fixes the term $4u$ of H, with no further terms zu, z^2u, $\bar\delta u$ or $\bar\delta zu$ at these orders in this prescription. Indeed the one-pair current moment is

$$Q_2(v, \bar\delta; \varepsilon) = \int_\varepsilon^1 r(1-r)\, e^{-2vf_{\bar\delta}(r)}\, \mathrm{d}r, \tag{B.3}$$

with $f_{\bar\delta}$ of Eq. (A.5) at $\kappa = 1$. At the marginal point its single logarithm is $-\ell$ (Eq. (A.35)), while derivatives in the charge or the tail parameter add powers of $\log r$ to the integrand and so produce higher powers of ℓ, not a new single logarithm (Appendix B.8). The tree coefficient of the fugacity in the essential variables is exactly z.

Inputs and periods. The five inputs a, p, b, c, f are listed in Sec. 2.3. They are fixed by elementary constants and three regulator periods: $\mathcal{P}_4$ and $\mathcal{R}_4$, Laurent coefficients of the angular Mellin function of the four-kink current moment and of its charge derivative, and $\mathcal{S}_5$, a Laurent coefficient of the angular Mellin function of the five-kink odd moment (Eq. (B.36)). The derivations of Appendices B.8 and B.9, and of Appendix A.9 for c, give

$$\begin{aligned} a &= -4\zeta(2), \qquad c = -16, \qquad p = 8 - 2\zeta(2) - 7\zeta(3) - 8\log 2, \\ b &= -21 + 38\zeta(2) + 60\zeta(3) + 32\log 2 - \mathcal{S}_5, \qquad f = 16\mathcal{P}_4 + 4\mathcal{R}_4 - 32 + 16\zeta(2) + 32\zeta(3), \end{aligned} \tag{B.4}$$

and Appendices B.5 and B.7 evaluate the periods,

$$\begin{aligned} \mathcal{P}_4 &= 4\zeta(2) + \tfrac{8}{3}\zeta(3) - 2 + 3\log 3 + \log^2 2 + 2\,\mathrm{Li}_2(-\tfrac{1}{2}), \\ \mathcal{R}_4 &= 24 - 20\zeta(2) - 30\zeta(3) - 16\log 2, \\ \mathcal{S}_5 &= 44\zeta(2) + \tfrac{152}{3}\zeta(3) - 3 + 12\log 3 + 4\log^2 2 + 8\,\mathrm{Li}_2(-\tfrac{1}{2}). \end{aligned} \tag{B.5}$$

In particular

$$\mathcal{S}_5 - 4\mathcal{P}_4 = 5 + 28\zeta(2) + 40\zeta(3). \tag{B.6}$$

With Eq. (B.5), Eq. (B.4) gives the values of b and f in Sec. 2.3. The regulator constants $\log 2$, $\log 3$, $\log^2 2$ and $\mathrm{Li}_2(-\frac{1}{2})$ are genuine parts of the individual inputs. In T_2 below, b and f enter as

$$\frac{b}{4} + \frac{f}{16} = -\frac{29}{4} + \frac{21}{2}\zeta(2) + 17\zeta(3) + 8\log 2 - \frac{\mathcal{S}_5 - 4\mathcal{P}_4}{4} + \frac{\mathcal{R}_4}{4}, \tag{B.7}$$

so the periods enter only through $\mathcal{R}_4$ and the combination (B.6), and the regulator constants cancel. The evaluation of $\mathcal{P}_4$ and $\mathcal{S}_5$ turns on one finite part,

$$J_L = \lim_{T\to\infty}\Big[\int_2^T \frac{\log(t-1)\log(t+1)}{t}\,\mathrm{d}t - \frac{1}{3}\log^3 T\Big] = -\frac{1}{3}\zeta(3), \tag{B.8}$$

which Appendix B.6 proves with explicit primitives and a real-branch Spence–Kummer identity, not by numerical recognition.

Normalized responses. With $\ell = \log\varepsilon$, the normalized low-order responses at $v = 1$, $\bar\delta = 0$ begin

$$\mathcal{C}_3 = 2\ell^2 + 5\ell + 3 - 2\zeta(2) + \cdots, \tag{B.9}$$

$$\mathcal{K}_4 = -\frac{4}{3}\ell^3 - 6\ell^2 + [-8 + 4\zeta(2)]\ell + \mathcal{P}_4 + 4\zeta(2) - 10 + \cdots, \tag{B.10}$$

$$\mathcal{C}_5 = \frac{10}{3}\ell^4 + \frac{62}{3}\ell^3 + \Big(\frac{95}{2} - 24\zeta(2)\Big)\ell^2 + [96 - 40\zeta(2) - \mathcal{S}_5]\ell + \cdots, \tag{B.11}$$

where the dots stand for terms that vanish as $\varepsilon \to 0$ and, in $\mathcal{C}_5$, for its finite part, which is not needed here. Their substitution into the subtraction recurrence (A.47), together with the finite parts and the charge and source derivatives of the lower responses, yields the inputs; the raw single logarithms alone do not. The finite-tail factor of Eq. (A.22) is needed for p (Appendix B.9).

Invariant assembly. Solving $H = 0$ and $F = 0$ order by order gives $u_* = \bar\delta/4 - c\,\bar\delta^2/64 + O(\bar\delta^3)$ and $z_* = -\bar\delta/4 + O(\bar\delta^2)$. In $D = 2u(1+z)(F_uH_z - F_zH_u)$ the derivative $H_z = fu^2 + \cdots$ is of order $\bar\delta^2$, so through order $\bar\delta^2$, with $F_z = 1 + au + \cdots$ and $H_u = 4 + 2cu + \cdots$,

$$D = -2u(1+z)F_zH_u + O(\bar\delta^3) = -8u\Big(1 + z + au + \frac{c}{2}u\Big) + O(\bar\delta^3). \tag{B.12}$$

In $T = 2uF_u + (1+z)H_z$, with $F_u = 1 + az + p\bar\delta + 2bu + \cdots$,

$$T = 2u\big(1 + az + p\bar\delta + 2bu\big) + fu^2 + O(\bar\delta^3). \tag{B.13}$$

Inserting the fixed point gives $D = -2\bar{\delta} + D_2\bar{\delta}^2 + \cdots$ and $T = \bar{\delta}/2 + T_2\bar{\delta}^2 + \cdots$ with

$$D_2 = \frac{1}{2} - \frac{a}{2} - \frac{c}{8}, \qquad T_2 = -\frac{a}{8} + \frac{b}{4} - \frac{c}{32} + \frac{f}{16} + \frac{p}{2}, \tag{B.14}$$

as in Sec. 2.3. With Eqs. (B.4) to (B.6) these are the closed forms $D_2 = \frac{5}{2} + 2\zeta(2)$ and $T_2 = 2 - 2\zeta(2) - 4\zeta(3)$ of Sec. 2.4, and Eq. (2.8) gives $C_3^{\mathrm{SR}} = D_2 - \frac{1}{16}$ and $C_4^{\mathrm{SR}} = 2T_2$ of Eq. (2.12). The highest insertion order needed for C_4^{SR} is five (the period $\mathcal{S}_5$); an insertion count is not a Feynman-loop count.

B.3 Fifth order: D_3

The determinant at order $\bar{\delta}^3$ needs the further sectors

$$\begin{aligned} F &= z + (1 + az + p\bar{\delta} + Az^2 + E\bar{\delta}z)u + (b + Gz)u^2 + \cdots, \\ H &= -\bar{\delta} + 4u + (c + fz + q\bar{\delta})u^2 + ku^3 + \cdots. \end{aligned} \tag{B.15}$$

The three-kink inputs A and E. They have convergent representations and closed forms,

$$\begin{aligned} A &= -8 - 4\zeta(2) - 2\zeta(3) + 16\log 2, \\ E &= -16 + 14\zeta(2) - 22\zeta(3) + 12\zeta(4) + 8\log 2 - 8\log^2 2 - 16\,\mathrm{Li}_4(\tfrac{1}{2}) \\ &\quad - 14\zeta(3)\log 2 + 4\zeta(2)\log^2 2 - \tfrac{2}{3}\log^4 2. \end{aligned} \tag{B.16}$$

For A one may evaluate

$$A = -4\int_0^{1/2} \frac{\log^2(1-y)}{y^2(1-y)^2}\,\mathrm{d}y, \tag{B.17}$$

and for E an independent convergent representation is

$$E = p + 4\int_0^{1/2} \frac{\log^2(1-y)\,[2\log y - \log(1-y)]}{y^2(1-y)^2}\,\mathrm{d}y. \tag{B.18}$$

Appendix B.10 reduces E to the weight-four constants of Eq. (B.16); the key step is

$$\int_0^1 \frac{\log t\,\log^2(1+t)}{t}\,\mathrm{d}t = \frac{2}{3}\int_0^1 \frac{\log^3(1+t)}{t}\,\mathrm{d}t - \frac{1}{4}\zeta(4). \tag{B.19}$$

Endpoint constants. Appendix B.10 also derives the finite parts of two source derivatives of raw moments,

$$\partial_v\partial_\eta A_{3,0} = -108 + 60\zeta(2) + 22\zeta(3) - 34\zeta(4) + 16\log 2, \qquad \partial_\eta Q_{4,0} = 6 - 12\zeta(2) - 54\zeta(3) - 25\zeta(4). \tag{B.20}$$

Source derivatives of this kind enter the subtraction recurrence through the Jacobian entries j_n and λ_{2m} of Eq. (A.42). The precise moment convention is given in Appendix B.10. In particular, the conversion from angular to radial variables must keep $\log(\varepsilon/R) = \ell - \log R$ in full (R the span of the kinks); dropping the radial logarithms changes finite parts.

The inputs G, q, k. The remaining inputs, $G = b_{5,v}$, $q = -4h_{4,\bar{\delta}}$ and $k = -4h_6$, are archived exact period expressions (`code/dual_d1/data/archived/C6_closed_form/input_expressions/`). Their values are

$$\begin{aligned} G &= 32.763409276837655798312474117 4\ldots, \\ q &= -103.560357927042746213210123654 5\ldots, \\ k &= 104.672392743212637331086084748 7\ldots. \end{aligned} \tag{B.21}$$

The exact expressions, not these decimals, enter the assembly.

Fixed point and determinant. Through the orders needed,

$$\begin{aligned} u_* &= \frac{\bar\delta}{4} - \frac{c}{64}\bar\delta^2 + \Big(\frac{c^2}{512} + \frac{f-k}{256} - \frac{q}{64}\Big)\bar\delta^3 + \cdots, \\ z_* &= -\frac{\bar\delta}{4} + \Big(\frac{a-b}{16} + \frac{c}{64} - \frac{p}{4}\Big)\bar\delta^2 + \cdots, \end{aligned} \tag{B.22}$$

and the determinant gives

$$D_3 = \frac{16A - 32E - 8G + 8b + c^2 + 4f - 4k + 32p - 8q}{64}. \tag{B.23}$$

With $\mathcal{R} = 2G + k + 2q - 2b - f$ and $c^2/64 = 4$, this is $D_3 = \frac{A}{4} - \frac{E}{2} + 4 + \frac{p}{2} - \frac{\mathcal{R}}{16}$. The exact certificate of D_3 (Appendix C.1; the table of Sec. 2.4) proves

$$\begin{aligned} \mathcal{R} = {} & 140 + 64\zeta(2) - 64\zeta(3) - 272\zeta(4) - 64\log 2 + 64\log^2 2 \\ & + 112\zeta(3)\log 2 - 32\zeta(2)\log^2 2 + \tfrac{16}{3}\log^4 2 + 128\,\mathrm{Li}_4(\tfrac{1}{2}). \end{aligned} \tag{B.24}$$

With Eq. (B.16) and the value of p, all regulator constants other than zeta values cancel, and $D_3 = \frac{21}{4} - 13\zeta(2) + 11\zeta(3) + 11\zeta(4) = 8.99403863655111357\ldots$, the closed form of Sec. 2.4. Equation (2.8) then gives C_5^{SR} of Eq. (2.12), $29.2378115809900396\ldots$.

B.4 Sixth order: T_3

To determine T_3, retain

$$\begin{aligned} F &= z + u(1 + az + p\bar\delta + Az^2 + E\bar\delta z + V\bar\delta^2) + u^2(b + Gz + R\bar\delta) + Ju^3 + \cdots, \\ H &= -\bar\delta + 4u + u^2(c + fz + q\bar\delta + Lz^2 + M\bar\delta z) + u^3(k + Nz) + \cdots. \end{aligned} \tag{B.25}$$

The six new inputs of Eq. (B.1),

$$V = \tfrac{1}{2}b_{3,\bar\delta\bar\delta}, \quad R = b_{5,\bar\delta}, \quad J = b_7, \quad L = -2h_{4,vv}, \quad M = -4h_{4,v\bar\delta}, \quad N = -4h_{6,v}, \tag{B.26}$$

are evaluated at the marginal point $v = 1$, $\bar\delta = 0$, not at the interacting fixed point. The term Nzu^3 is of order $\bar\delta^4$ in H at the fixed point but contributes at order $\bar\delta^3$ after a derivative in z, so it cannot be dropped.

input	largest new moment sector	value
V	three-kink second tail	-19.518741011574465808
R	five-kink tail	-1.148356985726247710
J	seven-kink odd response	32.280324973629037003
L	four-kink second charge	-7.677303759056569914
M	four-kink charge–tail	13.315217322864044765
N	six-kink charge derivative	210.030599248618268410

These are beta inputs of the prescription, not universal exponents. Their exact expressions are archived with those of G, q, k (`code/dual_d1/data/archived/C6_closed_form/input_expressions/`).

A convergent check of V. For three kinks, put $l = \log y$, $\bar l = \log(1-y)$ and

$$P_V(y) = -l^2\bar l + l\bar l^2 - \tfrac{1}{3}\bar l^3 + 2l^2\bar l^2 - 2l\bar l^3 + \tfrac{1}{2}\bar l^4. \tag{B.27}$$

The direct regulated calculation reduces to

$$V = \frac{2}{3}\log^3 2 - 2\int_0^{1/2}\Big[\frac{P_V(y)}{y^2(1-y)^2} - \frac{\log^2 y}{y}\Big]\,\mathrm{d}y. \tag{B.28}$$

The subtraction removes the endpoint singularity. Both the cubic-logarithm term $\frac{1}{3}S_3$ and the square $\frac{1}{2}S_2^2$ of the first tail insertion in Eq. (A.41) are needed. The finite hard-core endpoint contributes $2\int_1^\infty \log^2 w\, w^{-2}\,\mathrm{d}w = 4$, which cancels a -4 in the beta assembly (the coefficient of $\bar\delta^2$ in $2vj_1 = -4/(1+\bar\delta)$ at $v=1$, Eqs. (A.42) and (A.47)). This finite piece is derived with the regulator kept; it is not an adjustable counterterm.

Trace assembly. With the fixed point (B.22), the trace gives

$$C_6^{\mathrm{SR}} = \mathcal{B}_{\mathrm{old}}^{(6)} + 2V + R + \frac{3}{8}J - \frac{1}{8}L + \frac{1}{4}M + \frac{1}{16}N, \tag{B.29}$$

where the part fixed by the inputs of the lower orders is

$$\begin{aligned}\mathcal{B}_{\mathrm{old}}^{(6)} = \frac{A}{8} - \frac{E}{2} - \frac{G}{4} + \frac{a^2-ab}{8} + \frac{ac}{16} - \frac{ap}{2} - \frac{bc}{8} + \frac{c^2}{64} \\ - \frac{cf}{32} - \frac{cp}{8} - \frac{f}{32} - \frac{k}{32} - \frac{q}{8},\end{aligned} \tag{B.30}$$

with the archived value $\mathcal{B}_{\mathrm{old}}^{(6)} = -48.0003600840977427059\ldots$. The complete regulated expression gives $C_6^{\mathrm{SR}} = -58.665697474225308949717146128630 82\ldots$. The exact certificate at weights two to five (Appendix C.1) establishes the closed form of C_6^{SR} in Eq. (2.12), and $T_3 = C_6^{\mathrm{SR}}/4$ is the expression of Sec. 2.4.

Checks that do not use the new single logarithms. The lower flow predicts, for the normalized seven-kink response,

$$[\ell^6]\,\mathcal{C}_7 = \frac{244}{45}, \qquad [\ell^5]\,\mathcal{C}_7 = \frac{170}{3}, \qquad [\ell^4]\,\mathcal{C}_7 = 247 - \frac{316}{3}\zeta(2), \tag{B.31}$$

and the archived exact integrations reproduce these coefficients without using J. For the finite-tail correction to the five-kink response, the matching of Eq. (A.22) predicts

$$[\ell^6]\,\partial_{\bar\delta}\mathcal{C}_5 = -\frac{142}{45}, \qquad [\ell^5]\,\partial_{\bar\delta}\mathcal{C}_5 = -\frac{71}{5}, \qquad [\ell^4]\,\partial_{\bar\delta}\mathcal{C}_5 = \frac{14}{3}\zeta(2) - \frac{55}{3}, \tag{B.32}$$

which do not use R. The archive contains further higher-logarithm checks, exact derivatives of the primitives and direct finite-cutoff numerical integrations of the original kernels. Some period residuals in the higher-logarithm tests were checked numerically, not by exact certificates, and the verification reports keep that distinction.

Two archived full period evaluations used 80 and 100 decimal digits, matching points 100 and 120, and different continuation steps and series orders; they agreed in all 65 output digits of the first. This is a convergence check, not an interval bound; the closed form of C_6^{SR} rests on the rational certificate, not on numerical agreement.

B.5 Mellin representation, finite parts and the period $\mathcal{R}_4$

This and the following subsections give the low-insertion calculations behind Appendix B.2, with their subtractions and branch conventions. The normalized responses are those of Eq. (A.37), and the gap variables are those of Appendix A.7.

Separating the radial integral. Let F be homogeneous of degree d_F in m gaps, and put $k_F = m + d_F$. Define, with the translation factor of Appendix A.7,

$$I_F(\varepsilon) = \int_{r_i\ge\varepsilon,\ \sum r_i\le 1} \Big(1 - \sum_i r_i\Big) F(\boldsymbol{r})\ \mathrm{d}^m r. \tag{B.33}$$

First integrate ε up to $\min_i r_i$, then write $r_i = Rx_i$ with $\sum_i x_i = 1$. This gives

$$\int_0^\infty \varepsilon^{q-1} I_F(\varepsilon)\,\mathrm{d}\varepsilon = \frac{\mathcal{A}_F(q)}{q(q+k_F)(q+k_F+1)}, \qquad \mathcal{A}_F(q) = \int_{\sum x_i=1} F(\boldsymbol{x})(\min_i x_i)^q\,\mathrm{d}^{m-1}x. \quad \text{(B.34)}$$

Here q is the Mellin variable, not the input q. The identity is first used in a half-plane where it converges and is then continued after explicit subtraction; no divergent integral is evaluated naively at $q = 0$. The pole of order n at $q = 0$ gives the ε^0 terms of I_F through

$$q^{-n} \;\longleftrightarrow\; \frac{(-\ell)^{n-1}}{(n-1)!}, \quad \text{(B.35)}$$

so the coefficient of q^{-1} is the finite part; power divergences give poles away from $q = 0$.

The needed Laurent coefficients of the angular functions define the regulator periods:

$$\begin{aligned}
\mathcal{A}_{Q_4}(q)\big|_{v=1} &= \frac{8}{q^2} - \frac{6}{q} - 6 - 4\zeta(2) + \mathcal{P}_4\, q + O(q^2),\\
\partial_v \mathcal{A}_{Q_4}(q)\big|_{v=1} &= \frac{32}{q^3} - \frac{4}{q^2} + \frac{4+16\zeta(2)}{q} + \mathcal{R}_4 + O(q),\\
\mathcal{A}_{Z_4}(q) &= \frac{12}{q^2} - \frac{3}{q} - 5 - 6\zeta(2) + O(q),\\
\mathcal{A}_{A_5}(q) &= \frac{80}{q^3} - \frac{68}{q^2} - \frac{56+48\zeta(2)}{q} + \mathcal{S}_5 + O(q).
\end{aligned} \quad \text{(B.36)}$$

The subscript names the moment whose kernel is meant: $\Xi_4 W_4$ for Q_4, W_4 for Z_4 and W_5 for A_5 (the compendium writes G_5 for the last). For Q_4, $k_F = 4 - 4v$, so a charge derivative of Eq. (B.34) also differentiates its radial denominator; that contribution must be included (Appendix B.8). For example, at $v = 1$ ($k_F = 0$) the first line and Eq. (B.35) give the ε^0 part of Q_4, $-\frac{4}{3}\ell^3 - 7\ell^2 + [-8 + 4\zeta(2)]\ell + \mathcal{P}_4 + 4\zeta(2) - 8$; subtracting the ε^0 part of $Q_2 Z_2$, which is $2 - \ell^2$ by Eq. (A.35), gives $\mathcal{K}_4$ of Eq. (B.10).

Hard-cut angular functions. Define

$$\mathcal{H}_F(e) = \int_{x_i \ge e,\ \sum x_i = 1} F(\boldsymbol{x})\,\mathrm{d}^{m-1}x, \qquad \mathcal{A}_F(q) = q\int_0^{1/m} e^{q-1}\mathcal{H}_F(e)\,\mathrm{d}e. \quad \text{(B.37)}$$

The constant term ($e^0 \log^0 e$) of $\mathcal{H}_{A_5}$ is $\mathcal{S}_5$, and that of $\partial_v \mathcal{H}_{Q_4}$ is $\mathcal{R}_4$. Power divergences produce poles away from $q = 0$ and do not change these identifications. The period $\mathcal{P}_4$ is instead a finite part of $\int \mathcal{H}_{Q_4}(e)\,\mathrm{d}e/e$ (Appendix B.7).

Finite parts. Throughout, FP denotes the constant coefficient that remains after all displayed powers and logarithms of the endpoint cutoff are subtracted; for instance,

$$\mathrm{FP}\int_2^\infty f(t)\,\mathrm{d}t = \text{constant term of } \int_2^T f(t)\,\mathrm{d}t \text{ as } T \to \infty. \quad \text{(B.38)}$$

On the interval $(0, 1)$ the cutoffs are ϑ and $1 - \vartheta$. Finite parts are not invariant under a change of cutoff variable unless the corresponding conversion terms are kept; an example occurs in the evaluation of $\mathcal{P}_4$.

To extract the constant of a one-dimensional angular integral with moving endpoints, one uses three regions: the bulk, a lower endpoint region $w = et$ and an upper endpoint region $w = 1 - et$. The integrand, *including the Jacobian*, is expanded in each region, and its coefficient of $e^0 \log^0 e$ is kept; the desired constant is the sum of the three finite-part integrals. To justify this, split the original domain at fixed ϑ and $1 - \vartheta$, take $e \ll \vartheta$ first, and cancel the powers and logarithms of the overlaps before taking $\vartheta \to 0$; the constant that remains is the sum stated. This procedure keeps finite boundary contributions that an expansion of the bulk alone would miss. In symbolic software, a term such as $1/[e(t+1)]$ must be recognized as $e^{-1}/(t+1)$ before a Laurent coefficient is read off; reading a coefficient from an uncollected rational denominator can silently drop an endpoint term.

Elementary finite parts. With $l = \log w$ and $\bar{l} = \log(1-w)$, the nontrivial finite parts over $(0,1)$ needed below are (convergent integrals take their ordinary values):

integrand	FP $\int_0^1$	integrand	FP $\int_0^1$
$l\bar{l}$	$2-\zeta(2)$	$\bar{l}/w$	$-\zeta(2)$
$l\bar{l}/w$	$\zeta(3)$	$\bar{l}/(w-1)$	0
$l\bar{l}/(w-1)$	$-\zeta(3)$	$\bar{l}/(w-1)^2$	-1
$l\bar{l}/w^2$, $l\bar{l}/(w-1)^2$	$\zeta(2)-1$	$\bar{l}/(w-1)^3$	$\frac{1}{4}$
l	-1	l^2/w	0
l/w	0	l^2/w^2	-2
l/w^2	-1	$l^2/(w+1)$	$\frac{3}{2}\zeta(3)$
$l/(w-1)$	$\zeta(2)$	$l^2/(w+1)^2$	$\zeta(2)$
$l/(w-1)^2$	-1	$\bar{l}^2$, $\bar{l}^2/(w-1)$	2, 0
$l/(w+1)$	$-\frac{1}{2}\zeta(2)$	$\bar{l}^2/w$, $\bar{l}^2/w^2$	$2\zeta(3)$, $2\zeta(2)$
$l/(w+1)^2$	$-\log 2$	$1/(w+1)$, $1/(w+1)^2$	$\log 2$, $\frac{1}{2}$
$l/(w+1)^3$	$-\frac{1}{2}\log 2-\frac{1}{4}$	$1/w$, $1/(w-1)$	0, 0
		$1/w^2$, $1/(w-1)^2$	-1, -1
		$1/(w-1)^3$	$\frac{1}{2}$

For example,

$$\mathrm{FP}\int_0^1 \frac{\log w \log(1-w)}{w^2}\,\mathrm{d}w = \sum_{n=2}^{\infty}\frac{1}{n(n-1)^2} = \zeta(2)-1, \tag{B.39}$$

where the term $n=1$ of the expansion of $\log(1-w)$ gives a pure logarithmic divergence with zero finite constant. Likewise

$$\int_0^1 \frac{\log^2 w}{1+w}\,\mathrm{d}w = 2\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n^3} = \frac{3}{2}\zeta(3). \tag{B.40}$$

Repeated denominators follow by integration by parts, and pure powers and logarithms can be integrated directly; these procedures derive the whole table.

The period $\mathcal{R}_4$. On the angular simplex $r+t+w=1$ of three gaps, Eq. (A.33) simplifies to

$$W_4 = \frac{1}{r^2w^2}+\frac{2}{rtw}+\frac{1}{t^2}, \tag{B.41}$$

and the current kernel is $\Xi_4 W_4 = (r+w)W_4$. For a fixed outer gap r, let $\mathcal{M}_{\mathrm{out}}(r,e)$ be the integral of $\Xi_4 W_4$ over the other outer gap, and for a fixed middle gap r let $\mathcal{M}_{\mathrm{mid}}(r,e)$ be the integral over an outer gap. Elementary partial fractions give, with $\rho_r = e^{-1}-(1-r-e)^{-1}$ and $L_r = \log[(1-r-e)/e]$,

$$\begin{aligned}\mathcal{M}_{\mathrm{out}}(r,e) &= (1+r^{-1})\rho_r + \frac{(r+1)(r^2+1)}{r^2(1-r)}L_r,\\ \mathcal{M}_{\mathrm{mid}}(r,e) &= \frac{2\rho_r}{1-r}+\frac{4(r^2-r+1)}{r(1-r)^2}L_r+\frac{(1-r)(1-r-2e)}{r^2}.\end{aligned} \tag{B.42}$$

On the simplex $\log W_4 = 2\log[(1-r)/r]+2\log[(1-w)/w]-2\log t$, and $\partial_v W_4^v|_{v=1} = W_4\log W_4$. The symmetry between the two outer gaps then gives the exact one-dimensional reduction

$$\partial_v \mathcal{H}_{Q_4}(e,v)\big|_{v=1} = \int_e^{1-2e}\Big[4\mathcal{M}_{\mathrm{out}}(r,e)\log\frac{1-r}{r}-2\mathcal{M}_{\mathrm{mid}}(r,e)\log r\Big]\,\mathrm{d}r. \tag{B.43}$$

Its three regions are the bulk (at fixed r, renamed w), the lower region $r = et$ with $t \geq 1$, and the upper region $r = 1 - et$ with $t \geq 2$. With $l = \log w$ and $\bar{l} = \log(1 - w)$, the constant integrands are

$$B_4(w) = l\bar{l}\Big(4 + \frac{16}{w-1} - \frac{8}{(w-1)^2} - \frac{16}{w} - \frac{4}{w^2}\Big) + l\Big(-2 - \frac{8}{w-1} + \frac{4}{(w-1)^2} + \frac{8}{w} - \frac{2}{w^2}\Big) + \bar{l}^2\Big(-4 - \frac{16}{w-1} + \frac{8}{w} + \frac{4}{w^2}\Big) + \bar{l}\Big(\frac{8}{w-1} - \frac{4}{w}\Big), \tag{B.44}$$

$$L_4(t) = \Big(-8 + \frac{12}{t} + \frac{8}{t^2}\Big)\log t - 4, \qquad t \geq 1, \tag{B.45}$$

$$U_4(t) = \frac{16}{t}\log t \log(t-1) + \Big(8 - \frac{8}{t-1}\Big)\log t + \frac{8}{t}\log(t-1) + 4 - \frac{4}{t-1}, \qquad t \geq 2. \tag{B.46}$$

The finite-part table gives, group by group,

group of B_4	$l\bar{l}$	l	$\bar{l}^2$	$\bar{l}$
FP $\int_0^1$	$20 - 16\zeta(2) - 32\zeta(3)$	$-8\zeta(2)$	$-8 + 8\zeta(2) + 16\zeta(3)$	$4\zeta(2)$

so that

$$\mathrm{FP}\int_0^1 B_4(w)\,\mathrm{d}w = 12 - 12\zeta(2) - 16\zeta(3), \qquad \mathrm{FP}\int_1^\infty L_4(t)\,\mathrm{d}t = 4. \tag{B.47}$$

In the upper integral, the term with two logarithms is

$$J_U = \mathrm{FP}\int_2^\infty \frac{\log t \log(t-1)}{t}\,\mathrm{d}t = -\frac{\log^3 2}{3} - \log 2\,\mathrm{Li}_2(\tfrac{1}{2}) - \mathrm{Li}_3(\tfrac{1}{2}) = -\frac{7}{8}\zeta(3), \tag{B.48}$$

where the last step uses the values (B.56) at $\frac{1}{2}$. The remaining terms have the primitive

$$8t\log t - 4t + 8\,\mathrm{Li}_2(1-t) + 4\log^2 t + 8\,\mathrm{Li}_2(1/t) - 4\log(t-1), \tag{B.49}$$

whose constant at infinity is $-8\zeta(2)$ and whose value at $t = 2$ is $16\log 2 - 8$. Hence

$$\mathrm{FP}\int_2^\infty U_4(t)\,\mathrm{d}t = 8 - 8\zeta(2) - 14\zeta(3) - 16\log 2. \tag{B.50}$$

Adding the three regions gives

$$\mathcal{R}_4 = 24 - 20\zeta(2) - 30\zeta(3) - 16\log 2, \tag{B.51}$$

with no numerical reconstruction.

B.6 The trilogarithm lemma and the Spence–Kummer identity

Polylogarithms. For $|z| < 1$, $\mathrm{Li}_n(z) = \sum_{k\geq 1} z^k/k^n$, continued by $\mathrm{d}\,\mathrm{Li}_n(z)/\,\mathrm{d}z = \mathrm{Li}_{n-1}(z)/z$ with $\mathrm{Li}_1(z) = -\log(1-z)$. Real principal values are used for $z < 1$; where an argument exceeds one, the real part is indicated explicitly. The identities needed are

$$\mathrm{Li}_2(z) + \mathrm{Li}_2(1-z) = \zeta(2) - \log z \log(1-z), \tag{B.52}$$

$$\mathrm{Li}_2\Big(\frac{z}{z-1}\Big) = -\,\mathrm{Li}_2(z) - \frac{1}{2}\log^2(1-z), \tag{B.53}$$

$$\mathrm{Li}_n(z) + \mathrm{Li}_n(-z) = 2^{1-n}\,\mathrm{Li}_n(z^2), \tag{B.54}$$

$$\mathrm{Li}_3(z) + \mathrm{Li}_3(1-z) + \mathrm{Li}_3\Big(-\frac{z}{1-z}\Big) = \zeta(3) + \zeta(2)\log(1-z) - \frac{1}{2}\log z\log^2(1-z) + \frac{1}{6}\log^3(1-z). \tag{B.55}$$

The first two follow by differentiation and evaluation at an endpoint, the third from the power series, and the fourth by differentiation using the first two. At $z = \frac{1}{2}$ they give

$$\mathrm{Li}_2(\tfrac{1}{2}) = \frac{\zeta(2)}{2} - \frac{\log^2 2}{2}, \qquad \mathrm{Li}_3(\tfrac{1}{2}) = \frac{7}{8}\zeta(3) - \frac{\zeta(2)\log 2}{2} + \frac{\log^3 2}{6}. \tag{B.56}$$

The real-domain Spence–Kummer identity. For real $0 < x < y < 1$ define

$$\begin{aligned} f_1 &= \frac{x(1-y)^2}{y(1-x)^2}, & f_2 &= xy, & f_3 &= x/y, \\ f_4 &= \frac{x(1-y)}{y(1-x)}, & f_5 &= -\frac{x(1-y)}{1-x}, & f_6 &= \frac{1-y}{1-x}, \\ f_7 &= -\frac{1-y}{y(1-x)}, & f_8 &= x, & f_9 &= y. \end{aligned} \tag{B.57}$$

All positive arguments are smaller than one, and the two negative ones are on the real principal branch. The identity is

$$\mathrm{Li}_3(f_1)+\mathrm{Li}_3(f_2)+\mathrm{Li}_3(f_3)-2\sum_{i=4}^{9}\mathrm{Li}_3(f_i)+2\zeta(3) = \log^2 y\,\log\frac{1-y}{1-x}-2\zeta(2)\log y-\frac{1}{3}\log^3 y. \tag{B.58}$$

The branch choices are the real-domain specialization of the path formulation of Ref. [24].

A differential verification uses the five-term relation of the dilogarithm,

$$\mathrm{Li}_2(a)+\mathrm{Li}_2(b) = \mathrm{Li}_2(ab)+\mathrm{Li}_2\Big(\frac{a(1-b)}{1-ab}\Big)+\mathrm{Li}_2\Big(\frac{b(1-a)}{1-ab}\Big)+\log\frac{1-a}{1-ab}\log\frac{1-b}{1-ab}, \tag{B.59}$$

together with Eqs. (B.52) and (B.53); Eq. (B.59) itself follows by differentiation in a and b and evaluation at $a=0$. Differentiating Eq. (B.58) with respect to x and multiplying by $x(1-x)$ leaves the dilogarithm equality

$$\begin{aligned} &(1+x)\,\mathrm{Li}_2(f_1)+(1-x)\big[\mathrm{Li}_2(f_2)+\mathrm{Li}_2(f_3)\big]-2\big[\mathrm{Li}_2(f_4)+\mathrm{Li}_2(f_5)\big] \\ &\quad -2x\big[\mathrm{Li}_2(f_6)+\mathrm{Li}_2(f_7)\big]-2(1-x)\,\mathrm{Li}_2(x) = x\log^2 y. \end{aligned} \tag{B.60}$$

Substituting the rational f_i into Eq. (B.59), followed by Eqs. (B.52) and (B.53), verifies this equality. The integration constant is fixed by $x \to 0$, where Eq. (B.58) reduces exactly to the Landen relation (B.55) at $1-y$. This proves the real-domain identity without a numerically inferred special value.

At $x = \frac{1}{2}$, $y = \frac{2}{3}$ the nine arguments are

$$(f_1,\dots,f_9) = \Big(\frac{1}{3},\frac{1}{3},\frac{3}{4},\frac{1}{2},-\frac{1}{3},\frac{2}{3},-1,\frac{1}{2},\frac{2}{3}\Big). \tag{B.61}$$

Since $\mathrm{Li}_3(-1) = -\frac{3}{4}\zeta(3)$, Eq. (B.58) becomes

$$2\,\mathrm{Li}_3(\tfrac{1}{3})+\mathrm{Li}_3(\tfrac{3}{4})-4\,\mathrm{Li}_3(\tfrac{1}{2})-2\,\mathrm{Li}_3(-\tfrac{1}{3})-4\,\mathrm{Li}_3(\tfrac{2}{3})+\frac{7}{2}\zeta(3) = \frac{2}{3}\log^3\tfrac{2}{3}-2\zeta(2)\log\tfrac{2}{3}. \tag{B.62}$$

All its arguments are real and at most one, so no continuation across a branch cut of the trilogarithm is needed. The compendium's script `code/dual_d1/code/archived/LRI_thermal_closed_form/code/verify_closed_forms.py` also checks this specialization and the Landen eliminations that follow.

Lemma (a finite part with an exact zeta value). With the finite-part convention of Eq. (B.38),

$$J_L = \lim_{T\to\infty}\Big[\int_2^T \frac{\log(t-1)\log(t+1)}{t}\,\mathrm{d}t-\frac{1}{3}\log^3 T\Big] = -\frac{1}{3}\zeta(3). \tag{B.63}$$

Proof. Define the convergent integral

$$\Lambda(z) = \int_0^z \frac{\log(1-t)\log(1+t)}{t}\,\mathrm{d}t. \tag{B.64}$$

The substitution $t \mapsto 1/t$ in Eq. (B.63) gives the absolutely convergent representation

$$J_L = \Lambda(\tfrac{1}{2}) - \frac{1}{2}\log 2\,\mathrm{Li}_2(\tfrac{1}{4}) - \frac{1}{4}\mathrm{Li}_3(\tfrac{1}{4}) - \frac{1}{3}\log^3 2. \tag{B.65}$$

For an explicit reduction introduce

$$\Theta(z) = \int_0^z \frac{\log^2(1-t)}{t}\,\mathrm{d}t = \log z \log^2(1-z) + 2\log(1-z)\,\mathrm{Li}_2(1-z) - 2\,\mathrm{Li}_3(1-z) + 2\zeta(3); \tag{B.66}$$

the equality follows by differentiation and the limit $z \to 0$. Squaring $\log(1-t^2) = \log(1-t)+\log(1+t)$ gives

$$\Lambda(z) = \frac{1}{4}\Theta(z^2) - \frac{1}{2}\Theta(z) - \frac{1}{2}\Theta(-z). \tag{B.67}$$

In $\Theta(-\frac{1}{2})$ the real parts of Eq. (B.66) are taken. Equivalently, for $a > 0$ the manifestly real formula

$$\begin{aligned}\Theta(-a) = &-\log a \log^2(1+a) - 2\zeta(2)\log(1+a) - 2\log(1+a)\,\mathrm{Li}_2(-a)\\ &- 2\,\mathrm{Li}_3\Big(\frac{1}{1+a}\Big) + \frac{1}{3}\log^3(1+a) + 2\zeta(3)\end{aligned} \tag{B.68}$$

holds; it follows from Eq. (B.52) and from $\mathrm{Re}\,\mathrm{Li}_3(\beta) = \mathrm{Li}_3(1/\beta) - \frac{1}{6}\log^3\beta + 2\zeta(2)\log\beta$ for real $\beta > 1$. Let

$$\Pi = \log^3 3 - 3\log^2 3\log 2 + 6\log 3\log^2 2 - 5\log^3 2 - 6\zeta(2)\log 3 + 6\zeta(2)\log 2. \tag{B.69}$$

Substituting Eq. (B.66) and the formula for $\Theta(-a)$ into Eq. (B.65), and using Eqs. (B.52) to (B.54), gives

$$J_L = \mathrm{Li}_3(\tfrac{2}{3}) - \frac{1}{2}\mathrm{Li}_3(\tfrac{3}{4}) - \mathrm{Li}_3(-\tfrac{1}{2}) - \frac{3}{2}\zeta(3) - \frac{\Pi}{6}. \tag{B.70}$$

The dilogarithms are eliminated in favour of $\mathrm{Li}_2(\frac{1}{3})$,

$$\begin{gathered}\mathrm{Li}_2(-\tfrac{1}{2}) = -\mathrm{Li}_2(\tfrac{1}{3}) - \tfrac{1}{2}\log^2\tfrac{3}{2}, \qquad \mathrm{Li}_2(\tfrac{1}{4}) = \zeta(2) - \log^2 2 - 2\,\mathrm{Li}_2(\tfrac{1}{3}) - \log^2\tfrac{3}{2},\\ \mathrm{Li}_2(\tfrac{3}{4}) = \zeta(2) - \mathrm{Li}_2(\tfrac{1}{4}) + 2\log 2\,\log\tfrac{3}{4},\end{gathered} \tag{B.71}$$

and all coefficients of $\mathrm{Li}_2(\frac{1}{3})$ cancel. Next, Eq. (B.55) at $z = \frac{1}{3}$ and $z = \frac{1}{4}$ eliminates $\mathrm{Li}_3(\frac{1}{3})$ and $\mathrm{Li}_3(-\frac{1}{3})$ from Eq. (B.62). With $\mathrm{Li}_3(\frac{1}{4}) = 4[\mathrm{Li}_3(\frac{1}{2}) + \mathrm{Li}_3(-\frac{1}{2})]$ and Eq. (B.56) the result is

$$\mathrm{Li}_3(\tfrac{2}{3}) - \frac{1}{2}\mathrm{Li}_3(\tfrac{3}{4}) - \mathrm{Li}_3(-\tfrac{1}{2}) = \frac{7}{6}\zeta(3) + \frac{\Pi}{6}. \tag{B.72}$$

Substitution in Eq. (B.70) gives $J_L = -\zeta(3)/3$. Every functional identity is used with its branch specified, and the final cancellation is symbolic, not an integer-relation fit. □

B.7 Closed forms of the four- and five-insertion periods

The four-insertion period $\mathcal{P}_4$. The decomposition (B.41) allows the angular current kernel to be integrated directly. Put

$$L_e = \log\frac{1-2e}{e}, \qquad z_e = \frac{e}{1-e}, \qquad 0 < e < \frac{1}{3}. \tag{B.73}$$

The result is

$$\mathcal{H}_{Q_4}(e) = \frac{2L_e+1}{e} - \frac{4L_e}{1-e} - 2(1-e)L_e - 2 - 3e + 4\log^2 z_e - 4\zeta(2) + 8\,\mathrm{Li}_2(z_e), \tag{B.74}$$

with the small-e expansion

$$\mathcal{H}_{Q_4}(e) = \frac{-2\log e + 1}{e} + 4\log^2 e + 6\log e - 6 - 4\zeta(2) + O(e\log e). \tag{B.75}$$

The finite part that corresponds to the first line of Eq. (B.36) is therefore

$$\mathcal{P}_4 = \lim_{\vartheta\to 0}\Big[\int_{\vartheta}^{1/3} \frac{\mathcal{H}_{Q_4}(e)}{e}\,\mathrm{d}e + \frac{2\log\vartheta+1}{\vartheta} + \frac{4}{3}\log^3\vartheta + 3\log^2\vartheta - \big(6+4\zeta(2)\big)\log\vartheta\Big]. \tag{B.76}$$

An explicit antiderivative evaluates it. Change variables to $z = e/(1-e)$, so that the upper endpoint is $z = \frac{1}{2}$, and write $\lambda(z) = \log(1-z) - \log z$. The integrand becomes

$$A_P(z)\lambda(z) + B_P(z) + \frac{4\log^2 z - 4\zeta(2) + 8\,\mathrm{Li}_2(z)}{z(1+z)}, \tag{B.77}$$

with

$$A_P(z) = \frac{2}{1+z} + \frac{2}{(1+z)^2} - \frac{6}{z} + \frac{2}{z^2}, \qquad B_P(z) = \frac{2}{1+z} - \frac{3}{(1+z)^2} - \frac{2}{z} + \frac{1}{z^2}. \tag{B.78}$$

A primitive is $K_P = K_{P,\le 2} + K_{P,3}$, with

$$\begin{aligned} K_{P,\le 2}(z) &= 2\Big[(\log 2 - \log z)\log(1+z) - \mathrm{Li}_2\Big(\frac{1+z}{2}\Big) - \mathrm{Li}_2(-z)\Big] - 2\lambda(z)\Big(\frac{1}{z} + \frac{1}{1+z}\Big) \\ &\quad + \frac{1}{z} + \frac{3}{1+z} + 6\,\mathrm{Li}_2(z) + 3\log^2 z - 6\log z + 3\log(1-z^2), \end{aligned} \tag{B.79}$$

$$\begin{aligned} K_{P,3}(z) &= \frac{4}{3}\log^3 z - 4\log^2 z\log(1+z) - 8\log z\,\mathrm{Li}_2(-z) + 8\,\mathrm{Li}_3(-z) + 8\,\mathrm{Li}_3(z) \\ &\quad - 8\,\mathrm{Li}_2(z)\log(1+z) - 8\Lambda(z) - 4\zeta(2)\log\frac{z}{1+z}, \end{aligned} \tag{B.80}$$

which is verified directly by differentiation, with $\Lambda'(z) = \log(1-z)\log(1+z)/z$ (Λ as in Appendix B.6). The lower endpoint is $z = \vartheta/(1-\vartheta)$, not $z = \vartheta$, and there

$$K_P\Big(\frac{\vartheta}{1-\vartheta}\Big) = \frac{2\log\vartheta+1}{\vartheta} + \frac{4}{3}\log^3\vartheta + 3\log^2\vartheta - \big(6+4\zeta(2)\big)\log\vartheta + 6 - 2\,\mathrm{Li}_2(\tfrac{1}{2}) + o(1). \tag{B.81}$$

Consequently

$$\mathcal{P}_4 = K_P(\tfrac{1}{2}) - 6 + 2\,\mathrm{Li}_2(\tfrac{1}{2}). \tag{B.82}$$

The representation (B.65), Euler reflection and duplication reduce this to

$$\mathcal{P}_4 = 4\zeta(2) - 2 + 3\log 3 + \log^2 2 + 2\,\mathrm{Li}_2(-\tfrac{1}{2}) - 8J_L, \tag{B.83}$$

and the lemma (B.63) gives

$$\mathcal{P}_4 = 4\zeta(2) + \frac{8}{3}\zeta(3) - 2 + 3\log 3 + \log^2 2 + 2\,\mathrm{Li}_2(-\tfrac{1}{2}). \tag{B.84}$$

The logarithms and the dilogarithm are allowed in this regulator-dependent period; they cancel from the stability invariants (Appendix B.2).

The five-insertion period $\mathcal{S}_5$: the middle vertex. On the angular simplex of five kinks, fix the first and last kinks at 0 and 1 and write the intermediate positions as $0 < \alpha < \xi < \beta < 1$, with every consecutive gap at least e. Let $w = \beta - \alpha$ and $P_0(x) = x^2(1-x)^2$. Then

$$W_5 = \frac{w^2}{P_0(\alpha)P_0(\beta)}\,\frac{P_0(\xi)}{(\xi-\alpha)^2(\beta-\xi)^2}. \tag{B.85}$$

With $\xi = \alpha + t$, the rational identity

$$\frac{P_0(\alpha+t)}{t^2(w-t)^2} = 1 + \frac{\mu_1}{t} + \frac{\mu_2}{w-t} + \frac{\nu_1}{t^2} + \frac{\nu_2}{(w-t)^2} \tag{B.86}$$

has the coefficients

$$\nu_1 = \frac{P_0(\alpha)}{w^2}, \quad \nu_2 = \frac{P_0(\beta)}{w^2}, \quad \mu_1 = \frac{P_0'(\alpha)}{w^2} + \frac{2P_0(\alpha)}{w^3}, \quad \mu_2 = -\frac{P_0'(\beta)}{w^2} + \frac{2P_0(\beta)}{w^3}. \tag{B.87}$$

The t integration runs from e to $w-e$ and leaves

$$\frac{w^2(w-2e) + \mathcal{N}_1 \log[(w-e)/e] + \mathcal{N}_2\big[e^{-1} - (w-e)^{-1}\big]}{P_0(\alpha)P_0(\beta)}, \tag{B.88}$$

where $\mathcal{N}_2 = P_0(\alpha) + P_0(\beta)$ and $\mathcal{N}_1 = P_0'(\alpha) - P_0'(\beta) + 2\mathcal{N}_2/w$. The integration over α, from e to $1-e-w$, is again elementary, and leaves

$$\mathcal{H}_{A_5}(e) = \int_{2e}^{1-2e} I_5(w,e)\,\mathrm{d}w. \tag{B.89}$$

With

$$\begin{gathered} L_0 = \log\frac{w-e}{e}, \qquad \rho_0 = \frac{1}{e} - \frac{1}{w-e}, \qquad \rho_- = \frac{1}{e} - \frac{1}{1-e-w}, \qquad \rho_+ = \frac{1}{w+e} - \frac{1}{1-e}, \\ L_- = \log\frac{1-e-w}{e}, \qquad L_+ = \log\frac{1-e}{w+e}, \end{gathered} \tag{B.90}$$

the exact remaining kernel is

$$\begin{aligned} I_5(w,e) = {}& \rho_-\Big[\frac{2(w-2e)}{(w-1)^2} - \frac{4L_0}{w-1} + 2\rho_0\Big] + \rho_+\Big[\frac{2(w-2e)}{(w+1)^2} - \frac{4L_0}{w+1} + 2\rho_0\Big] \\ & + L_-\Big[\frac{4(w^2-3w+1)(w-2e)}{w(w-1)^3} + \Big(-\frac{8}{w} + \frac{8}{w^2(w-1)^2}\Big)L_0 + 4\rho_0\Big] \\ & + L_+\Big[\frac{4(w^2+3w+1)(w-2e)}{w(w+1)^3} + \Big(-\frac{8}{w} - \frac{8}{w^2(w+1)^2}\Big)L_0 + 4\rho_0\Big]. \end{aligned} \tag{B.91}$$

Its equality with the α integral of Eq. (B.88) can be checked by rational partial fractions in α; the compendium's code also compares both sides numerically at finite positive cutoffs, before any subtraction or expansion. This makes the reduction reproducible without a sector-decomposition package.

The five-insertion period $\mathcal{S}_5$: bulk and endpoint finite parts. The three regions are obtained by expanding $I_5(w,e)$, $e\,I_5(et,e)$ and $e\,I_5(1-et,e)$ and keeping the coefficient of $e^0\log^0 e$. The lower and upper ranges begin at $t=2$, because the middle separation satisfies $w \geq 2e$ and the outer complement $1-w \geq 2e$. With $l = \log w$ and $\bar{l} = \log(1-w)$, the bulk integrand is

$$\begin{aligned} B_5(w) = {}& l^2\Big(\frac{16}{w+1} + \frac{8}{(w+1)^2} - \frac{8}{w} + \frac{8}{w^2}\Big) + l\bar{l}\Big(-\frac{16}{w-1} + \frac{8}{(w-1)^2} + \frac{8}{w} + \frac{8}{w^2}\Big) \\ & + l\Big(\frac{4}{w+1} - \frac{4}{(w+1)^2} + \frac{4}{(w+1)^3} - \frac{4}{(w-1)^2}\Big) + \bar{l}\Big(\frac{4}{w-1} - \frac{4}{(w-1)^2} - \frac{4}{(w-1)^3} - \frac{4}{w}\Big) \\ & - 6 - \frac{2}{w+1} + \frac{4}{(w+1)^2} + \frac{6}{w-1} - \frac{4}{(w-1)^2} + \frac{2}{(w-1)^3} - \frac{4}{w} - \frac{6}{w^2}. \end{aligned} \tag{B.92}$$

The lower region $w = et$ gives

$$L_5(t) = -\frac{8}{t}\log(t-1)\log(t+1) + \Big(4 - \frac{4}{t+1} - \frac{8}{t}\Big)\log(t-1) + \Big(-4 + \frac{4}{t-1}\Big)\log(t+1) - 4 + \frac{4}{t-1}. \tag{B.93}$$

The term $-4\log(t-1)/(t+1)$ is essential; omitting it changes the finite part. The upper region $w = 1-et$ gives

$$U_5(t) = \Big(4 - \frac{12}{t} - \frac{8}{t^2}\Big)\log(t-1) - 6 + \frac{10}{t-1} - \frac{8}{t}. \tag{B.94}$$

These are the coefficients of $e^0\log^0 e$ after the Jacobians of the endpoint regions are included.

The finite-part table evaluates the bulk integral group by group,

group of B_5	l^2	$l\bar{l}$	l	$\bar{l}$	rational
FP $\int_0^1$	$24\zeta(3)+8\zeta(2)-16$	$24\zeta(3)+16\zeta(2)-16$	$-2\zeta(2)+2\log 2+3$	$4\zeta(2)+3$	$7-2\log 2$

so that

$$\mathrm{FP}\int_0^1 B_5(w)\,\mathrm{d}w = 48\zeta(3)+26\zeta(2)-19. \tag{B.95}$$

For the upper region a primitive is

$$(4t-2)\log(t-1)-10t+4-6\log^2 t-12\,\mathrm{Li}_2(1/t)+\frac{8}{t}\log(t-1); \tag{B.96}$$

its finite constant at infinity is zero and its value at $t=2$ is $-16-6\zeta(2)$. Thus

$$\mathrm{FP}\int_2^\infty U_5(t)\,\mathrm{d}t = 16+6\zeta(2). \tag{B.97}$$

The lower region contains the only nontrivial weight-three endpoint integral, $-8J_L$, from its first term. For the remaining terms one primitive is

$$\begin{aligned} K_L(t) = {}& 4\big[(t-1)\log(t-1)-(t+1)\log(t+1)+2\big]-2\log^2(t+1)-4\,\mathrm{Li}_2\Big(\frac{2}{t+1}\Big) \\ & +4\log 2\log(t-1)-4\,\mathrm{Li}_2\Big(-\frac{t-1}{2}\Big)-4\log^2 t-8\,\mathrm{Li}_2(1/t)-4t+4\log(t-1), \end{aligned} \tag{B.98}$$

whose finite part at infinity is $4\zeta(2)+2\log^2 2$. Its value at $t=2$, followed by Euler reflection and the inversion formula of the dilogarithm at negative argument, gives

$$\mathrm{FP}\int_2^\infty L_5(t)\,\mathrm{d}t = -8J_L+12\zeta(2)+12\log 3+4\log^2 2+8\,\mathrm{Li}_2(-\tfrac{1}{2}). \tag{B.99}$$

Combining Eqs. (B.95) to (B.99) with the lemma (B.63) gives

$$\mathcal{S}_5 = 44\zeta(2)+\frac{152}{3}\zeta(3)-3+12\log 3+4\log^2 2+8\,\mathrm{Li}_2(-\tfrac{1}{2}), \tag{B.100}$$

and subtracting four times Eq. (B.84) gives Eq. (B.6) without any numerical inference. With Eq. (B.51), this completes the closed-form evaluation of the three periods.

B.8 Counterterms and the inputs b, c and f

The period evaluation is one part of the calculation of the exponents. This subsection records the response functions, finite terms and counterterm insertions that attach the periods to the beta inputs.

Projected responses. The two essential counterterms can be extracted by projecting the finite-interval ordered exponential onto the odd spin-flip vertex and onto the even current operator. At the endpoint the odd projection can equivalently be neutralized by a compensating vertex at infinity, with its free two-point normalization divided out; the current projection is the corresponding normalized dipole response. At finite $\bar{\delta}$ the calculation uses the regulated finite-interval responses and their kernel derivatives; it does not assume a charged conformal limit of the auxiliary theory. Displaying the terms needed, the odd and even responses are

$$\begin{aligned} \Gamma_g &= g+\Gamma_{gh}\,gh+\Gamma_{gh^2}\,gh^2+\mathcal{C}_3\,g^3+\Gamma_{g^3h}\,g^3h+\mathcal{C}_5\,g^5+\cdots, \\ \Gamma_h &= h+\sqrt{2}\,b_0\,g^2Q_2+\Gamma_{hg^2}\,hg^2+\sqrt{2}\,b_0\,g^4\mathcal{K}_4+\cdots, \end{aligned} \tag{B.101}$$

with the normalized responses of Eq. (A.37), not raw moments. Contractions involving the current coupling are generated by differentiating the exact current-resummed charges, keeping the endpoint factors of Eq. (A.18). At $b_0 = 1$, $\bar{\delta} = 0$ the mixed coefficients needed are

$$\begin{aligned}
\Gamma_{gh} &= -2\sqrt{2}\,(\ell+1),\\
\Gamma_{hg^2} &= 4\ell^2 + 10\ell + 10 - 4\zeta(2),\\
\Gamma_{gh^2} &= 4\ell^2 + 10\ell + 12 - 2\zeta(2),\\
\frac{\Gamma_{g^3h}}{\sqrt{2}} &= -\frac{20}{3}\ell^3 - 28\ell^2 + \big[-48 + 24\zeta(2)\big]\ell + \Gamma_{40}, \qquad \Gamma_{40} = -62 + 36\zeta(2) + 30\zeta(3) + 16\log 2.
\end{aligned} \tag{B.102}$$

These finite constants belong to the kink-only prescription, not to one that regulates current–kink separations independently. For example, Γ_{40} multiplies the counterterm of h at second order in the renormalized fugacity g_R and contributes to the fifth-order single logarithm. (The compendium writes F_G, F_H, F_{gh}, … for these responses and coefficients.)

Lower counterterms. The two-kink responses of Eq. (A.35) give, at $h_R = 0$,

$$h_{\text{bare}} = \sqrt{2}\,g_R^2\ell + O(g_R^4), \qquad g_{\text{bare}} = g_R + g_R^3(2\ell^2 - \ell) + O(g_R^5), \tag{B.103}$$

where g_R and h_R are the renormalized couplings. The odd subtraction can be checked directly: $\mathcal{C}_3 + \Gamma_{gh}\,(\sqrt{2}\ell) + (2\ell^2 - \ell)$ has no logarithm. Charge differentiation of the one-pair integral generates only higher powers of ℓ, not a new charge-dependent single logarithm, and its finite-$\bar{\delta}$ derivative likewise gives the cubic logarithm of Appendix B.9. This is the one-pair origin of the absent terms stated below Eq. (B.2).

Fourth and fifth insertions. At $h_R = 0$ and $\bar{\delta} = 0$ the counterterms are

$$\begin{aligned}
h_{\text{bare}} &= \sqrt{2}\,g_R^2\ell + \sqrt{2}\,g_R^4\Big(\frac{4}{3}\ell^3 - 2\ell^2 - 4\ell\Big) + \cdots,\\
g_{\text{bare}} &= g_R + g_R^3(2\ell^2 - \ell) + g_R^5\Big[\frac{10}{3}\ell^4 - \frac{22}{3}\ell^3 - \Big(\frac{13}{2} + 8\zeta(2)\Big)\ell^2 - b\,\ell\Big] + \cdots.
\end{aligned} \tag{B.104}$$

To make the subtraction reproducible, set

$$h_2^{\text{ct}} = \sqrt{2}\,\ell, \qquad g_3^{\text{ct}} = 2\ell^2 - \ell, \qquad h_4^{\text{ct}} = \sqrt{2}\Big(\frac{4}{3}\ell^3 - 2\ell^2 - 4\ell\Big). \tag{B.105}$$

Then the coefficient of g_R^5 in the bare fugacity is

$$g_5^{\text{ct}} = -\Big[\mathcal{C}_5 + \Gamma_{gh}(h_4^{\text{ct}} + g_3^{\text{ct}}h_2^{\text{ct}}) + 3\,\mathcal{C}_3\,g_3^{\text{ct}} + \Gamma_{gh^2}(h_2^{\text{ct}})^2 + \Gamma_{g^3h}\,h_2^{\text{ct}}\Big]_{\text{logs}}, \tag{B.106}$$

where $[\,\cdot\,]_{\text{logs}}$ keeps the terms of order ε^0 with a positive power of ℓ. All quantities on the right are specified in Eqs. (B.9), (B.11) and (B.102). The coefficient of the single logarithm gives

$$b = -21 + 38\zeta(2) + 60\zeta(3) + 32\log 2 - \mathcal{S}_5. \tag{B.107}$$

The fourth-insertion charge counterterm gives $c = -16$ (the value $h_4 = 4$ of Appendix A.9).

The charge derivative that enters the trace. Write the fourth-order auxiliary charge counterterm as $h_4^{\text{ct}} = \sqrt{2}\,b_0\,c_4^{\text{ct}}(v_0)$ with $v_0 = b_0^2$. Its derivative at $v_0 = 1$ is

$$c_4^{\text{ct}\,\prime} = -\frac{8}{3}\ell^4 + \frac{16}{3}\ell^3 + (8 + 4\zeta(2))\ell^2 + \big[4\mathcal{P}_4 + \mathcal{R}_4 - 8 + 4\zeta(2) + 8\zeta(3)\big]\ell. \tag{B.108}$$

Consequently

$$f = 16\mathcal{P}_4 + 4\mathcal{R}_4 - 32 + 16\zeta(2) + 32\zeta(3). \tag{B.109}$$

The lower endpoint integral needed in this derivative is

$$\begin{aligned} J_{\mathrm{end}}(v) &= 2\int_0^1 r^{1-2v}\left[(1-r)\log(1-r) + r\log r\right]\mathrm{d}r, \\ J_{\mathrm{end}}(1) &= -2\zeta(2), \qquad J'_{\mathrm{end}}(1) = -4\big(\zeta(2)+\zeta(3)\big), \end{aligned} \tag{B.110}$$

and it gives

$$\partial_v \Gamma_{hg^2}\big|_{v=1} = -\frac{8}{3}\ell^3 + 2\ell^2 + 24\ell + 44 - 12\zeta(2) - 8\zeta(3). \tag{B.111}$$

Equation (B.108) can be reconstructed algebraically. Let $Q_2 \to -\ell - 1$ and $\partial_v Q_2 \to \ell^2 - 2$ denote the ε^0 parts of the one-pair current moment and of its charge derivative, and $\partial_v g_3^{\mathrm{ct}} = -\frac{4}{3}\ell^3 + 3\ell^2 + 4\zeta(2)\ell$. Differentiating Eq. (B.34) at $k_F = 4 - 4v = 0$ gives

$$\partial_v \mathcal{M}_{Q_4} = \frac{\partial_v \mathcal{A}_{Q_4}}{q^2(q+1)} + \frac{4(2q+1)\,\mathcal{A}_{Q_4}}{q^3(q+1)^2}, \tag{B.112}$$

where $\mathcal{M}_{Q_4}$ is the Mellin transform of Q_4. Its poles give the ε^0 part of $\partial_v Q_4$ through Eq. (B.35). Subtracting the derivative of $Q_2 Z_2$ forms $\partial_v \mathcal{K}_4$; with $Z_2 = \int_\varepsilon^1 (1-r) r^{-2v}\,\mathrm{d}r$ and $Q_2 = \int_\varepsilon^1 (1-r) r^{1-2v}\,\mathrm{d}r$ at $\bar\delta = 0$, the ε^0 part of $\partial_v(Q_2 Z_2)$ at $v = 1$ is $2\ell^3 - 8\ell$. Then

$$c_4^{\mathrm{ct}\,\prime} = -\partial_v \mathcal{K}_4\big|_{\mathrm{logs}} - 2(\partial_v Q_2)\, g_3^{\mathrm{ct}} - 2Q_2\, \partial_v g_3^{\mathrm{ct}} - (\partial_v \Gamma_{hg^2})\,\ell + \Gamma_{hg^2}\,\ell^2. \tag{B.113}$$

Equations (B.36) and (B.112) fix every term of this expression; the compendium checks it symbolically in its code.

B.9 Charge and finite-$\bar\delta$ derivatives at three insertions: a and p

An incomplete-beta representation. For three kinks with gaps r and t, take the kernel $r^{-2p_{12}} t^{-2p_{23}} (r+t)^{2p_{13}}$ with general powers (the subscripts name the kink pairs). Its Mellin integral is

$$\mathcal{M}_3(q; p_{12}, p_{23}, p_{13}) = \frac{\mathrm{B}_{1/2}(q+1-2p_{12},\, 1-2p_{23}) + \mathrm{B}_{1/2}(q+1-2p_{23},\, 1-2p_{12})}{q(q+k_3)(q+k_3+1)}, \tag{B.114}$$

with $k_3 = 2 - 2p_{12} - 2p_{23} + 2p_{13}$ and the incomplete beta function $\mathrm{B}_x(a,b) = \int_0^x y^{a-1}(1-y)^{b-1}\,\mathrm{d}y$, continued meromorphically from its domain of convergence. The formula follows by writing $r = Ry$ and $t = R(1-y)$, splitting at $y = \frac{1}{2}$ and doing the radial integral. The kernel W_3 at $v = 1$, $\bar\delta = 0$ corresponds to $p_{12} = p_{23} = p_{13} = 1$.

The charge derivative and a. The charge derivative varies all three powers together. The normalized three-kink response then has

$$\partial_v \mathcal{C}_3\big|_{v=1} = -\frac{8}{3}\ell^3 - 3\ell^2 + (4 - 4\zeta(2))\ell + 18 - 14\zeta(2) - 15\zeta(3) - 8\log 2 + \cdots. \tag{B.115}$$

Combining it with the lower mixed response and counterterms gives

$$a = -4\zeta(2). \tag{B.116}$$

Equation (B.114) is a compact exact representation from which the logarithmic and finite terms of Eq. (B.115) can be regenerated.

The finite-tail deformation of the kernel and p. For separated charges the Gaussian weight expands as

$$\exp\Big[\frac{2v}{\bar\delta}\sum_{i<j}q_iq_j(r_{ij}^{\bar\delta}-1)\Big]=W^v\Big[1+v\,\bar\delta\sum_{i<j}q_iq_j\log^2 r_{ij}+O(\bar\delta^2)\Big], \tag{B.117}$$

where r_{ij} is the separation of kinks i and j and W^v the weight at $\bar\delta=0$. Thus the three-kink Mellin derivative is

$$\partial_{\bar\delta}\mathcal{M}_3|_{\bar\delta=0}=\frac{1}{4}(\partial_{p_{13}}^2-\partial_{p_{12}}^2-\partial_{p_{23}}^2)\mathcal{M}_3\big|_{p_{12}=p_{23}=p_{13}=1}. \tag{B.118}$$

The positive sign belongs to the like-charge outer pair, the negative signs to the two neighbouring opposite-charge pairs. Expanding Eq. (B.118) and normalizing by the two-kink moment gives

$$\begin{gathered}\partial_{\bar\delta}\mathcal{C}_3|_{\bar\delta=0}=-\ell^4-\ell^3-2\zeta(2)\ell^2+(p-4)\ell+O(1),\qquad \partial_{\bar\delta}\Gamma_{gh}|_{\bar\delta=0}=\sqrt{2}\,(2-\ell^2),\\ p=8-2\zeta(2)-7\zeta(3)-8\log 2.\end{gathered} \tag{B.119}$$

The $\mathrm{B}_{1/2}$ formula, the explicit differential operator and the lower moments fix this single logarithm independently of any fit to thermal exponents. The compendium's three-vertex residue script evaluates these derivatives as a separate check.

The finite-$\bar\delta$ current counterterm at order g_R^2 is

$$h_{\mathrm{bare}}|_{\bar\delta g_R^2}=-\frac{\sqrt{2}}{3}\,\bar\delta\,g_R^2\ell^3. \tag{B.120}$$

Substitution into the odd response gives

$$g_{\mathrm{bare}}|_{\bar\delta g_R^3}=\bar\delta\,g_R^3\Big[-\frac{1}{3}\ell^4+\frac{5}{3}\ell^3+2\zeta(2)\ell^2-p\,\ell\Big]. \tag{B.121}$$

This auxiliary fugacity is not yet the essential wall fugacity. Multiplication by the factor of Eq. (A.22) changes it to

$$Y_{\mathrm{bare}}|_{\bar\delta g_R^3}=\bar\delta\,g_R^3\Big[-\frac{4}{3}\ell^4+\frac{13}{6}\ell^3+2\zeta(2)\ell^2-p\,\ell\Big]. \tag{B.122}$$

The difference in the terms ℓ^4 and ℓ^3 is the finite-$\bar\delta$ matching, and it is indispensable.

B.10 Analytic three-kink and endpoint inputs

This subsection gives the analytic inputs of Appendix B.3. These reductions of lower inputs do not assume the higher-kink identity (B.24) that the certificate proves. Here $\partial_\eta A_3$ denotes the integrated first source derivative of the raw odd three-kink moment, not the derivative of an already normalized CFT correlator; the normalization and derivative factors of the subtraction recurrence must still be applied. At $\bar\delta=0$ the endpoint factors of Eq. (A.18) are

$$U_n^{\mathrm{odd}}(\eta)=\prod_i[X_i(1-X_i)]^{\eta q_i},\qquad U_n^{\mathrm{even}}(\eta)=\prod_i[X_i/(1-X_i)]^{\eta q_i}, \tag{B.123}$$

and

$$\partial_\eta A_3=\partial_\eta\int W_3^vU_3^{\mathrm{odd}}\Big|_{\eta=0},\qquad \partial_\eta Q_4=\partial_\eta\int \Xi_4W_4^vU_4^{\mathrm{even}}\Big|_{\eta=0}, \tag{B.124}$$

with the integrals over $\Omega_3(\varepsilon)$ and $\Omega_4(\varepsilon)$; the endpoints of the interval are not cut. The charge-curvature input has the convergent representation

$$A=-4\int_0^{1/2}\frac{\log^2(1-t)}{t^2(1-t)^2}\,\mathrm{d}t=-8-4\zeta(2)-2\zeta(3)+16\log 2, \tag{B.125}$$

obtained by partial fractions and integration by parts.

The mixed three-kink coefficient E. The representation (B.18) reads

$$E = p - 4\,\mathsf{I}_3 + 8\,\mathsf{I}_{12}, \qquad \mathsf{I}_3 = \int_0^{1/2} \frac{\log^3(1-y)}{y^2(1-y)^2}\,\mathrm{d}y, \qquad \mathsf{I}_{12} = \int_0^{1/2} \frac{\log y \log^2(1-y)}{y^2(1-y)^2}\,\mathrm{d}y, \tag{B.126}$$

and it can be integrated exactly. The result is the expression of E in Eq. (B.16).

Proof. Put $t = y/(1-y)$. Then $0 < t < 1$, $\log(1-y) = -\log(1+t)$, and the rational measure becomes $(1 + 2/t + 1/t^2)\,\mathrm{d}t$. Define

$$\mathsf{B}_3 = \int_0^1 \frac{\log^3(1+t)}{t}\,\mathrm{d}t, \qquad \mathsf{B}_{12} = \int_0^1 \frac{\log t \log^2(1+t)}{t}\,\mathrm{d}t. \tag{B.127}$$

The elementary reductions by integration by parts are

$$\begin{aligned} \int_0^1 \Big(1 + \frac{2}{t} + \frac{1}{t^2}\Big) \log^3(1+t)\,\mathrm{d}t &= -6\log^2 2 + 12\log 2 - 6 + \frac{3}{4}\zeta(3) + 2\mathsf{B}_3, \\ \int_0^1 \Big(1 + \frac{2}{t} + \frac{1}{t^2}\Big) \log t \log^2(1+t)\,\mathrm{d}t &= 2\mathsf{B}_{12} + 2\zeta(2) - 4\log^2 2 + 8\log 2 - 6 - \frac{3}{2}\zeta(3). \end{aligned} \tag{B.128}$$

Thus

$$E = -16 + 14\zeta(2) - 22\zeta(3) - 8\log^2 2 + 8\log 2 + 16\mathsf{B}_{12} - 8\mathsf{B}_3. \tag{B.129}$$

To evaluate B_3, set $u = (1+t)^{-1}$ and expand $(1-u)^{-1}$. A primitive needed for the logarithmic part is

$$\int \frac{\log^3 u}{1-u}\,\mathrm{d}u = -\log^3 u \log(1-u) - 3\log^2 u\,\mathrm{Li}_2(u) + 6\log u\,\mathrm{Li}_3(u) - 6\,\mathrm{Li}_4(u). \tag{B.130}$$

Evaluation between $u = \frac{1}{2}$ and 1, with Eq. (B.56), gives

$$\mathsf{B}_3 = 6\zeta(4) - 6\,\mathrm{Li}_4(\tfrac{1}{2}) - \frac{21}{4}\zeta(3)\log 2 + \frac{3}{2}\zeta(2)\log^2 2 - \frac{1}{4}\log^4 2. \tag{B.131}$$

For B_{12} introduce

$$\mathsf{K}(\alpha,\beta) = \int_0^1 t^{\alpha-1}(1+t)^{\beta}\,\mathrm{d}t. \tag{B.132}$$

Splitting the complete integral over $(0,\infty)$ at one and inverting t in the second half proves, first in a common domain of convergence and then by meromorphic continuation,

$$\mathsf{K}(\alpha,\beta) + \mathsf{K}(-\alpha-\beta,\beta) = \mathrm{B}(\alpha,-\alpha-\beta), \tag{B.133}$$

with B Euler's beta function. The cubic regular part of the right side, after its pole, is

$$\frac{7}{4}\zeta(4)(\alpha^2\beta + \alpha\beta^2) + \zeta(4)\beta^3, \tag{B.134}$$

which follows from the expansion

$$\mathrm{B}(\alpha,-\alpha-\beta) = \frac{\beta}{\alpha(\alpha+\beta)} \exp \sum_{n\geq 2} \frac{(-1)^n\zeta(n)}{n}\big[\alpha^n + (-\alpha-\beta)^n - (-\beta)^n\big]; \tag{B.135}$$

the term of degree four in the exponential, including its quadratic $\zeta(2)^2$ contribution, supplies the cubic regular part, with $\zeta(2)^2 = \frac{5}{2}\zeta(4)$. The coefficient of β^3 on the left of Eq. (B.133) is

$$\frac{\mathsf{B}_3}{3} - \frac{\mathsf{B}_{12}}{2} + \frac{1}{2}\int_0^1 \frac{\log^2 t \log(1+t)}{t}\,\mathrm{d}t = \frac{\mathsf{B}_3}{3} - \frac{\mathsf{B}_{12}}{2} + \frac{7}{8}\zeta(4). \tag{B.136}$$

Therefore

$$\mathsf{B}_{12} = \frac{2}{3}\mathsf{B}_3 - \frac{1}{4}\zeta(4). \tag{B.137}$$

Substituting Eqs. (B.131) and (B.137) into Eq. (B.129) gives the closed form of E. □

An independent quadrature of Eq. (B.126), which does not use the functional identity, gives

$$E = -21.663876674184610081020147908761053559049257064774\ldots, \tag{B.138}$$

in agreement with the closed form to more than eighty decimal places. This precision refers to a convergent one-dimensional check, not to the full quadrature of D_3.

The three-kink endpoint derivative. For the uncut interval endpoints and the consecutive-kink hard core, the finite part of the charge-differentiated source derivative of the odd three-kink moment is

$$\partial_v\partial_\eta A_{3,0} = -108 + 60\zeta(2) + 22\zeta(3) - 34\zeta(4) + 16\log 2. \tag{B.139}$$

Its reduction is explicit. Put

$$j(r) = (1-r)\log(1-r) - r\log r. \tag{B.140}$$

For the two gaps r and t with span $R = r + t$, the integral over the first position gives the odd endpoint factor

$$\Phi_{\mathrm{odd}}(r,t) = 2j(R) - j(r) - j(t) - 2(1-R). \tag{B.141}$$

Consequently

$$\partial_v\partial_\eta A_3(\varepsilon) = 2\int_{2\varepsilon}^{1} \big[j(R) - (1-R)\big] I_{R,v}(R,\varepsilon)\,\mathrm{d}R - 2\int_{\varepsilon}^{1-\varepsilon} j(r)\, I_{r,v}(r,\varepsilon)\,\mathrm{d}r. \tag{B.142}$$

Both inner integrals, $I_{R,v}$ at fixed span and $I_{r,v}$ at fixed gap, have elementary dilogarithmic primitives. Define

$$\begin{aligned} J_0(e) &= \frac{2}{e} - \frac{2}{1-e} - 4\log e + 4\log(1-e), \\ \Phi_{\log}(y) &= -\frac{\log y + 1}{y} + \frac{\log y}{1-y} - \log y + \log(1-y) + \log^2 y - 2\log y\log(1-y) - 2\,\mathrm{Li}_2(y), \\ J_1(e) &= -4\big[\Phi_{\log}(1-e) - \Phi_{\log}(e)\big]. \end{aligned} \tag{B.143}$$

Then

$$I_{R,v} = R^{-1}\big[-2\log R\, J_0(\varepsilon/R) + J_1(\varepsilon/R)\big], \tag{B.144}$$

and indeed $\Phi'_{\log}(y) = \log y/[y^2(1-y)^2]$. For the second integral let

$$\begin{aligned} \Psi_0(t) &= t + 2\log t - \frac{1}{t}, \\ \Psi_1(t) &= t\big[\log(1+t) - \log t\big] - 2\,\mathrm{Li}_2(-t) - \log^2 t + \frac{\log t + 1 - \log(1+t)}{t} + \log t. \end{aligned} \tag{B.145}$$

The derivative of Ψ_1 is $(1 + 2/t + 1/t^2)[\log(1+t) - \log t]$, and

$$I_{r,v} = \frac{2}{r}\Big[\Psi_1\Big(\frac{1-r}{r}\Big) - \Psi_1\Big(\frac{\varepsilon}{r}\Big) - \log r\Big\{\Psi_0\Big(\frac{1-r}{r}\Big) - \Psi_0\Big(\frac{\varepsilon}{r}\Big)\Big\}\Big]. \tag{B.146}$$

The finite part of Eq. (B.142) consists of two bulk terms and the lower-span region $R = \varepsilon t$; the second integral has no additional finite boundary term. The pieces are

piece	finite contribution
first bulk	$-144 + 64\zeta(2) + 32\zeta(3) - 24\zeta(4)$
second bulk	$60 - 16\zeta(2) - 40\zeta(3) - 10\zeta(4)$
lower-span boundary	$-24 + 16\log 2 + 12\zeta(2) + 30\zeta(3)$

and their sum is Eq. (B.139). The first bulk is

$$2\ \mathrm{FP}\int_0^1 \frac{j(r) - (1-r)}{r}\big[-4\log^2 r + 8\log r + 8 + 8\zeta(2)\big]\,\mathrm{d}r. \tag{B.147}$$

For the second bulk set $l = \log r$, $\bar{l} = \log(1-r)$, $\tau = (1-r)/r$ and

$$\Upsilon = -2\tau(\bar{l} + l) + 4\,\mathrm{Li}_2(1-r) - 2\bar{l}^2 + 2l^2 + \frac{2(\bar{l} + l + 1)}{\tau} + 2\bar{l} + 2. \tag{B.148}$$

The integrand is $-2(\tau\bar{l}-l)\Upsilon$; its part with logarithms only integrates to $12+8\zeta(2)-24\zeta(3)-16\zeta(4)$, and its dilogarithm part to $48-24\zeta(2)-16\zeta(3)+6\zeta(4)$. The lower-span contribution is

$$2\ \mathrm{FP}\int_0^{1/2}\frac{-2\log e\,J_0(e)-J_1(e)}{e}\,\mathrm{d}e. \tag{B.149}$$

Euler's identity (B.52) reduces its integrand, with $l=\log e$ and $\bar{l}=\log(1-e)$, to

$$\frac{8(\bar{l}+1)}{e^2}-\frac{8(\bar{l}+1)}{e(1-e)}+\frac{16(l-\bar{l})}{e}+\frac{8(\bar{l}^2+l^2)}{e}-\frac{16\zeta(2)}{e}+\frac{32\,\mathrm{Li}_2(e)}{e}, \tag{B.150}$$

whose six finite integrals over $(0,\frac{1}{2})$ are, in order,

$$-24+16\log 2,\quad 4\zeta(2),\quad 8\zeta(2),\quad 2\zeta(3)-\tfrac{16}{3}\log^3 2,\quad 16\zeta(2)\log 2,\quad 28\zeta(3)-16\zeta(2)\log 2+\tfrac{16}{3}\log^3 2. \tag{B.151}$$

This gives the boundary entry of the table directly. The compendium checks all primitives and rational-logarithm moment operations symbolically (`code/dual_d1/code/archived/LRI_1d_complete/stages/D3_exact/reproducibility/`).

The four-kink endpoint finite part. The current endpoint response has the expansion

$$\begin{aligned}\partial_\eta Q_4(\varepsilon)=&-\frac{2\zeta(2)}{\varepsilon}+\ell^3-(1+8\zeta(2))\ell^2-(2+32\zeta(2)+28\zeta(3))\ell\\&+6-12\zeta(2)-54\zeta(3)-25\zeta(4)+o(1),\end{aligned} \tag{B.152}$$

so that

$$\partial_\eta Q_{4,0}=6-12\zeta(2)-54\zeta(3)-25\zeta(4). \tag{B.153}$$

Let $\mathsf{f}(t)=t\log t+(1-t)\log(1-t)$, with continuous values at the endpoints. For the gaps r,t,w and span $R=r+t+w$, the integral of the endpoint logarithm over the first position is

$$\Phi_4=2\mathsf{f}(R)+\mathsf{f}(r)+\mathsf{f}(w)-\mathsf{f}(R-r)-\mathsf{f}(R-w). \tag{B.154}$$

The rational four-kink kernel is W_4 of Eq. (A.33). At fixed span put $y=r/R$ and $e=\varepsilon/R$; the integration over the other angular gap is elementary. With

$$\mathcal{A}(y)=\frac{(y+1)(y^2+1)}{y^2(1-y)}=-1+\frac{4}{1-y}+\frac{2}{y}+\frac{1}{y^2},\qquad \mathsf{F}_R(y)=\mathsf{f}(R)+\mathsf{f}(Ry)-\mathsf{f}\big(R(1-y)\big), \tag{B.155}$$

and the function $\mathcal{M}_{\mathrm{out}}(y,e)=(1+1/y)[e^{-1}-(1-y-e)^{-1}]+\mathcal{A}(y)\log[(1-y-e)/e]$ of Eq. (B.42), one obtains the exact two-dimensional formula

$$\partial_\eta Q_4(\varepsilon)=2\int_{3\varepsilon}^1\frac{\mathrm{d}R}{R}\int_{\varepsilon/R}^{1-2\varepsilon/R}\mathcal{M}_{\mathrm{out}}(y,\varepsilon/R)\,\mathsf{F}_R(y)\,\mathrm{d}y, \tag{B.156}$$

which is also integrated directly at finite cutoff as a separate check.

Angular finite part and the radial logarithms. Let $\varkappa(R)=2\log R-\log(1-R)-1$ and $\mathsf{F}_1=2\mathsf{f}(R)$. Near $y=0$, $\mathsf{F}_R(y)=Ry[\log y+\varkappa(R)]+O(y^2)$; near $y=1$, $\mathsf{F}_R(y)\to\mathsf{F}_1$. Define

$$\mathcal{H}(e,R)=\int_e^{1-2e}\mathcal{M}_{\mathrm{out}}(y,e)\,\mathsf{F}_R(y)\,\mathrm{d}y,\qquad J_{\mathrm{FP}}(R)=\mathrm{FP}\int_0^1\mathcal{A}(y)\,\mathsf{F}_R(y)\,\mathrm{d}y. \tag{B.157}$$

Differentiating $\mathcal{H}$ with respect to $\log e$, including the moving boundaries, gives its part of order e^0,

$$\partial_{\log e}\mathcal{H}\big|_{e^0}=\frac{3R}{2}\log^2 e+(2R\varkappa+4\mathsf{F}_1)\log e-J_{\mathrm{FP}}-R+2\mathsf{F}_1. \tag{B.158}$$

Hence

$$\mathcal{H}\big|_{e^0} = \frac{R}{2}\log^3 e + (R\varkappa + 2\mathsf{F}_1)\log^2 e + (-J_{\mathrm{FP}} - R + 2\mathsf{F}_1)\log e + C_0(R), \tag{B.159}$$

where the constant is

$$C_0(R) = \mathrm{FP}\int_0^1 [-\mathcal{B}(y) + \mathcal{A}(y)\log(1-y)]\mathsf{F}_R(y)\,\mathrm{d}y + R(1-\varkappa) - (4 + 2\zeta(2))\mathsf{F}_1, \tag{B.160}$$

with $\mathcal{B}(y) = (1 + 1/y)/(1 - y)$. These expressions keep both angular endpoints.

It is essential to insert $\log e = \ell - \log R$ in the *whole* polynomial; integrating only $C_0(R)$ would miss finite terms that come from powers of $\log e$. The radial moments are

$$\begin{aligned}
\mathsf{U}(y) &= \int_0^1 \mathsf{F}_R(y)\frac{\mathrm{d}R}{R} = -2\,\mathrm{Li}_2(y) - \log y \log(1-y), \\
\mathsf{V}(y) &= \int_0^1 \log R\,\mathsf{F}_R(y)\frac{\mathrm{d}R}{R} = \zeta(3) - \mathsf{U}(y) + \mathrm{Li}_3(y) - \mathrm{Li}_3(1-y),
\end{aligned} \tag{B.161}$$

which follow, for example, from

$$\int_0^1 \mathsf{f}(cR)\frac{\mathrm{d}R}{R} = \mathsf{f}(c) - \mathrm{Li}_2(c), \qquad \int_0^1 \log R\,\mathsf{f}(cR)\frac{\mathrm{d}R}{R} = -\mathsf{f}(c) + \mathrm{Li}_2(c) + \mathrm{Li}_3(c). \tag{B.162}$$

The remaining bulk integral is

$$2\ \mathrm{FP}\int_0^1 \Big[-\mathcal{B}\,\mathsf{U} + \mathcal{A}\big\{\log(1-y)\,\mathsf{U} + \mathsf{V}\big\}\Big]\,\mathrm{d}y = 12 - 26\zeta(3) - 29\zeta(4). \tag{B.163}$$

One integration by parts removes the trilogarithms. A primitive of $\mathcal{A}$ is

$$\Pi_{\mathcal{A}}(y) = -y + 2\log y - 4\log(1-y) - \frac{1}{y}, \tag{B.164}$$

and the finite endpoint term for $\mathcal{D}(y) = \zeta(3) + \mathrm{Li}_3(y) - \mathrm{Li}_3(1-y)$ is

$$\big[\Pi_{\mathcal{A}}\mathcal{D}\big]_0^1 = 1 + \zeta(2) - 4\zeta(3). \tag{B.165}$$

The resulting rational-logarithm and dilogarithm moments are elementary derivatives of beta integrals.

The other endpoint constants contribute $4+16\zeta(2)+20\zeta(4)$. The shift from the radial logarithms is

$$\Delta_{\mathrm{rad}} = 2\int_0^1 \Big[-\frac{2\mathsf{F}_1}{R}\log R + \Big(\varkappa + \frac{2\mathsf{F}_1}{R}\Big)\log^2 R - \frac{1}{2}\log^3 R\Big]\,\mathrm{d}R = -10 - 28\zeta(2) - 28\zeta(3) - 16\zeta(4), \tag{B.166}$$

which follows from

$$\begin{gathered}
\int_0^1 \mathsf{f}(R)\frac{\log R}{R}\,\mathrm{d}R = \zeta(2) + \zeta(3), \qquad \int_0^1 \mathsf{f}(R)\frac{\log^2 R}{R}\,\mathrm{d}R = -2\big(\zeta(2) + \zeta(3) + \zeta(4)\big), \\
\int_0^1 \varkappa(R)\log^2 R\,\mathrm{d}R = -8 - 2\zeta(2) - 2\zeta(3).
\end{gathered} \tag{B.167}$$

Adding Eq. (B.163), the other endpoint constants and Δ_{rad} gives Eq. (B.153), and collecting the remaining logarithms gives Eq. (B.152).

C The dual of the chain: exact certificates, the seventh and eighth orders

This appendix gives the derivations of the compendium (written by OpenAI GPT-6 astra; Sec. 1.2) behind Secs. 2.4 and 2.5, edited to the notation of this note. As in Appendices A and B, everything here holds within the matched two-coupling prescription of Sec. 2.1: the certificates and the rational part of C_8^{SR} prove reductions of the archived expressions of that prescription, and the numerical values are evaluations within it (Sec. 2.1, "What 'proved within the prescription' covers"). Appendix C.1 sets up the exact integration and the word algebra in which the certificates of C_5^{SR} to C_7^{SR} are written, and Appendix C.2 completes the seventh order. Appendices C.3 to C.7 treat the eighth order: its trace reduction, the exact rational part, the four computational stages of the full value, the assembly with the zeta-value candidate, and the numerical evaluation of periods. Appendices C.8 and C.9 record the general formulas that were tested and the relation to the other end of the long-range window; Appendices C.10 and C.11 give two results that use only published data of the dual; Appendix C.12 prints the polynomial B_{old} of Eq. (2.10).

Notation. As in Sec. 2.1, $\bar\delta = 1 - s$, $x = \Delta_\sigma = \bar\delta/2$, ε is the hard core and $\ell = \log\varepsilon$; $u = g^2$ and $v = B^2 = 1 + z$ are the essential couplings, with $\beta_u = 2uF$ and $\beta_z = (1+z)H$. The beta inputs are Taylor coefficients of F and H at the marginal reference point $v = 1$, $\bar\delta = 0$, not at the fixed point. Their names $(a, p, b, c, f, \dots, J_9, K_v)$ are the compendium's and denote coefficients only. Several of these letters also have unrelated uses here, distinguished by subscripts or context: $\Omega_n(\varepsilon)$ is the hard-core domain, W_n a wall weight and $W_N\mathcal{Z}$ a span of MZVs, Q_{2m} a current moment, R a sum of gaps and R_w a regularized constant, S_j an insertion, P_6, P_8, $P_{w,k}$ periods and polynomials, L a logarithmic variable, and B, B_j a boundary derivative and its face integrals (Appendices C.4 and C.5), not the essential charge. The input N_δ keeps the compendium's name; the δ in it is $\bar\delta$. For a regulated response I we write $I_0 = [\varepsilon^0\ell^0]I$ and $I_1 = [\varepsilon^0\ell^1]I$, and more generally $(I)_m = [\varepsilon^0\ell^m]I$. A subscript v or $\bar\delta$ after a comma denotes a derivative at the reference point, as in $b_{3,v\bar\delta} = \partial_v\partial_{\bar\delta}b_3$, while a final subscript 0 or 1 after a comma is the projection I_0 or I_1, as in $Q_{2,0}$, $\mathcal{K}_{8,1}$ and $A_{9,1} = (A_9)_1$, the coefficient written $A_9[\varepsilon^0\ell^1]$ in Sec. 2.6; thus $(\partial_v A_7)_1$ is the coefficient of $\varepsilon^0\ell^1$ in $\partial_v A_7$. The symbol ∂_η is the derivative with respect to the endpoint source η of Sec. 2.1, item (ii), at $\eta = 0$. The other symbols of the dual are those of Appendix A.1: the ordered moments A_{2m+1}, Z_{2m} and Q_{2m} with the current weight Ξ_{2m} (Eq. (A.31)), the vacuum-normalized odd and current responses $\mathcal{C}_{2m+1}$ and $\mathcal{K}_{2m}$ (Eq. (A.37)), the charge and tail insertions $S_j = \sum_{i<k} q_iq_k\log^j(X_k - X_i)$ (Eq. (A.41)), the Jacobian entries j_n and λ_{2m} (Eq. (A.42)) and the subtraction recurrence (Eq. (A.47)). The formal projection π_0 is defined in Appendix C.4; a superscript (0) denotes it, as in $U^{(0)} = \pi_0 U$.

C.1 Exact integration, regularized periods and the certificates

The numerical tables of the compendium are not the definitions of its results. The primary exact data are rational functions of the cutoff multiplying iterated integrals. This subsection explains how they are integrated and reduced, including the constants that a symbol calculation does not see.

Keeping the hard core during exact integration. After the integral over the overall translation, use cumulative coordinates and shift $X_j = j\varepsilon + \tilde X_j$. The remaining simplex is $0 < \tilde X_1 < \cdots < \tilde X_{n-1} < 1 - (n-1)\varepsilon$. At $v = 1$, $\bar\delta = 0$ the kernel is rational, and the charge and tail derivatives insert polynomials in logarithms. Partial fractions, integration by parts and shuffle products reduce the successive integrations to hyperlogarithms, defined with the outermost letter first by

$$G(a_1, \dots, a_n; \Lambda) = \int_0^\Lambda \frac{\mathrm{d}t'}{t' - a_1}\, G(a_2, \dots, a_n; t'), \qquad G(;\Lambda) = 1 \tag{C.1}$$

(the compendium writes T for the argument Λ; here T is the stability trace). For zero letters the tangential-base convention $G(0^n;\Lambda) = \log^n \Lambda/n!$ is used. The total differential is

$$\mathrm{d}G(a_1,\dots,a_n;\Lambda) = \sum_{i=1}^{n} G(a_1,\dots,\widehat{a_i},\dots,a_n;\Lambda)\ \mathrm{d}\log\frac{a_{i-1}-a_i}{a_{i+1}-a_i}, \qquad a_0=\Lambda, \quad a_{n+1}=0, \tag{C.2}$$

with coincident letters taken as limits. For affine letters it gives an exact recurrence in rational kernels. Antiderivatives are checked by differentiation, and their constants are fixed by the specified base point; differentiation alone does not fix a constant.

For a word w of nonnegative letters write $-w$ for the word with every letter negated. At infinity

$$G(-w;\Lambda) = \sum_{n\ge 0} \Lambda^{-n} P_{w,n}(\log\Lambda), \qquad P_{w,0}(L) = \sum_{j=0}^{|w|} \frac{L^j}{j!}\, R_{w_{j+1}\cdots w_{|w|}}, \tag{C.3}$$

with the constants R defined below. The inverse-power coefficients follow from the differential equation. A rational prefactor can grow as a power of Λ, so enough inverse powers must be kept before a finite coefficient is taken. In the same way, positive powers of ε are kept whenever a vacuum denominator can lower their degree.

The rational shuffle algebra. For a word $w = (a_1,\dots,a_n)$ of nonnegative letters define the regularized constant at infinity

$$R_w = \mathrm{Reg}_{\Lambda\to\infty}\, G(-a_1,\dots,-a_n;\Lambda), \qquad R_\varnothing = 1, \tag{C.4}$$

where Reg takes the coefficient of $\Lambda^0(\log\Lambda)^0$. The hyperlogarithms themselves have no positive powers of Λ, so the constants obey the shuffle product,

$$R_w R_{w'} = \sum_{w''\in w \amalg w'} R_{w''}, \tag{C.5}$$

with shuffle multiplicities. The basic values are $R_0 = R_1 = 0$ and $R_a = -\log a$ for $a > 0$. A subscript ab that denotes the single integer product satisfies $R_{ab} = R_a + R_b$; it must not be confused with the two-letter word (a,b).

Lemma (repeated-letter values). For $b > 0$, $a \ge 0$ and $n \ge 2$,

$$R_{a,b^{n-1}} = \frac{(-\log b)^n}{n!} + \mathrm{Li}_n(1-a/b), \tag{C.6}$$

where b^{n-1} is the letter b repeated $n-1$ times.

Proof. The inner repeated-letter integral is $\log^{n-1}(1+t'/b)/(n-1)!$. Set $\vartheta = \log(1+t'/b)$ in the outer integration. The kernel splits into a constant 1 and $(1-a/b)/(e^\vartheta - (1-a/b))$. The first term integrates to the logarithmic polynomial. The remaining convergent geometric-series integral is $\mathrm{Li}_n(1-a/b)$, extended along the nonsingular real domain; its integrable limit at $a = 0$ is $\zeta(n)$. □

Hence

$$\zeta(n) = -\frac{2^{n-1}}{2^{n-1}-1}\, R_{2,1^{n-1}}, \qquad \mathrm{Li}_n(r) = R_{a,b^{n-1}} - R_{b^n} \quad (1-r = a/b > 0). \tag{C.7}$$

These identities, together with the prime factorization of rational logarithms, give a deterministic lift of the exact beta expressions to rational vectors of words. For words whose last letter is positive, uniform scaling gives

$$R_{\kappa w} = \sum_{j=0}^{|w|} R_{\kappa^j}\, R_{w_{j+1}\cdots w_{|w|}}, \qquad \kappa > 0, \tag{C.8}$$

where κw multiplies every letter by κ and κ^j is the word of j letters κ. The restriction on the last letter prevents silently rescaling a trailing-zero tangential logarithm.

Confluence fixes the full constants. *Lemma (convergent confluence).* Let $e = (e_1, \dots, e_n)$ be a binary word with $e_1 = 0$ and $e_n = 1$. Choose $a_i \geq 0$ for a zero bit and $a_i > 0$ for a one bit, and set $\lambda_i(\Lambda) = -a_i$ if $e_i = 0$ and $\lambda_i(\Lambda) = \Lambda + a_i$ if $e_i = 1$. Then

$$\lim_{\Lambda\to\infty} G(\lambda_1(\Lambda), \dots, \lambda_n(\Lambda); \Lambda) = G(e_1, \dots, e_n; 1). \tag{C.9}$$

Proof. Rescale every integration variable by Λ. The two kinds of kernels become $\mathrm{d}v/(v + a_i/\Lambda)$ and $-\mathrm{d}v/(1 - v + a_i/\Lambda)$, whose absolute values are bounded by $\mathrm{d}v/v$ and $\mathrm{d}v/(1-v)$. The product of these bounds is integrable on the ordered simplex because the binary word begins with 0 and ends with 1. Dominated convergence gives the limit of the actual integral, including its constant. Fixed zero intercepts do not invalidate the bound. □

Applying Eq. (C.2) to the affine letters, every nonconstant logarithmic derivative has a fixed nonpositive pole, and deleting a letter produces a finite recursive expansion

$$G(\lambda_1(\Lambda), \dots, \lambda_n(\Lambda); \Lambda) = \sum_w f_w\, G(-w; \Lambda), \qquad f_w \in \mathbb{Z}. \tag{C.10}$$

The original convergent word vanishes at $\Lambda \to 0$, while pure zero subwords keep their tangential logarithm; these conditions fix the constants. Taking Reg yields the exact relations $\sum_w f_w R_w = G(e; 1)$. For example,

$$-R_{a,b} - R_{b,a} + R_{a+b,b} + R_{a+b,a} = -\zeta(2) \qquad (a, b > 0), \tag{C.11}$$

and the zero-intercept case is $-R_{0,b} + R_{b,b} = -\zeta(2)$. This zero-letter extension is needed by the target of C_6^{SR}; treating all letters as strictly positive would not suffice.

The weight-five constants. Multiple zeta values (MZVs) are written with decreasing indices,

$$\zeta(n_1, \dots, n_r) = \sum_{k_1 > \cdots > k_r \geq 1} \frac{1}{k_1^{n_1} \cdots k_r^{n_r}}, \qquad n_1 \geq 2. \tag{C.12}$$

For convergent words the right-hand sides of Eq. (C.9) are MZVs, with a sign set by the number of one letters. Through weight four their reduction uses shuffle, stuffle, duality and $\zeta(2)^2 = \frac{5}{2}\zeta(4)$. At weight five the required height-one identity follows from

$$1 - \frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-u-v)} = u \int_0^1 t'^{-u-1}\left[(1-t')^{-v} - 1\right] \mathrm{d}t' = \sum_{m,n\geq 1} \zeta(m+1, 1^{n-1})\, u^m v^n, \tag{C.13}$$

where u and v are formal variables here. The Taylor expansion of $\log\Gamma$ and the coefficient of u^3v^2 give $\zeta(4,1) = 2\zeta(5) - \zeta(2)\zeta(3)$. The convergent stuffle and shuffle products give respectively

$$\zeta(2)\zeta(3) = \zeta(2,3) + \zeta(3,2) + \zeta(5), \qquad \zeta(2)\zeta(3) = \zeta(2,3) + 3\zeta(3,2) + 6\zeta(4,1), \tag{C.14}$$

hence

$$\zeta(3,2) = 3\zeta(2)\zeta(3) - \frac{11}{2}\zeta(5), \qquad \zeta(2,3) = \frac{9}{2}\zeta(5) - 2\zeta(2)\zeta(3). \tag{C.15}$$

Duality supplies the remaining binary words of weight five. No assumption of numerical independence is needed. The iterated-integral and MZV framework is standard [25, 27]; the programs of the compendium implement the explicit conventions above.

The certificates and their logical scope. The proposed closed expression is subtracted from the exact archived expression of the renormalization-group (RG) assembly, and the difference is lifted to the word algebra. The following finite identities then hold in the free rational vector space of words (C_7^{SR} is in Appendix C.2):

target	weight	nonzero words	certificate rows
$D_3 - D_3^{\text{closed}}$	2	14	10
	3	54	47
	4	102	419
$C_6^{\text{SR}} - C_6^{\text{SR closed}}$	2	26	21
	3	169	142
	4	608	3 879
	5	1 142	52 339

The totals, 170 words and 476 rows for D_3 and 1 945 words and 56 381 rows for C_6^{SR}, are those of Sec. 2.4. For each weight the target X is written explicitly as

$$\mathsf{X} = \sum_j r_j (\mathsf{B}_j \sqcup\!\sqcup \langle v_j \rangle), \qquad r_j \in \mathbb{Q}, \tag{C.16}$$

where $\langle v_j \rangle$ is a word and every B_j is one of the analytically established zero relations above (shuffle, stuffle, duality, the confluence relations and the height-one identity). The verifier regenerates these relations from their labels and checks the rational coefficients word by word. Since the relations evaluate to zero, so does the target.

The electronic certificate is part of the proof of the reduction, not optional numerical support: it contains all coefficients, and none is inferred from decimal agreement. Both certificates are plain JSON embedded in standalone Python verifiers (for D_3, `code/dual_d1/code/archived/D3_analytic_proof/verify_D3_proof_standalone.py`). Modular elimination was used to discover the certificates, but the final verification uses exact rational arithmetic and no numerical period. The proof does not require the chosen relations to span all MZV identities. Both certificates replay successfully (Sec. 2.8). In addition, a standard-library checker (`code/dual_d1/code/archived/C7_closed_form_proof/code/verify_C6_input_map.py`) parses the archived beta expressions as a restricted syntax tree, lifts the individual beta coefficients directly into the shuffle algebra and verifies the trace assembly, the archived C_6^{SR} expression, its difference from the closed form and all 1 945 target coefficients, so that the certificate is tied to the RG inputs. It does not reintegrate the original kink kernels.

C.2 The seventh order and the weight-six certificate

By Eq. (2.8), $C_7^{\text{SR}} = 4D_4 - T_3 - T_2^2 + \frac{1}{2} C_3^{\text{SR}} C_5^{\text{SR}}$ needs a new determinant coefficient but no new trace coefficient. This follows from expanding the characteristic polynomial, not from extrapolating a truncated beta function. The closed form of D_4 is in Sec. 2.4, and

$$D_4 = -2.534629799538250115260606412 4\ldots, \qquad C_7^{\text{SR}} = 51.068988612928108831 18\ldots. \tag{C.17}$$

New derivatives. At this order the required jets are

$$\begin{aligned} F &= z + u(1 + az + p\bar{\delta} + Az^2 + E\bar{\delta}z + V\bar{\delta}^2 + \alpha z^3 + \beta z^2\bar{\delta} + \gamma z\bar{\delta}^2) \\ &\quad + u^2(b + Gz + R\bar{\delta} + \Phi z^2 + \Psi z\bar{\delta}) + u^3(J + \Omega z) + \cdots, \\ H &= -\bar{\delta} + 4u + u^2(c + fz + q\bar{\delta} + Lz^2 + Mz\bar{\delta} + Q\bar{\delta}^2) + u^3(k + Nz + P\bar{\delta}) + Ku^4 + \cdots, \end{aligned} \tag{C.18}$$

and the determinant combination is

$$D_4 = B_{\text{old}}^{D_4} - \frac{3\alpha}{32} + \frac{\beta}{4} - \frac{\gamma}{2} + \frac{\Phi}{16} - \frac{\Psi}{8} - \frac{\Omega}{32} - \frac{Q}{8} - \frac{P}{16} - \frac{3K}{128}. \tag{C.19}$$

The polynomial $B_{\text{old}}^{D_4}$ involves only earlier inputs; it is regenerated by `code/dual_d1/code/archived/C7_closed_form_proof/code/derive_D4.py`. Differentiation must precede the fixed-point

substitution, since a term that is small at the fixed point can contribute to a derivative at the required order. The derivatives that define the new inputs are listed in Appendix C.3.

The two largest new integrations were the eight-kink current response and the charge derivative of the seven-kink odd response. The latter enters

$$\Omega = \partial_v b_7\big|_{v=1,\bar\delta=0} = -234.42145528896368752651444525 23\ldots. \tag{C.20}$$

Every term of b_7 in the subtraction recurrence, Eq. (A.47), is differentiated, including the endpoint terms and the explicit factors v, and so is every product in the normalized response $\mathcal{C}_7 = A_7 - A_5Z_2 + A_3(Z_2^2 - Z_4) - Z_2^3 + 2Z_2Z_4 - Z_6$; the raw integral $\partial_v A_7$ alone is not Ω. An early partial calculation of C_7^{SR} had used an incorrect numerical value of the second-tail input; the rebuilt value is $Q = -107.06548149554763940457268131 65\ldots$, and with all other inputs fixed $C_7^{\mathrm{SR}} = 21.76630670180764789\ldots - \Omega/8$. The certificate below is tied to the rebuilt input expressions.

Higher-logarithm checks. The rebuilt numerical integration passed the following higher-logarithm tests, which do not use the new single-logarithm coefficient:

$$\begin{gathered}[\ell^8]\partial_v\mathcal{C}_7 = 0, \quad [\ell^7]\partial_v\mathcal{C}_7 = -\frac{6416}{315}, \quad [\ell^6]\partial_v\mathcal{C}_7 = -\frac{1622}{9}, \quad [\ell^5]\partial_v\mathcal{C}_7 = \frac{3848\zeta(2) - 8824}{15}, \\ [\ell^7]\mathcal{K}_8 = -\frac{1088}{315}, \qquad [\ell^6]\mathcal{K}_8 = -\frac{664}{15}, \qquad [\ell^5]\mathcal{K}_8 = 96\zeta(2) - \frac{3656}{15}.\end{gathered} \tag{C.21}$$

A direct finite-cutoff test compares the whole of $\partial_v A_7$ at $\varepsilon = 1/10$ with a separate cubature of the original kernels, not with a fitted asymptotic polynomial.

What the certificate establishes. All 25 archived beta inputs of C_3^{SR} to C_7^{SR} are parsed directly from the archived expressions and assembled into D_4 and C_7^{SR}. Subtracting the proposed zeta-value result and lifting logarithms and polylogarithms to the word algebra gives a target with the homogeneous components

weight	target words	certificate rows	final residual words
2	37	33	0
3	337	284	0
4	1 810	2 880	0
5	7 428	54 363	0
6	15 600	531 318	0
total	25 212	588 878	0

The weight-six identity includes the products of lower-weight periods; a reduction modulo products would not suffice. In the free rational word space the certificate has the form

$$\mathsf{X}_6 = \sum_j r_j\, \mathsf{B}_j \sqcup\!\sqcup \mathsf{M}_j, \qquad r_j \in \mathbb{Q}, \tag{C.22}$$

where X_6 is the weight-six target, every B_j is a proved zero relation and every multiplier M_j is expanded with its full shuffle multiplicities, so that evaluation annihilates the right-hand side. The verifier does not use the Lyndon-polynomial representation employed during discovery; it expands ordinary words with exact integers after clearing denominators.

All 16 convergent binary boundary values of weight six are derived from 31 rational equations, using the shuffle and stuffle products, duality and the height-one identity (C.13). Examples are

$$\begin{aligned}\zeta(5,1) &= \tfrac{3}{4}\zeta(6) - \tfrac{1}{2}\zeta(3)^2, \qquad \zeta(4,2) = -\tfrac{4}{3}\zeta(6) + \zeta(3)^2, \\ \zeta(3,3) &= -\tfrac{1}{2}\zeta(6) + \tfrac{1}{2}\zeta(3)^2, \quad \zeta(2,4) = \tfrac{25}{12}\zeta(6) - \zeta(3)^2.\end{aligned} \tag{C.23}$$

No assumption of numerical independence of MZVs is used. The final certificates and the target suffice for verification; the large discovery matrices and redundant intermediate certificates are not part of the proof input.

C.3 The eighth order: trace reduction and recurrence

The integer powers of x in $\Delta_{\mathcal{E}}$ (the even powers of $\sqrt{x}$) come entirely from the half-trace $T/2$, so $C_8^{\mathrm{SR}} = 8T_4$ (Eq. (2.8)). No new determinant coefficient D_5 is needed, but T_4 needs new derivatives that D_4 does not fix. The jets required for this trace, keeping only the sectors that can contribute after differentiation, extend Eq. (C.18) by ten coefficients:

$$\begin{aligned} F &= z + u(1 + az + p\bar\delta + Az^2 + E\bar\delta z + V\bar\delta^2 + \alpha z^3 + \beta\bar\delta z^2 + \gamma\bar\delta^2 z + W\bar\delta^3) \\ &\quad + u^2(b + Gz + R\bar\delta + \Phi z^2 + \Psi\bar\delta z + S\bar\delta^2) + u^3(J + \Omega z + U\bar\delta) + J_9 u^4 + \cdots, \\ H &= -\bar\delta + 4u + u^2(c + fz + q\bar\delta + Lz^2 + M\bar\delta z + Q\bar\delta^2 + L_3 z^3 + M_{21}\bar\delta z^2 + M_{12}\bar\delta^2 z) \\ &\quad + u^3(k + Nz + P\bar\delta + N_2 z^2 + N_\delta \bar\delta z) + u^4(K + K_v z) + \cdots. \end{aligned} \tag{C.24}$$

These are not the exact beta functions beyond the displayed order. For example, the term $K_v z u^4$ of H must be kept, because H_z contributes at order $\bar\delta^4$.

Derivative dictionary. All derivatives are taken at $v = 1$, $\bar\delta = 0$. The normalized auxiliary current beta function is $\beta_h/(\sqrt{2}b_0) = -g^2 + h_4 g^4 + h_6 g^6 + h_8 g^8 + \cdots$ (so $h_2 = -1$), and b_3, b_5, b_7, b_9 are the odd fugacity coefficients of the subtraction recurrence. The lower jets are

$$\begin{aligned} &a = b_{3,v}, && p = b_{3,\bar\delta}, && A = \tfrac{1}{2}b_{3,vv}, && E = b_{3,v\bar\delta}, && V = \tfrac{1}{2}b_{3,\bar\delta\bar\delta}, \\ &\alpha = \tfrac{1}{6}b_{3,vvv}, && \beta = \tfrac{1}{2}b_{3,vv\bar\delta}, && \gamma = \tfrac{1}{2}b_{3,v\bar\delta\bar\delta}, \\ &b = b_5, && G = b_{5,v}, && R = b_{5,\bar\delta}, && \Phi = \tfrac{1}{2}b_{5,vv}, && \Psi = b_{5,v\bar\delta}, \\ &J = b_7, && \Omega = b_{7,v}, \\ &c = -4h_4, && f = -4h_{4,v}, && q = -4h_{4,\bar\delta}, && L = -2h_{4,vv}, && M = -4h_{4,v\bar\delta}, \\ &Q = -2h_{4,\bar\delta\bar\delta}, && k = -4h_6, && N = -4h_{6,v}, && P = -4h_{6,\bar\delta}, && K = -4h_8, \end{aligned} \tag{C.25}$$

and the ten additional inputs of C_8^{SR} are

$$\begin{aligned} &W = \tfrac{1}{6}b_{3,\bar\delta\bar\delta\bar\delta}, && S = \tfrac{1}{2}b_{5,\bar\delta\bar\delta}, && U = b_{7,\bar\delta}, && J_9 = b_9, && L_3 = -\tfrac{2}{3}h_{4,vvv}, \\ &M_{21} = -2h_{4,vv\bar\delta}, && M_{12} = -2h_{4,v\bar\delta\bar\delta}, && N_2 = -2h_{6,vv}, && N_\delta = -4h_{6,v\bar\delta}, && K_v = -4h_{8,v}. \end{aligned} \tag{C.26}$$

A mixed first derivative carries no factor $\frac{1}{2}$; Taylor factors appear only for repeated derivatives, as displayed.

Finite-response recurrence through ninth order. The eight-kink current response is

$$\mathcal{K}_8 = Q_8 - Q_6 Z_2 + Q_4(Z_2^2 - Z_4) + Q_2(-Z_2^3 + 2Z_2 Z_4 - Z_6), \tag{C.27}$$

as in Eq. (A.37), and the nine-kink odd response, the coefficient of g^8 in the ratio of the odd generating function to the vacuum one (Appendix A.7) at zero source ($A_1 = 1$), is

$$\mathcal{C}_9 = A_9 - A_7 Z_2 + A_5(Z_2^2 - Z_4) + A_3(-Z_2^3 + 2Z_2 Z_4 - Z_6) + Z_2^4 - 3Z_2^2 Z_4 + Z_4^2 + 2Z_2 Z_6 - Z_8. \tag{C.28}$$

The Jacobian entries needed next are

$$j_7 = -2\partial_v \mathcal{C}_{7,0} + \partial_\eta \mathcal{C}_{7,0}, \qquad \lambda_6 = -2\mathcal{K}_{6,0} - 4v\,\partial_v \mathcal{K}_{6,0} + 2v\,\partial_\eta \mathcal{K}_{6,0}, \tag{C.29}$$

with the same rule as j_3, j_5 and λ_2, λ_4 in Eq. (A.42). The compendium defines j_7 in this form but uses λ_6 without writing it; Eq. (C.29) gives λ_6 by that rule. The new counterterm recurrences, which continue Eq. (A.47), are

$$\begin{aligned} h_8 &= \mathcal{K}_{8,1} - 2Q_{2,0}b_7 - 4\mathcal{K}_{4,0}b_5 - 6\mathcal{K}_{6,0}b_3 - \lambda_2 h_6 - \lambda_4 h_4 + \lambda_6, \\ b_9 &= \mathcal{C}_{9,1} - 3\mathcal{C}_{3,0}b_7 - 5\mathcal{C}_{5,0}b_5 - 7\mathcal{C}_{7,0}b_3 - 2vj_1 h_8 - 2vj_3 h_6 - 2vj_5 h_4 + 2vj_7. \end{aligned} \tag{C.30}$$

Products and derivatives are formed before the finite parts are extracted. An unavailable raw scalar is an error, not a zero coefficient, and the compendium's full scalar-aware convolution code enforces this.

Solving $F = H = 0$ with Eq. (C.24) through $\bar{\delta}^4$ and expanding $T = 2uF_u + (1+z)H_z$ gives Eq. (2.10): C_8^{SR} is B_{old} plus a fixed linear combination of the ten new inputs. The polynomial B_{old} involves only the 25 earlier inputs; it is printed in Appendix C.12. Its value is $-861.663567235479779201497190992 3\ldots$, and its rational projection is $\frac{1153}{6}$. Replacing it by its rational projection before the assembly would lose genuine terms.

C.4 The rational part of the eighth order

The formal weight-zero map π_0 acts on the graded representation of an integrated quantity: it removes the positive-weight constant periods. It is not a numerical limit in which the real zeta values vanish, and variable logarithms cannot be discarded before integration; for example,

$$\int_0^\infty \frac{\log(1+t')}{(1+t')^2}\,\mathrm{d}t' = 1. \tag{C.31}$$

Products, coefficient convolutions and power divergences must already have been treated consistently when π_0 is applied.

Four raw residues. Define

$$\begin{aligned} \mathcal{R}_6 &= \pi_0(\partial_v\partial_{\bar{\delta}}Q_6)_1, & \mathcal{R}_8 &= \pi_0(\partial_v Q_8)_1, \\ \mathcal{R}_{\mathrm{tail}} &= \pi_0(\partial_{\bar{\delta}}A_7)_1 - \pi_0(\partial_{\bar{\delta}}Z_6)_1, & \mathcal{R}_9 &= \pi_0(A_9)_1 - \pi_0(Z_8)_1 + 2\pi_0(\partial_\eta A_7)_0, \end{aligned} \tag{C.32}$$

the residues of the six-kink and eight-kink current moments, of the seven-kink tail and of the nine-kink fugacity. Before any of them was inserted, the exact trace and subtraction reduction gave

$$\pi_0 C_8^{\mathrm{SR}} = \frac{5345}{12} - \frac{1}{2}\mathcal{R}_6 - \frac{1}{8}\mathcal{R}_8 + \frac{3}{4}\mathcal{R}_{\mathrm{tail}} + \frac{1}{4}\mathcal{R}_9. \tag{C.33}$$

The completed values are

$$\mathcal{R}_6 = \frac{399343}{2}, \qquad \mathcal{R}_8 = -757993, \qquad \mathcal{R}_{\mathrm{tail}} = -\frac{910891}{6}, \qquad \mathcal{R}_9 = \frac{1423087}{3}, \tag{C.34}$$

and after the first two are inserted the known term is $-\frac{111389}{24}$, which gives the sum $\pi_0 C_8^{\mathrm{SR}} = 88$ of Sec. 2.5. None of the raw integrals was inferred from that answer or from the earlier conjecture 160. For reconstruction, the last five raw scalars are

raw scalar	rational value	raw scalar	rational value
$\pi_0(\partial_{\bar{\delta}}A_7)_1$	$-\frac{468683}{3}$	$\pi_0(Z_8)_1$	$\frac{23294}{3}$
$\pi_0(\partial_{\bar{\delta}}Z_6)_1$	$-\frac{8825}{2}$	$\pi_0(\partial_\eta A_7)_0$	$-\frac{184413}{2}$
$\pi_0(A_9)_1$	666540		

The rational inputs and the full-Jacobian check. A separate finite-response reconstruction gives

$$U^{(0)} = \frac{16534}{3}, \qquad J_9^{(0)} = -\frac{10192}{3}, \qquad N_\delta^{(0)} = -20848, \qquad K_v^{(0)} = \frac{77920}{3}, \tag{C.35}$$

and the other six new rational inputs are

$$W^{(0)} = 168, \quad S^{(0)} = -1751, \quad L_3^{(0)} = 128, \quad M_{21}^{(0)} = -384, \quad M_{12}^{(0)} = 2720, \quad N_2^{(0)} = 3712. \tag{C.36}$$

A separate literal full-Jacobian calculation then yields $T^{(0)}(\bar{\delta}) = \frac{\bar{\delta}}{2} + 2\bar{\delta}^2 + \frac{13}{2}\bar{\delta}^3 + 11\bar{\delta}^4 + O(\bar{\delta}^5)$, which reproduces $\pi_0 C_8^{\mathrm{SR}} = 8 \times 11$ without using the residue assembly above.

Boundary differentiation and its checks. If $M(\varepsilon)=\sum_{p,m} c_{p,m}\varepsilon^p\ell^m$ and $B=-\varepsilon\partial_\varepsilon M$, then

$$[\varepsilon^p\ell^0]B=-p\,c_{p,0}-c_{p,1},\qquad [\varepsilon^0\ell^1]M=-[\varepsilon^0\ell^0]B. \tag{C.37}$$

Differentiating a consecutive-gap lower limit removes one integration. For $n=2m$ kinks the reflection weights of the $2m-1$ faces are $(2,\dots,2,1)$; for odd n the faces are all paired. These factors and the minus sign in Eq. (C.37) are essential. The negative powers of ε are predicted from lower moments. For $Q_{6,v\bar\delta}$ the predictions are $(c_{-2,0},c_{-2,1},c_{-1,0},c_{-1,1})^{(0)}=(-7,-32,2193,555)$; for $Q_{8,v}$ they are

power p	$c_{p,0}^{(0)}$	$c_{p,1}^{(0)}$
-3	$\frac{2}{3}$	2
-2	$-\frac{104}{3}$	$-\frac{52}{3}$
-1	$-\frac{80255}{12}$	$-\frac{20429}{6}$

All comparisons in the rational calculation are exact. They are generated from lower moments and do not use the target finite-power residue. The tail and ninth-order groups also pass complete comparisons of their negative-power logarithmic polynomials, and further direct finite-cutoff and alternative endpoint-representation tests are kept in their stage records.

C.5 The full eighth order: the four computational stages

The rational result does not give the full coefficient. Restoring the constant periods requires undoing every earlier projection, keeping the integer-offset logarithms and lower-endpoint constants, and keeping the full counterterm contributions.

The six inherited inputs. Six of the ten new inputs have recorded numerical evaluations only:

input	full numerical value	rational projection
W	$-37.708111514952555\dots$	168
S	$878.183733747745103\dots$	-1751
L_3	$-0.638692810583406\dots$	128
M_{21}	$-171.235576565588699\dots$	-384
M_{12}	$-422.715057969894520\dots$	2720
N_2	$189.742012494084223\dots$	3712

They enter the assembly as inherited numerical records: decimal values from one recorded 100-digit evaluation, with no surviving exact expression and no second evaluation (Sec. 2.5). This is one reason why a complete analytic proof for C_8^{SR} is a further task. The other four new inputs were computed in four stages, F1 (N_δ), F2 (U), F3 (K_v) and F4 (J_9). Their definitions and subtraction factors are fixed by Eq. (2.10) and the recurrence; they are not adjustable parameters.

F1a: the six-kink mixed logarithm. Define

$$P_6=(Q_{6,v\bar\delta})_1\big|_{v=1,\bar\delta=0}=[\varepsilon^0\ell^1]\int_{\Omega_6(\varepsilon)}\Xi_6 W_6\,(S_2+2S_1S_2)\,\mathrm{d}^6X, \tag{C.38}$$

where $\Omega_6(\varepsilon)$ is the ordered hard-core domain of six kinks, W_6 the six-kink weight, $\Xi_6=\sum_{j=1}^3(X_{2j}-X_{2j-1})$ the current weight, and $(S_2+2S_1S_2)W_6=\partial_v\partial_{\bar\delta}W_6$ the mixed insertion, with both terms and no Taylor factor. Early quadrature estimates around $1261.9448\dots$ were not accurate enough for an integer-relation search with many constants. The final computation instead integrated the

complete cutoff-boundary source with all constant periods kept. With the five gaps written r_i and $R = \sum_i r_i$, the face integrals are

$$B_j = \varepsilon \int_{\substack{r_i \geq \varepsilon \ (i\neq j) \\ R<1}} (1-R)\, \Xi_6 W_6 \,(S_2 + 2S_1 S_2) \prod_{i\neq j} \mathrm{d}r_i, \qquad r_j = \varepsilon, \tag{C.39}$$

with $B = 2B_1 + 2B_2 + B_3$ (the reflection weights of the five faces) and $P_6 = -[\varepsilon^0 \ell^0]B$ by Eq. (C.37). The final record includes all 21 first-stage blocks and all 64 terminal outputs; exact collection gives 66 404 period terms, and the rational projection is $\frac{399343}{2} = \mathcal{R}_6$. The value of P_6 is given in Sec. 2.5. The two fresh evaluations at 70 and 100 working digits differ in the matching point, the step size, the Taylor order and the asymptotic order. A complete four-kink mixed control agrees exactly with the archived expression; a comparison of the full stage in moving-letter and in ordinary-word form agrees in 196 169 rational coefficients; and all 23 816 rational prefactors are reconstructed exactly from their partial fractions.

On a face all separations carry a factor ε, and the scale is restored by

$$S_1 = -3\ell + s_1, \qquad S_2 = -3\ell^2 + 2s_1\ell + s_2, \tag{C.40}$$

where s_1 and s_2 are the insertions of the rescaled configuration (the six alternating charges give $\sum_{i<k} q_i q_k = -3$). The logarithm-free boundary coefficient uses $s_2 + 2s_1 s_2$, but its variable and constant logarithms remain. The full positive-weight divergence differences at the powers ε^{-2} and ε^{-1} are numerically consistent with zero at about 10^{-88} and 10^{-87}; these are not exact period certificates.

F1b: vacuum normalization. The six-kink current response is $\mathcal{K}_6 = Q_6 - Q_4 Z_2 + Q_2 Z_2^2 - Q_2 Z_4$, so that

$$(\partial_v \partial_{\bar\delta} \mathcal{K}_6)_1 = P_6 + D_{\mathrm{norm}}, \qquad D_{\mathrm{norm}} = [\varepsilon^0 \ell^1]\, \partial_v \partial_{\bar\delta} (-Q_4 Z_2 + Q_2 Z_2^2 - Q_2 Z_4)\big|_{v=1,\bar\delta=0}. \tag{C.41}$$

The three product contributions are

$$-1300.763962200486633\ldots, \qquad -108 \text{ (exact)}, \qquad -1261.404579171677298\ldots, \tag{C.42}$$

which give

$$D_{\mathrm{norm}} = -2670.168541372163931149878165554559\ldots, \qquad \pi_0 D_{\mathrm{norm}} = -\frac{8719}{2}. \tag{C.43}$$

All 13 Leibniz terms are included, with their repeated-factor multiplicities. The terms with zero cutoff powers alone would give $-1497.3871181439\ldots$, so the products of suppressed and divergent powers are substantial. The input map uses 32 complete scalar coefficients; the 16 two-kink ones can be rederived by elementary integration, and the four-kink finite expressions are inherited exact inputs, not integrated afresh in this step.

F1c: finite-response correction. Define

$$D_{\mathrm{ct}} = \partial_v \partial_{\bar\delta} \big[-2Q_{2,0} b_5 - 4\mathcal{K}_{4,0} b_3 - \lambda_2 h_4 + \lambda_4\big]\big|_{v=1,\bar\delta=0}, \tag{C.44}$$

the derivative of the counterterm terms of h_6 in the recurrence. The complete result is

$$D_{\mathrm{ct}} = -134.559964098286854425679630067592\ldots, \qquad \pi_0 D_{\mathrm{ct}} = -190100. \tag{C.45}$$

With $k_4 = \mathcal{K}_{4,0}$ and $e_4 = \partial_\eta \mathcal{K}_{4,0}$, the explicit coupling factors of λ_4 give

$$(\lambda_4)_{v\bar\delta} = -6k_{4,v\bar\delta} - 4k_{4,vv\bar\delta} + 2e_{4,\bar\delta} + 2e_{4,v\bar\delta}; \tag{C.46}$$

the extra terms would be lost if the prefactors were treated as constants. Two new complete four-kink finite parts are

$$\begin{aligned}(Q_{4,vv\bar\delta})_0 &= 17290.158097306598287876627149758 44\ldots,\\ (\partial_\eta Q_{4,v\bar\delta})_0 &= 34463.595640455961750271581755932 35\ldots.\end{aligned} \tag{C.47}$$

Since $N_\delta = -4h_{6,v\bar\delta}$ and $h_{6,v\bar\delta} = P_6 + D_{\mathrm{norm}} + D_{\mathrm{ct}}$, the three F1 stages combine into

$$N_\delta = -4P_6 + 11218.914021881803142302231182488606\ldots, \tag{C.48}$$

where the ellipsis abbreviates an archived exact period expression, not an unknown contribution. The rational check is $\frac{399343}{2} - \frac{8719}{2} - 190100 = 5212$, hence $N_\delta^{(0)} = -20848$.

F2: the seven-kink tail coefficient. The derivative of the b_7 recurrence with respect to $\bar\delta$ isolates

$$U = (\partial_{\bar\delta} A_7)_1 - (\partial_{\bar\delta} Z_6)_1 + 2(\partial_\eta \partial_{\bar\delta} A_5)_0 - 2(\partial_\eta \partial_{\bar\delta} Z_4)_0 + D_{\mathrm{lower}}, \tag{C.49}$$

where D_{lower} collects the remaining lower-order terms of the differentiated recurrence. The lower shift terms, the vacuum tail and the endpoint terms are kept unprojected. Collecting everything but the first term into D_{known},

$$\begin{aligned}(\partial_{\bar\delta} A_7)_1 &= 315459.45115269022844857671653830079 7\ldots,\\ D_{\mathrm{known}} &= -316756.13911298175494023311031164286 6\ldots,\end{aligned} \tag{C.50}$$

and their sum is the value of U in Sec. 2.5, $U = -1296.68796029152649165639377334206 9\ldots$, with $U^{(0)} = \frac{16534}{3}$. The seven-kink expression has 66 476 terms and the final U 81 495; their exact scalar relation and the rational projection are checked, and full evaluations at 80 and 110 digits cover 57 346 periods through weight seven. A separately derived leading logarithm,

$$[\varepsilon^0 \ell^8]\partial_{\bar\delta} A_7 = -\frac{2362}{315}, \qquad [\varepsilon^0 \ell^9]\partial_{\bar\delta} A_7 = 0, \tag{C.51}$$

agrees with the leading-logarithm recursion of the lower flow; this is not a replay of the whole subleading logarithmic polynomial. The historical coverage record of F2 names 258 intermediate files that are not among the surviving files. The complete period expression of U and its numerical replay survive, but an independent replay from the full kernels would require regenerating those source stages.

F3: the eight-kink logarithm and six-kink finite responses. Define the full, unprojected scalars

$$P_8 = (Q_{8,v})_1, \qquad X_{vv} = (Q_{6,vv})_0, \qquad X_E = (\partial_v \partial_\eta Q_6)_0. \tag{C.52}$$

The finite-response calculation of $K_v = -4h_{8,v}$ gives

$$K_v = -4P_8 + 16X_{vv} - 8X_E + \mathcal{S}_3, \qquad \mathcal{S}_3 = 24144.92587771278969209842737284526 3\ldots, \tag{C.53}$$

with

$$\begin{aligned}X_{vv} &= 8059.73610218661612575272790112098 5\ldots,\\ X_E &= -88077.17236355213705219767553829123 6\ldots,\\ P_8 &= 214801.26113259864188196280525361862 7\ldots,\\ K_v &= -1486.96210927882340612774291736359 3\ldots.\end{aligned} \tag{C.54}$$

The complete period expression of P_8 has 120 411 terms, and all 37 original source blocks are included through the weighted boundary calculation. Its rational projection is $-757993 = \mathcal{R}_8$, a separate check of the rational calculation.

For certain six-kink finite endpoint responses, scaling gives an overall factor ε multiplying the last integral; proper rational terms then grow only logarithmically after integration and vanish as $\varepsilon \log^k \varepsilon$, so only the part with polynomial growth contributes to the finite constant. This shortcut is not applied to P_8, whose scaling differs and whose proper rational terms contribute to the residue; both endpoints and the corresponding regularized constants are kept. The complete expression of $Q_{6,vv}$ at $\varepsilon = 1/10$ equals $425.5449632473382428\ldots$ and agrees with direct quadrature of the original kernels. The full source of P_8 passes the six rational divergence checks of the table in Appendix C.4, and a positive-weight divergence comparison is numerically below 10^{-85}. Exact source coverage and numerical period checks are distinct kinds of check.

F4: the ninth-order fugacity. The last new combination is

$$J_9 = A_{9,1} - Z_{8,1} + 2(\partial_\eta A_7)_0 - 2(\partial_\eta Z_6)_0 + D_{9,\text{lower}}, \tag{C.55}$$

with the full inputs

quantity	numerical value
$A_{9,1}$	$-613817.497451118368344265\ldots$
$Z_{8,1}$	$11059.024476469114411035\ldots$
$(\partial_\eta A_7)_0$	$-14142.938944879185749695\ldots$
$(\partial_\eta Z_6)_0$	$305.264828822013665772\ldots$
$D_{9,\text{lower}}$	$653499.204958990556939720\ldots$

They give $J_9 = -273.72451599932464651407791039864 7\ldots$ and $J_9^{(0)} = -\frac{10192}{3}$; the value is a cancellation between terms of order 6×10^5. The complete expression has 84 928 terms. Some base-integral stages were restored from rational-stage files only after an exact audit of the omitted constant terms, and the endpoint-offset logarithms were restored separately. An intermediate error in an endpoint-vacuum multiplicity was corrected, and the affected outputs were regenerated before the accepted evaluations; only the corrected scalar enters the assembly. The independent re-derivation of Sec. 2.6 differs from this record in the first entry of the table, $A_{9,1}$, and agrees on the other four pieces of J_9.

C.6 The full numerical assembly and the zeta-value candidate

The six inherited inputs combined with B_{old} give

$$B_{\text{done}} = B_{\text{old}} + 4W + 2S + \tfrac{3}{32}L_3 - \tfrac{1}{4}M_{21} + \tfrac{1}{2}M_{12} - \tfrac{1}{16}N_2 = 563.4040661247776631183091472046 16\ldots, \tag{C.56}$$

and Eq. (2.10) becomes the scalar identity

$$C_8^{\text{SR}} = B_{\text{done}} + \frac{N_\delta}{8} + \frac{3U}{4} + \frac{K_v}{32} + \frac{J_9}{4}. \tag{C.57}$$

With Eq. (C.48) it reduces to $C_8^{\text{SR}} = C_{\text{fixed}} - \frac{1}{2}P_6$, with the C_{fixed} and the record of Eq. (2.14) given in Sec. 2.5. Every additive ingredient has a numerical value; there is no missing F3 or F4 term, and the exact P_6 integration removes the earlier 10^{-4} discrepancy of the quadrature estimates of P_6 (Appendix C.7). A separate literal-polynomial implementation solves $F = H = 0$ with Eq. (C.24) through $\bar{\delta}^4$, differentiates the full beta functions and obtains the same $8T_4$. The exact equality of this result, of the literal trace polynomial and of $C_{\text{fixed}} - \frac{1}{2}P_6$, with the supplied decimal strings treated as fractions, checks the assembly; it does not certify the numerical integration errors of those strings.

Recognition. The first fresh evaluation, at 70 digits, was used for an integer-relation search in the basis of Sec. 2.5, $\{\zeta(2), \zeta(3), \zeta(4), \zeta(5), \zeta(2)\zeta(3), \zeta(6), \zeta(3)^2, \zeta(7), \zeta(2)\zeta(5), \zeta(3)\zeta(4)\}$. The relation found for ($C_8^{\mathrm{SR}} - 88$, basis) is

$$(3,\ 534,\ -330,\ -339,\ -14676,\ 7128,\ 8630,\ -5616,\ 36504,\ -18144,\ -4608), \tag{C.58}$$

which is Eq. (2.15). The second evaluation, at 100 digits and with the candidate held fixed, differs from it by about 3.0854×10^{-86}; a separate search with the rational intercept free recovers 88. A looser earlier search had produced a different relation that failed at higher precision; its rejected record is kept as a warning against overinterpreting integer relations. The highest-weight part of the candidate, $-12168\zeta(7) + 6048\zeta(2)\zeta(5) + 1536\zeta(3)\zeta(4)$, is compatible with the maximal weight seven (Appendix C.8); compatibility proves neither this coefficient identity nor a period-class theorem to all orders. With the even zeta values written out, the same candidate is

$$\begin{aligned} C_8^{\mathrm{SR}} \stackrel{?}{=} 88 - \frac{89\pi^2}{3} + \frac{113\pi^4}{90} - \frac{1726\pi^6}{567} + \Big(110 - 396\pi^2 + \frac{256\pi^4}{15}\Big)\zeta(3) + (4892 + 1008\pi^2)\zeta(5) \\ + 1872\zeta(3)^2 - 12168\zeta(7), \end{aligned} \tag{C.59}$$

algebraically identical to Eq. (2.15).

Status. The record is numerical: its digits are cross-checked only through P_6, to about 57 decimal places; six of its inputs are single inherited records; and it carries no error bound (Sec. 2.5). Equation (2.15) is conjectured: it has no certificate, and the record is its only high-precision support, so its agreement with the record is not independent evidence for the record's digits. The independent re-derivation of Sec. 2.6 gives $C_8^{\mathrm{SR}} = 242.8 \pm 2.4$ (one standard error), 1.9 to 4.0 standard errors below the record depending on the error estimator, with the difference in $A_{9,1}$. The magnetization-scheme calculation of Appendix I, which shares no input with the compendium, gives 247.38118 ± 0.13, 2.5×10^{-5} from the record; it supports the leading digits of the record but not the coefficients of Eq. (2.15). A proof of Eq. (2.15) would follow the route stated in Sec. 2.5; its verifier would also have to regenerate each source relation and check the complete rational vector, not only its primitive quotient or its numerical value. No such certificate exists.

C.7 Numerical evaluation of periods and error control

The numerical value of a regulated period is not obtained by substituting a large cutoff into a divergent integral; it is the constant term of a controlled expansion in logarithms and inverse powers. For a fixed nonnegative letter a the differential system is triangular,

$$\frac{\mathrm{d}}{\mathrm{d}\Lambda} G(-a, -w; \Lambda) = \frac{G(-w; \Lambda)}{\Lambda + a}, \tag{C.60}$$

where $(-a, -w)$ is the word $-w$ with the letter $-a$ prefixed. The empty word is one, all-zero words have the tangential value $\log^n \Lambda / n!$, and words beginning with a zero letter need their regular and singular parts treated explicitly.

For $a > 0$ and a regular Taylor expansion $G(-w; \Lambda) = \sum_n c_{w,n} \Lambda^n$, the coefficients $c_{a,w,n}$ of $G(-a, -w; \Lambda)$ near the origin obey

$$c_{a,w,n} = \frac{1}{n} \sum_{j=0}^{n-1} (-1)^j a^{-j-1} c_{w,n-1-j}, \qquad n \geq 1. \tag{C.61}$$

The local Taylor expansion is continued along the positive axis in steps smaller than the distance to the nearest singularity. At a large enough matching point the expansion (C.3), $G(-w; \Lambda) =$

$\sum_{k\ge 0}\Lambda^{-k}P_{w,k}(\log\Lambda)$, is used. For the word (a,w) the leading polynomial satisfies $P'_{a,w,0}=P_{w,0}$, with the constant fixed by matching, and the higher polynomials solve

$$(\partial_L-k)P_{a,w,k}(L)=\sum_{j=0}^{k}(-a)^j P_{w,k-j}(L),\qquad k\ge 1. \tag{C.62}$$

The regularized period is $R_w=P_{w,0}(0)$. Ordering the suffixes depth first lowers the peak memory without changing the recurrence. A C++ implementation with the GMP library supplies the arithmetic; a separate Python implementation with decimal arithmetic checks sampled periods and closed controls.

Every accepted run records its working precision, matching point, Taylor order, asymptotic order, step fraction, input hashes and backend. Repeating a run with more printed digits only is not an independent accuracy test; the comparisons of the compendium change several settings at once. Their stability and the controls on lower integrals support the quoted values, but no rigorous global error bound for the high-order assembly exists (Sec. 2.5, "How many digits").

The superseded Mellin quadrature. For a cutoff moment $M(\varepsilon)$ the Mellin transform $\mathcal{M}(\varsigma)=\int_0^{\varepsilon_{\max}}\varepsilon^{\varsigma-1}M(\varepsilon)\,\mathrm{d}\varepsilon$ converges in a right half-plane and is continued analytically; a term $\varepsilon^0\log\varepsilon$ contributes $-\varsigma^{-2}$. Sector decomposition with analytic power and logarithm moments evaluates the continued transform without integrating divergent powers. The compendium's six-kink runs of this kind covered all ordered-gap sectors with compensated summation, but its finest runs of P_6 still differed at about 10^{-4}. They are kept only as historical diagnostics and are not used in the high-precision test of the candidate; the exact boundary calculation of Appendix C.5 replaces them.

C.8 General-formula conjectures and what survived

Matching a finite series and deriving the next coefficient from an independent integral are different tests. The compendium used the latter to reject several attractive extrapolations; none of the general generating formulas it tried is validated.

Exact recursion versus an all-orders solution. Define

$$Q_{\mathrm{root}}(\bar\delta)=\frac{T(\bar\delta)^2/4-D(\bar\delta)}{2\bar\delta}=1+\sum_{m\ge1}q_m\bar\delta^m,\qquad \sqrt{Q_{\mathrm{root}}}=1+\sum_{m\ge1}s_m\bar\delta^m. \tag{C.63}$$

Then, with $T_1=\frac12$ and $s_0=1$,

$$\begin{gathered} q_m=\frac18\sum_{i=1}^{m}T_iT_{m+1-i}-\frac12 D_{m+1},\qquad s_m=\frac12\Big(q_m-\sum_{i=1}^{m-1}s_is_{m-i}\Big),\\ C^{\mathrm{SR}}_{2m}=2^{m-1}T_m,\qquad C^{\mathrm{SR}}_{2m+1}=-2^{m+1}s_m. \end{gathered} \tag{C.64}$$

These equations are exact algebra; they do not determine uncalculated invariants. For instance, C^{SR}_9 needs the new determinant coefficient D_5, which T_4 does not fix.

Filtered MZV structure. Give $\zeta(n_1,\dots,n_r)$ the weight $\sum_i n_i$, with weights adding under products, and let $W_N\mathcal{Z}$ be the rational span of MZVs of weight at most N. The structural conjecture that remains viable is

$$C^{\mathrm{SR}}_n\in W_{n-1}\mathcal{Z}. \tag{C.65}$$

It is not uniform transcendentality, and it does not specify the rational coefficients. The closed forms through C^{SR}_7 obey it, and the candidate (2.15) is consistent with it; its unproved status must

not be upgraded into a theorem to all orders. The geometry of ordered points on the line (genus zero) motivates the period class [26, 25]. A conditional angular-counting argument suggests

$$[u^k z^j \bar{\delta}^r]F \in W_{2k+j+2r-1}\mathcal{Z}, \qquad [u^k z^j \bar{\delta}^r]H \in W_{2k+j+2r-2}\mathcal{Z}. \tag{C.66}$$

The individual inputs of the prescription contain the regulator constants $\log 2$, $\log 3$, $\mathrm{Li}_2(-\frac{1}{2})$ and $\mathrm{Li}_4(\frac{1}{2})$, which are not written as MZVs and which cancel only from the invariants (Sec. 2.3; $\mathrm{Li}_4(\frac{1}{2})$ enters through the input E, Appendix B.3); the compendium does not say how the bound is to be read for them. If subtraction and matching preserve these bounds and close in the two essential couplings, the fixed-point recursion divides only by the rational determinant -4 of the leading Jacobian of (F, H) in (u, z), and the bounds imply Eq. (C.65). Proving that every required renormalization operation preserves them is the missing step. At weight eight the motivic MZV algebra admits a new depth-two direction, representable by $\zeta(5,3)$, beyond products of single zeta values [27]. Under the conjectured bound it could first enter C_9^{SR}, and it need not have a nonzero coefficient; this is not a claim of numerical independence of the corresponding real constants.

Specific guesses and direct tests. For the inverse curve $y = 1 - \Delta_{\mathcal{E}}$ write $x = y^2\mathcal{G}(y)/4$. A specified trial function $\widehat{\mathcal{G}}$ predicts, by Lagrange inversion,

$$\widehat{C}_n = -\frac{2^n}{n}[y^{n-1}]\,\widehat{\mathcal{G}}(y)^{-n/2}. \tag{C.67}$$

Two rational–digamma trials, with scales $(1,2,3)$ and $(1,2,4)$, fit C_1^{SR} to C_6^{SR} exactly but predicted $C_7^{\mathrm{SR}} \simeq 17.91784$ and 247.18113; the completed value $51.06899\ldots$ rejects both fixed generators. A later implicit Gamma family used

$$\mathcal{K}_{ab}(y,t) = 1 - \frac{\Gamma(1+ay)^2}{\Gamma(1+ay+bt)\,\Gamma(1+ay-bt)}, \qquad \mathcal{S} = y^2 - 4t^2 + t^2\Big[P(y) + \sum P_{ab}(y)\,\mathcal{K}_{ab}\Big], \tag{C.68}$$

with $t = \sqrt{x}$. It fits all seven supplied coefficients but predicts the wrong rational part of C_8^{SR}. Quadratic log-Gamma and Borel–Laplace variants also fit their inputs but fail the same direct test. The rational-invariant Padé prescription $\widehat{C}_8 = C_6^{\mathrm{SR}\,2}/C_4^{\mathrm{SR}}$ fails after clearing denominators and taking weight zero:

fixed hypothesis	prediction tested	outcome
geometric rational trace	$\pi_0 C_8^{\mathrm{SR}} = 160$	false: 88
implicit Gamma fit	$\pi_0 C_8^{\mathrm{SR}} = -\frac{55777}{64}$	false
quadratic log-Gamma and Borel–Gamma fits	$\pi_0 C_8^{\mathrm{SR}} = -\frac{78669187}{917504}$	false
rational-invariant Padé	$\pi_0 C_8^{\mathrm{SR}} = 26^2/4 = 169$	false
rational–digamma fits	$C_7^{\mathrm{SR}} \simeq 17.92$ or 247.18	false
filtered MZV class and weight bound	constants allowed at each order	unproved; not contradicted
determinant-only rational guess	$D_m^{(0)} = (3^m + 2^m + 7)/8$ $(m \geq 2)$	next value $D_5^{(0)} = \frac{141}{4}$ untested

The rational trace guess $T_m^{(0)} = (3^m - 1)/4$ is excluded by $T_4^{(0)} = 11 \neq 20$; this alone does not exclude its determinant-only companion. Adjusting a new parameter after learning C_8^{SR} is a new fit, not a successful prior prediction.

Gamma-product and Euler-kernel obstructions. The logarithm of a finite product of Gamma functions with rational linear arguments consists of single-zeta terms, whereas the inverse series contains

$$[y^5]\log\mathcal{G}(y)\big|_{\text{weight }5} = -\frac{17}{2}\zeta(5) + 2\zeta(2)\zeta(3). \tag{C.69}$$

This excludes the simple Gamma-product ansatz as a formal (motivic) identity, but not every nonlinear implicit Gamma equation. A broader single-product trial,

$$K_{\rho,e}(y) = \sum_{n\geq 1} \frac{1}{n(n+ey)} \prod_{k<n} \rho(y/k), \qquad \mathcal{G}_{\max}(y) = 1 + \frac{y^2}{2} K_{\rho,e}(y), \tag{C.70}$$

was tested with an arbitrary analytic ρ, $\rho(0) = 1$. Formal matching of the product terms requires both

$$8e^2 - 16e + 23 = 0, \qquad 864e^4 - 4176e^3 + 16504e^2 - 21884e + 14095 = 0, \tag{C.71}$$

whose resultant is $401375222784 \neq 0$. This exact exclusion applies to the stated family, not to all possible spectral kernels; the code derives the elimination equations rather than loading them as empirical numerical constraints.

C.9 The two ends are not a coefficient-by-coefficient duality

Both perturbative descriptions concern the same long-range family. The expansion of Sec. 2 is about $s = 1$, with $x \to 0$; the expansion about the generalized free field (GFF) with a quartic coupling is about $s = \frac{1}{2}$, with

$$\varepsilon_{\mathrm{G}} = 2s - 1 = 1 - 4x \to 0. \tag{C.72}$$

The strong–weak duality compares two descriptions at the same s; it is not a symmetry that exchanges the two ends [4]. A finite GFF expansion is not controlled at $\varepsilon_{\mathrm{G}} = 1$ and cannot be substituted there to compute a coefficient C_n^{SR}.

In one dimension $\Delta_{\mathcal{E}} = 1 - \nu^{-1}$, and the GFF expansion gives

$$\Delta_{\mathcal{E}} = \frac{1}{2} - \frac{1-4x}{6} + \frac{\pi + 4\log 2}{9}(1-4x)^2 + O\big((1-4x)^3\big), \tag{C.73}$$

where the three-loop diagram must be the corrected tetrahedron value, not the superseded one [23, 28]. With $t = \sqrt{x}$ and $f(t) = \Delta_{\mathcal{E}}(t^2)$, this gives the local derivatives at the GFF end

$$f(\tfrac{1}{2}) = \frac{1}{2}, \qquad f'(\tfrac{1}{2}) = \frac{2}{3}, \qquad f''(\tfrac{1}{2}) = \frac{4}{3} + \frac{32}{9}(\pi + 4\log 2). \tag{C.74}$$

They are not relations derived from C_1^{SR} to C_7^{SR}. Only under extra convergence and termwise-continuation assumptions could one write, for the GFF coefficients a_m of $\varepsilon_{\mathrm{G}}^m$,

$$a_m = (-1)^m \sum_{n\geq 1} \frac{C_n^{\mathrm{SR}}}{2^n} \binom{n/2}{m}, \qquad m \geq 1, \tag{C.75}$$

and even then each a_m mixes infinitely many orders of the expansion about $s = 1$. For an asymptotic series alone the identity is not justified: a flat term $e^{-1/t^2} h(t)$ changes the data at the GFF end without changing any formal coefficient C_n^{SR}. This is a logical example, not a proposed physical correction.

An independent global exclusion. A specified global Gamma trial fails the value at the GFF end outright. Its one-parameter repair enforces the value and the slope, and matching the curvature fixes its last parameter; the resulting third derivative is less than 927. With $A_{\mathrm{G}} = \pi + 4\log 2$, the corrected cubic coefficient b_{G} of the GFF expansion of $\nu^{-1} = 1 - \Delta_{\mathcal{E}}$ (the coefficient of $\varepsilon_{\mathrm{G}}^3$; the compendium calls it the cubic mass coefficient) and the third derivative are

$$b_{\mathrm{G}} = \frac{\mathfrak{a}}{81} - \frac{A_{\mathrm{G}}^2 + \pi^2/3}{108}, \qquad f'''(\tfrac{1}{2}) = \frac{64 A_{\mathrm{G}}}{3} + 384 b_{\mathrm{G}}, \tag{C.76}$$

where, in the corrected integral normalization [28],

$$\mathfrak{a} = 2\int_{\mathbb{R}^2} \frac{\mathrm{d}u\ \mathrm{d}v}{\sqrt{|u(1-u)v(1-v)(u-v)|}} \tag{C.77}$$

(u and v are integration variables here). Reflection and inversion divide the integral into three equal diagonal and six equal off-diagonal sectors; with their integrals $\mathcal{D}$ and $\mathcal{O}$, $\mathfrak{a} = 6\mathcal{D} + 12\mathcal{O}$. The off-diagonal substitution $v \to 1/v$ gives $\mathcal{O} \geq \pi^2$. For independent arcsine variables U_1, U_2 on $[0,1]$, $\mathcal{D} = \pi^2\,\mathbb{E}|U_1 - U_2|^{-1/2}$ and $\mathbb{E}|U_1 - U_2| = 4/\pi^2$, so Jensen's inequality gives

$$\mathcal{D} \geq \frac{\pi^3}{2}, \qquad \mathfrak{a} \geq 3\pi^3 + 12\pi^2 > 211. \tag{C.78}$$

This implies $f'''(\frac{1}{2}) > 992$, which contradicts the repaired global trial; the archived proof gives exact rational interval enclosures for the strict inequalities. The argument excludes that global function family; without further analyticity or uniqueness assumptions it does not by itself exclude a formal rule for the coefficients C_n^{SR}. The rational tests of C_8^{SR} in Appendix C.8 do not need that global assumption.

C.10 Further consequences of the shadow relation

The results of this subsection are separate from the thermal series. They use the published shadow identity and leading data of operator product expansion (OPE) coefficients of the dual [4], not the integrals of many kinks. Unit-normalized primaries obey $\langle \mathcal{O}_i(X)\mathcal{O}_j(0)\rangle = \delta_{ij}|X|^{-2\Delta_i}$, and χ denotes the renormalized, unit-normalized primary of dimension $\Delta_\chi = 1 - x$ obtained from the current of Eq. (2.1), not the unsubtracted current. Here $x = \Delta_\sigma = \bar{\delta}/2$ as throughout.

The gamma-function factor. A scalar shadow transform is proportional to

$$\mathsf{S}_x f(X) = \mathcal{N}_x \int \frac{f(Y)\ \mathrm{d}Y}{|X-Y|^{2(1-x)}}, \tag{C.79}$$

defined by continuation of homogeneous distributions with contact subtractions. For three-point functions at separated points the factor that depends on the pair follows from

$$\int \mathrm{d}X\, e^{-\mathrm{i}pX}|X|^{-\alpha} = 2^{1-\alpha}\sqrt{\pi}\,\frac{\Gamma((1-\alpha)/2)}{\Gamma(\alpha/2)}\,|p|^{\alpha-1}. \tag{C.80}$$

Putting one spectator at zero and one at infinity and convolving the remaining power law gives

$$q(x;\Delta_i,\Delta_j) = \frac{\Gamma\big((1-x+\Delta_i-\Delta_j)/2\big)\Gamma\big((1-x-\Delta_i+\Delta_j)/2\big)}{\Gamma\big((x+\Delta_i-\Delta_j)/2\big)\Gamma\big((x-\Delta_i+\Delta_j)/2\big)}. \tag{C.81}$$

The published relation has the form $c_{\chi ij}/c_{\sigma ij} = \mathcal{N}(\bar{\delta})\,q(x;\Delta_i,\Delta_j)$, with a normalization independent of the spectator pair (Ref. [4], Eq. (5.116)). For the leading operator $\mathcal{O}_+$, of dimension Δ_+ (Eq. (2.6)), the published data give $c_{\sigma\chi+}/c_{\sigma\sigma+} \sim 1/\sqrt{\bar{\delta}}$, while $q(x;x,\Delta_+) \sim 1/(\sqrt{2}\pi\sqrt{\bar{\delta}})$; hence $\mathcal{N}(0) = \sqrt{2}\pi$. Applying the relation twice to an even primary $\mathcal{O}$ of dimension Δ gives

$$\frac{c_{\chi\chi\mathcal{O}}}{c_{\sigma\sigma\mathcal{O}}} = \mathcal{N}(\bar{\delta})^2 P(x,\Delta), \qquad P(x,\Delta) = \frac{\Gamma\big((1-2x+\Delta)/2\big)\Gamma\big((2-2x-\Delta)/2\big)}{\Gamma\big((2x-\Delta)/2\big)\Gamma\big((2x-1+\Delta)/2\big)}. \tag{C.82}$$

The isolated dimension-two primary. At the endpoint the unique primary at dimension two that is even under parity and under the Ising symmetry is $\rho \to :\chi^2:/\sqrt{2}$. Its published leading data are $c_{\chi\chi\rho} = \sqrt{2} + \cdots$ and $c_{\sigma\sigma\rho} = \bar{\delta}/\sqrt{2} + \cdots$, so the double-shadow ratio diverges as $2/\bar{\delta}$. A generic

displacement $\Delta = 2 + a'\bar{\delta} + \cdots$ leaves Eq. (C.82) finite; the required pole lies at $\Delta = 2 - \bar{\delta} + e$ with $e = o(\bar{\delta})$, where

$$P\big(\tfrac{\bar{\delta}}{2}, 2 - \bar{\delta} + e\big) = \frac{\Gamma(\frac{3}{2} - \bar{\delta} + \frac{e}{2})\,\Gamma(-\frac{e}{2})}{\Gamma(-1 + \bar{\delta} - \frac{e}{2})\,\Gamma(\frac{1}{2} + \frac{e}{2})} \sim \frac{\bar{\delta}}{e}. \tag{C.83}$$

Matching $2/\bar{\delta} \sim 2\pi^2\bar{\delta}/e$ gives

$$\Delta_\rho = 2 - \bar{\delta} + \pi^2\bar{\delta}^2 + O(\bar{\delta}^3) = 2 - 2x + 4\pi^2x^2 + O(x^3). \tag{C.84}$$

For the same solution $q(x; x, \Delta_\rho) = \pi\bar{\delta} + O(\bar{\delta}^2)$, hence

$$c_{\sigma\chi\rho} = \pi^2\bar{\delta}^2 + O(\bar{\delta}^3). \tag{C.85}$$

The sign assumes $c_{\chi\chi\rho} > 0$ and the stated sign of χ. The integer-power remainders assume analytic continuation of the isolated even branch; without that regularity the last justified term is followed by $o(\bar{\delta}^2)$. Conjugation by $\hat{\sigma}_3$ implements $g \mapsto -g$, which explains why an isolated even analytic branch has no odd powers of g. No new fully renormalized four-point integral was used.

A weighted statement, not a resolved higher spectrum. Let $N = 2, 4, 6, \ldots$, and let α label the branches that approach dimension N, with v_α the value of $c_{\chi\chi\alpha}$ at $\bar{\delta} = 0$ and $u_\alpha = \lim_{\bar{\delta}\to 0} c_{\sigma\sigma\alpha}/\bar{\delta}$. For regular branches visible to the leading $\chi\chi$ OPE write $\Delta_\alpha = N - \bar{\delta} + \gamma_\alpha^{[2]}\bar{\delta}^2 + o(\bar{\delta}^2)$. The shadow residue implies

$$v_\alpha\gamma_\alpha^{[2]} = \pi^2 N(N-1)\,u_\alpha, \tag{C.86}$$

with a zero u_α interpreted by continuity. The published mixed-correlator weight and the free χ block coefficient are

$$A_N = \sum_\alpha u_\alpha v_\alpha = \frac{4(N-1)[(N-2)!]^2}{N(2N-2)!}, \qquad a_N = \sum_\alpha v_\alpha^2 = \frac{2N(N-1)[(N-1)!]^2}{(2N-2)!}; \tag{C.87}$$

the second follows by matching $1 + z_c^2 + [z_c/(1-z_c)]^2$ to $1 + \sum_N a_N z_c^N\, {}_2F_1(N, N; 2N; z_c)$, with z_c the cross ratio. Therefore

$$\frac{\sum_\alpha v_\alpha^2\gamma_\alpha^{[2]}}{\sum_\alpha v_\alpha^2} = \frac{2\pi^2}{N(N-1)}. \tag{C.88}$$

For $N = 2$ the primary is unique. For higher N this is only an average weighted by the OPE: it does not assign the same correction to every degenerate branch, and it does not determine branches invisible to the leading OPE.

C.11 The logarithmic endpoint at $s = 1$

At $\bar{\delta} = 0$, with $b_0 = 1$, the published beta functions of the dual are those of Eq. (A.26) [4]. Choose the marginally irrelevant branch $g > 0$, $h < 0$, and write its critical separatrix as $h = -g/\sqrt{2} + a_2g^2 + O(g^3)$. The cubic beta functions give $\beta_h - h'(g)\beta_g = -6a_2g^3 + O(g^4)$, hence $a_2 = 0$. With $L = \log(r/r_0)$ (not the input L),

$$\frac{\mathrm{d}g}{\mathrm{d}L} = -2g^2 - 2g^3 + O(g^4), \qquad g(L) = \frac{1}{2L} - \frac{\log L}{4L^2} + \frac{C}{L^2} + O\Big(\frac{\log^2 L}{L^3}\Big). \tag{C.89}$$

Let $M_\sigma^2 = \lim_{r\to\infty}\langle\sigma(r)\sigma(0)\rangle$ on this trajectory. The wall gas gives $G_\sigma(r) - 1 = -4g^2\log(r/\varepsilon) + \cdots$, because a kink pair changes the spin product only when one wall lies between the insertions; so $\gamma_\sigma = 2g^2 + \cdots$, and

$$\log\frac{G_\sigma(r)}{M_\sigma^2} = 2\int_L^\infty \gamma_\sigma(g(u'), h(u'))\,\mathrm{d}u' + \log F_\sigma(g(L), h(L)). \tag{C.90}$$

The finite matching function satisfies $F_\sigma = 1 + O(g^2)$: at $g = 0$ the spin is conserved for every h, and a nonzero spin-flip correction needs two flips, so it cannot change the term in $\log L$. Inserting the running coupling and exponentiating gives

$$\frac{G_\sigma(r)}{M_\sigma^2} = 1 + \frac{1}{L} - \frac{\log L}{2L^2} + \frac{K_{\mathrm{match}}}{L^2} + O\Big(\frac{\log^2 L}{L^3}\Big). \tag{C.91}$$

The coefficient $-\frac{1}{2}$ is fixed by the cubic flow; K_{match} depends on the matching convention and on the definition of r_0. This extends the leading endpoint logarithm of Ref. [4] (its Appendix E). The difference $G_\sigma(r) - M_\sigma^2$ is best called the plateau-subtracted correlator: in the symmetric mixture $\langle\sigma\rangle = 0$ while the plateau is nonzero, so it need not equal the usual connected subtraction. Neither the results of Appendices C.10 and C.11 nor the thermal coefficients rely on a stronger four-point statement that the compendium's earliest record had argued from a formal principal-value cancellation in a current integral; such an identity alone does not determine composite-operator counterterms or the fully renormalized four-point function.

C.12 The polynomial B_{old} of the eighth order

The coefficient names are those of the derivative dictionary of Appendix C.3. All terms are fixed by the 25 inputs of C_3^{SR} to C_7^{SR}; the ten new inputs enter only through the linear terms of Eq. (2.10). This is the unprojected expression: its rational projection is $\frac{1153}{6}$, and its value is $-861.663567235479779201497190992 3\ldots$.

$$\begin{aligned} B_{\mathrm{old}} = {} & -\gamma - \frac{5c^3}{1024} - \frac{3\Omega}{16} - \frac{\Psi}{2} - \frac{Q}{4} - \frac{M}{16} - \frac{P}{16} \\ & - \frac{\alpha}{16} - \frac{a^3}{16} - \frac{K}{64} - \frac{N}{64} + \frac{\beta}{4} + \frac{\Phi}{8} + \frac{f^2}{64} \\ & + \frac{3L}{64} + \frac{Ap}{2} + \frac{Ea}{2} - Ep - Va - \frac{9Jc}{64} - \frac{5ac^2}{256} \\ & - \frac{3Aa}{16} - \frac{3Ac}{64} - \frac{3a^2c}{64} - \frac{3Nc}{128} - \frac{Gp}{2} - \frac{Eb}{4} - \frac{Lp}{4} \\ & - \frac{Ra}{4} - \frac{Rc}{4} - \frac{Vc}{4} - \frac{bq}{4} - \frac{pq}{4} - \frac{Gb}{8} - \frac{Ja}{16} \\ & - \frac{Lb}{16} - \frac{Mc}{16} - \frac{bk}{16} - \frac{fq}{16} - \frac{kp}{16} - \frac{af}{64} - \frac{fk}{64} \\ & + \frac{a^2p}{4} + \frac{Ab}{8} + \frac{Ec}{8} + \frac{aq}{8} + \frac{La}{16} + \frac{a^2b}{16} + \frac{cq}{16} \\ & + \frac{ak}{32} + \frac{c^2p}{32} + \frac{cf}{256} + \frac{3Ga}{16} + \frac{3Gc}{32} + \frac{3Lc}{64} + \frac{3bf}{64} \\ & + \frac{5bc^2}{128} + \frac{5ck}{256} + \frac{5c^2f}{512} + \frac{acp}{8} + \frac{3abc}{64}. \end{aligned} \tag{C.92}$$

The machine-readable version is the field `old_part` of `code/dual_d1/data/archived/C8_full_work/full_trace.json`. The separate full-Jacobian program recomputes this expression from the beta functions; it does not read an expected value of C_8^{SR} to obtain it.

D The two-coupling closure and the prescription

This appendix gives two formal arguments about the dual of Sec. 2.1, both written by OpenAI GPT-6 astra (via ChatGPT) and supplied by the author on 15 October 2026 (Secs. 1.2 and 5), merged and edited to the notation of this note: the *closure argument*, on the bulk counterterms of the dual, and the *prescription argument*, on what the matched two-coupling prescription computes. Neither has been refereed. This note counts both as formal, unrefereed arguments (Secs. 2.1 and 2.9). Where an argument states a proposition or a theorem, it is given below as the statement that the argument establishes formally, coefficientwise in g, h and $\bar\delta$; it is not a proved result of this note. Appendix D.15 lists what the review checked.

Throughout, $\bar\delta = 1 - s$ and $x = \Delta_\sigma = \bar\delta/2$. The dual, its operators $\mathcal{O}_g$ and $\mathcal{O}_h$ and the charge $b = b_0 - \sqrt{2}h$ are those of Eq. (2.1). The bare current $\chi_B = \frac{\mathrm{i}}{\sqrt{2}}\partial\phi$ is the χ of Eq. (2.1); as in Appendix A.3, the subscript distinguishes it from the renormalized current primary, and neither is the generalized free field χ of Secs. 3 and 4. The regulator, the current resummation, the finite-tail matching, the vacuum normalization, and the finite-response Jacobian and recurrence are given as the compendium states them in Appendices A.4, A.5, A.7 and A.9, whose notation is used here (in particular ∂_η is the derivative with respect to the endpoint source η at $\eta = 0$). Where the prescription argument re-derives one of these objects, its derivation is given here and the formula is referred to there.

D.1 The question, three claims and the common framework

The thermal series of Sec. 2 rests on two assumptions (Sec. 2.1): the renormalization of the dual involves only the two couplings (g, b), and the matched prescription extracts their β functions. Three claims must be kept apart.

(1) *The full Wilsonian action at every finite cutoff contains only two interactions.* This is false in an ordinary fixed-field cutoff scheme (Appendix D.6).

(2) *Only two nontrivial independent bulk counterterm structures are needed in the formal perturbation theory about the endpoint.* This is the restricted statement of the closure argument (Appendix D.2).

(3) *The archived subtraction algorithm implements that renormalization, so that its computed coefficients are those of the dual.* This follows neither from operator counting nor from the exact period certificates of Sec. 2.4. The prescription argument addresses the analytic recipe behind it (Appendix D.3); whether the historical integrations follow that recipe is a separate question (Sec. 2.6 and Appendix H.5).

The distinction between (1) and (2) is not semantic: finite irrelevant interactions can contribute to higher-order marginal β functions even when no independent irrelevant counterterm is required (Appendix D.9).

Claim (2) does not imply claim (3). The argument "there are two counterterms, and two response equations determine two numbers" is insufficient, because for arbitrary responses the two numbers can absorb an endpoint or insertion anomalous dimension. Schematically, the single-logarithm equation of two responses could be

$$R_1 = \mathcal{J}\beta + \Gamma_\partial R_0 + \mathcal{A}_{\mathrm{probe}}, \tag{D.1}$$

where R_0 and R_1 are the finite and single-logarithm parts of the responses, $\mathcal{J} = D_c R_0$ is the Jacobian of their finite parts (Appendix D.12; its tree-level value is the identity), Γ_∂ an endpoint anomalous dimension and $\mathcal{A}_{\mathrm{probe}}$ a renormalization of the probe. Inverting $\mathcal{J}$ does not show that the last two terms vanish. The prescription argument argues that both are absent for the specific normalized, separated-probe prescription of Sec. 2.1, and only then inverts $\mathcal{J}$.

Common framework. Both arguments work coefficientwise in a joint formal series in g, h and $\bar{\delta}$ about $g = h = \bar{\delta} = 0$ (and, where needed, in the reference charge about $b_0 = 1$), at zero magnetic field. They use a fixed compact target coordinate and a reference normalization of the Wick products. Distances in the Gaussian formulas are dimensionless; a reference inverse length κ is restored by replacing r by κr. An infrared regulator, or a smooth compactly supported switching function of the interaction, is kept throughout. The subject is ultraviolet collisions, not an infrared limit and not the renormalization of arbitrary composite insertions.

The published analysis identifies two symmetry-invariant marginal perturbations at $\bar{\delta} = 0$ but distinguishes this explicitly from an all-orders renormalizability result for its non-scaling perturbation at $\bar{\delta} > 0$ [4, Secs. 3.2.3, 4 and 6]; the compendium likewise qualifies its higher coefficients by the two-coupling and matching assumptions.

D.2 The closure statement

The closure argument, a formal and unrefereed argument written by OpenAI GPT-6 astra (see the head of this appendix; its scope is in Appendix D.14 and its checks in Appendix D.15), establishes formally the following statement, which it states as a proposition.

> *Formal bulk counterterm class.* For the matrix-valued Gaussian model of Eq. (2.1), impose its diagonal shift symmetry, internal reflection and parity (Appendix D.4). Work coefficientwise in g, h and $\bar{\delta}$ about the marginal endpoint, using the Wick products of the deformed covariance (Appendix D.7), with a fixed infrared cutoff. Then local symmetry-preserving subtractions can be constructed recursively such that every required bulk action counterterm for constant couplings lies in
>
> $$\operatorname{span}\{\mathbf{1}, \mathcal{O}_g, \mathcal{O}_h\}, \tag{D.2}$$
>
> modulo total derivatives. Only two nontrivial independent bulk coupling counterterms are therefore required. Field-insertion and endpoint counterterms are not included in this count.

The argument has three parts: the exact selection rules and the classification of local operators (Appendix D.5), power counting (Appendix D.7), and the recursive subtraction of all partial coincidences (Appendix D.8). Since h and $b = b_0 - \sqrt{2}h$ are two coordinates on one coupling direction (Appendix D.11), the statement supplies a counterterm-class argument for the two essential coordinates (g, b). The closure argument sums it up as follows: in the zero-field, symmetry-preserving formal expansion about the marginal endpoint, two nontrivial independent bulk ultraviolet counterterm directions suffice, in addition to vacuum and external-insertion or endpoint renormalization; a finite-cutoff Wilsonian action nevertheless contains irrelevant terms, and their finite contributions must be incorporated in any correct high-order calculation.

The statement uses the standard local extension construction explicitly. It is not a claim that any procedure that cancels two chosen scalar divergences implements that construction; that is the subject of the prescription argument.

D.3 The prescription statement

The prescription argument, a formal and unrefereed argument written by OpenAI GPT-6 astra (see the head of this appendix; its scope is in Appendix D.14 and its checks in Appendix D.15), establishes formally the following statement, which it states as a theorem. Its hypotheses are:

(a) the reference Gaussian and matrix operator algebra of Appendix D.4, at zero magnetic field;

(b) a coefficientwise expansion in g, h and $\bar{\delta}$ and, when needed, in the reference charge about $b_0 = 1$;

(c) the diagonal $O(2)$ symmetry (the $U(1)$ of Sec. 2.1 and the internal reflection) is preserved;

(d) the interaction is restricted to a finite interval, with a hard core on successive kink positions only;

(e) current products are defined by exact Gaussian current resummation, retaining the specified Wick-ordering and endpoint factors;

(f) every response is normalized by the same vacuum expectation of that interval operator;

(g) the reference-theory probe of each response lies outside the interaction support, and any projection to an infinitely distant probe is taken only afterwards.

> *Formal normalized-interval prescription.* Within local, scaling-degree-preserving position-space renormalization, the bulk counterterms are $\mathbf{1}$, $\mathcal{O}_g$ and $\mathcal{O}_h$, and all ultraviolet counterterms localized at either endpoint of the interval are scalar multiples of $\mathbf{1}$. Consequently the normalized responses require no additional endpoint or probe counterterm. If their tree-level two-by-two response matrix is invertible, their single-logarithm equation determines the local logarithmic β function. Relative to any other local normalization in the same class, the result is related by a regular finite change of the two nontrivial bulk couplings, so the fixed-point stability eigenvalues are identical.

As defined in Sec. 2.1, the matched two-coupling prescription has the form of hypotheses (d) to (f): (d) is its item (i), (e) its items (ii) and (iii), and (f) its item (iv). The prescription argument identifies its two responses with the far-field projections of separated reference probes, as hypothesis (g) requires (Appendix D.10); that projection is stated rather than shown in detail (Appendix D.15). Identical eigenvalues mean identical invariants T and D (Sec. 2.2), and so an identical series $\Delta_{\mathcal{E}} = 1 + \sum_n C_n^{\mathrm{SR}} x^{n/2}$.

The statement concerns the prescription as an analytic recipe, not whether a particular computer output implemented every part of it without error. The complete normalized moments, the first endpoint-source derivatives and the exact normal-ordering map are essential hypotheses, not optional approximations; the argument compares them with the compendium's formulas (Appendices D.11 and D.12). It uses standard local distribution-extension and finite-renormalization results [29, 30]; the model-specific steps are the collision bounds, the endpoint classification, the spectator factorization and the audit of the current and normal-ordering factors. These steps do not use Lorentzian locality of the nonlocal Gaussian action: they concern the explicit Euclidean Wick kernels, which are smooth away from coincidence diagonals.

The order of the steps is what makes the argument non-circular: the local renormalization is built for arbitrary spectators first (Appendix D.8), the endpoint cancellation and the absence of probe renormalization follow (Appendix D.10), and the invertibility of the response matrix is used only at the end, for uniqueness (Appendix D.13).

D.4 The reference theory, its symmetries and the regulated responses

Reference theory. At $\bar{\delta} = 0$ the field normalization and compactification are chosen so that

$$\langle \phi(X)\phi(0)\rangle_0 = -2\log(\kappa|X|), \qquad V_\pm = \kappa :e^{\pm \mathrm{i} b_0 \phi}:, \qquad \chi_B = \frac{\mathrm{i}}{\sqrt{2}}\partial\phi, \qquad b_0 = 1. \tag{D.3}$$

Thus $V_\pm$ and χ_B have reference dimension one. The deformed covariance, before any expansion in $\bar{\delta}$, is

$$C_{\bar{\delta}}(r) = \frac{2(\kappa^{-\bar{\delta}} - |r|^{\bar{\delta}})}{\bar{\delta}} = -2 f_{\bar{\delta}}(|r|), \qquad f_{\bar{\delta}}(r) = \frac{r^{\bar{\delta}} - \kappa^{-\bar{\delta}}}{\bar{\delta}}, \tag{D.4}$$

which at $\kappa = 1$ is the covariance of Sec. 2.1. The zero mode is removed, and complete charged observables carry the appropriate neutralizing probe. With $\hat{\sigma}_\pm = (\hat{\sigma}_1 \pm \mathrm{i}\hat{\sigma}_2)/2$, the interactions are $\mathcal{O}_g = \hat{\sigma}_+ V_+ + \hat{\sigma}_- V_-$ and $\mathcal{O}_h = \hat{\sigma}_3 \chi_B$, and the evolution over an interval I is

$$\mathcal{D}_I(c) = \mathrm{P} \exp \int_I \mathrm{d}X \, [g\mathcal{O}_g + h\mathcal{O}_h], \qquad c = (g, h). \tag{D.5}$$

Exact symmetries. With A a matrix of the qubit, the global symmetry includes

$$\phi \mapsto \phi + \alpha/b_0, \quad A \mapsto e^{-\mathrm{i}\alpha\hat{\sigma}_3/2} A e^{\mathrm{i}\alpha\hat{\sigma}_3/2}; \qquad \phi \mapsto -\phi, \quad A \mapsto \hat{\sigma}_1 A \hat{\sigma}_1. \tag{D.6}$$

The first is the $U(1)$ of Sec. 2.1, the second the internal reflection; together they form the diagonal $O(2)$. Parity combines $X \mapsto -X$, $\phi(X) \mapsto -\phi(-X)$, the reversal of the path ordering and matrix transposition. These transformations preserve the regulated matrix expansion [4, Eqs. (3.31)–(3.35)]. At zero magnetic field the reflection is not broken. The internal reflection, not spatial parity alone, is what excludes a local $\hat{\sigma}_3$ counterterm at a single endpoint (Appendix D.7).

The regulated responses. The interaction is restricted to $I = [0, L]$, and the distinct-kink integrations run over

$$0 < X_1 < \cdots < X_n < L, \qquad X_{i+1} - X_i \geq \varepsilon, \tag{D.7}$$

with no hard core between a kink and an endpoint; $\ell = \log \varepsilon$ (or $\log(\varepsilon\mu)$ at a reference scale μ). The compendium works at $L = \kappa^{-1} = 1$ (Eq. (A.14)). At a finite regulator it is useful first to keep a smoothed Gaussian covariance, for which the matrix change-of-frame identity of Appendix D.11 is exact, and then to pass to normal-ordered products. Changes of the smoothing at separations of order ε are local regulator changes, handled by the forest construction of Appendix D.8. The coincident Wick convention used in the archive is $C(0) \to C_{\bar{\delta}}(\varepsilon)$.

For a reference probe $\mathcal{X}$ separated from I, the normalized response is

$$\mathcal{R}_{\mathcal{X}}(\varepsilon; c) = \frac{\frac{1}{2}\operatorname{tr}\langle \mathcal{X}\, \mathcal{D}_{I,\varepsilon}(c)\rangle_0}{\frac{1}{2}\operatorname{tr}\langle \mathcal{D}_{I,\varepsilon}(c)\rangle_0}, \tag{D.8}$$

with the same interval and regulator in numerator and denominator. The probes used for the two coordinates are the reference vertex-and-matrix and current-and-matrix probes, normalized by their free two-point functions. They are *not* the fully interacting, renormalized primaries inserted inside the interacting interval.

D.5 Matrix algebra and the local operators below threshold

Matrix algebra excludes higher-charge vertices. The exact relations

$$\hat{\sigma}_+^2 = \hat{\sigma}_-^2 = 0, \qquad \hat{\sigma}_+\hat{\sigma}_- = \frac{\mathbf{1} + \hat{\sigma}_3}{2}, \qquad \hat{\sigma}_-\hat{\sigma}_+ = \frac{\mathbf{1} - \hat{\sigma}_3}{2} \tag{D.9}$$

force the sequence of nonzero flips to alternate. Inserting diagonal $\hat{\sigma}_3$ factors, as the current interaction does, does not change this, since $\hat{\sigma}_\pm \hat{\sigma}_3^k \hat{\sigma}_\pm = 0$. Consequently every nonzero contiguous matrix word M has vertex charge $Q \in \{0, +1, -1\}$ and satisfies $[\hat{\sigma}_3, M] = 2QM$: an even number of flips gives $Q = 0$ and a diagonal matrix, an odd number gives $Q = \pm 1$ and a matrix proportional to $\hat{\sigma}_\pm$. This holds for words of every length, not only for a two-vertex product. In particular, no ordering of the interactions generates a bulk counterterm of charge two.

Equivalently, the complete shift-invariant local Wick algebra generated by these fields consists of

$$P_{\mathbf{1}}(\partial\phi, \partial^2\phi, \ldots)\, \mathbf{1}, \qquad P_3(\partial\phi, \partial^2\phi, \ldots)\, \hat{\sigma}_3, \qquad P_+(\partial\phi, \ldots)\, \hat{\sigma}_+ V_+, \qquad P_-(\partial\phi, \ldots)\, \hat{\sigma}_- V_-, \tag{D.10}$$

with $P_{\mathbf{1}}, P_3, P_\pm$ polynomials in the derivatives of ϕ. This is the perturbative local algebra: disorder or boundary-condition-changing fields outside it are not adjoined to the theory. An undifferentiated polynomial in ϕ is not an extra shift-invariant local operator (ϕ itself is the unobservable zero-mode field); expanding a vertex in powers of ϕ and giving those powers independent couplings would break the exact diagonal shift symmetry.

Classification. The reference dimension of a derivative monomial is

$$d\Big(\prod_{j\geq 1}(\partial^j\phi)^{n_j}\Big) = \sum_{j\geq 1} j n_j, \tag{D.11}$$

and a charged vertex sector adds one. These are dimensions at $\bar{\delta} = 0$; they do not claim that the deformed vertex is a scaling operator at $\bar{\delta} \neq 0$. Through dimension one the complete list of candidates is that of Table 5. The two even entries of dimension one also have the required parity.

candidate	reference dimension	internal reflection
$\mathbf{1}$	0	even
$\hat{\sigma}_3$	0	odd
$\partial\phi$	1	odd
$\hat{\sigma}_3\partial\phi$	1	even
$\hat{\sigma}_+V_+ + \hat{\sigma}_-V_-$	1	even
$\hat{\sigma}_+V_+ - \hat{\sigma}_-V_-$	1	odd

Table 5: The $U(1)$-invariant local operators of the algebra (D.10) through reference dimension one, with their behaviour under the internal reflection of Eq. (D.6).

The only nontrivial symmetry-invariant local action densities at dimension one are therefore $\mathcal{O}_g$ and $\mathcal{O}_h = \frac{\mathrm{i}}{\sqrt{2}}\hat{\sigma}_3\partial\phi$. The identity changes the vacuum normalization. A source for $\hat{\sigma}_3$ (a magnetic field) would be an additional relevant coupling; it is excluded by the zero-field symmetry of the problem considered.

At dimension zero the invariant sector is $\mathbb{R}\,\mathbf{1}$: a neutral constant matrix is diagonal, and the internal reflection equates its two entries ($\hat{\sigma}_3$ is odd, and the bare $\hat{\sigma}_\pm$ are charged). This is the input of the endpoint classification of Appendix D.7.

D.6 The product $\mathcal{O}_g\mathcal{O}_g$: no literal two-operator closure

At a symmetric pair of points with $r > 0$, Wick's theorem with Eq. (D.3) gives

$$\begin{aligned}\mathcal{O}_g(X+\tfrac{r}{2})\,\mathcal{O}_g(X-\tfrac{r}{2}) &= r^{-2}\Big[\mathbf{1} + \mathrm{i}r\hat{\sigma}_3\partial\phi - \frac{r^2}{2}{:}(\partial\phi)^2{:} + O(r^3)\Big] \\ &= \frac{\mathbf{1}}{r^2} + \frac{\sqrt{2}}{r}\mathcal{O}_h(X) + {:}\chi_B^2{:}(X) + O(r).\end{aligned} \tag{D.12}$$

Only $\hat{\sigma}_+\hat{\sigma}_-$ and $\hat{\sigma}_-\hat{\sigma}_+$ contribute; they multiply $r^{-2}{:}e^{\pm\mathrm{i}(\phi(X+r/2)-\phi(X-r/2))}{:}$, and $\phi(X+\frac{r}{2}) - \phi(X-\frac{r}{2}) = r\partial\phi + O(r^3)$. The midpoint convention removes a spurious first derivative of the expansion point. The coefficient of ${:}\chi_B^2{:}$ is one, not zero. Integrating a short-distance shell $\varepsilon < r < \varepsilon e^{\mathrm{d}\ell}$ in a standard real-space shell prescription therefore produces, besides the two lower-dimensional terms, an irrelevant term proportional to

$$g^2\varepsilon\;\mathrm{d}\ell\int \mathrm{d}X\;{:}\chi_B^2{:}(X). \tag{D.13}$$

Its detailed Wilsonian coefficient depends on the scheme; its appearance as an allowed term of the operator product expansion does not.

The current-squared channel is finite as $\varepsilon \to 0$ in this two-insertion integral (its radial integrand is r^0). This is why its generation does not refute the closure of the ultraviolet counterterm class (Appendix D.2). It does refute the unqualified statement that no other operator coefficient is generated at finite cutoff, that is, claim (1) of Appendix D.1. Some such terms can be removed by redundant transformations in particular schemes, but that is not the same as their absence from a fixed-field Wilsonian action. How these finite terms enter the marginal β functions is the subject of Appendix D.9.

D.7 Power counting in the bulk and at the endpoints

Bulk clusters. Consider $N \geq 2$ insertions of reference dimension one collapsing toward one interior point. Removing the centre coordinate leaves $N-1$ relative coordinates, written as a scale ρ and shape variables, with radial measure $\rho^{N-2}\,\mathrm{d}\rho$. If the product produces a local field $\mathcal{A}$ of reference dimension $d_{\mathcal{A}}$, its coefficient scales as $\rho^{d_{\mathcal{A}}-N}$, up to logarithms introduced by subtractions, and the radial integral has the form

$$\int_0^{\rho_0} \mathrm{d}\rho\, \rho^{N-2}\rho^{d_{\mathcal{A}}-N}(\log\rho)^M = \int_0^{\rho_0} \mathrm{d}\rho\, \rho^{d_{\mathcal{A}}-2}(\log\rho)^M. \tag{D.14}$$

The threshold does not depend on N:

$$d_{\mathcal{A}} = 0:\ \text{power divergence}; \qquad d_{\mathcal{A}} = 1:\ \text{logarithmic divergence}; \qquad d_{\mathcal{A}} \geq 2:\ \text{integrable}, \tag{D.15}$$

and for any finite M the logarithms do not change it. By Table 5, symmetry leaves $\mathbf{1}$, $\mathcal{O}_g$ and $\mathcal{O}_h$ below the threshold.

Alternating vertices. The same follows directly from Wick's theorem. For n vertex insertions with charges $q_i = \pm 1$ and total charge Q, at $b_0 = \kappa = 1$,

$$\prod_i V_{q_i}(X_i) = \prod_{i<j} |X_i - X_j|^{2q_iq_j} : \exp\Big(\mathrm{i}\sum_i q_i\phi(X_i)\Big):, \qquad 2\sum_{i<j} q_iq_j = Q^2 - n. \tag{D.16}$$

For a cluster $X_i = X + \rho r_i$ the normal product is organized exactly as

$$\prod_i V_{q_i}(X_i) = \mathcal{W}(\rho, \boldsymbol{r}) : \exp\Big[\mathrm{i}b_0 Q\phi(X) + \mathrm{i}b_0 \sum_{k\geq 1} \frac{\rho^k m_k}{k!}\partial^k\phi(X)\Big]:, \qquad m_k = \sum_i q_i r_i^k, \tag{D.17}$$

where the prefactor $\mathcal{W}$ has radial degree $Q^2 - n$ at the reference point. A descendant of derivative level k adds a factor ρ^k, so the radial power is

$$\rho^{n-2}\rho^{Q^2-n+k} = \rho^{Q^2+k-2}. \tag{D.18}$$

For even (neutral) clusters, $Q = 0$, only $k \leq 1$ diverges; for odd clusters, $Q = \pm 1$, only $k = 0$; all higher derivatives are integrable. The operator content follows without reference to any probe: summing the two orientations of an even cluster gives the identity at level zero and a multiple of $\hat{\sigma}_3\partial\phi$ at level one, while an odd cluster gives $\mathcal{O}_g$ at level zero and its first-derivative descendants already have dimension two. Current insertions obey the same bound, because χ_B has reference dimension one and its contractions use one or two derivatives of the logarithmic covariance. Simultaneous collapse is not the whole argument, since the shape variables still have partial-collision singularities; these are treated in Appendix D.8.

Endpoint clusters: the degree differs by one. At a fixed endpoint of I the centre of the cluster also approaches the endpoint. There are then N, not $N-1$, scaled coordinates, and

$$\int_0 \mathrm{d}\rho\, \rho^{N-1}\rho^{d_{\mathcal{A}}-N}(\log\rho)^M = \int_0 \mathrm{d}\rho\, \rho^{d_{\mathcal{A}}-1}(\log\rho)^M. \tag{D.19}$$

An endpoint divergence therefore requires $d_{\mathcal{A}} \le 0$. The invariant dimension-zero sector is $\mathbb{R}\,\mathbf{1}$ (Appendix D.5), hence

every primitive ultraviolet counterterm at an endpoint is a scalar multiple of $\mathbf{1}$. (D.20)

This holds separately at the left and at the right endpoint; their scalar coefficients need not be assumed equal. The classification is made in the original invariant (g, h) frame. In the frame in which the current is resummed, the endpoint vertices are dressed and can carry non-integer auxiliary charges; they must be transformed together with the bulk charge and the Wick factors, and treating them as unrelated new endpoint interactions would not describe the same theory. The estimates (D.14) and (D.19) apply after proper collision subgraphs have been subtracted; they must not be applied separately to unprepared products of a power-divergent counterterm and an omitted remainder.

The long-range deformation does not enlarge the threshold. At fixed reference scale ($\kappa = 1$) the deformation of the covariance (D.4) is

$$f_{\bar\delta}(r) = \frac{r^{\bar\delta} - 1}{\bar\delta} = \sum_{j\ge 0} \frac{\bar\delta^{\,j} \log^{j+1} r}{(j+1)!} = \log r + \frac{\bar\delta}{2}\log^2 r + \frac{\bar\delta^2}{6}\log^3 r + \cdots . \tag{D.21}$$

At each fixed order in $\bar\delta$, a Gaussian vertex coefficient is the power-law coefficient at $\bar\delta = 0$ times a finite polynomial in logarithms. Spatial derivatives of $\log^j r$ have the same inverse powers of the distance as the same derivatives of $\log r$, multiplied by logarithms and constants, so they do not raise the superficial degree of divergence at that order. This is visible before the outgoing operator is expanded: the Wick normal product has the exact form

$$:\exp\Big(\mathrm{i}\sum_i q_i\phi(X_i)\Big):_{\bar\delta} = :\exp\Big(\mathrm{i}Q\phi(X) + \mathrm{i}\sum_{k\ge 1}\frac{\rho^k m_k}{k!}\partial^k\phi(X)\Big):_{\bar\delta}, \tag{D.22}$$

with the moments m_k of Eq. (D.17). The deformation changes the contraction coefficients and the normal-ordering convention, but not the list of possible outgoing charges or derivative levels. The normal ordering is varied consistently with the contractions, rather than differentiating the contraction and forgetting the definition of the local field. Consequently, for every fixed coefficient of $g^{n_1}h^{n_2}\bar\delta^{\,n_3}$, the coefficient $t_{\mathcal{A}}$ of an outgoing field obeys the reference scaling-degree bound

$$\operatorname{sd} t_{\mathcal{A}} \le N - d_{\mathcal{A}}, \tag{D.23}$$

with possibly higher powers of logarithms, so that the degree of divergence is $\omega_{\mathcal{A}} = \operatorname{sd} t_{\mathcal{A}} - (N-1) \le 1 - d_{\mathcal{A}}$ in the bulk and $\omega_{\mathcal{A}} \le -d_{\mathcal{A}}$ at an endpoint. Derivatives in the charge, that is an expansion in the resummed charge $b^2 - 1$, likewise insert only logarithms, provided the expansion is taken order by order about the reference value and is not a statement uniform in an arbitrary fixed b.

Order of limits. The bound (D.23) concerns a formal coefficient before all powers of $\bar\delta$ are summed. It does not justify interchanging that infinite sum with $\varepsilon \to 0$. At fixed $\bar\delta > 0$, $r^{\bar\delta}$ tends to zero as $r \to 0$, while each fixed Taylor coefficient in Eq. (D.21) grows logarithmically; these are different ultraviolet organizations. The existence of coefficientwise counterterms establishes neither convergence, nor Borel summability, nor a unique identification of the formal series with a theory at fixed $\bar\delta$; neither argument asserts such an interchange.

D.8 The local forest construction

This construction supplies the renormalized family *before* any response condition is used, which is what keeps the prescription argument from being circular. It is the recursive position-space (Epstein–Glaser) construction, with the matrix multiplication kept in its original order.

The extension principle. A distribution in m relative variables, defined away from a coincidence point and of finite scaling degree sd, has an extension of the same scaling degree. If $\mathrm{sd} < m$ the extension is unique; otherwise its local ambiguity is a finite sum of derivatives, of order at most $\lfloor \mathrm{sd} - m \rfloor$, of the Dirac distribution at the coincidence point. A finite logarithmic scaling anomaly does not change that bound [29, 32]; Euclidean recursive versions are also available [31]. The arguments apply these results to the explicit finite-order Gaussian and operator-product coefficient distributions of the dual; the cited works do not establish the nonperturbative impurity theory.

Structure at separated points. At separated points every fixed perturbative coefficient is a finite sum of products of rational powers of differences and finite powers of logarithms, multiplying multilocal normal-ordered exponentials and derivative polynomials with Pauli matrices. Only coincidence loci can be singular, and all contractions with spectators a fixed distance away are smooth functions of the cluster coordinates.

The expansion is operator valued and made before projection. The collision of a cluster is expanded with Eq. (D.17) into the local operators of Eq. (D.10), as an operator statement, with no reference to the two probes used in the computation. To see that its coefficients do not depend on distant spectators, take external charges Q_α at points y_α a fixed positive distance from the cluster. In a chamber where the signed distances are fixed, their Wick factors expand as

$$\prod_{i,\alpha} |y_\alpha - X - \rho r_i|^{2q_i Q_\alpha} = \prod_{\alpha} |y_\alpha - X|^{2QQ_\alpha} \exp\Big[-2 \sum_{k\geq 1} \frac{\rho^k m_k}{k} \sum_{\alpha} \frac{Q_\alpha}{(y_\alpha - X)^k}\Big]. \tag{D.24}$$

The level-zero and level-one terms are exactly the Wick correlators of the local operators of Eq. (D.17) with those spectators; derivative spectators follow by differentiation. Spectator dependence therefore multiplies universal local collision coefficients and does not enter their definition.

A finite Taylor jet suffices. The expansion into local output fields is in general infinite, but only a finite Taylor jet is needed: neutral clusters need the constant and first-derivative jets, and charged clusters the leading vertex jet. The next Taylor remainder gains at least two powers of ρ in the neutral channel and one in the charged channel. Tested against separated smooth spectators, it obeys the improved bound in the remaining shape sectors, so the argument does not interchange an unproved infinite operator-product sum with a divergent integral.

Induction over clusters. Suppose that the products of fewer than N insertions have been extended at all their coincidence loci. The N-point configuration space is subdivided by relative cluster scales. In one dimension a collapsing subset of an ordered chamber is consecutive, and compatible scale hierarchies consist of nested or disjoint clusters, that is, forests. On sectors where some clusters have already collapsed, the lower-order local subtractions are inserted. This defines the renormalized product away from the total diagonal. Overlapping candidate subclusters occur in different hierarchy sectors, so they are never subtracted twice in one forest, nor treated as independent simultaneous subtractions.

For the remaining total collision of a cluster γ, let $t_{\gamma,\mathcal{A}}$ be the prepared coefficient distribution of its output field $\mathcal{A}$, in m relative coordinates. By Eq. (D.23) its excess of scaling degree obeys

$$\omega_{\gamma,\mathcal{A}} \leq \begin{cases} 1 - d_{\mathcal{A}} & \text{(bulk)}, \\ -d_{\mathcal{A}} & \text{(fixed endpoint)}. \end{cases} \tag{D.25}$$

Choose a smooth compactly supported function w equal to one near zero, at a reference scale μ. For $\omega \geq 0$ a test function f is replaced by

$$(\mathcal{T}_\omega f)(\xi) = f(\xi) - w(\xi) \sum_{|\alpha|\leq\omega} \frac{\xi^\alpha}{\alpha!}\, \partial^\alpha f(0). \tag{D.26}$$

The extra vanishing order makes $t_{\gamma,\mathcal{A}}(\mathcal{T}_\omega f)$ integrable; for $\omega < 0$ nothing is subtracted. Finite local normalization constants are then added to specify the chosen extension. The subtractions are local by Eq. (D.26), and their coefficients are independent of spectators by Eq. (D.24). In channels with $d_\mathcal{A} \geq 2$ the scaling-degree-preserving extension is unique once the subdivergences have been removed, so no new irrelevant bulk counterterm is needed. Only fields of dimension at most one can carry a counterterm ambiguity, and Table 5 leaves $\mathbf{1}$, $\mathcal{O}_g$ and $\mathcal{O}_h$. Derivatives of the Dirac distribution in a dimension-zero channel act on switching functions or produce total derivatives; they are not extra bulk interactions with constant couplings. Counterterms of the identity give the vacuum normalization.

Stability under counterterm insertions. An inserted counterterm is an effective vertex of dimension at most one. With N' effective vertices of dimensions $d_i \leq 1$, the new degree of divergence is bounded by

$$\omega_\mathcal{A} \leq \sum_{i=1}^{N'} d_i - d_\mathcal{A} - (N' - 1) \leq 1 - d_\mathcal{A} \tag{D.27}$$

in the bulk, with one further unit of improvement at an endpoint. All proper collisions therefore remain subtractable within the same class. Finite powers of logarithms from $\bar{\delta}$ derivatives and from earlier subtractions do not change the power bound. This completes the induction at every finite order, rather than assuming only simultaneous collapse.

Symmetric extensions. Symmetric extensions can be chosen: the original kernels are invariant, and averaging an extension over the compact internal symmetry and the finite group of reflection and parity preserves its scaling bound and its factorization, once the lower extensions have been chosen invariantly. No anomalous symmetry breaking is forced by this finite-cutoff matrix and Gaussian construction.

The hard-core regulator. For a coefficient regulated by the hard core, the difference between its integral and the Taylor-subtracted one is a finite sum of local Taylor jets,

$$\langle t_{\gamma,\mathcal{A};\varepsilon}, f\rangle - \langle t_{\gamma,\mathcal{A};\varepsilon}, \mathcal{T}_\omega f\rangle = \sum_{|\alpha|\leq\omega} c_{\gamma,\mathcal{A},\alpha}(\varepsilon,\mu)\, \partial^\alpha f(0), \tag{D.28}$$

where the coefficients are regulated moments of $t_{\gamma,\mathcal{A};\varepsilon}\, w(\xi)\, \xi^\alpha/\alpha!$. Their divergent and finite parts define the cutoff counterterms recursively. Replacing the cutoff for an already collapsed effective vertex changes only a shrinking collision region, and the next step of the same Taylor construction absorbs its local difference. Thus no exact finite-cutoff factorization of the exclusion region of a collapsed kink is assumed.

Identity terms and sharp endpoints. The identity and its first test-function derivatives are vacuum and switching-function counterterms. For a constant coupling on the interval they give a bulk vacuum energy and scalar endpoint factors; the endpoint Taylor ambiguities have order zero and multiply only $\mathbf{1}$. The construction can first be formulated with a smooth switching function and then taken to the sharp-edge limit. The renormalization is local in the coupling profile, and at finite smoothing width a profile is not assumed equal to its square. In the sharp limit all nontrivial bulk terms have the common plateau coupling, while the additional ultraviolet terms at either edge obey Eq. (D.19) and are scalar. Finite higher-dimensional regulator remainders are kept in the prepared integrals; dropping them by hand would not be this construction.

No new nonlocal Gaussian counterterm. A primitive ultraviolet counterterm is supported on a collision diagonal. In momentum space it is therefore polynomial in the external momenta, not

a new nonanalytic kernel such as $|p|^{1+\bar{\delta}}$ or one of its logarithmic Taylor coefficients. The only local quadratic derivative term that could occur has reference dimension at least two and is excluded from the required counterterms by the same power counting. This concerns divergent counterterms; finite self-energy effects and anomalous dimensions are not excluded.

D.9 What the closure statement does not mean

Irrelevant operators feed the marginal β functions. A simple formal example has a marginal coupling g and an irrelevant dimensionless coupling ϑ, with (in the $\mu\partial_\mu$ convention)

$$\beta_g = e_1 g^3 + e_2 g\vartheta + \cdots , \qquad \beta_\vartheta = \vartheta - e_3 g^2 + \cdots . \tag{D.29}$$

An invariant perturbative trajectory has $\vartheta = e_3 g^2 + O(g^4)$, so that

$$\beta_g^{\text{reduced}} = (e_1 + e_2 e_3) g^3 + O(g^5). \tag{D.30}$$

Setting $\vartheta = 0$ blindly loses a cubic contribution, although ϑ need not be an independent continuum parameter. In the dual such finite effects, among them the $:\chi_B^2:$ term of Eq. (D.12), occur inside the fully integrated multi-kink kernels and their finite subtractions. The absence of an independent counterterm does not license truncating every intermediate operator product to its singular terms when finite higher-order coefficients are computed. The coefficient of $:\chi_B^2:$ at level two is locally integrable in the expansion about the endpoint; all such finite terms are retained in the full cutoff integrals and finite responses of the prescription. The prescription argument would fail for an algorithm that replaced every cluster by its two marginal channels and discarded the integrable remainder before the higher integrations.

The stability matrix is tangent to the renormalized family. Suppose a full Wilsonian description has coordinates (q^a, ϑ^i), with q the two retained couplings, and a two-parameter invariant surface $\vartheta^i = \Theta^i(q)$ with slope $\Theta' = \partial\Theta/\partial q$. At its fixed point write the full linearization in blocks as

$$M_{\text{full}} = \begin{pmatrix} M_{qq} & M_{q\vartheta} \\ M_{\vartheta q} & M_{\vartheta\vartheta} \end{pmatrix} . \tag{D.31}$$

The reduced matrix and the invariance condition are

$$M_{\text{red}} = M_{qq} + M_{q\vartheta}\Theta', \qquad M_{\text{full}} \begin{pmatrix} \mathbf{1} \\ \Theta' \end{pmatrix} = \begin{pmatrix} \mathbf{1} \\ \Theta' \end{pmatrix} M_{\text{red}}. \tag{D.32}$$

The two eigenvalues of M_{red} are genuine tangent eigenvalues of the full flow; the block M_{qq} alone is in general not the correct matrix. This identity explains what a correctly renormalized two-coupling description computes; it does not construct an infinite-dimensional Wilsonian invariant manifold.

The statement is formal, not uniform in a finite charge displacement. Even at $\bar{\delta} = 0$, resumming a charge displacement $b^2 = 1 + z$ before expanding can change the ultraviolet power counting. For an even cluster of n flips the Wick prefactor scales as $\rho^{(1+z)(Q^2-n)} = \rho^{-n(1+z)}$, so the level-two channel has radial behaviour $\rho^{n-2}\rho^{-n(1+z)}\rho^2 = \rho^{-nz}$. It is integrable for $nz < 1$, while each of its fixed Taylor coefficients in z is an integrable polynomial in $\log\rho$. A formal all-order statement about coefficients is therefore not a uniform statement for an arbitrary fixed b and an arbitrarily large number of insertions. The analogous non-uniformity in $\bar{\delta}$ is the order of limits of Appendix D.7.

Identity, endpoints and composite insertions. The identity counterterm cancels vacuum divergences and may depend on the couplings. It drops out of normalized bulk correlators, but it cannot be ignored before the normalization is formed correctly. Sharp interval endpoints and external operator insertions can require their own local renormalizations. Such constants are not new translation-invariant couplings of the action, but they can affect an extraction prescription based on integrated responses on a finite interval; the endpoint sources created by the current resummation belong to this separate bookkeeping. This is the gap that the prescription argument addresses (Appendices D.10 to D.13).

D.10 Endpoint cancellation and the separated probes

Factorization and cancellation. In correlators with separated probes, the operator-valued forest construction gives the asymptotic factorization

$$\mathcal{D}_{I,\varepsilon}(c_B) = \mathcal{V}(\varepsilon, c_B, L)\, \mathcal{Z}_0(\varepsilon, c_B)\, \mathcal{Z}_L(\varepsilon, c_B)\, \mathcal{D}_I^{\mathrm{loc}}(c_R) + o(1), \tag{D.33}$$

after the same nontrivial local bulk redefinition $c_B = \mathcal{B}(c_R, \varepsilon\mu, \bar{\delta})$ at each order. Here c_B and c_R are the bare and renormalized couplings, $\mathcal{D}_I^{\mathrm{loc}}$ is the locally renormalized evolution, $\mathcal{V}$ is the bulk identity normalization, and $\mathcal{Z}_0$, $\mathcal{Z}_L$ are the scalar factors of the two endpoints (Eq. (D.20)). A finite scalar factor can be moved to either side without changing the conclusion. Equation (D.33) holds coefficientwise, after the proper subdivergences are prepared; it is not a uniform nonperturbative limit in operator norm.

For any separated reference probe in the numerator and the vacuum in the denominator, the three scalar factors cancel:

$$\mathcal{R}_{\mathcal{X},\varepsilon}(\mathcal{B}(c_R)) \longrightarrow \frac{\mathrm{tr}\langle \mathcal{X}\, \mathcal{D}_I^{\mathrm{loc}}(c_R)\rangle_0}{\mathrm{tr}\langle \mathcal{D}_I^{\mathrm{loc}}(c_R)\rangle_0}. \tag{D.34}$$

The cancellation happens before any differentiation with respect to parameters, so derivatives of coupling-dependent scalar endpoint normalizations cancel as well.

Only the ultraviolet normalization factors cancel. The finite endpoint-source interactions created by the current resummation do not; they remain in the finite responses and in the Jacobian of Appendix D.12. Ignoring them would be a different, unjustified prescription.

The first endpoint logarithm appears at two kinks. On an interval of length L with hard core ε, the two-kink vacuum and current integrals at $\bar{\delta} = 0$, $b_0 = 1$ are

$$Z_2(L, \varepsilon) = \int_\varepsilon^L \frac{L - r}{r^2}\, \mathrm{d}r = \frac{L}{\varepsilon} + \log\frac{\varepsilon}{L} - 1, \tag{D.35}$$

$$Q_2(L, \varepsilon) = \int_\varepsilon^L \frac{L - r}{r}\, \mathrm{d}r = L \log\frac{L}{\varepsilon} - L + \varepsilon, \tag{D.36}$$

which at $L = 1$ are the moments of Appendix A.7. In Z_2 the term L/ε is the bulk identity normalization and the logarithm, which does not scale with L, is an endpoint identity contribution; it is not zero. Both multiply any distant numerator by the same identity coefficient and cancel when it is divided by the vacuum moment. In Q_2 the logarithm is proportional to L: it multiplies the bulk operator $\mathcal{O}_h$ and survives the normalization. With the operator product $\mathcal{O}_g(X + r)\mathcal{O}_g(X) = r^{-2}\mathbf{1} + \sqrt{2}r^{-1}\mathcal{O}_h(X) + O(1)$ it gives the standard leading $\beta_h = -\sqrt{2}g^2$. The two integrals exhibit, rather than assume, the difference between an endpoint logarithm and a bulk one.

Separated reference probes need no ultraviolet counterterm. Fix a probe at y, a positive distance from I. There is no collision diagonal involving y and an integrated point, so its fixed free normal ordering suffices. Counterterms inside I act on any spectator through the universal local expansion (D.24). A qubit matrix attached to the probe may be transported freely through the

free exterior; the colliding interval product still multiplies it by the same scalar endpoint coefficient, and transporting the matrix creates no new Gaussian contraction.

At $\bar\delta > 0$ the projection to a distant probe is normalized by the free two-point function, as in [4, Eq. (4.18)]. For a vertex probe the remaining position-dependent external Wick factors contain $C(y-X)-C(y)$. On a fixed compact interval these tend to zero uniformly, like $O(y^{\bar\delta-1})$ for $\bar\delta < 1$, or coefficientwise like y^{-1} times finite polynomials in logarithms. For a neutral current probe the monopole term cancels and the first moment survives after the free normalization: it is the total length Ξ_{2m} of the reversed domains, which weights the current moments Q_{2m} (Appendix A.7), plus the exact endpoint current term. This yields the archived odd responses A_{2m+1} and current moments Q_{2m}. The limit is taken after the local ultraviolet subtractions and introduces no further cutoff renormalization of the probe.

An interacting primary *inside* the interval is different: it needs its own insertion renormalization, and it cannot replace the exterior reference probe in the same two-response formula.

D.11 The current resummation and the normal-ordering factors

From the longitudinal coupling to the charge. For smooth regulated fields, the change of frame with $R_h(X) = \exp[ih\hat\sigma_3\phi(X)/\sqrt{2}]$, $\mathcal{D}_h(L,0) = R_h(L)\widetilde{\mathcal{D}}(L,0)R_h(0)^{-1}$ (Appendix A.4), removes the derivative interaction from the interior and shifts the kink charge to $b = b_0 - \sqrt{2}h$. It follows either by differentiating the identity or by integrating the piecewise-constant spin history $m(X) = \pm 1$ by parts, $\int m\,\partial\phi = m\phi|_{\partial I} - \int m'\phi$. It is not a change of the Gaussian integration variable and needs no path-integral Jacobian. The endpoint matrices remain and must not be omitted [4, Sec. 3.2.1]. With the reference normalization fixed, h and b are therefore two coordinates on the same coupling direction, not independent third and fourth parameters; at $\bar\delta \neq 0$ the classical scaling must also be included in passing to a dimensionless charge. This is why the closure statement in (g,h) is a statement about the two essential coordinates (g,b).

Wick-ordering factors. Wick ordering supplies a local factor at each kink [4, Eqs. (3.20)–(3.23)],

$$g_h = g\exp\big[(\sqrt{2}b_0h - h^2)\,C(\varepsilon)\big], \tag{D.37}$$

with $C = C_{\bar\delta}$ of Eq. (D.4). The endpoint charges have magnitude $h/\sqrt{2}$. Their two self-contractions give the common scalar $e^{-h^2C(\varepsilon)/2}$, identical in numerator and denominator. The finite endpoint-to-endpoint and endpoint-to-kink contractions must be kept. The endpoint-to-kink source parameter is the $\eta = \sqrt{2}hb = \sqrt{2}b_0h - 2h^2$ of Eq. (A.17), and for alternating histories with $q_i = (-1)^{i+1}$ the finite endpoint factors are the U_n^{odd} and U_n^{even} of Eq. (A.18), with $f_{\bar\delta}(L - X_i)$ in place of $f_{\bar\delta}(1 - X_i)$ on an interval of length L. At $L = \kappa^{-1} = 1$ the finite endpoint-pair covariance vanishes ($C_{\bar\delta}(\kappa^{-1}) = 0$); a general L retains it as part of the finite response, not as a new bulk coupling. The first source derivative may be combined by reflection, as in the archive; that replacement cannot be made before squaring for a second source derivative.

The essential charge and fugacity. The dimensionless essential charge and fugacity are

$$B = (b_0 - \sqrt{2}h)\,\varepsilon^{\bar\delta/2}, \qquad Y = g_h\,\kappa\varepsilon\,e^{b^2C(\varepsilon)/2} = g\,\kappa\varepsilon\,e^{b_0^2C(\varepsilon)/2}, \tag{D.38}$$

where the second equality is the exact cancellation

$$(\sqrt{2}b_0h - h^2) + \tfrac{1}{2}(b_0 - \sqrt{2}h)^2 = \tfrac{1}{2}b_0^2. \tag{D.39}$$

With Eq. (D.4), $Y = g\kappa\varepsilon\exp[b_0^2(\kappa^{-\bar\delta} - \varepsilon^{\bar\delta})/\bar\delta]$; this is precisely the finite-tail matching (A.21) of the compendium (expanded in Eq. (A.22)), where g is called g_{aux} and Y is renamed g after matching (Sec. 2.1). The known explicit scale factors must be differentiated when the essential β functions

are formed. On the reference section $h = 0$, with the couplings identified at the reference matching scale and $v = B^2$, the chain rule applied to Eq. (D.38) gives

$$\frac{\beta_Y}{Y} = v - 1 + \frac{\beta_g^{\log}}{g}, \qquad \frac{\beta_v}{v} = -\bar{\delta} - \frac{4\beta_h^{\log}}{\sqrt{2}b_0}, \tag{D.40}$$

where $\beta^{\log}$ denotes the logarithmic part extracted from the responses. This explains the classical-scaling terms $v - 1$ and $-\bar{\delta}$, rather than appending them in an unrelated scheme. With $u = Y^2$ and $z = v - 1$ these are F and H of Sec. 2.1 (Eq. (A.23)); for instance $\beta_h^{\log}/(\sqrt{2}b_0) = -g^2 + \cdots$ gives $H = -\bar{\delta} + 4u + \cdots$.

An exact check at $g = 0$. With $J_I = \int_I \chi_B$,

$$\frac{\mathrm{tr}\langle \hat{\sigma}_3 \chi_B(y)\, e^{h\hat{\sigma}_3 J_I}\rangle_0}{\mathrm{tr}\langle e^{h\hat{\sigma}_3 J_I}\rangle_0} = h\,\langle \chi_B(y) J_I\rangle_0, \tag{D.41}$$

since the qubit traces give $e^{hJ_I} \mp e^{-hJ_I}$, and for a Gaussian field

$$\langle \chi_B(y)\, e^{\pm hJ_I}\rangle_0 = \pm h\,\langle \chi_B(y) J_I\rangle_0\,\langle e^{hJ_I}\rangle_0. \tag{D.42}$$

Every Gaussian endpoint normalization factor cancels. The normalized far current response is exactly hL, so an endpoint anomalous factor cannot generate a spurious pure-h quantum β function.

D.12 The response Jacobian

Dividing the integrated odd and current numerators by the same integrated vacuum series gives, at $h = 0$ and on the unit interval, the two normalized responses

$$R_g = g + \mathcal{C}_3 g^3 + \mathcal{C}_5 g^5 + \cdots, \qquad R_h = \sqrt{2}b_0(Q_2 g^2 + \mathcal{K}_4 g^4 + \mathcal{K}_6 g^6 + \cdots), \tag{D.43}$$

and away from that section R_h also contains the tree term h. The coefficients $\mathcal{C}_{2m+1}$ and $\mathcal{K}_{2m}$ are the vacuum-normalized moments of Eq. (A.37), for example $\mathcal{C}_3 = A_3 - Z_2$ and $\mathcal{K}_4 = Q_4 - Q_2 Z_2$. They are the coefficients of the normalized correlators (D.8), not two functions selected after integration. The products are formed before $[\varepsilon^0 \ell^0]$ or $[\varepsilon^0 \ell^1]$ is extracted, since a term of order ε can multiply one of order $1/\varepsilon$; the common-scalar cancellation (D.34) requires this, and it is not a dispensable numerical detail.

The projections $\mathcal{C}_{n,0}$, $\mathcal{C}_{n,1}$, $\mathcal{K}_{2m,0}$, $\mathcal{K}_{2m,1}$ and $Q_{2,0}$ (finite part and single-logarithm coefficient) are those of Appendix A.7, and $v = b_0^2$ on the reference section. Exact resummation gives, on the normalized moment part at $h = 0$, the source chain rule $\partial_h = \sqrt{2}b_0(-2\partial_v + \partial_\eta)$ of Eq. (A.19), because h enters only through the charge $b^2 = v$ and the endpoint source η. The local factor (D.37) has no finite constant at $\kappa\varepsilon = 1$ (there $C(\varepsilon) = 0$); its logarithms are accounted for by the matching. Differentiating also the explicit current prefactor $\sqrt{2}b$, with $\partial_h(\sqrt{2}b) = -2$, gives the finite source and current entries

$$\begin{aligned} j_1 &= -\frac{2}{1+\bar{\delta}}, \qquad j_{2m+1} = -2\partial_v \mathcal{C}_{2m+1,0} + \partial_\eta \mathcal{C}_{2m+1,0}, \\ \lambda_{2m} &= -2\mathcal{K}_{2m,0} - 4v\partial_v \mathcal{K}_{2m,0} + 2v\partial_\eta \mathcal{K}_{2m,0}, \end{aligned} \tag{D.44}$$

with $\mathcal{K}_2 = Q_2$; for $m = 1, 2$ these are the entries of Eq. (A.42). Here j_1 is the case $m = 0$, with $\mathcal{C}_1 = A_1$: the one-kink response has no charge dependence, and j_1 is its endpoint integral $\int_0^1 [f_{\bar{\delta}}(X) + f_{\bar{\delta}}(1-X)]\,\mathrm{d}X = 2(\frac{1}{1+\bar{\delta}} - 1)/\bar{\delta}$. The coupling and source derivatives thus produce exactly the compendium's finite-response matrix $\mathcal{J} = D_c R_0$ of Eq. (A.44), continued to the orders needed: ∂_g of the finite parts of R_g and R_h gives its first column, and ∂_h, through Eq. (D.44), its second. Its free value is the identity, so it is formally invertible; the invertibility is used only *after* locality and the endpoint cancellation have been established.

D.13 Uniqueness of the two-response solution, the recurrence and scheme independence

Uniqueness. Keep the known engineering and normal-ordering factors separate for the moment, and write the normalized bare responses, in the common reference coordinates c, as

$$R(c;\varepsilon) = R_0(c) + \ell R_1(c) + \ell^2 R_2(c) + \cdots + o(1), \qquad \mathcal{J} = D_c R_0. \tag{D.45}$$

The absence of residual power divergences follows from the operator-valued local construction and the scalar normalization, not from dropping such terms in the raw moments. The independently constructed local scheme has a bare map

$$c_B = \mathcal{B}(\tilde{c},\ell), \qquad \mathcal{B}(\tilde{c},0) = \mathcal{G}(\tilde{c}), \qquad \mathcal{G}(\tilde{c}) = \tilde{c} + O(\tilde{c}^2), \tag{D.46}$$

whose coefficients are finite formal coefficients: they are the finite local Taylor moments of Eq. (D.28), recursively including forests. Local finite normalizations can be chosen so that $\mathcal{G}$ is the identity, and no response condition is needed for that choice; for an arbitrary local normalization $\mathcal{G}$ is kept explicit. At fixed bare couplings, with $\ell = \log(\varepsilon\mu)$, the local logarithmic renormalization-group identity is

$$\partial_\ell \mathcal{B} + D_{\tilde{c}}\mathcal{B}\,\tilde{\beta}^{\log} = 0. \tag{D.47}$$

Every normalized spectator response is finite in the renormalized couplings, so its ℓ derivative at fixed $\tilde{c}$ vanishes. At first order in ℓ this gives

$$R_1(\mathcal{G}(\tilde{c})) = DR_0(\mathcal{G}(\tilde{c}))\, D\mathcal{G}(\tilde{c})\, \tilde{\beta}^{\log}(\tilde{c}), \tag{D.48}$$

and therefore

$$\beta^{\log}_{\text{archive}}(\mathcal{G}(\tilde{c})) = D\mathcal{G}(\tilde{c})\, \tilde{\beta}^{\log}(\tilde{c}), \qquad \beta^{\log}_{\text{archive}} = \mathcal{J}^{-1} R_1. \tag{D.49}$$

The known explicit classical-scaling and normal-ordering maps are then applied on both sides, including their scale derivatives as in Eq. (D.40); this gives the corresponding relation for the full essential vector fields. The charge is not held fixed when a change of scale physically changes B.

The argument is not circular. It constructs the local renormalization and the finiteness for all spectators in Appendices D.7 to D.10, independently of the response equations; Equation (D.48) then makes the local β function a solution, and the invertibility of $\mathcal{J}$ makes it the *unique* solution in the specified coordinates. No independent anomalous term of Eq. (D.1) remains: the endpoint factors cancel before differentiation, and the exterior reference probe does not collide with an integrated operator. If either fact were false, the uniqueness argument would not apply.

The recurrence at every required power. With the auxiliary coefficients b_{2m+1} and h_{2m} of Appendix A.9 ($\beta_g = b_3 g^3 + b_5 g^5 + \cdots$, $\beta_h/(\sqrt{2}b_0) = -g^2 + h_4 g^4 + \cdots$), solving Eq. (D.49) on the section $h = 0$ means solving the system $\mathcal{J}\beta = R_1$ of Eq. (A.46), with R_1 of Eq. (A.45) continued to the orders needed. Order by order in g it is triangular: the coefficient of g^{2m+1} in the first row contains b_{2m+1} with coefficient one and otherwise only lower coefficients, and the coefficient of g^{2m} in the second row does the same for h_{2m} (the factor $\sqrt{2}b_0$ of $\mathcal{J}_{12}$ times the factor $\sqrt{2}b_0$ of β_h gives the $2v$ of the recurrence; in the second row the factor $\sqrt{2}b_0$ of $\mathcal{J}_{21}$ cancels against that of R_1). The solution is the compendium's recurrence: Eq. (A.47) through b_7, and its continuation to h_8 and b_9 in Appendix C.3 (Eq. (C.30)). All finite and source terms are retained; it is not a new fit to the final C_n^{SR}. It was reproduced by the prescription argument's symbolic checker and re-derived independently by the review (Appendix D.15).

The lowest order, also the check of signs of Appendix A.9, shows what the finite endpoint term does. At $\bar{\delta} = 0$ and $v = 1$ the archived exact normalization gives

$$\mathcal{C}_3 = 2\ell^2 + 5\ell + 3 - 2\zeta(2) + o(1), \qquad j_1 = -2, \tag{D.50}$$

so the first line of the recurrence, $b_3 = \mathcal{C}_{3,1} + 2vj_1$, gives $b_3 = 5 - 4 = 1$, while $h_2 = -1$ follows from Eq. (D.36). This is the leading $F = z + u + \cdots$ of Sec. 2.1. Discarding the finite endpoint derivative j_1 would give $b_3 = 5$, a wrong cubic coefficient, even though the ultraviolet endpoint factor itself has cancelled.

Scheme independence. For the full essential vector fields, Eq. (D.49) is the usual transformation under a finite change of coordinates. At a fixed point c_*,

$$M_{\text{archive}}(\mathcal{G}(c_*)) = D\mathcal{G}(c_*)\, M_{\text{local}}(c_*)\, D\mathcal{G}(c_*)^{-1}; \tag{D.51}$$

the term with second derivatives of $\mathcal{G}$ vanishes because the β function itself vanishes there. The trace, the determinant and the two scaling eigenvalues therefore coincide coefficientwise. The finite map is regular in the formal $\bar{\delta}$ expansion, so substituting the fixed-point series in $\sqrt{\bar{\delta}}$ does not introduce a hidden change of the physical parameter s. This gives the formal scheme statement at any finite computed order. It needs no separately evaluated integral on a ring, and none is claimed: a ring in the compact boson theory also has a monodromy sector that must be treated correctly, and deleting the endpoints of an existing interval integral is not automatically a second prescription [4].

Other observables. For any other separated observable the same local forest construction and the same bare map apply: Eq. (D.24) identifies its local collision coefficients before projection, and Eq. (D.34) removes its scalar endpoint factors. There is therefore no spectator-dependent bulk counterterm. Physical operator insertions inside the support are renormalized separately as composite operators, and their mixing is not a third bulk coupling.

D.14 What the arguments establish and what they do not

For the operator content. The closure argument improves the status of the operator-content assumption: at the finite orders used for the expansion about the endpoint, no third symmetry-preserving bulk operator that needs an ultraviolet counterterm is expected or required in the formal local subtraction construction, and the same argument applies at each finite order rather than stopping at a two-insertion test. It does not certify the archived numerical β coefficients. The exact certificates of Sec. 2.4 prove identities between supplied integral expressions; they do not show that the finite-response prescription assembled the correct local counterterms for all collision configurations.

For the prescription. To connect the archive with the closure statement, the closure argument named a concrete audit in three steps: (i) construct the collision subtractions of the regulated, path-ordered kernels as operator-valued products, including every proper cluster and not only the projections that define the two response coordinates; (ii) show that the archive's vacuum normalization, current resummation, endpoint prescription and finite-$\bar{\delta}$ matching implement those subtractions, or differ from them by an explicit finite redefinition of the two essential couplings and separately specified operator and endpoint normalizations; (iii) check an independent family of non-projected correlators after inserting the same counterterms, whose coefficients in the $\mathbf{1}$, $\mathcal{O}_g$ and $\mathcal{O}_h$ channels must not depend on the distant spectator configuration, with locally integrable remainders of derivative order two and higher. In this note's assessment (Appendix D.15), the prescription argument supplies steps (i) and (ii) at the formal level (Appendices D.8 to D.13). The prescription argument addresses the analytic content of step (iii) through the spectator expansion (D.24) and the paragraph "Other observables" of Appendix D.13; the computation itself has not been done. A numerical check of the prescription with a second regulator and different responses, within the same two-coupling description, is the magnetization-scheme calculation of Sec. 2.7, which reproduces C_3^{SR} to C_7^{SR} and the record of C_8^{SR} (Sec. 6). Given (i) and (ii), regular finite coupling

redefinitions transform the stability matrix by similarity, so its eigenvalues do not depend on the chosen two-coupling coordinates.

What is not established. Neither argument proves the lattice duality, the identification of the dual and of its thermal branch with the critical chain, or the infinite-volume limit; these keep the scope of the published proposal [3, 4]. Neither proves convergence or resummability at fixed $\bar{\delta}$, nor a non-perturbative renormalizability statement. Neither recomputes C_3^{SR} to C_7^{SR} in a second regulator or audits the arithmetic of the historical high-order integrations, whose 21 stage programs are missing (Appendix H.5). Their fidelity to the recipe is tested only numerically: by the independent re-derivation (Sec. 2.6) and, through the invariants they produce, by the magnetization-scheme calculation in a second regulator (Sec. 2.7); the exact certificates (Sec. 2.4) concern the later reduction of the archived expressions to zeta values. Neither bears on the conjectured zeta-value form of C_8^{SR} (Eq. (2.15)) or on the statistical claims of the NRG or Monte Carlo comparisons, for which no raw data were supplied to them. The statement of the prescription argument concerns the mathematically specified recipe: implementations must still be checked to follow it, and its formula audit covers the normalization, the current and endpoint identity, the Wick factors, the finite-response Jacobian and its recurrence. The arguments are also narrower than a proof of the fixed-s duality, and they avoid the opposite mistake of treating every generated irrelevant Wilsonian coefficient as a new independent tuning parameter.

Status in this note. Both arguments are formal and unrefereed; this note uses them as such. The status word "proved closed form within the prescription" (Sec. 1.3) is unchanged by them: they bear on whether the prescription computes the coefficients of the dual, not on the reduction of the archived expressions to zeta values. The evidence that does not use the dual at all is the NRG test of C_3^{SR} to C_5^{SR} (Sec. 2.9).

D.15 Assessment and checks by the review

Claude Code assessed both arguments at the author's request on 15 October 2026 (Sec. 5); the caveat "Two-coupling closure" of Sec. 2.9 is the summary that the author approved afterwards.

The closure argument. Verdict: narrowed; the operator content is closed formally, and the matching of the archived prescription was left open. The argument is sound as a standard perturbative power-counting (Epstein–Glaser) argument, and no gap was found that changes its conclusion. It is a careful sketch: the uniform scaling bounds in the shape sectors and the sufficiency of a finite Taylor jet are asserted, not estimated in detail, and the extension step is invoked from the cited theorems. Checked independently: the classification through dimension one under the shift symmetry, the internal reflection and parity (Table 5); the charge selection rule; the radial-power identity; the product $\mathcal{O}_g\mathcal{O}_g$ (D.12); the coefficientwise logarithmic deformation (D.21); symmetric extensions by group averaging; and the frame change between h and b, identical to the compendium's. The assessment found that the archive's prescription did not obviously implement the local subtraction, and named the risks: (i) the β functions are solved from exactly two projected responses, a square system that would absorb a mis-assigned endpoint or insertion logarithm silently; (ii) all responses live on one unit interval, with the current resummed into endpoint sources and no endpoint hard core, so endpoint and bulk logarithms are not separated by construction; (iii) the finite-$\bar{\delta}$ matching is the exact normal-ordering factor, coefficientwise a regular redefinition, but its equivalence to the local scheme was not shown; (iv) the current resummation and (v) the vacuum normalization were judged low risk. The prescription argument answers this open link: (i) by the scalar endpoint factors, the separated probes and the uniqueness argument (Appendices D.10 and D.13); (ii) by the endpoint power counting and the two-kink integrals (D.35) and (D.36); (iii) by the normal-ordering audit (Appendix D.11).

The prescription argument. Verdict: the link to the prescription is closed at the formal level, at the same level of rigour as the closure argument. Checked: the endpoint power counting (one fewer free coordinate, so endpoint divergences only at reference dimension zero or below); the endpoint classification (the invariant dimension-zero sector is the identity, so endpoint ultraviolet factors are scalars that cancel in the normalized responses); that the finite endpoint-source terms survive and are the compendium's j and λ terms ($b_3 = 1$, against 5 without j_1); the cancellation (D.39) and $\eta = \sqrt{2}b_0 h - 2h^2$; the two-kink integrals at general L and ε; and the recurrence for b_3 to b_9 and h_4 to h_8, re-derived from $\mathcal{J}\beta = R_1$ and equal to the compendium's formulas. The logic is not circular: local renormalization for arbitrary spectators is constructed first, and the invertibility of the response system is used only for uniqueness. Stated rather than shown in detail: the sharp-edge limit from smooth switching, the recursion relating the hard-core regulator to the local subtraction when exclusion regions do not factorize (Eq. (D.28)), and the projection to a distant probe.

Code. `code/closure_d1/public/check_closure.py`, written by the review and run by the package's `test` and `replay` targets (standard library and sympy, a few seconds), checks the finite algebra:

- all $3 + 3^2 + \cdots + 3^9 = 29\,523$ words of length one to nine in $\hat{\sigma}_+$, $\hat{\sigma}_-$, $\hat{\sigma}_3$, in exact integer arithmetic: the 2 035 nonzero words all have $Q \in \{0, \pm 1\}$ and $[\hat{\sigma}_3, M] = 2QM$, with no violation;
- the radial-power identity $2\sum_{i<j} q_i q_j = Q^2 - n$ for alternating charges, $n = 1$ to 11;
- the coefficients 1, $\sqrt{2}$, 1 of the midpoint product (D.12) (to 10^{-15});
- a rerun of the prescription argument's own checker, `code/closure_d1/prescription/code/check_prescription.py` (sympy): all 22 of its checks are true and equal to its shipped record. They cover the matrix words, the scalar invariant matrix at dimension zero, the bulk and endpoint radial powers, the cancellation (D.39) and η, the source chain rule of λ_{2m}, the integral j_1, every coefficient equation of the response system through g^9, both two-kink integrals at general L and ε, the spectator expansion at zeroth and first order, a toy response-generator identity, and the cancellation of a common scalar factor;
- independently of that checker, the recurrence $b_3, \ldots, b_9$, $h_4, \ldots, h_8$ solved order by order from $\mathcal{J}\beta = R_1$ (sympy), which equals the compendium's formulas exactly and gives $b_3 = 1$ at $v = 1$, $\mathcal{C}_{3,1} = 5$, $j_1 = -2$.

The export tests on the Intel and AMD nodes of the symmetry cluster give the same table.

What the checks establish. They establish the finite algebra on which the arguments rest: the selection rule for words up to length nine (the general rule is argued for every length independently of the enumeration), the pair product, the radial powers, the normal-ordering identities, the two-kink integrals, and that the compendium's recurrence is exactly what the response system $\mathcal{J}\beta = R_1$ implies. They do not check the analytic induction, the uniform bounds in the shape sectors, the sharp-edge limit, the hard-core recursion or the far-field projection; they recompute no archived β coefficient; and they are not a numerical test of the prescription. The closure argument describes its own script of finite checks (the same algebra, a bound on the logarithmic degree through order nine and an exact example of the tangent-stability identity); the script was not supplied and is not part of the release, and the review's checker above replaces its first three items.

E The closed form of c_2: derivation

This appendix gives the derivation of the closed form of c_2 in the closed-form study ("the study", written by OpenAI GPT-6 astra; Sec. 1.2), edited to the notation of this note.

Section 3 states the steps and their results; here each step is derived, in the order in which the results depend on one another. Appendices E.1 to E.9 reduce the coefficient to the elliptic finite part $\mathcal{I}$ of Eq. (3.5) and give Eq. (3.7); Appendices E.10 to E.14 evaluate $\mathcal{I}$ in closed form; and Appendix E.15 describes the numerical checks of that evaluation. The status is that of Table 2: the evaluation of $\mathcal{I}$ is proved, and c_2 is a closed form conditional on one unproved step, the uniform-remainder exchange (Appendix E.8 and Sec. 3.6) on which the identification of Eq. (3.9) with the physical coefficient rests.

Notation. We write c_2 for the study's C_2, h_{MS} for the fourth-order coefficient of the anomalous dimension in minimal subtraction (the study's h_0), κ for the parameter of a finite operator redefinition (the study's q), and $\mathsf{P}, \mathsf{Q}, \mathsf{T}$ for its three ${}_4F_3(1)$ constants. Auxiliary symbols are renamed where the study's symbol would clash with another one in this note: the Möbius scale Λ (study: a), the factor Υ (Y), the kernel combination $\Sigma(x, y; z)$ (S), the weights Ω_1, Ω_0 (U, W), the constant matrices $\mathsf{J}, \mathsf{L}, \mathsf{C}_\pm, \mathsf{B}_-, \mathsf{H}, \mathsf{D}$ ($J, L, C_\pm, B_-, H, D$), the charge factors $\mathcal{W}_{\boldsymbol{e}}, \mathcal{P}_{\boldsymbol{e}}$ ($W_{\boldsymbol{e}}, P_{\boldsymbol{e}}$), the moment combinations m_U, m_V (f, g) and d_O (d), the polynomial ϖ (p), the normalized integral $\hat{K} = 2K/\pi$ (F), the hypergeometric functions Φ_A, Φ_D (A, D_h), the period vectors $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{W}$ (X_v, Y_v, Z_v, W_v), the derivative combinations $\mathcal{D}_1, \mathcal{D}_0$ (D_1, D_0), and the Mellin integrals j_0, j_1, j_2 (J_0, J_1, J_2). The complex beta integral is written throughout as

$$\mathcal{B}(a, b) = \mathrm{AC}\int_{\mathbb{C}} |z|^{2a-2}|1-z|^{2b-2}\,\mathrm{d}^2 z = \pi\frac{\Gamma(a)\Gamma(b)\Gamma(1-a-b)}{\Gamma(1-a)\Gamma(1-b)\Gamma(a+b)}, \tag{E.1}$$

which the study writes as $J(a-1, b-1)$ in its lower-order steps.

E.1 Observable, regulator and continuation

The crossover theory is that of Sec. 3: the Ising CFT times a generalized free scalar χ, with unit-normalized fields, $\langle\sigma(x)\sigma(0)\rangle = |x|^{-1/4}$ and $\langle\epsilon(x)\epsilon(0)\rangle = |x|^{-2}$, and the Ising data listed there. An insertion at infinity means $O(\infty) = \lim_{|x|\to\infty}|x|^{2\Delta_O}O(x)$, so that $\langle O(0)O(\infty)\rangle = 1$. The fusion rules are $\sigma\times\sigma = \mathbb{1} + \epsilon$ and $\epsilon\times\epsilon = \mathbb{1}$, each family including all its Virasoro descendants. The generalized free field is fixed by Wick's theorem and

$$\langle\chi(x)\chi(y)\rangle = |x-y|^{-15/4+2\delta}, \tag{E.2}$$

so that $\Delta_\chi = 15/8 - \delta$ and the perturbing operator $\sigma\chi$ has dimension $2-\delta$. The bare action is

$$S = S_{\mathrm{Ising}} + S_\chi + \int g_0(x)\,\sigma\chi(x)\,\mathrm{d}^2x + \int t_0(x)\,\epsilon(x)\,\mathrm{d}^2x, \tag{E.3}$$

where g_0 has mass dimension δ and the thermal source t_0 is tuned to the critical surface. The thermal source cannot be left out of the definition of the prescription, because the short-distance product contains

$$\sigma\chi(x)\,\sigma\chi(0) \supset |x|^{-4+2\delta}\mathbb{1} + \tfrac{1}{2}|x|^{-3+2\delta}\epsilon(0). \tag{E.4}$$

The second term produces a thermal power divergence. It is retained by analytic continuation, as appropriate to the critically tuned theory.

Observable. The observable is the dimension $\Delta_{\mathcal{E}}$ of the critical thermal eigenoperator continued from ϵ,

$$\Delta_{\mathcal{E}} = 1 + c_2\delta^2 + O(\delta^3); \tag{E.5}$$

the absence of a linear term is derived in Appendix E.4. In terms of s, $\Delta_{\mathcal{E}} = 1 - 1.01656753703313038819\ldots\,(7/4-s)^2 + O((7/4-s)^3)$, the coefficient being $c_2/4$ (Table 1). Since $1/\nu = 2 - \Delta_{\mathcal{E}}$, also $\nu = 1 + c_2\delta^2 + O(\delta^3)$.

Regulator and continuation. All complex integrals use $\mathrm{d}^2 z = \mathrm{d}(\mathrm{Re}\, z)\,\mathrm{d}(\mathrm{Im}\, z)$. The renormalized coupling is localized by the source profile $f(w) = e^{-|w|^2}$; its width and the renormalization scale μ are set to one after the renormalization identities have been derived. The Gaussian controls the infrared. Ultraviolet integrals are first defined in a domain of convergence and then continued meromorphically in δ and, where needed, in auxiliary exponents; AC denotes this continuation. In particular,

$$\mathrm{AC}\int_0^R r^{p-1+b\delta}\,\mathrm{d}r = \frac{R^{p+b\delta}}{p+b\delta}. \tag{E.6}$$

For $p \neq 0$ the value at $\delta = 0$ need not vanish; for $p = 0$ the integral has a logarithmic pole. A circular finite part, FP, is defined with rotationally symmetric exclusions about each specified singularity, and the subtracted local terms are restored in the same continuation. This convention fixes the angular contacts used below.

E.2 Renormalization-group reduction

This subsection explains why a fourth-order residue determines the quadratic coefficient. In pole-only analytic minimal subtraction let

$$\beta(g,\delta) = \mu\frac{\mathrm{d}g}{\mathrm{d}\mu} = -\delta g + b_0 g^3 + O(g^5), \qquad \gamma_{\mathcal{E}}(g) = h_{\mathrm{MS}} g^4 + O(g^6). \tag{E.7}$$

The interacting fixed point has $g_*^2 = \delta/b_0 + O(\delta^2)$, and the dimension there is $\Delta_{\mathcal{E}} = 1 + \gamma_{\mathcal{E}}(g_*)$, so $c_2 = h_{\mathrm{MS}}/b_0^2$. The thermal operator renormalizes multiplicatively, $\mathcal{E}_0 = Z_{\mathcal{E}}\mathcal{E}_R$, with $\gamma_{\mathcal{E}} = \beta\,\partial_g \log Z_{\mathcal{E}}$. There is no energy pole at order g^2: the second-order integral is finite at $\delta = 0$ because the first-order shift vanishes (Appendices E.4 and E.5). Therefore

$$\log Z_{\mathcal{E}} = \frac{z_{4,1}}{\delta} g^4 + O(g^6), \qquad \gamma_{\mathcal{E}} = \Big(-\delta g + \cdots\Big)\frac{4 z_{4,1} g^3}{\delta} = -4 z_{4,1} g^4 + O(g^6), \tag{E.8}$$

which gives $h_{\mathrm{MS}} = -4 z_{4,1}$ and the relation $c_2 = -4 z_{4,1}/b_0^2$ of Sec. 3.2.

The factor four relies on there being a single renormalization factor. The source is localized at the origin, so in the correlator with the energies at 0 and ∞ the only local ultraviolet factor is the $Z_{\mathcal{E}}$ at 0; since $f(\infty) = 0$, the energy at infinity supplies no second source-dependent ultraviolet factor.

The restriction to pole-only subtraction is substantive. The split of c_2 between a δg^2 term and a g^4 term of $\gamma_{\mathcal{E}}$ depends on the scheme (Sec. 3.2): the finite redefinition $\mathcal{E}_R \to e^{\kappa g^2}\mathcal{E}_R$ moves weight from one to the other. The finite lower-order source contact and the fourth-order pole must therefore be matched in one and the same convention, which is what Appendices E.5 to E.9 do.

E.3 Exact Ising correlators

For $2n = 2, 4$ distinct spin positions z_i, sum over the neutral charges $e_i = \pm 1$ with $\sum_i e_i = 0$, and define

$$\mathcal{W}_{\boldsymbol{e}} = \prod_{i<j} |z_i - z_j|^{e_i e_j/2}, \qquad \mathcal{P}_{\boldsymbol{e}} = 1 - \frac{1}{4}\Big(\sum_i e_i z_i\Big)\Big(\sum_i \frac{e_i}{z_i}\Big). \tag{E.9}$$

The Euclidean correlators needed are [13]

$$S_{2n} = \langle \sigma_1 \cdots \sigma_{2n}\rangle = \Big[2^{-n}\sum_{\boldsymbol{e}} \mathcal{W}_{\boldsymbol{e}}\Big]^{1/2}, \qquad H_{2n} = \langle \epsilon(0)\epsilon(\infty)\sigma_1 \cdots \sigma_{2n}\rangle = \Big[2^{-n}\sum_{\boldsymbol{e}} \mathcal{W}_{\boldsymbol{e}}|\mathcal{P}_{\boldsymbol{e}}|^2\Big]^{1/2}. \tag{E.10}$$

The positive Euclidean square root is fixed by the identity OPE. These formulas follow by combining two independent Ising copies, $\sigma^{(1)}\sigma^{(2)} = \sqrt{2}\cos(\varphi/2)$ with $\langle\varphi(z)\varphi(w)\rangle = -2\log|z-w|$, and representing the product of energies by currents. Charge neutrality gives the displayed sum; taking the square root removes the second copy. For energies at finite positions x, y, $\mathcal{P}_e$ is replaced by

$$\frac{1}{(x-y)^2} + \frac{1}{4}\Big(\sum_i \frac{e_i}{x-z_i}\Big)\Big(\sum_i \frac{e_i}{y-z_i}\Big). \tag{E.11}$$

For two spins (energies at 0 and ∞) this gives

$$S_2(a,b) = |a-b|^{-1/4}, \qquad H_2(a,b) = \frac{|a+b|^2}{4|ab|\,|a-b|^{1/4}}. \tag{E.12}$$

E.4 The vanishing first-order shift and the exact b_0

First order. The first-order energy shift is proportional to the integral $I_2(\delta)$ of Sec. 3, whose integrand is $|1-z|^{-4+2\delta}$ times $H_2(z,1)/S_2(z,1) - 1$ from Eq. (E.12). The combined integrand is absolutely integrable for $0 < \mathrm{Re}\,\delta < 1/2$. The term -1 (the subtracted spin two-point function) gives a scaleless integral, which vanishes in the continuation. With $|1+z|^2 = 2(1+|z|^2) - |1-z|^2$, the rest is three complex beta integrals (E.1):

$$I_2(\delta) = \tfrac{1}{2}\mathcal{B}(\tfrac{1}{2}, \delta-1) + \tfrac{1}{2}\mathcal{B}(\tfrac{3}{2}, \delta-1) - \tfrac{1}{4}\mathcal{B}(\tfrac{1}{2}, \delta) = \frac{\pi\delta^2}{4(1-\delta)^2}\frac{\Gamma(\delta)\Gamma(\frac{1}{2}-\delta)}{\Gamma(1-\delta)\Gamma(\frac{1}{2}+\delta)}. \tag{E.13}$$

Expanding the Gamma ratios gives $I_2(\delta) = \frac{\pi}{4}[\delta + (2+4\log 2)\delta^2 + O(\delta^3)]$. Thus $I_2(0) = 0$, which reproduces the leading cancellation ($c_1 = 0$), while $I_2'(0) = \pi/4$. The finite derivative is retained in the source matching of Appendix E.5; it is not a separate contribution to be added to $c_2 = h_{\mathrm{MS}}/b_0^2$.

The exact normalization of b_0. In the same field normalization the leading coefficient of the beta function is

$$b_0 = -\pi I_\beta, \qquad I_\beta = \mathrm{FP}\int_{\mathbb{C}} |z|^{-15/4}\big[S_4(0,z,1,\infty) - |z|^{-1/4}\big]\,\mathrm{d}^2 z, \tag{E.14}$$

with S_4 from Eq. (E.10). The quadratic map $z = 4w/(1+w)^2$, followed by $u = w^2$, with its multiplicity and Jacobian, reduces the integral to

$$I_\beta = \frac{1}{64}\,\mathrm{AC}\int_{\mathbb{C}} (|u|^{-3} + |u|^{-5/2})|1-u|^{3/2}\,\mathrm{d}^2 u. \tag{E.15}$$

Inversion, $u \mapsto 1/u$, interchanges the two terms. Equation (E.1) therefore gives

$$I_\beta = \frac{1}{32}\mathcal{B}(-\tfrac{1}{2}, \tfrac{7}{4}) = -\frac{9\pi}{8}\Big[\frac{\Gamma(3/4)}{\Gamma(1/4)}\Big]^2, \tag{E.16}$$

which is the exact $b_0 = \frac{9\pi^2}{8}[\Gamma(\frac{3}{4})/\Gamma(\frac{1}{4})]^2$ of Sec. 3.2. All power terms in this reduction use Eq. (E.6).

E.5 The localized source and the fourth-order logarithm

The connected kernel. With the energies at 0 and ∞, write $S_{ij} = S_2(w_i, w_j)$ and $H_{ij} = H_2(w_i, w_j)$ from Eq. (E.12), and S_4, H_4 for the four-spin functions of Eq. (E.10). The fourth-order coefficient of a logarithm is its fourth-order moment minus one half the square of its second-order moment. Applied to the logarithm of the vacuum-normalized energy two-point function, for the fixed generalized-free pairing (12)(34), this gives the connected kernel $\mathcal{F} = H_4 - S_4 - H_{12}H_{34} + S_{12}S_{34}$ of Sec. 3; its last two terms are fixed by the logarithm. The three Wick pairings have equal integrals by permutation symmetry, so $3/4! = 1/8$ and

$$A_4(\delta) = \frac{1}{8}\,\mathrm{AC}\!\int_{\mathbb{C}^4}\prod_{i=1}^{4} \mathrm{d}^2 w_i\, e^{-\sum_i |w_i|^2}\,|w_{12}|^{-15/4+2\delta}|w_{34}|^{-15/4+2\delta}\,\mathcal{F}(w_1,w_2,w_3,w_4), \tag{E.17}$$

with $w_{ij} = w_i - w_j$. Connectedness alone does not make Eq. (E.17) an ordinary ultraviolet-convergent integral.

Why the finite Gaussian two-point term is needed. Locality of the coupling counterterm gives

$$g_0(w) = \mu^\delta\Big[g\,f(w) + \frac{b_0}{2\delta}g^3 f(w)^3 + O(g^5)\Big]. \tag{E.18}$$

The power f^3 follows from the three perturbing insertions in the local coupling logarithm. It cannot be replaced by f inside a finite lower-order integral. Define the second-order integral with two Gaussian weights,

$$A_{a,b}(\delta) = \frac{1}{2}\,\mathrm{AC}\!\int_{\mathbb{C}^2} \mathrm{d}^2 w_1\,\mathrm{d}^2 w_2\, e^{-a|w_1|^2 - b|w_2|^2}\,|w_{12}|^{-15/4+2\delta}(H_{12} - S_{12}). \tag{E.19}$$

Inserting Eq. (E.18) into the second-order term gives two equivalent placements of the cubic counterterm, whose total fourth-order contribution is $b_0 A_{3,1}(\delta)/\delta$. Hence $z_{4,1} = \mathrm{Res}_{\delta=0}\, A_4 + b_0 B_{31}$ with $B_{31} = A_{3,1}(0)$, as in Sec. 3. There is one ultraviolet external-operator factor: the localized source produces collisions at the finite energy insertion and is suppressed near infinity, so there is no extra factor of two from the second energy.

Evaluation of B_{31}. Set $w_1 = \Lambda z$ and $w_2 = \Lambda$ in Eq. (E.19), with Λ the overall complex scale. The integral over Λ gives

$$A_{a,b}(\delta) = \frac{\pi}{2}\Gamma(\delta)\,\mathrm{AC}\!\int_{\mathbb{C}} \mathrm{d}^2 z\,(a|z|^2 + b)^{-\delta} F_\delta(z), \qquad F_\delta(z) = |1-z|^{-4+2\delta}\Big(\frac{|1+z|^2}{4|z|} - 1\Big). \tag{E.20}$$

By Appendix E.4, $\mathrm{AC}\!\int F_\delta(z)\,\mathrm{d}^2 z = I_2(\delta) = \frac{\pi}{4}\delta + O(\delta^2)$, so the vanishing two-point residue leaves a finite contribution when it is multiplied by the scale factor $\Gamma(\delta)$. For smooth radial test functions h with the required decay, and by continuation for the logarithmic weights below, the circular finite part about $z = 1$ obeys the distributional identity

$$\mathrm{FP}\!\int_{\mathbb{C}} \mathrm{d}^2 z\, F_0(z)\, h(|z|) = \frac{\pi}{2}\int_0^\infty \frac{|r-1|}{(1+r)^3} h(r)\,\mathrm{d}r - \frac{\pi}{4}h(1). \tag{E.21}$$

For $r \neq 1$ the first term follows from the angular integral. The second is the local term from the circular exclusion about $z = 1$. Its coefficient is also fixed by $I_2(0) = 0$, since the bulk radial integral for $h = 1$ is $\pi/4$. It must be retained when the source weight is differentiated. Expanding Eq. (E.20) at $\delta = 0$ gives

$$A_{a,b}(0) = \frac{\pi^2}{8} - \frac{\pi}{2}\,\mathrm{FP}\!\int_{\mathbb{C}} \mathrm{d}^2 z\, F_0(z)\log(a|z|^2 + b). \tag{E.22}$$

At $(a,b)=(3,1)$, inversion of the part $r>1$ and an integration by parts reduce the remaining finite part to an ordinary integral:

$$B_{31}=\frac{\pi^2}{8}-\frac{\pi^2}{2}\int_0^1\frac{\mathrm{d}r}{(1+r)^2}\Big(\frac{1}{1+3r^2}-\frac{r^2}{r^2+3}\Big),$$
$$\int_0^1\frac{\mathrm{d}r}{(1+r)^2}\Big(\frac{1}{1+3r^2}-\frac{r^2}{r^2+3}\Big)=\frac{\pi\sqrt{3}}{48}+\frac{3\log 3}{16},\tag{E.23}$$

which gives $B_{31}=\frac{\pi^2}{96}(12-\pi\sqrt{3}-9\log 3)$, the value quoted in Sec. 3. It is the combination $\mathrm{Res}_{\delta=0}\,A_4+b_0B_{31}$, not this source-dependent finite number alone, that determines the exponent.

E.6 The regulated conformal map and the elliptic energy integral

Keeping the source factor through the coordinate change. Put the four spins at the reference positions $(0,z,1,\infty)$ and the energies at (x,y). The Möbius map

$$M(t)=\Lambda\frac{t-x}{t-y},\qquad (w_1,w_2,w_3,w_4)=(M(0),M(z),M(1),M(\infty)),\tag{E.24}$$

returns the energies to $(0,\infty)$; Λ is again an overall complex scale. Its complex Jacobian determinant is

$$\det\frac{\partial(w_1,w_2,w_3,w_4)}{\partial(\Lambda,x,y,z)}=-\frac{\Lambda^3(x-y)^2}{y^2(y-z)^2(y-1)^2}.\tag{E.25}$$

Hats denote correlators with the spins at these reference positions. In particular $\widehat{H}_4=\langle\epsilon(x)\epsilon(y)\sigma(0)\sigma(z)\sigma(1)\sigma(\infty)\rangle$, and $\widehat{H}_{12},\widehat{H}_{34}$ each include both energy insertions; their energy two-point function is not divided out. The transformed connected kernel is

$$\mathcal{G}(x,y;z)=\frac{\widehat{H}_4(x,y;z)}{|x-y|^2}-\frac{\widehat{S}_4(z)}{|x-y|^4}-\widehat{H}_{12}(x,y)\widehat{H}_{34}(x,y)+\frac{|z|^{-1/4}}{|x-y|^4},\tag{E.26}$$

where, from Eqs. (E.10) and (E.11),

$$\widehat{H}_{12}(x,y)=|z|^{-1/4}\Big|\frac{1}{(x-y)^2}+\frac{z^2}{4xy(x-z)(y-z)}\Big|,\qquad \widehat{H}_{34}(x,y)=\Big|\frac{1}{(x-y)^2}+\frac{1}{4(x-1)(y-1)}\Big|.\tag{E.27}$$

Define the exact Gaussian factors

$$R(x,y,z)=\Big|\frac{x}{y}\Big|^2+\Big|\frac{z-x}{z-y}\Big|^2+\Big|\frac{1-x}{1-y}\Big|^2+1,$$
$$\Upsilon(y,z)=|y(y-z)(y-1)|,\qquad W(x,y,z)=\frac{|x-y|^2}{\Upsilon(y,z)R(x,y,z)}.\tag{E.28}$$

Combining the conformal weights, the Wick kernels and the Jacobian gives the full density, apart from the factor $1/8$,

$$|\Lambda|^{-2+4\delta}e^{-|\Lambda|^2R}\,|z|^{-15/4+2\delta}\Big(\frac{|x-y|^2}{\Upsilon}\Big)^{2\delta}\mathcal{G}.\tag{E.29}$$

The scale integral $\int_{\mathbb{C}}|\Lambda|^{-2+4\delta}e^{-R|\Lambda|^2}\,\mathrm{d}^2\Lambda=\pi\Gamma(2\delta)R^{-2\delta}$ therefore yields

$$A_4(\delta)=\frac{\pi}{8}\Gamma(2\delta)J(\delta),\qquad J(\delta)=\mathrm{AC}\!\int_{\mathbb{C}}\mathrm{d}^2z\,|z|^{-15/4+2\delta}\int_{\mathbb{C}^2}\mathrm{d}^2x\,\mathrm{d}^2y\,W(x,y,z)^{2\delta}\,\mathcal{G}(x,y;z).\tag{E.30}$$

This is an equality before the endpoint $\delta\to 0$ is taken, and hence an equality of the prescribed meromorphic continuations. The source factor $W^{2\delta}$ (the W of Sec. 3) is responsible for the matching contact of Appendix E.7.

An explicit algebraic form of the lifted Ising kernel. The energy positions are uniformized by the elliptic curve E_z, the double cover of the t-plane branched at the zeros of $f_z(t) = t(t-z)(t-1)$ and at infinity, with Abel coordinate $\mathrm{d}u = \mathrm{d}t/\sqrt{f_z(t)}$. The complete elliptic integrals use the parameter convention

$$K(z) = \int_0^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{1 - z\sin^2\theta}}, \qquad E(z) = \int_0^{\pi/2} \sqrt{1 - z\sin^2\theta}\, \mathrm{d}\theta, \tag{E.31}$$

continued consistently away from the real unit interval. The periods of E_z are $4K(z)$ and $4\mathrm{i}K(1-z)$; the torus area $\mathcal{A}(z)$, the circular mean parameter $q(z)$, the Ising block weights $w_\pm, w_0$ and the constants $c_\pm, c_0$ are those defined after Eq. (3.5). In particular $\widehat{S}_4 = w_+ + w_-$; changing the sign of $\sqrt{1-z}$ interchanges the two block labels and leaves the sum invariant.

The passage from the exact Ising correlators to the elliptic kernel is an algebraic identity. Set $s_z = (1+z)/3$ and

$$\Sigma(x,y;z) = -\frac{x+y-2s_z}{2} + \frac{f_z(x)+f_z(y)}{2(x-y)^2}, \qquad D^2(x,y;z) = \frac{|f_z(x)f_z(y)|}{|x-y|^4}. \tag{E.32}$$

Substituting Eq. (E.10) and its finite-energy variant (E.11) into Eq. (E.26) gives

$$\mathcal{G} = \frac{w_+|\Sigma + c_+|^2 + w_-|\Sigma + c_-|^2 - w_0|\Sigma + c_0|^2 - (w_+ + w_- - w_0)D^2}{|f_z(x)f_z(y)|}. \tag{E.33}$$

Every quantity in this identity is explicit; no unknown conformal-block coefficient remains.

The uniformization is $x = 4\wp(u) + s_z$, $y = 4\wp(v) + s_z$, with $\wp$ the Weierstrass function of E_z. The addition formulas give

$$\Sigma(x,y;z) = \wp(u-v) + \wp(u+v), \qquad D^2(x,y;z) = |\wp(u-v) - \wp(u+v)|^2; \tag{E.34}$$

for example $f_z(x) = 16\wp'(u)^2$ and $\wp(u-v) - \wp(u+v) = \wp'(u)\wp'(v)/(\wp(u)-\wp(v))^2$. The denominator of Eq. (E.33) is cancelled by $\mathrm{d}^2x\,\mathrm{d}^2y = |f_z(x)f_z(y)|\,\mathrm{d}^2u\,\mathrm{d}^2v$. Each energy plane has two preimages on the curve, so its integral is one half of the integral over the cover, and the two energies give an overall factor $1/4$.

Circular torus integration. In the circular torus prescription

$$\mathrm{FP}\int_{E_z} \wp(u)\,\mathrm{d}^2u = -\mathcal{A}\,q. \tag{E.35}$$

One way to obtain Eq. (E.35) is from the periodic completion $\zeta_{\mathrm{W}}(u) - qu - (\pi/\mathcal{A})\bar{u}$ of the Weierstrass zeta function ζ_{W}. For the half-period $\omega_1 = 2K(z)$, with quasi-period $\eta_1 = \zeta_{\mathrm{W}}(\omega_1)$, its periodicity gives $q = \eta_1/\omega_1 - \pi\bar{\omega}_1/(\omega_1\mathcal{A})$, and the elliptic-integral expression for η_1/ω_1 gives the formula for q after Eq. (3.5). Integrating the u derivative of the completion over a torus with a circular hole gives Eq. (E.35), because the spin-two principal part has zero circular boundary average.

Write $\wp_\mp = \wp(u \mp v)$. Subtracting the squared kernels before integrating gives

$$|\wp_- + \wp_+ + c|^2 - |\wp_- - \wp_+|^2 = 4\,\mathrm{Re}(\wp_-\bar{\wp}_+) + 2\,\mathrm{Re}\big[c(\bar{\wp}_- + \bar{\wp}_+)\big] + |c|^2. \tag{E.36}$$

The map $(u,v) \mapsto (u-v, u+v)$ is a finite covering of E_z^2 and preserves the normalized Haar measure. With the circular finite parts understood,

$$\mathrm{FP}\int_{E_z^2} \wp_-\bar{\wp}_+\ \mathrm{d}^2u\,\mathrm{d}^2v = \mathcal{A}^2|q|^2, \qquad \mathrm{FP}\int_{E_z^2} \wp_-\ \mathrm{d}^2u\,\mathrm{d}^2v = \mathrm{FP}\int_{E_z^2} \wp_+\ \mathrm{d}^2u\,\mathrm{d}^2v = -\mathcal{A}^2 q, \tag{E.37}$$

and therefore

$$\frac{1}{4}\,\mathrm{FP}\int_{E_z^2} \mathrm{d}^2u\,\mathrm{d}^2v\,\big(|\wp_- + \wp_+ + c|^2 - |\wp_- - \wp_+|^2\big) = \frac{\mathcal{A}^2}{4}|c - 2q|^2. \tag{E.38}$$

Applying Eq. (E.38) to Eq. (E.33), with $c = c_\pm, c_0$, gives

$$\mathcal{T}(z) = \frac{\mathcal{A}(z)^2}{4}\Big[w_+|c_+ - 2q|^2 + w_-|c_- - 2q|^2 - w_0|c_0 - 2q|^2\Big], \tag{E.39}$$

and $\mathcal{I} = \mathrm{FP}\int \mathrm{d}^2 z\,|z|^{-15/4}\mathcal{T}(z)$ is the reduced period integral of Eq. (3.5). This operation defines the flat torus finite part. It is not yet the limit of the source-weighted integral $J(\delta)$ of Eq. (E.30); the comparison follows.

E.7 The finite contact

The four same-branch contacts. Fix a nondegenerate modulus z. A single energy approaching a branch point is locally integrable, and so are energies approaching distinct branch points. Coincident energies away from a branch point leave a spin-two term with a radial leading regulator and zero scalar contact. The remaining case has both energies approaching the same branch point. In Abel coordinates u, v that vanish at that point, including the chart at infinity, the exact source factor (E.28) has the leading form

$$W = \frac{|u-v|^2|u+v|^2|v|^2}{4(|u|^4+3|v|^4)}\big[1 + O(|u|+|v|)\big]. \tag{E.40}$$

With $\alpha_1 = u - v$ and $\alpha_2 = u + v$, the leading density, including the cover and linear-coordinate factors, is

$$\frac{\widehat{S}_4 - w_0}{4}\,\mathrm{Re}\frac{1}{\alpha_1^2\bar{\alpha}_2^2}\,\mathrm{d}^2\alpha_1\,\mathrm{d}^2\alpha_2. \tag{E.41}$$

Now put $\alpha_1 = \alpha_2\zeta$ and $\alpha_2 = re^{\mathrm{i}\theta}$. Then

$$W = r^2|\zeta|^2\frac{|1-\zeta|^2}{|1+\zeta|^4+3|1-\zeta|^4} + o(r^2). \tag{E.42}$$

The radial integral up to an auxiliary radius ρ is $\rho^{4\delta}/(4\delta)$. Its zeroth-order angular coefficient is zero, but the first derivative of $W^{2\delta}$ survives and produces, per branch point, the contact $A_{\mathrm{br}}[\widehat{S}_4 - w_0]$ with

$$A_{\mathrm{br}} = \frac{\pi}{4}\,\mathrm{FP}\int_{\mathbb{C}} \mathrm{d}^2\zeta\,\mathrm{Re}(\zeta^{-2})\log\frac{|1-\zeta|^2}{|1+\zeta|^4+3|1-\zeta|^4}. \tag{E.43}$$

The term $2\log|\zeta|$ of $\log W$ has zero angular contribution. The auxiliary radius ρ also drops out, because it multiplies a vanishing zeroth-order residue. There are four branch points (0, z, 1 and ∞); the sheet factors are already included, so there is no further multiplicity.

The contact and the lower Gaussian integral. The contact is related exactly to the integral $A_{a,b}$ of Eq. (E.19). Let $B_{M1} = A_{M,1}(0)$ (so that B_{31} is the case $M = 3$), and let $A_{\mathrm{br}}(M)$ be Eq. (E.43) with 3 replaced by $M \geq 0$. The distributional change of variables $\tilde{w} = (1+\zeta)/(1-\zeta)$ gives

$$\mathrm{FP}\int_{\mathbb{C}} \mathrm{d}^2\zeta\,\mathrm{Re}(\zeta^{-2})\,h\Big(\Big|\frac{1+\zeta}{1-\zeta}\Big|^2\Big) = 8\,\mathrm{FP}\int_{\mathbb{C}} \mathrm{d}^2 z\,F_0(z)\,h(|z|). \tag{E.44}$$

Indeed, the transformed bulk kernel is $4\,\mathrm{Re}[(\tilde{w}-1)^{-2}(\bar{\tilde{w}}+1)^{-2}]$. Its angular integral, in the variable $\varrho = |\tilde{w}|^2$, is $4\pi|\varrho-1|/(1+\varrho)^3$, with local term $-2\pi h(1)$; this is eight times Eq. (E.21). Differentiating $A_{\mathrm{br}}(M)$ and $4B_{M1}$ in M, and using Eq. (E.44) and inversion, gives equal derivatives. The endpoint values $A_{\mathrm{br}}(0) = \pi^2/4$ and $B_{01} = \pi^2/8$ (from Eq. (E.22)) fix the constant, $A_{\mathrm{br}}(M) = 4B_{M1} - \pi^2/4$, which at $M = 3$ is the relation $A_{\mathrm{br}} = 4B_{31} - \pi^2/4$ of Sec. 3. These $M = 0$ values are the prescribed continued limits; no ordinary infrared convergence is asserted for the unweighted limit.

The fixed-modulus comparison. For $z \notin \{0, 1, \infty\}$ the comparison is therefore

$$\lim_{\delta\to 0} \mathrm{AC} \int \mathrm{d}^2x\, \mathrm{d}^2y\, W^{2\delta}\mathcal{G} = \mathcal{T}(z) + 4A_{\mathrm{br}}\big[\widehat{S}_4(z) - w_0(z)\big], \tag{E.45}$$

with $\mathcal{T}(z)$ the torus integrand of Appendix E.6. The source-dependent first derivative of the regulator is necessary even though the corresponding pointwise zeroth-order angular residue vanishes.

The external-primary cancellation. Extending Eq. (E.45) to the modulus boundaries uses a stronger fact than the integrated equality $I_2(0) = 0$. For fixed noncoincident a, b, c, d (energies at 0 and ∞), the exact simultaneous OPE gives

$$\begin{aligned} H_4(\lambda a, \lambda b, c, d) &= \lambda^{-5/4}\frac{|a-b|^{3/4}|c-d|^{3/4}}{4|ab|} + \lambda^{-1/4}H_2(a,b)H_2(c,d) + O(\lambda^{3/4}), \\ S_4(\lambda a, \lambda b, c, d) &= \lambda^{-1/4}S_2(a,b)S_2(c,d) + O(\lambda^{3/4}). \end{aligned} \tag{E.46}$$

Consequently

$$\mathcal{F}(\lambda a, \lambda b, c, d) = \lambda^{-5/4}\frac{|a-b|^{3/4}|c-d|^{3/4}}{4|ab|} + O(\lambda^{3/4}) : \tag{E.47}$$

the entire $\lambda^{-1/4}$ energy-primary term cancels in the connected logarithm. Multiplying by the internally contracted generalized-free pair changes the leading degree to $-5 + 2\delta$ in four real coordinates, which produces $\mathrm{d}\lambda\, \lambda^{-2+2\delta}$. Its Mellin denominator is $-1+2\delta$, which does not vanish at the endpoint. The absent term would have given $\mathrm{d}\lambda\, \lambda^{-1+2\delta}$, a logarithm. The cancellation persists for nonzero δ, since the same generalized-free kernel multiplies all terms of $\mathcal{F}$. If the shrinking pair is not internally contracted in the fixed pairing, the remaining generalized-free fields raise the output dimension beyond this threshold.

E.8 Modulus boundaries, subtractions and the qualification

Thresholds at the modulus boundaries. In the fixed pairing the contracted degeneration is $z \to 0$; the pair degenerations at $z \to 1$ and $z \to \infty$ lack this internal generalized-free contraction. At $r = |z| \to 0$, let Δ be the exchanged Ising dimension. Including the measures of all energy coordinates that shrink with the pair gives the scalar thresholds

shrinking cluster	radial measure	logarithmic threshold
two spins, neither energy	$\mathrm{d}r\, r^{\Delta-3+2\delta}$	$\Delta = 2$
two spins and one energy	$\mathrm{d}r\, r^{\Delta-2+2\delta}$	$\Delta = 1$
two spins and both energies	$\mathrm{d}r\, r^{\Delta-3+2\delta}$	$\Delta = 2$

The disconnected identity terms cancel in the connected expression; an identity-channel power term with denominator $-1 + 2\delta$ is retained and analytically continued. The energy primary of the middle row cancels by Eq. (E.47). At dimension two, the even Ising identity and energy families contain descendants of nonzero spin but no scalar. The remaining thermal powers are restored by Eq. (E.6); their regular denominators must not be dropped.

Spin residues and coupling logarithms. The regulator also has to preserve the vanishing spin residues. In the joint corner $x = z\xi$, $y = z\eta$ its leading term is

$$W \longrightarrow \frac{|\xi-\eta|^2}{|\eta(\eta-1)|\big(|\xi/\eta|^2 + |(1-\xi)/(1-\eta)|^2 + 2\big)}. \tag{E.48}$$

This is independent of $\arg z$, so a leading term of nonzero spin keeps zero angular residue for nonzero δ. The corners with zero and with one energy have the same property. The only proper

scalar coupling logarithm left is a cluster of three insertions of $\sigma\chi$, which is already represented by the counterterm (E.18). Two distinct triples among four perturbing fields overlap, so they cannot both be separate proper nested or disjoint members of one forest. Thermal pairs inside a triple are included in the continued lower-order coupling projection.

The global comparison and its unproved step. The global comparison uses local OPE and Taylor subtractions in each nested sector, in the distribution-extension framework of Ref. [29]: subtract the nonintegrable powers and spin terms, restore the powers by their Mellin continuation and the spin terms by their circular finite parts, and then interchange the regulator limit with the subtracted remainder integral. This last step requires an integrable dominating bound, uniform near $\delta = 0$, including the degenerating tangent shapes at intersections of sectors. The study gives the channel classification above and a general subtraction argument for this step. An audit stage that is part of the study itself (same authorship; not the review of Sec. 1.2) checked the exact algebra, the local contacts and representative simultaneous limits, but did not reconstruct every sectorwise subtraction or prove the uniform remainder bound. This is the one unproved step of Sec. 3.6, and the identification of Eq. (3.9) with the physical coefficient is a formal perturbative result in the critically tuned analytic minimal-subtraction prescription, conditional on it. The study adds that the qualification is not evidence for a missing contact, and that neither a fixed-modulus check nor a high-precision evaluation of $\mathcal{I}$ can replace the bound: they establish different parts of the method.

E.9 Assembly

Under the global comparison just described, Eq. (E.45) is integrated over z with the weight $|z|^{-15/4}$. The term $4A_{\mathrm{br}}[\widehat{S}_4 - w_0]$ then gives $4A_{\mathrm{br}}I_\beta$, with the coupling integral of Eq. (E.14), $I_\beta = \mathrm{AC}\int \mathrm{d}^2z\,|z|^{-15/4}[\widehat{S}_4(z) - |z|^{-1/4}] = -b_0/\pi$. With Eq. (E.30), in which $\Gamma(2\delta) = 1/(2\delta) + O(1)$, and the contact relation of Appendix E.7,

$$\begin{aligned} J(0) &= \mathcal{I} + 4A_{\mathrm{br}}I_\beta = \mathcal{I} - \frac{16b_0B_{31}}{\pi} + \pi b_0, \\ \operatorname*{Res}_{\delta=0} A_4 &= \frac{\pi}{16}J(0) = \frac{\pi}{16}\mathcal{I} - b_0B_{31} + \frac{\pi^2 b_0}{16}, \end{aligned} \tag{E.49}$$

and $z_{4,1} = \mathrm{Res}_{\delta=0}\,A_4 + b_0B_{31}$ is Eq. (3.7). The finite source constant B_{31} cancels between the bare residue and its source counterterm. The remaining $-\pi^2/(4b_0)$ in c_2 is the finite matching contribution; its size is comparable to that of the period term (Sec. 3). The evaluation that follows concerns $\mathcal{I}$ alone; it supplies an explicit value once this renormalization comparison has been specified.

E.10 Complex moments and their reduction to interval periods

From here to Appendix E.14 the target is $\mathcal{I}$ in $c_2 = -\pi\mathcal{I}/(4b_0^2) - \pi^2/(4b_0)$, with $b_0 = 9\pi^2/(8R_\Gamma^2)$ and $R_\Gamma = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$. The continuation and finite-part prescriptions remain those of Eq. (3.5). In particular, an integral with a negative endpoint exponent denotes its meromorphic continuation, not an ordinary improper integral.

Interval periods and complex moments. With $K(t)$ of parameter t, put

$$\begin{aligned} \mathbf{k}(t) &= \begin{pmatrix} K(t)^2 \\ K(t)K(1-t) \\ K(1-t)^2 \end{pmatrix}, \qquad \mathsf{J} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \\ \mathbf{P}(a,b) &= \mathrm{AC}\int_0^1 t^{a-1}(1-t)^{b-1}\mathbf{k}(t)\,\mathrm{d}t, \qquad \mathbf{p}(a,b) = \frac{4}{\pi^2}\mathbf{P}(a,b). \end{aligned} \tag{E.50}$$

The vector $\mathbf{k}$ reverses under $t \mapsto 1-t$; in particular the third component of $\mathbf{p}(a,b)$ is the first component of $\mathbf{p}(b,a)$. Write

$$\mathbf{p}(a,b) = \big(\mathcal{U}(a,b),\, \mathcal{V}(a,b),\, \mathcal{U}(b,a)\big)^{\mathsf{T}}. \tag{E.51}$$

The complex area moment is

$$\mathcal{M}(a,b) = \mathrm{AC}\int_{\mathbb{C}} \mathrm{d}^2 t\, |t|^{2a-2}|1-t|^{2b-2}\mathcal{A}(t)^2, \tag{E.52}$$

and $\mathcal{B}(a,b)$ is Eq. (E.1). The single-valued period pairing uses the elliptic connection formulas [34] and the bilinear contour construction of Ref. [33]. In the present normalization it reads

$$\mathcal{M}(a,b) = 64\,\mathbf{P}(a,b)^{\mathsf{T}}\mathsf{L}(b)\,\mathbf{P}(2-a-b,b), \qquad \mathsf{L}(b) = \begin{pmatrix} 0 & 0 & \sin\pi b \\ 0 & 2\sin\pi b & -2\cos\pi b \\ \sin\pi b & -2\cos\pi b & -\sin\pi b \end{pmatrix}. \tag{E.53}$$

Contour conventions. On $0<t<1$ both $K(t)$ and $K(1-t)$ are real and positive. They are continued from this interval to the slit plane, with the upper and lower boundary values used separately. For real $t>1$ they obey

$$K(t \pm \mathrm{i}0) = t^{-1/2}\big[K(1/t) \pm \mathrm{i}K(1-1/t)\big], \qquad K(1-t) = t^{-1/2}K(1-1/t), \tag{E.54}$$

so that

$$\mathbf{k}(t \pm \mathrm{i}0) = t^{-1}\mathsf{C}_{\pm}\mathbf{k}(1/t), \qquad \mathsf{C}_{\pm} = \begin{pmatrix} 1 & \pm 2\mathrm{i} & -1 \\ 0 & 1 & \pm\mathrm{i} \\ 0 & 0 & 1 \end{pmatrix}. \tag{E.55}$$

These are the symmetric-square connection matrices of the elliptic equation. Moreover $\mathcal{A}(t)^2 = 64\,\mathbf{k}(t)^{\mathsf{T}}\mathsf{J}\,\overline{\mathbf{k}(t)}$. To obtain Eq. (E.53), take a primitive of the holomorphic one-form $t^{a-1}(1-t)^{b-1}\mathbf{k}(t)\,\mathrm{d}t$ in this single-valued pairing and apply Stokes' theorem on the cut plane. The two banks of the cut $[1,\infty)$ give

$$\frac{e^{\mathrm{i}\pi b}\mathsf{J}\mathsf{C}_- - e^{-\mathrm{i}\pi b}\mathsf{J}\mathsf{C}_+}{2\mathrm{i}} = \mathsf{L}(b). \tag{E.56}$$

The substitution $t = 1/\tau$ on that cut changes the power to $\tau^{1-a-b}(1-\tau)^{b-1}$ and so gives the second interval period $\mathbf{P}(2-a-b,b)$. This accounts for the parameter $2-a-b$ and for every entry of $\mathsf{L}(b)$. The contour identity is first applied with convergent endpoint regulators and then continued meromorphically; small-circle terms at nonintegrable powers are continued with the same prescription, not discarded. This is the standard single-valued period-pairing step of the calculation, not a separate numerical integration of the four-point correlator.

E.11 Landen substitution and the six complex moments

The quadratic substitution and the area Wronskian are

$$z = \frac{4w}{(1+w)^2}, \quad t = w^2, \qquad \mathcal{A}\Big(\frac{4w}{(1+w)^2}\Big) = \frac{|1+w|^2}{2}\mathcal{A}(w^2), \qquad |\partial_t\mathcal{A}|^2 = \frac{1}{4}\partial_t\partial_{\bar t}(\mathcal{A}^2) + \frac{2\pi^2}{|t(1-t)|^2}. \tag{E.57}$$

They are inserted in the three-term integrand of Eq. (3.5) as follows. The derivative of the area and Legendre's relation give

$$q(z) = \frac{z(1-z)}{2}\frac{\partial_z\mathcal{A}(z)}{\mathcal{A}(z)} + \frac{1-2z}{12}. \tag{E.58}$$

For $t = w^2$, abbreviate (in this subsection only) $A = \mathcal{A}(t)$ and $A_t = \partial_t A$. Using $\sqrt{1-z} = (1-w)/(1+w)$ on a consistent sheet, the three factors of the integrand become

$$\begin{aligned}\mathcal{A}(z)(c_+ - 2q) &= -t(1-w)(1+\bar{w})A_t,\\ \mathcal{A}(z)(c_- - 2q) &= -(1-w)(1+\bar{w})(A/2 + tA_t),\\ \mathcal{A}(z)(c_0 - 2q) &= \frac{t(1+\bar{w})}{1+w}\big[A/2 - (1-t)A_t\big].\end{aligned} \tag{E.59}$$

The exchange of the two square-root sheets interchanges the first two and leaves their sum unchanged. The Jacobian $|\,\mathrm{d}z/\,\mathrm{d}w|^2 = 16|1-w|^2/|1+w|^6$ and the multiplicities of the degree-two maps $w \mapsto z$ (w and $1/w$ have the same image) and $w \mapsto t$ give

$$\mathcal{I} = \frac{1}{256}\,\mathrm{AC}\!\int_{\mathbb{C}} \mathrm{d}^2 t \left\{ \frac{|1-t|^{7/2}}{|t|}|A_t|^2 + \frac{|1-t|^{7/2}}{|t|^{5/2}}|A/2 + tA_t|^2 - \frac{|1-t|^2}{|t|}|A/2 - (1-t)A_t|^2 \right\}. \tag{E.60}$$

Under inversion $\mathcal{A}(1/t) = |t|\mathcal{A}(t)$, so the first two integrals are equal. Put

$$\Omega_1 = |t|^{-1}|1-t|^{7/2}, \qquad \Omega_0 = |t|^{-1}|1-t|^2. \tag{E.61}$$

The two remaining integrals are evaluated by integration by parts. For the first, the Wronskian identity gives

$$\mathrm{AC}\!\int \Omega_1 |A_t|^2\,\mathrm{d}^2 t = \mathrm{AC}\!\int \Big[\frac{1}{4}(\partial_t\partial_{\bar{t}}\Omega_1)A^2 + \frac{2\pi^2\Omega_1}{|t(1-t)|^2}\Big]\,\mathrm{d}^2 t, \qquad \frac{1}{4}\partial_t\partial_{\bar{t}}\Omega_1 = \frac{|t|^{-3}|1-t|^{3/2}}{64}|2+5t|^2, \tag{E.62}$$

with $|2+5t|^2 = 14 + 35|t|^2 - 10|1-t|^2$. In the third integral the cross term is also integrated by parts. The resulting coefficient of A^2 is

$$\frac{\Omega_0}{4} + \frac{1}{4}\Big\{\partial_t\big[\Omega_0(1-t)\big] + \partial_{\bar{t}}\big[\Omega_0(1-\bar{t})\big]\Big\} + \frac{1}{4}\partial_t\partial_{\bar{t}}\big[\Omega_0|1-t|^2\big] = \frac{|t|^{-3}|1-t|^2}{16}|1+t|^2, \tag{E.63}$$

with $|1+t|^2 = 2 + 2|t|^2 - |1-t|^2$, and its Wronskian term is $2\pi^2\mathcal{B}(-\frac{1}{2}, 2) = 0$, a zero of the reciprocal Gamma function $1/\Gamma(1-b)$ in Eq. (E.1). These integrations by parts are justified by first replacing the coefficient of $|A_t|^2$ by an auxiliary weight $|t|^{2a}|1-t|^{2b}$ with $\mathrm{Re}\,a > 0$, $\mathrm{Re}\,b > 0$ and $\mathrm{Re}(a+b) < 1$, where the boundary terms vanish, and then continuing to the stated exponents. They therefore preserve the finite-part boundary prescription. The result, written so as to keep every endpoint contribution, is

$$\begin{aligned}\mathcal{I} &= \frac{\mathcal{D}_1}{128} - \frac{\mathcal{D}_0}{1024},\\ \mathcal{D}_1 &= -\frac{5}{32}\mathcal{M}(-\tfrac{1}{2}, \tfrac{11}{4}) + \frac{35}{64}\mathcal{M}(\tfrac{1}{2}, \tfrac{7}{4}) + \frac{7}{32}\mathcal{M}(-\tfrac{1}{2}, \tfrac{7}{4}) + 2\pi^2\mathcal{B}(-\tfrac{1}{2}, \tfrac{7}{4}),\\ \mathcal{D}_0 &= \frac{\mathcal{M}(\frac{1}{2}, 2) + \mathcal{M}(-\frac{1}{2}, 2)}{2} - \frac{1}{4}\mathcal{M}(-\tfrac{1}{2}, 3).\end{aligned} \tag{E.64}$$

The derivative reduction thus gives six complex area moments and one elementary complex beta function, which evaluates exactly to $\mathcal{B}(-\frac{1}{2}, \frac{7}{4}) = -32b_0/\pi$.

The vanishing of $\mathcal{D}_0$. Behind $\mathcal{D}_0 = 0$ is an endpoint identity. The symmetric-square elliptic equation gives

$$\mathbf{P}(-\tfrac{1}{2}, 2) + \mathbf{P}(\tfrac{1}{2}, 2) = (-4, 0, 0)^{\mathsf{T}}, \tag{E.65}$$

an identity of continued periods that includes the Mellin boundary term at $t = 0$. It has an explicit differential certificate. For any component y of $\mathbf{k}(t)$ define

$$\mathcal{C}[y] = -8t^{3/2}(1-t)^2 y'' + 4\sqrt{t}(1-t)(5t-3)y' + \frac{2(1-t)(3t-1)}{\sqrt{t}}y. \tag{E.66}$$

The symmetric-square differential equation, $y''' + 3\rho_1 y'' + (\rho_1' + 2\rho_1^2 + 4\rho_0)y' + (2\rho_0' + 4\rho_1\rho_0)y = 0$ with $\rho_1 = (1-2t)/[t(1-t)]$ and $\rho_0 = -1/[4t(1-t)]$, gives, by direct substitution,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{C}[y] = t^{-3/2}(1-t^2)\,y. \tag{E.67}$$

At $t=0$ the continued constant boundary term is zero. At $t=1$ the three boundary values of $\mathcal{C}[y]$ are $-4, 0, 0$, from $K(t) \sim \frac{1}{2}\log[16/(1-t)]$ and $K(1-t) \sim \pi/2$. This proves Eq. (E.65) with its finite endpoint term. To see how it enters, let $\mathbf{u} = \mathbf{P}(-\frac{1}{2}, 2)$ and $\mathbf{v} = \mathbf{P}(\frac{1}{2}, 2)$. Since $\mathsf{L}(2)(\mathbf{u}+\mathbf{v}) = 0$, $\mathsf{L}(3) = -\mathsf{L}(2)$ and $\mathbf{P}(-\frac{1}{2}, 3) = \mathbf{u} - \mathbf{v}$ (by Eq. (E.70) below), the pairing gives

$$\mathcal{M}(-\tfrac{1}{2}, 3) = 4\mathcal{M}(\tfrac{1}{2}, 2) = 4\mathcal{M}(-\tfrac{1}{2}, 2), \qquad \mathcal{D}_0 = 0. \tag{E.68}$$

Reduction to four interval periods. Abbreviate

$$\mathbf{X} = \mathbf{P}(-\tfrac{1}{2}, \tfrac{7}{4}), \quad \mathbf{Y} = \mathbf{P}(\tfrac{1}{2}, \tfrac{7}{4}), \quad \mathbf{Z} = \mathbf{P}(-\tfrac{1}{4}, \tfrac{7}{4}), \quad \mathbf{W} = \mathbf{P}(\tfrac{3}{4}, \tfrac{7}{4}). \tag{E.69}$$

The contiguous identity

$$\mathbf{P}(a, b+1) = \mathbf{P}(a, b) - \mathbf{P}(a+1, b) \tag{E.70}$$

and $\mathsf{L}(\frac{11}{4}) = -\mathsf{L}(\frac{7}{4})$ turn the part of $\mathcal{D}_1$ without the beta function into

$$10(\mathbf{X}-\mathbf{Y})^{\mathsf{T}}\mathsf{L}(\tfrac{7}{4})(\mathbf{Z}-\mathbf{W}) + 35\,\mathbf{Y}^{\mathsf{T}}\mathsf{L}(\tfrac{7}{4})\mathbf{Z} + 14\,\mathbf{X}^{\mathsf{T}}\mathsf{L}(\tfrac{7}{4})\mathbf{W} = (2\mathbf{X}+5\mathbf{Y})^{\mathsf{T}}\mathsf{L}(\tfrac{7}{4})(5\mathbf{Z}+2\mathbf{W}). \tag{E.71}$$

Consequently

$$\mathcal{I} = \frac{1}{128}(2\mathbf{X}+5\mathbf{Y})^{\mathsf{T}}\mathsf{L}(\tfrac{7}{4})(5\mathbf{Z}+2\mathbf{W}) - \frac{\pi b_0}{2}. \tag{E.72}$$

The constant here is the complex-beta term of $\mathcal{D}_1$. It is separate from the finite regulator-matching term $-\pi^2/(4b_0)$ of Eq. (3.7).

E.12 Reduction to symmetric elliptic moments

Use normalized period vectors and set

$$\mathbf{a} = 2\mathbf{p}(-\tfrac{1}{2}, \tfrac{7}{4}) + 5\mathbf{p}(\tfrac{1}{2}, \tfrac{7}{4}), \quad \mathbf{b} = 5\mathbf{p}(-\tfrac{1}{4}, \tfrac{7}{4}) + 2\mathbf{p}(\tfrac{3}{4}, \tfrac{7}{4}), \quad \mathsf{H} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 2 \\ 1 & 2 & -1 \end{pmatrix}, \quad \mathsf{L}(\tfrac{7}{4}) = -\frac{\mathsf{H}}{\sqrt{2}}. \tag{E.73}$$

Equations (3.7) and (E.72) give

$$c_2 = -\frac{R_\Gamma^2}{9} + \frac{\pi R_\Gamma^4}{10368\sqrt{2}}\,\mathbf{a}^{\mathsf{T}}\mathsf{H}\,\mathbf{b}. \tag{E.74}$$

Contour reflection. In the same boundary convention, contour reflection of these particular interval periods gives

$$\mathbf{p}(-\tfrac{1}{2}, \tfrac{7}{4}) = \mathsf{D}\,\mathbf{p}(\tfrac{3}{4}, \tfrac{7}{4}), \qquad \mathbf{p}(\tfrac{1}{2}, \tfrac{7}{4}) = \mathsf{D}\,\mathbf{p}(-\tfrac{1}{4}, \tfrac{7}{4}), \qquad \mathsf{D} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -2 & -1 \\ -1 & -1 & 0 \\ 0 & -2 & -1 \end{pmatrix}. \tag{E.75}$$

These relations use the same connection matrices and continued endpoint contours as Eq. (E.53); they are not identities obtained by naive substitution in divergent real integrals. They can be derived without assuming D. For real $a, b > 0$ with $a' = 2-a-b > 0$, integrate $t^{a-1}(1-t)^{b-1}\mathbf{k}(t)\,\mathrm{d}t$

along the real axis, closed in the upper half-plane; the small and large arcs vanish in this domain of convergence. On the negative interval set $t = -\tau/(1-\tau)$; its connection matrix is

$$\mathsf{B}_- = \begin{pmatrix} 1 & 0 & 0 \\ -\mathrm{i} & 1 & 0 \\ -1 & -2\mathrm{i} & 1 \end{pmatrix}. \tag{E.76}$$

The three real intervals give the vector identity

$$\mathbf{P}(a,b) = e^{\mathrm{i}\pi a}\mathsf{B}_-\mathbf{P}(a,a') + e^{-\mathrm{i}\pi b}\mathsf{C}_+\mathbf{P}(a',b), \tag{E.77}$$

which continues to the parameters of Eq. (E.75). Let $\mathsf{A}_c = e^{\mathrm{i}\pi a}\mathsf{B}_-$ and $\mathsf{E}_c = e^{-\mathrm{i}\pi b}\mathsf{C}_+$. The left side and the two interval vectors are real, so eliminating $\mathbf{P}(a,a')$ between the imaginary and real parts gives

$$\mathbf{P}(a,b) = [\operatorname{Re}\mathsf{E}_c - \operatorname{Re}\mathsf{A}_c(\operatorname{Im}\mathsf{A}_c)^{-1}\operatorname{Im}\mathsf{E}_c]\mathbf{P}(a',b). \tag{E.78}$$

For $a = \pm\frac{1}{2}$ and $b = \frac{7}{4}$, $e^{\mathrm{i}\pi a} = \pm\mathrm{i}$ and $e^{-\mathrm{i}\pi b} = (1+\mathrm{i})/\sqrt{2}$, and the bracket evaluates exactly to D for either sign. This proves Eq. (E.75), and hence $\mathbf{a} = \mathsf{D}\,\mathbf{b}$.

Symmetric moments. Define

$$U(a) = \mathcal{U}(a,a), \qquad V(a) = \mathcal{V}(a,a), \qquad m_U = \tfrac{5}{2}U(-\tfrac{1}{4}) - 4U(\tfrac{3}{4}), \qquad m_V = \tfrac{5}{2}V(-\tfrac{1}{4}) - 4V(\tfrac{3}{4}). \tag{E.79}$$

For $a = -\frac{1}{4}$ the weight in $\mathbf{b}$ is $[t(1-t)]^{a-1}\varpi(t)$, where

$$\varpi(t) = (5+2t)(1-t)^2, \quad \frac{\varpi(t)+\varpi(1-t)}{2} = \frac{5}{2} - 4t(1-t), \quad \varpi(t) - \varpi(1-t) = (1-2t)\{5+2t(1-t)\}. \tag{E.80}$$

For the components $[\mathbf{b}]_i$ of $\mathbf{b}$ it follows directly that $([\mathbf{b}]_1 + [\mathbf{b}]_3)/2 = m_U$ and $[\mathbf{b}]_2 = m_V$.

The antisymmetric moment. The remaining antisymmetric moment has an exact Gamma evaluation:

$$O(a) := \mathcal{U}(a,a+1) - \mathcal{U}(a+1,a) = -\frac{2^{1-2a}}{\sqrt{\pi}}\Big[\frac{\Gamma(a)}{\Gamma(a+\frac{1}{2})}\Big]^3. \tag{E.81}$$

The following derivation also fixes its sign. Set $\hat{K}(t) = 2K(t)/\pi$ and $x = 4t(1-t)$ on $0 < t < \frac{1}{2}$, and introduce the Meijer function

$$G_0(x) = G^{3,0}_{3,3}\left(x \,\middle|\, \begin{matrix} \frac{1}{2},\frac{1}{2},\frac{1}{2} \\ 0,0,0 \end{matrix}\right), \qquad \int_0^1 x^{s-1}G_0(x)\,\mathrm{d}x = \frac{\Gamma(s)^3}{\Gamma(s+\frac{1}{2})^3}. \tag{E.82}$$

The elliptic symmetric-square equation $[\vartheta^3 - x(\vartheta+\frac{1}{2})^3]Y = 0$, with $\vartheta = x\partial_x$, together with its endpoint normalization, gives

$$\hat{K}(1-t)^2 - \hat{K}(t)^2 = \frac{2}{\sqrt{\pi}}G_0(4t(1-t)). \tag{E.83}$$

The normalization can be checked directly, without an unspecified constant. The triple pole at $s = 0$ of the Mellin transform (E.82) gives

$$\frac{2}{\sqrt{\pi}}G_0(x) = \frac{1}{\pi^2}\log^2\frac{64}{x} - 1 + O(x\log^2 x), \tag{E.84}$$

which agrees with $\hat{K}(1-t)^2 - \hat{K}(t)^2$ because $t = x/4 + O(x^2)$, $\hat{K}(t) = 1 + O(t)$ and $\hat{K}(1-t) = \pi^{-1}\log(16/t) + O(t\log t)$. The coefficients of the squared logarithm, the logarithm and the constant fix the three local solutions of the third-order equation. Folding the odd moment about $\frac{1}{2}$ gives

$$O(a) = \int_0^{1/2}[t(1-t)]^{a-1}(1-2t)\{\hat{K}(t)^2 - \hat{K}(1-t)^2\}\,\mathrm{d}t = -\frac{2\cdot 4^{-a}}{\sqrt{\pi}}\int_0^1 x^{a-1}G_0(x)\,\mathrm{d}x, \tag{E.85}$$

which proves Eq. (E.81) where the integrals converge, and then by meromorphic continuation. With the decomposition of ϖ,

$$d_O := [\mathbf{b}]_1 - [\mathbf{b}]_3 = 5O(-\tfrac{1}{4}) + 2O(\tfrac{3}{4}) = \frac{576\sqrt{2}}{\sqrt{\pi}R_\Gamma^3}, \qquad \mathbf{b} = \begin{pmatrix} m_U + d_O/2 \\ m_V \\ m_U - d_O/2 \end{pmatrix}. \tag{E.86}$$

Direct matrix multiplication, with no further integral identity, gives

$$\frac{(\mathsf{D}\mathbf{b})^{\mathsf{T}}\mathsf{H}\,\mathbf{b}}{\sqrt{2}} = \frac{d_O^2}{4} - (m_U + m_V)(m_U + 3m_V), \tag{E.87}$$

and therefore

$$c_2 = \frac{16}{R_\Gamma^2} - \frac{R_\Gamma^2}{9} - \frac{\pi R_\Gamma^4}{10368}(m_U + m_V)(m_U + 3m_V). \tag{E.88}$$

The term $16/R_\Gamma^2$ comes from the evaluated antisymmetric period. Its origin differs from that of both constants of the residue and complex-beta reductions.

E.13 Ordinary hypergeometric evaluation of the moments

The generalized hypergeometric series is [34]

$${}_pF_q\left(\begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix}; x\right) = \sum_{n=0}^{\infty} \frac{(a_1)_n \cdots (a_p)_n}{(b_1)_n \cdots (b_q)_n} \frac{x^n}{n!}, \qquad (a)_n = \frac{\Gamma(a+n)}{\Gamma(a)}, \tag{E.89}$$

the convention of $\mathsf{P}, \mathsf{Q}, \mathsf{T}$ in Sec. 3.5. Define

$$\Phi_A(a) = {}_4F_3\left(\begin{matrix} \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, a \\ 1, 1, a + \frac{1}{2} \end{matrix}; 1\right), \qquad \Phi_D(a) = {}_4F_3\left(\begin{matrix} \frac{1}{2}, a, a, a \\ a + \frac{1}{2}, a + \frac{1}{2}, a + \frac{1}{2} \end{matrix}; 1\right). \tag{E.90}$$

Both series have parametric excess one at every nonresonant parameter used below. The symmetric moments are

$$U(a) = 2^{1-2a}\left[\sqrt{\pi}\frac{\Gamma(a)}{\Gamma(a+\frac{1}{2})}\Phi_A(a) + \frac{1}{\sqrt{\pi}}\Big(\frac{\Gamma(a)}{\Gamma(a+\frac{1}{2})}\Big)^3\Phi_D(a)\right], \tag{E.91}$$

$$V(a) = 2^{1-2a}\left[\sqrt{\pi}\frac{\Gamma(a-\frac{1}{2})}{\Gamma(a)}\Phi_A(1-a) + \frac{\tan \pi a}{\sqrt{\pi}}\Big(\frac{\Gamma(a)}{\Gamma(a+\frac{1}{2})}\Big)^3\Phi_D(a)\right]. \tag{E.92}$$

These are meromorphic identities in a. At $a = -\frac{1}{4}$ and $a = \frac{3}{4}$ no limiting cancellation of Gamma poles is needed in the displayed summands.

The square moment. The quadratic transformation and Clausen's identity give

$$\hat{K}(t)^2 = {}_3F_2\left(\begin{matrix} \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \\ 1, 1 \end{matrix}; x\right), \qquad x = 4t(1-t), \quad 0 < t < \tfrac{1}{2}. \tag{E.93}$$

Combine this with Eq. (E.83) and fold the interval. Since $[t(1-t)]^{a-1}\,\mathrm{d}t = 4^{-a}x^{a-1}(1-x)^{-1/2}\,\mathrm{d}x$,

$$U(a) = 2\cdot 4^{-a}\left[\int_0^1 x^{a-1}(1-x)^{-1/2}\,{}_3F_2\left(\begin{matrix} \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \\ 1, 1 \end{matrix}; x\right)\,\mathrm{d}x + \frac{1}{\sqrt{\pi}}\int_0^1 x^{a-1}(1-x)^{-1/2}G_0(x)\,\mathrm{d}x\right]. \tag{E.94}$$

The first integral is the Euler beta integral for $\Phi_A(a)$. Expanding $(1-x)^{-1/2}$ and applying the Mellin transform (E.82) to each term gives $\Phi_D(a)$ for the second. This proves Eq. (E.91).

The mixed moment. Its Mellin–Barnes representation follows from the inverse Mellin formula, on $0 < x < 1$,

$$\hat{K}(t_x)\hat{K}(1-t_x) = \frac{1}{2\pi\mathrm{i}\sqrt{\pi}} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \frac{\Gamma(s)^2\Gamma(\frac{1}{2}-s)}{\Gamma(s+\frac{1}{2})^2\Gamma(1-s)} x^{-s}\,\mathrm{d}s, \qquad t_x = \frac{1-\sqrt{1-x}}{2}, \quad 0 < c < \tfrac{1}{2}. \tag{E.95}$$

One way to verify it: its Mellin kernel $m(s)$ satisfies $m(s+1)/m(s) = s^3/(s+\frac{1}{2})^3$, so it solves the same symmetric-square equation. The double pole at zero has principal part

$$\frac{\Gamma(s)^2\Gamma(\frac{1}{2}-s)}{\sqrt{\pi}\,\Gamma(s+\frac{1}{2})^2\Gamma(1-s)} = \frac{1}{\pi}\Big[\frac{1}{s^2} + \frac{6\log 2}{s} + O(1)\Big]. \tag{E.96}$$

The resulting local behaviour is $\pi^{-1}\log(64/x)$ with no squared logarithm, exactly that of the left side, and these Frobenius data determine the identity. Folding the interval in $V(a)$ and inserting Eq. (E.95) adds the Euler beta factor $B(a-s,\frac{1}{2}) = \sqrt{\pi}\,\Gamma(a-s)/\Gamma(a+\frac{1}{2}-s)$. Consequently

$$V(a) = 2\cdot 4^{-a}\frac{1}{2\pi\mathrm{i}} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \frac{\Gamma(s)^2\Gamma(\frac{1}{2}-s)\Gamma(a-s)}{\Gamma(s+\frac{1}{2})^2\Gamma(1-s)\Gamma(a+\frac{1}{2}-s)}\,\mathrm{d}s, \qquad 0 < c < \operatorname{Re} a < \tfrac{1}{2}. \tag{E.97}$$

Close the contour to the right, with its clockwise orientation. The two pole families, $s = \frac{1}{2}+n$ and $s = a+n$, give respectively

$$\sqrt{\pi}\frac{\Gamma(a-\frac{1}{2})}{\Gamma(a)}\Phi_A(1-a), \qquad \frac{\Gamma(a)^2\Gamma(\frac{1}{2}-a)}{\sqrt{\pi}\,\Gamma(a+\frac{1}{2})^2\Gamma(1-a)}\Phi_D(a). \tag{E.98}$$

The residues of $\Gamma(\frac{1}{2}-s)$ and $\Gamma(a-s)$ have the opposite sign to ordinary left-pole residues; the clockwise closure cancels that sign. Gamma reflection reduces the second prefactor to $(\tan\pi a/\sqrt{\pi})[\Gamma(a)/\Gamma(a+\frac{1}{2})]^3$. Restoring $2\cdot 4^{-a}$ proves Eq. (E.92) and allows its continuation to the required values of a. Both pole families are part of the derivation.

The six-function form. At the two values of a in Eq. (E.79) exactly six ordinary ${}_4F_3(1)$ values occur. They enter only through

$$X = 5\Phi_A(-\tfrac{1}{4}) + 2\Phi_A(\tfrac{3}{4}), \qquad Y = 5\Phi_A(\tfrac{5}{4}) - 6\Phi_A(\tfrac{1}{4}), \qquad Z = 5\Phi_D(-\tfrac{1}{4}) + 2\Phi_D(\tfrac{3}{4}). \tag{E.99}$$

Gamma recursion in Eqs. (E.91) and (E.92) gives

$$m_U = -\frac{4\sqrt{2\pi}}{R_\Gamma}X - \frac{64\sqrt{2}}{\sqrt{\pi}R_\Gamma^3}Z, \qquad m_V = \frac{\sqrt{2\pi}R_\Gamma}{3}Y + \frac{64\sqrt{2}}{\sqrt{\pi}R_\Gamma^3}Z. \tag{E.100}$$

In particular Z cancels in $m_U + m_V$, but it contributes to $m_U + 3m_V$. Equivalently,

$$c_2 = \frac{16}{R_\Gamma^2} - \frac{R_\Gamma^2}{9} - \frac{\pi^2}{15552}(12X - R_\Gamma^2 Y)\Big[R_\Gamma^2(4X - R_\Gamma^2 Y) - \frac{128Z}{\pi}\Big]. \tag{E.101}$$

This six-function expression is an intermediate identity. The next step eliminates three independent combinations by exact telescoping, including their boundary values.

E.14 Telescoping proof of the three-function closed form

Let $\mathsf{P},\mathsf{Q},\mathsf{T}$ be the convergent series of Sec. 3.5. The closed form (3.9) follows from Eq. (E.101) by the three exact reductions

$$X=\frac{9}{2}\mathsf{P}-\frac{2R_\Gamma}{\pi^{3/2}},\qquad Y=\frac{45}{64}\mathsf{Q}-\frac{24}{\pi^{3/2}R_\Gamma},\qquad Z=\frac{72}{25}\mathsf{T}+\frac{R_\Gamma^3}{8\sqrt{\pi}},\tag{E.102}$$

which are proved here with every telescoping endpoint retained. Take the base summands

$$a_n=\frac{(\frac12)_n^3(\frac34)_n}{(n!)^3(\frac54)_n},\qquad b_n=\frac{(\frac12)_n^3(\frac14)_n}{(n!)^3(\frac34)_n},\qquad d_n=\frac{(\frac12)_n(\frac34)_n^3}{n!\,(\frac54)_n^3}.\tag{E.103}$$

The summands of X,Y,Z in Eq. (E.99) are respectively

$$-\frac{12n+7}{4n-1}a_n,\qquad \frac{3(12n-1)}{4n+3}b_n,\qquad \Big[2-5\Big(\frac{4n+1}{4n-1}\Big)^3\Big]d_n.\tag{E.104}$$

Let p_n,q_n,t_n denote the summands of $\mathsf{P},\mathsf{Q},\mathsf{T}$, and define the certificates

$$\begin{aligned}H_X(n)&=-\frac{4n^2(2n-3)(4n+1)^2}{(2n-1)^2(4n-1)},\qquad H_Y(n)=-\frac{3n(16n^3+44n^2+34n+15)}{2(n+1)^2},\\ H_Z(n)&=\frac{2n(4n+1)(64n^3+16n^2-20n+43)}{(4n-1)^3}.\end{aligned}\tag{E.105}$$

Direct rational simplification proves

$$\begin{aligned}-\frac{12n+7}{4n-1}a_n-\frac{9}{2}p_n&=H_X(n+1)a_{n+1}-H_X(n)a_n,\\ \frac{3(12n-1)}{4n+3}b_n-\frac{45}{64}q_n&=H_Y(n+1)b_{n+1}-H_Y(n)b_n,\\ \Big[2-5\Big(\frac{4n+1}{4n-1}\Big)^3\Big]d_n-\frac{72}{25}t_n&=H_Z(n+1)d_{n+1}-H_Z(n)d_n.\end{aligned}\tag{E.106}$$

For example, divide the first identity by a_n and use

$$\frac{a_{n+1}}{a_n}=\frac{(n+\frac12)^3(n+\frac34)}{(n+1)^3(n+\frac54)},\qquad \frac{p_n}{a_n}=\frac{(2n+1)(4n+1)}{(2n-1)^2(n+1)}.\tag{E.107}$$

It becomes an identity of rational functions of n, so checking it depends neither on numerical precision nor on summing the series. The other two identities are checked in the same way.

Every $H(0)$ vanishes, but the upper boundary terms do not:

$$\begin{aligned}\lim_{n\to\infty}\frac{(H_X,H_Y,H_Z)}{n^2}&=(-8,-24,8),\\ \lim_{n\to\infty}n^2a_n=\frac{R_\Gamma}{4\pi^{3/2}},\qquad \lim_{n\to\infty}n^2b_n&=\frac{1}{\pi^{3/2}R_\Gamma},\qquad \lim_{n\to\infty}n^2d_n=\frac{R_\Gamma^3}{64\sqrt{\pi}}.\end{aligned}\tag{E.108}$$

Summing Eq. (E.106) from $n=0$ to N and letting $N\to\infty$ gives exactly the three Gamma terms of Eq. (E.102). Substituting Eq. (E.102) into Eq. (E.101), the boundary constants cancel from both factors, and the result is exactly the closed form (3.9).

E.15 Numerical evaluation and the scope of the checks

Default evaluation. The study evaluates Eq. (3.9) at 80 working digits, and checks it against the six-function form (E.101) and against the endpoint series below, with 400 terms in each half-interval series; Sec. 3.5 quotes their agreement. With N_{h} terms in each half-interval series, the absolute difference $|\Delta c_2|$ between the endpoint-series value and the three-function value is

N_{h}	$\lvert\Delta c_2\rvert$	N_{h}	$\lvert\Delta c_2\rvert$	N_{h}	$\lvert\Delta c_2\rvert$	N_{h}	$\lvert\Delta c_2\rvert$
24	5.2×10^{-7}	64	2.8×10^{-19}	144	1.5×10^{-43}	320	2.5×10^{-80}
40	6.0×10^{-12}	96	5.1×10^{-29}	224	9.4×10^{-68}	400	2.5×10^{-80}

and it reaches the 80-digit arithmetic floor at 320 terms. This is numerical consistency, not an error bound. The values are in `code/c2_closed_d2/outputs/raw/three_4f3_80dps.json` and `code/c2_closed_d2/outputs/raw/convergence.json`.

Endpoint-series evaluation. An evaluation independent of the ${}_4F_3$ identities uses the Frobenius expansions

$$\begin{aligned} \hat K(t) &= \sum_{n\ge0}\phi_n t^n, \qquad \phi_n=\Big[\frac{(\frac12)_n}{n!}\Big]^2,\\ G(t) &= 4\sum_{n\ge0}\phi_n(\mathsf h_{2n}-\mathsf h_n)t^n, \qquad K(1-t)=\tfrac12\{\log(16/t)\hat K(t)-G(t)\},\end{aligned} \tag{E.109}$$

with the harmonic numbers $\mathsf h_0=0$, $\mathsf h_n=\sum_{k=1}^n 1/k$. Each period is split at $t=\frac12$, its second half is reflected, and $(1-t)^{b-1}\hat K^2$, $(1-t)^{b-1}\hat K G$ and $(1-t)^{b-1}G^2$ are expanded in power series of t, with coefficient arrays $\alpha_n,\beta_n,\gamma_n$. For each coefficient the only Mellin integrals required are, with $p=a+n$ and $\ell_{32}=\log 32$,

$$\begin{aligned} j_0(p)&=\mathrm{AC}\int_0^{1/2}t^{p-1}\,\mathrm dt=\frac{2^{-p}}{p}, \qquad j_1(p)=\mathrm{AC}\int_0^{1/2}t^{p-1}\log(16/t)\,\mathrm dt=2^{-p}\Big(\frac{\ell_{32}}{p}+\frac{1}{p^2}\Big),\\ j_2(p)&=\mathrm{AC}\int_0^{1/2}t^{p-1}\log^2(16/t)\,\mathrm dt=2^{-p}\Big(\frac{\ell_{32}^2}{p}+\frac{2\ell_{32}}{p^2}+\frac{2}{p^3}\Big).\end{aligned} \tag{E.110}$$

These hold first at $\operatorname{Re}p>0$ and continue analytically to the nonzero negative p needed here. The three unnormalized half-period sums (for $K(t)^2$, $K(t)K(1-t)$ and $K(1-t)^2$) are

$$\sum_n\frac{\pi^2}{4}\alpha_n j_0, \qquad \sum_n\frac{\pi}{4}(\alpha_n j_1-\beta_n j_0), \qquad \sum_n\frac14(\alpha_n j_2-2\beta_n j_1+\gamma_n j_0), \tag{E.111}$$

and adding the reflected half gives the full vector. Because the expansion is restricted to $0\le t\le\frac12$, this is an exponentially convergent numerical check of the continued periods that uses neither the six- nor the three-function formula. Agreement is a reproducible numerical consistency check, not an interval enclosure of the final digits.

A separate endpoint-subtracted quadrature. A second implementation checks the same moments by quadrature, with the elliptic integrals evaluated through Carlson's symmetric integral,

$$K(t)=R_F(0,1-t,1), \qquad K(1-t)=R_F(0,t,1), \qquad R_F(x,y,z)=\frac12\int_0^\infty\frac{\mathrm dv}{\sqrt{(v+x)(v+y)(v+z)}}, \tag{E.112}$$

for positive arguments, continued to the endpoint limits shown. This representation does not use the Frobenius coefficients. To make the negative moment at $a=-\frac14$ integrable, each symmetric

moment is folded onto $0 < t < \frac{1}{2}$ and its leading endpoint terms are subtracted. With $u = -\log t$ and $\ell = \log 16 + u$, define

$$\begin{aligned} R_U(u;a) &= (1-e^{-u})^{a-1}\{K(e^{-u})^2 + K(1-e^{-u})^2\} - \frac{\pi^2+\ell^2}{4}, \\ R_V(u;a) &= (1-e^{-u})^{a-1}K(e^{-u})K(1-e^{-u}) - \frac{\pi\ell}{4}. \end{aligned} \tag{E.113}$$

The continued Mellin integrals $j_k(a)$ then give

$$\begin{aligned} U(a) &= \frac{4}{\pi^2}\Big[\int_{\log 2}^{\infty} e^{-au} R_U(u;a)\,\mathrm{d}u + \frac{\pi^2 j_0(a) + j_2(a)}{4}\Big], \\ V(a) &= \frac{8}{\pi^2}\Big[\int_{\log 2}^{\infty} e^{-au} R_V(u;a)\,\mathrm{d}u + \frac{\pi j_1(a)}{4}\Big]. \end{aligned} \tag{E.114}$$

The remainders are $O(e^{-u}u^2)$, so the integrals converge for $\operatorname{Re} a > -1$, including $a = -\frac{1}{4}$ and $a = \frac{3}{4}$. The implementation uses Gauss–Legendre quadrature on the intervals with endpoints $\log 2, 2, 5, 10, 20, 40, 80, 120, 200$, evaluates the integrand at 100 decimal digits, and restores the analytic leading terms exactly as in Eq. (E.114). The finite upper limit $u = 200$ leaves a rapidly decaying tail. Inserted in Eq. (E.88), these moments reproduce the closed form to 1.40×10^{-61} in c_2 (record value 1.4045×10^{-61}; Sec. 3.5). This is an observed difference; the script supplies no rigorous bound on the quadrature error or on the cutoff (record: `code/c2_closed_d2/outputs/raw/independent_c2_80dps.json`).

Exact bounds on the series tails. The three final series also admit direct remainder inequalities. For any of them let v_n be the absolute value of its nth summand, $n \geq 1$. Its positive ratio is the rational function

$$r_n = \frac{v_{n+1}}{v_n} = \frac{\prod_{j=1}^{4}(n+u_j)}{(n+1)\prod_{j=1}^{3}(n+l_j)}, \tag{E.115}$$

with the upper parameters u_j and lower parameters l_j of that series. Set $\phi_c(n) = n/2 + c$ and $E_c(n) = \phi_c(n) - 1 - \phi_c(n+1)r_n$. For P and T take $(c_-, c_+) = (0, 1)$, and for Q take $(1, 2)$. Then $E_{c_-}(n) \leq 0$ and $E_{c_+}(n) \geq 0$ for all integers $n \geq 1$; these signs are verified exactly by replacing n with $m + 1$ and checking that the numerator and denominator of the suitably signed rational expression have nonnegative coefficients. Multiplying by v_n and summing telescopically gives

$$\Big(\frac{N}{2} + c_-\Big)v_N \leq \sum_{n=N}^{\infty} v_n \leq \Big(\frac{N}{2} + c_+\Big)v_N, \tag{E.116}$$

where the upper boundary term vanishes because $v_n = O(n^{-3})$. All terms of P and Q are positive; for T the terms with $n \geq 1$ are negative. These signs decide which tail is added to or subtracted from a partial sum. With $N = 20000$, propagation of these bounds gives the envelope of c_2 quoted in Sec. 3.5. The inequalities for the mathematical series are exact, but the arithmetic that evaluates their endpoints is ordinary arbitrary-precision floating point, so the envelope is an independent check, not a certified rounding interval.

Scope of the checks. The study's exact checks cover the three rational telescoping identities with their nonzero endpoint terms, the complex coordinate Jacobians, the lower-order Gamma reductions and the scheme covariance of c_2 under $\mathcal{E}_R \to e^{\kappa g^2}\mathcal{E}_R$. A separate check of the analytic bridges verifies the Landen coefficient reduction, the cut-jump and reflection matrices, and the differential certificate used for $\mathcal{D}_0 = 0$. The study's numerical checks test the elliptic-cover identity (E.33) at noncoincident points, the Gaussian finite term B_{31}, the contact relation, and representative simultaneous OPE limits; the source-contact checks test the dependence of the finite contact on the smooth infrared cutoff and its cancellation against the local counterterm. None of these checks

evaluates every global collision sector. The discrepancies quoted above are neither statistical error bars nor statements about the physical regulator comparison of Appendix E.8. The checks made by the review of the study are those of Secs. 3.5 and 3.7.

E.16 Further conditions stated by the study

Beyond Secs. 3.6 and 6, the study states three conditions on what it establishes.

- *Endpoint terms are part of the value.* The evaluation requires the finite source contact, the interval-period pairing with its continued endpoint terms, and the nonzero endpoint terms of the telescoping proof; every integral with a negative endpoint exponent is a meromorphic continuation, not an ordinary improper integral. The endpoint values $A_{\mathrm{br}}(0) = \pi^2/4$ and $B_{01} = \pi^2/8$ of the contact relation are prescribed continued limits; no ordinary infrared convergence is asserted for them.
- *The digits are checked, not certified, within the study.* Its long decimal value rests on high-precision evaluations by different routes (the six- and three-function forms, the endpoint series and the Carlson quadrature); it is not presented as a certified rounding theorem. A certified enclosure would combine interval arithmetic with the analytic tail bounds above, or bound both the Carlson quadrature error and its tail beyond $u = 200$. (The integer-interval evaluation to 51 decimal places mentioned in Sec. 3.5 is part of the cylinder base study (Appendix F.14), not of this study.)
- *The local checks do not bear on the global step.* The fixed-modulus contact calculation and the high-precision agreement of the special-function evaluations establish different parts of the method; neither supplies the uniform remainder bound of Appendix E.8.

F c_2 from fourth-order perturbation theory on the cylinder: method

This appendix gives the method of the base study of c_2 with the truncated cylinder Hamiltonian (written by OpenAI GPT-6 astra; Sec. 1.2), edited to the notation of this note.

Section 3.8 summarises the setup and gives the continuum analysis that produced the quoted value $c_2 = -4.0670(24)$, whose error is a linear sum of systematic diagnostics, not a standard deviation. That analysis was done later, with $N \leq 1024$ and with fit and selection rules fixed in code before the final numbers were computed (its error rule was revised once, after the first result had been seen; Sec. 3.8); it evaluates the quantities defined here. The appendix gives the normalization and the exact second-order coefficient (Appendices F.1 to F.3), the fourth-order Rayleigh–Schrödinger formula, its Hilbert space, regulator and counterterms (Appendices F.4 to F.7), the spectral generating functions that evaluate it (Appendices F.8 and F.9), the large-N tails and the regulator limit (Appendix F.10), the checks (Appendices F.11 to F.14), and the explicit formulas behind them (Appendices F.15 to F.19). The calculation is numerical: no step below bounds the remaining spectral or regulator extrapolation. The base study's checks of the closed form (Appendix F.14) certify the value of the closed expression, Eq. (3.9), not its identification with the physical coefficient, which remains conditional on the step of Sec. 3.6.

Notation: δ is that of Table 1 ($d = 2$), and c_2 is the coefficient the base study writes C_2. As in Sec. 3.8, $|\mathcal{E}\rangle$ is the state of the Ising energy field ϵ at the ultraviolet point, $E_{\mathcal{E}}$ its energy, and a subscript $\mathcal{E}$ on a state-dependent quantity refers to it.

F.1 Continuum setup and normalization

The short-range description. The theory is the crossover description of Sec. 3: the diagonal Ising CFT, with central charge $c = \frac{1}{2}$, tensored with a generalized free scalar χ, with

$$\Delta_\sigma = \frac{1}{8}, \qquad \Delta_\epsilon = 1, \qquad \Delta_\chi = \frac{15}{8} - \delta, \qquad C_{\sigma\sigma\epsilon} = \frac{1}{2}, \qquad C_{\epsilon\epsilon\epsilon} = 0, \tag{F.1}$$

and unit-normalized plane two-point functions, $\langle\sigma(x)\sigma(0)\rangle = |x|^{-1/4}$, $\langle\epsilon(x)\epsilon(0)\rangle = |x|^{-2}$ and $\langle\chi(x)\chi(0)\rangle = |x|^{-2\Delta_\chi}$. The higher correlators of χ are Wick-factorized. The perturbation is $\sigma\chi$, of dimension $p = 2 - \delta$. Critical thermal tuning is understood: the relevant energy coupling generated by the perturbation is removed along the critical trajectory. The limit is $\delta \to 0^+$, from the long-range side.

The cylinder Hamiltonian. On the radial cylinder of unit radius, with the unperturbed vacuum energy subtracted,

$$H_0 = D_{\text{Ising}} + D_\chi, \qquad V = \int_0^{2\pi} \mathrm{d}\theta\, \sigma(\theta)\chi(\theta), \qquad V_\epsilon = \int_0^{2\pi} \mathrm{d}\theta\, \epsilon(\theta), \tag{F.2}$$

where D_{Ising} and D_χ are the dilatation operators and V_ϵ is the integrated thermal operator (the base study writes $\mathcal{E}$ for it). The angular integral projects onto equal total spin and contributes a factor 2π at every vertex, so a chain of four vertices carries $(2\pi)^4$. These factors are part of the coupling convention and must not be changed independently of b_0.

Fixed point and coefficient extraction. The beta function is $\beta(g) = -\delta g + b_0 g^3 + O(g^5)$, with the exact b_0 of Sec. 3.2. This normalization agrees with the lower-order crossover calculation [12], and it is checked independently by the magnetic gap (Appendix F.12). The infrared fixed point has $g_*^2 = \delta/b_0 + O(\delta^2)$. Let $|d\rangle = L_{-1}|\mathcal{E}\rangle$ be the left-derivative state at the ultraviolet point. Since the energy field has holomorphic weight $\frac{1}{2}$, its norm is $2 \cdot \frac{1}{2} = 1$; its unperturbed dimension is two and its spin one. Following the energy eigenstates continuously connected to these states, define

$$X = 2(E_\mathcal{E} - E_0) - (E_d - E_0) = 2E_\mathcal{E} - E_d - E_0. \tag{F.3}$$

At a conformal fixed point $E_\mathcal{E} - E_0 = \Delta_\mathcal{E}$ and $E_d - E_0 = \Delta_\mathcal{E} + 1$, hence $X = \Delta_\mathcal{E} - 1$; at the ultraviolet point $X = 0$. In a consistently matched Hamiltonian scheme,

$$X = a_1\delta g^2 + h g^4 + O(g^6, \delta^2 g^2, \delta g^4), \tag{F.4}$$

and substituting g_* gives the relation $c_2 = a_1/b_0 + h_0/b_0^2$ of Sec. 3.8, with h_0 the value of h once the regulator is removed (Eq. (F.5) below). Only the leading fixed-point coupling is needed at this order, because there is no term $\delta^0 g^2$. The split between the two terms depends on the prescription (Sec. 3.2): the Hamiltonian h is not the pole-only coefficient h_{MS} of minimal subtraction, and dropping a_1/b_0 would be wrong in this convention. A shift Δh_0 moves c_2 by $\Delta h_0/b_0^2 \simeq 0.62156\,\Delta h_0$ ($b_0^2 \simeq 1.60885$); this is the conversion by which errors in h_0 are quoted in c_2.

Why the coefficients are computed at $\delta = 0$. Inferring a curvature from dimensions computed at finite δ would magnify an error in a dimension by δ^{-2}. The method instead computes the g^4 coefficient at $\delta = 0$ and the analytic δ derivative of the g^2 coefficient. The remaining fits are numerical fits in auxiliary cutoffs and in the regulator length, never in the physical δ. What the calculation shares with the closed-form route, and what it does not use, is listed in Sec. 3.8.

Notation and the three separate limits.

symbol	meaning
δ	physical distance from the short-range end; the fourth-order sums are evaluated at $\delta = 0$
$\Delta_{\chi,0} = \frac{15}{8}$	endpoint dimension of χ (not an oscillator index)
$r = 1, 2, 3$	order of the positive, polynomial-improved oscillator regulator (Appendix F.6)
$\lambda > 0$	length of that smooth ultraviolet regulator on the unit cylinder
N	cubical coefficient cutoff, $0 \le n, m, \ell < N$; it is not the number of Hilbert-space states
N_{mix}	two-index cutoff of the mixed thermal sums (4096 in the base study)
N_{col}	cutoff of the collapsed sums (200 000 and 800 000 in the base study)

A third auxiliary cutoff, besides N_{mix} and N_{col}, is the two-index cutoff of the leading-tail calculation (Appendix F.18). The auxiliary cutoffs are not interchangeable with N. The numerical limits are taken in the order

$$h_0 = \lim_{\lambda \to 0^+} \Big[\lim_{N \to \infty} h_N(\lambda) \Big], \tag{F.5}$$

with N_{mix}, N_{col} and the tail cutoff removed together with N at fixed λ (in the analysis of Sec. 3.8 they are held at common values for all points and their shifts under a change are propagated); $h_N(\lambda)$ is the assembled fourth-order quantity of Eq. (F.42) with the chain sums truncated at N. None of these limits samples the physical δ.

F.2 Exact second-order spectral continuation

Vacuum and thermal sums. Set $p = 2 - \delta$ and

$$A_n = \frac{(p)_n}{n!}, \qquad A_{-1} = A_{-2} = 0. \tag{F.6}$$

Expanding the exact cylinder correlators and projecting on angles gives

$$E_0^{(2)} = -4\pi^2 \sum_{n=0}^{\infty} \frac{A_n^2}{p + 2n}, \tag{F.7}$$

$$E_{\mathcal{E}}^{(2)} = -\pi^2 \sum_{n=0}^{\infty} \frac{(A_n + A_{n-1})^2}{p - 1 + 2n}. \tag{F.8}$$

The denominators are measured relative to the external state. The sums converge in an ordinary domain, for example $\frac{1}{2} < p < 1$ away from level crossings, and are then continued meromorphically; they are not convergent positive sums at the physical endpoint. With $\tilde{A}_n = (p-1)_n/n!$ (the base study's c_n),

$$A_n + A_{n-1} = \frac{2n + p - 1}{p - 1} \tilde{A}_n, \tag{F.9}$$

so Eq. (F.8) reduces to a linear combination of two Gauss sums,

$$\sum_{n \ge 0} \tilde{A}_n^2 = \frac{\Gamma(3 - 2p)}{\Gamma(2 - p)^2}, \qquad \sum_{n \ge 0} n \tilde{A}_n^2 = (p-1)^2 \frac{\Gamma(2 - 2p)}{\Gamma(2 - p)^2}. \tag{F.10}$$

With $\Gamma(3 - 2p) = (2 - 2p)\Gamma(2 - 2p)$ the required combination vanishes exactly:

$$E_{\mathcal{E}}^{(2)} = 0 \qquad \text{as a meromorphically continued function.} \tag{F.11}$$

The vacuum sum is a well-poised hypergeometric sum. Written as $p^{-1}\, {}_3F_2(p, p, \frac{p}{2}; 1, 1 + \frac{p}{2}; 1)$, it reduces to the standard well-poised summation, followed by Gamma reflection and recursion:

$$-E_0^{(2)} = \pi^2 \frac{\Gamma(p/2)^2\, \Gamma(1 - p)}{\Gamma(1 - p/2)^2\, \Gamma(p)}. \tag{F.12}$$

This fixes the analytic continuation used here; it does not assign a finite part to an endpoint cutoff sum.

The derivative state and the slope of the observable. Define

$$D_n = A_{n+1} + 3A_n + 3A_{n-1} + A_{n-2}. \tag{F.13}$$

The absolute second-order correction of the derivative state is

$$E_d^{(2)} = -\frac{\pi^2}{4(p-1)} \sum_{n\geq 0} \tilde{A}_n D_n. \tag{F.14}$$

For $\Sigma_j = \sum_{n\geq 0} \tilde{A}_n A_{n+j}$ and $\mathcal{G} = \Gamma(2-2p)/\Gamma(2-p)^2$ the continued shifted sums are

$$\frac{\Sigma_1}{\mathcal{G}} = \frac{p(1-p)}{2-p}, \qquad \frac{\Sigma_0}{\mathcal{G}} = 1-p, \qquad \frac{\Sigma_{-1}}{\mathcal{G}} = p-1, \qquad \frac{\Sigma_{-2}}{\mathcal{G}} = \frac{p(p-1)}{2-p}. \tag{F.15}$$

Consequently $\Sigma_1 + 3\Sigma_0 + 3\Sigma_{-1} + \Sigma_{-2} = 0$, and $E_d^{(2)} = 0$ as well. The coefficient of g^2 in X is therefore $-E_0^{(2)}$ at $p = 2-\delta$:

$$a(\delta) = \pi^2 \frac{\Gamma(1-\delta/2)^2\,\Gamma(\delta-1)}{\Gamma(\delta/2)^2\,\Gamma(2-\delta)} = -\frac{\pi^2}{4}\delta - \frac{\pi^2}{2}\delta^2 + O(\delta^3), \tag{F.16}$$

so that

$$a_1 = -\frac{\pi^2}{4}, \qquad \frac{a_1}{b_0} = -1.945279657979090345 63\ldots, \tag{F.17}$$

which is the finite matching term of Eq. (3.7). A complementary check (`code/c2_cylinder_d2/code/hamiltonian/slope_check.py`) differentiates the regulated Pochhammer coefficients, the regulator factors and the resolvent denominators analytically in δ; it does not finite-difference dimensions at several values of δ.

F.3 A finite endpoint-regulator anomaly

Setting $p = 2$ first and multiplying each second-order resolvent term by $e^{-\tau(E_n - E_a)}$, where E_n is the energy of the intermediate state, E_a that of the external state and $\tau > 0$ a Euclidean time cutoff, direct geometric summation gives

$$E_0^{(2)}(\tau) = -\frac{\pi^2}{2\sinh^2\tau}, \qquad E_{\mathcal{E}}^{(2)}(\tau) = -\frac{\pi^2\cosh\tau}{2\sinh^2\tau}, \tag{F.18}$$

hence

$$E_{\mathcal{E}}^{(2)}(\tau) - E_0^{(2)}(\tau) = -\frac{\pi^2}{4\cosh^2(\tau/2)} \xrightarrow{\tau\to 0} -\frac{\pi^2}{4}. \tag{F.19}$$

The analytically continued endpoint gap, however, is $a(0) = 0$. Even a finite difference of cutoff sums can therefore have the wrong finite part. This is the failure of an attempted endpoint prescription, not an error in the closed-form study; it motivates the explicitly matched Hermitian regulator of Appendix F.6. General fourth-order difficulties of Hamiltonian cutoffs, including nonlocal cutoff effects and disconnected cancellations, are discussed in Refs. [35, 36]. An earlier explicit-basis calculation that fixed the counterterms by trial matching to finite spectral gaps gave strongly cutoff-dependent fourth-order values; it is not used.

F.4 Fourth-order Rayleigh–Schrödinger perturbation theory

Reduced resolvents and the normalization subtraction. For an external state $|a\rangle$, work in its conserved-spin sector and use

$$Q_a = 1 - |a\rangle\langle a|, \qquad R_a = \frac{Q_a}{E_a^{(0)} - H_0}, \qquad |w_a\rangle = R_a V|a\rangle, \qquad |z_a\rangle = Q_a V|w_a\rangle. \tag{F.20}$$

The relevant external levels are nondegenerate within their symmetry sectors at the endpoint. Since $V_{aa} = 0$, the second- and fourth-order energy coefficients are

$$E_a^{(2)} = \langle a|VR_aV|a\rangle, \tag{F.21}$$

$$E_a^{(4)} = \langle z_a|R_a|z_a\rangle - E_a^{(2)}\langle w_a|w_a\rangle. \tag{F.22}$$

The first term, $\langle a|VR_aVR_aVR_aV|a\rangle$, is the chain. The second comes from the perturbative normalization of the state and is essential. It must use the same regulator as the chain; assigning unrelated finite parts to the two separately can destroy their cancellation.

Because V is linear in χ, four insertions that begin and end in the χ vacuum pass through the χ occupations $0 \to 1 \to 0 \to 1 \to 0$ or $0 \to 1 \to 2 \to 1 \to 0$. The explicit-basis implementation includes the factors $\sqrt{2}$ for double occupation of a mode; the generating-function implementation (Appendix F.8) includes the same multiplicities through the three Wick pairings. Finite-box tests compare the two (Appendix F.11).

Counterterms enter at the same perturbative order. Write the matched Hamiltonian as

$$H = H_0 + gV + g^2W_2 + g^3W_3 + g^4W_4 + \cdots . \tag{F.23}$$

For any insertion W of order g^2 define its mixed and quadratic fourth-order contributions

$$\begin{aligned} L_a(W) &= \langle w_a|(W - W_{aa})|w_a\rangle + 2\,\mathrm{Re}\,\langle Q_aWa|R_a|z_a\rangle, \\ \mathcal{Q}_a(W) &= \langle Q_aWa|R_a|Q_aWa\rangle, \end{aligned} \tag{F.24}$$

where $|Q_aWa\rangle$ means $Q_aW|a\rangle$ (not an independent state). For every state-dependent quantity Y write $Y_X = 2Y_{\mathcal{E}} - Y_d - Y_0$. For $W_2 = tV_\epsilon + kD_{\mathrm{Ising}}$ the quadratic contribution of the thermal term to X is exactly

$$\mathcal{Q}_X(tV_\epsilon) = \frac{32\pi^2}{3}\,t^2 : \tag{F.25}$$

a free Majorana thermal deformation gives the second-order gap coefficients $8\pi^2$ and $16\pi^2/3$ for $|\mathcal{E}\rangle$ and $|d\rangle$, and $2 \cdot 8\pi^2 - 16\pi^2/3 = 32\pi^2/3$. Quadratic terms involving D_{Ising} vanish because it is diagonal; its *mixed* fourth-order contribution $L_X(D_{\mathrm{Ising}})$ does not. The observable X was chosen so that the direct local fourth-order terms cancel (Appendix F.7); the mixed terms of Eq. (F.24) cannot be omitted on that basis.

F.5 Hilbert space, matrix elements and oscillators

Ising states. The validation implementation uses an orthonormal Majorana basis of Neveu–Schwarz (NS) and Ramond (R) states. In the NS sector the occupied chiral modes are positive half-integers, in the R sector positive integers. Chiral R energies carry the vacuum shift $\frac{1}{16}$, so the full R shift is $\frac{1}{8}$, and the diagonal Ising theory retains NS left and right occupations of equal parity. The vacuum, energy, derivative and spin states are

$$|0\rangle = (\varnothing,\varnothing)_{\mathrm{NS}}, \qquad |\mathcal{E}\rangle = (\tfrac{1}{2},\tfrac{1}{2})_{\mathrm{NS}}, \qquad |d\rangle = (\tfrac{3}{2},\tfrac{1}{2})_{\mathrm{NS}}, \qquad |\sigma\rangle = (\varnothing,\varnothing)_{\mathrm{R}}. \tag{F.26}$$

Products of chiral spin-field form factors give the full Ising σ matrix elements, and the angular integration then enforces conservation of total spin. Appendix F.15 gives the product formula and the fermion signs; the representation is also available in Ref. [37].

Generalized-free oscillators. For $\tau > 0$ the unit-cylinder correlator is

$$\langle \chi(\tau,\theta)\chi(0,0)\rangle = \big[2\cosh\tau - 2\cos\theta\big]^{-\Delta_\chi}. \tag{F.27}$$

Expanding its two chiral factors gives modes $m, n \geq 0$ with

$$\omega_{mn} = \Delta_\chi + m + n, \qquad J_{mn} = m - n, \qquad \gamma_{mn} = \sqrt{\frac{(\Delta_\chi)_m (\Delta_\chi)_n}{m!\, n!}}, \tag{F.28}$$

energy ω_{mn}, spin J_{mn} and oscillator coefficient γ_{mn} (the base study's c_{mn}), so that $D_\chi = \sum_{m,n} \omega_{mn} a^\dagger_{mn} a_{mn}$. These modes implement the radial Hamiltonian of the isotropic theory; no identification with a spatially long-range quantum spin chain is assumed.

F.6 The regulator family

At the endpoint set $\Delta_{\chi,0} = \frac{15}{8}$ and define

$$P_r(x) = \sum_{j=0}^{r} \frac{x^j}{j!}, \qquad q = e^{-\lambda} \tag{F.29}$$

(q is unrelated to the circular mean of Eq. (3.5)). The oscillator coefficient is replaced by

$$\tilde\gamma_{mn} = q^{(\Delta_{\chi,0}+m+n)/2} \sqrt{\frac{(\Delta_{\chi,0})_m (\Delta_{\chi,0})_n}{m!\, n!}\, P_r\big(\lambda(m + \tfrac{\Delta_{\chi,0}}{2})\big)\, P_r\big(\lambda(n + \tfrac{\Delta_{\chi,0}}{2})\big)}. \tag{F.30}$$

All factors under the square root are positive. Applying the same real coefficient to the creation and annihilation parts gives a Hermitian interaction at fixed $\lambda > 0$. The un-improved regulator is $r = 0$; the analysis uses $r = 1, 2, 3$. Excluding a common factor $q^{\Delta_{\chi,0}/2}$, the chiral contraction series is

$$H(z) = \sum_{n\geq 0} H_n z^n, \qquad H_n = \frac{(\Delta_{\chi,0})_n}{n!}\, q^n\, P_r\big(\lambda(n + \tfrac{\Delta_{\chi,0}}{2})\big). \tag{F.31}$$

Since $e^{-x} P_r(x) = 1 + O(x^{r+1})$, low-frequency regulator artifacts are suppressed. This improvement does not remove the need for local counterterms. At fixed λ the χ modes are exponentially damped and the remaining Ising collision singularities of the sums are integrable. The cubical spectral cutoff N is therefore a numerical truncation of a regulated theory, not the physical ultraviolet prescription.

F.7 Local counterterms and the improved observable

Thermal and spin-two terms from the short-distance kernel. The Ising operator product expansion contains

$$\sigma(z)\sigma(0) = |z|^{-1/4}\Big[1 + \frac{1}{2}|z|\,\epsilon(0) + \frac{1}{4} z^2 T(0) + \frac{1}{4}\bar z^2 \bar T(0) + \cdots\Big], \tag{F.32}$$

with T, $\bar T$ the stress tensor. The ϵ term gives a thermal power subtraction. The regulator distinguishes cylinder time from angle at short distances, so the spin-two terms generate a finite anisotropic subtraction. With the normalization of Eq. (F.2) the counterterm is

$$W_2 = t_r(\lambda)\, V_\epsilon - \kappa_r\, D_{\mathrm{Ising}}. \tag{F.33}$$

It is derived from the regulated kernel, not by setting the fourth-order answer or by fitting an energy gap to a desired value. For a local coordinate $z = \tau + \mathrm{i}x$ and unit regulator length define

$$\mathfrak{h}_{\alpha,r}(z) = \sum_{j=0}^{r} \frac{(\alpha)_j}{j!} (z+1)^{-\alpha-j}. \tag{F.34}$$

The planar regulated χ kernel is $|\mathfrak{h}_{\Delta_{\chi,0},r}(z)|^2$. Its thermal moment is

$$c_{t,r} = \frac{1}{2}\int_{\tau>0} \mathrm{d}\tau\,\mathrm{d}x\,(\tau^2+x^2)^{3/8}\,|\mathfrak{h}_{\Delta_{\chi,0},r}(\tau+\mathrm{i}x)|^2, \qquad t_r(\lambda) = \frac{c_{t,r}}{\lambda} + \cdots . \tag{F.35}$$

For the un-improved regulator

$$\begin{aligned} c_{t,0} &= {}_3F_2\left(\begin{matrix}-\frac{3}{8},1,1\\ \frac{3}{2},\frac{3}{2}\end{matrix};1\right) = 0.76212879108984270790\ldots, \\ \kappa_0 &= \frac{\pi^2}{4}\,\frac{\Delta_{\chi,0}-\rho_\chi}{\Delta_{\chi,0}-1}, \qquad \rho_\chi = \frac{\Gamma(\Delta_{\chi,0}+\frac{1}{2})}{\Gamma(\frac{3}{2})\Gamma(\Delta_{\chi,0})}, \qquad \kappa_0 = 1.20829489911072968942\ldots. \end{aligned} \tag{F.36}$$

For $r>0$ the difference of the spin-two moments of the improved and un-improved kernels converges and determines $\kappa_r - \kappa_0$. Appendix F.16 gives the compact finite-domain integrals used for these constants, so that they can be recomputed without any saved number.

Removing a leading cylinder artifact. The holomorphic cylinder kernel obeys

$$\left[2\sinh(z/2)\right]^{-\Delta_{\chi,0}} = z^{-\Delta_{\chi,0}} - \frac{\Delta_{\chi,0}}{24} z^{2-\Delta_{\chi,0}} + \cdots . \tag{F.37}$$

Applying the same regulator to this expansion produces a thermal term of order λ. For $r \geq 1$ the subtracted moment converges:

$$c_{\mathrm{cyl},r} = -\frac{\Delta_{\chi,0}}{24}\int_{\tau>0} \mathrm{d}^2z\,|z|^{3/4}\Big[\mathrm{Re}\big(\mathfrak{h}_{\Delta_{\chi,0}-2,r}(z)\,\overline{\mathfrak{h}_{\Delta_{\chi,0},r}(z)}\big) - \mathrm{Re}(z^2)\,|z|^{-2\Delta_{\chi,0}}\Big]. \tag{F.38}$$

The calculation therefore uses

$$\boxed{t_r(\lambda) = \frac{c_{t,r}}{\lambda} + c_{\mathrm{cyl},r}\,\lambda.} \tag{F.39}$$

The second term vanishes in the continuum. It is an explicitly derived regulator improvement, not an adjustable finite fourth-order constant, and it is checked separately by the protected magnetic coefficient (Appendix F.12). The constants are

r	$c_{t,r}$	κ_r	$c_{\mathrm{cyl},r}$
1	1.96478977995214	1.78455951314437	−0.11068091195498
2	3.31217153117521	2.20281564903759	−0.05396644709415
3	4.74456159798846	2.54264875945082	−0.03541592348370

They are ordinary quadrature values, refined as described in Appendix F.16, not interval enclosures.

Why the direct fourth-order local terms cancel in X. After the auxiliary spectral cutoff has been removed at fixed λ, short-distance power counting and the separate Ising and χ parity selection rules restrict the nonvanishing relevant or marginal local counterterms of even order to the identity, the thermal operator and the parity-even Ising spin-two term. Odd-spin total derivatives do not contribute. In particular χ^2 has dimension $\frac{15}{4} > 2$ at the endpoint and is not an additional relevant or marginal subtraction. The direct fourth-order expectations of these operators in X vanish:

operator in W_4	$2(W_4)_{\mathcal{E}\mathcal{E}} - (W_4)_{dd} - (W_4)_{00}$
$\nVdash$	$2-1-1=0$
V_ϵ	0 (each diagonal element vanishes)
D_{Ising}	$2\cdot 1 - 2 - 0 = 0$

(F.40)

The diagonal thermal elements vanish by Ising fusion ($C_{\epsilon\epsilon\epsilon}=0$) and remain zero for the descendant. This cancellation does not remove the mixed effects of W_2, which are included explicitly.

At third order the coupling counterterm $W_3 = \mu_3(\lambda)V$ contributes $2\,\mathrm{Re}\,\langle a|W_3 R_a V|a\rangle = 2\mu_3(\lambda)E_a^{(2)}$, and thus $2\mu_3(\lambda)X_2(\lambda)$ to X, where $X_2(\lambda)$ is the second-order observable at finite λ and $\delta = 0$ (Appendix F.9). The coefficient μ_3 grows logarithmically, whereas $X_2(\lambda)$ tends to zero as a positive power of λ, so this product supplies no unknown finite constant to the limiting h. The analysis uses the convenient choice

$$\mu_3(\lambda) = -b_0 \log\lambda, \qquad \Delta X_4 = -2b_0 \log\lambda\, X_2(\lambda). \tag{F.41}$$

Omitting this term, or shifting the reference scale of the logarithm, is a sensitivity test at finite regulator.

This is an argument about the local short-distance expansion of a specified, smoothly regulated theory, made *after* the numerical spectral limit. It does not license arbitrary finite counterterms at a fixed hard Hilbert-space cutoff, and the numerical calculation does not supply a uniform treatment of every subleading remainder.

The assembled fourth-order quantity. Combining all terms,

$$\begin{aligned} h(\lambda) = X_{4,\mathrm{chain}} + X_{4,\mathrm{norm}} + t_r(\lambda)\, L_X(V_\epsilon) - \kappa_r\, L_X(D_{\mathrm{Ising}}) \\ + \frac{32\pi^2}{3} t_r(\lambda)^2 - 2b_0 \log\lambda\, X_2(\lambda), \end{aligned} \tag{F.42}$$

where every sum on the right must first be converged at fixed λ, and $h_0 = \lim_{\lambda\to 0} h(\lambda)$. Here $X_{4,\mathrm{chain}}$ is the chain term of Eq. (F.22) combined into X (Appendix F.8), $X_{4,\mathrm{norm}}$ its normalization term and $L_X(D_{\mathrm{Ising}})$ the mixed spin-two term (both Appendix F.9), and $L_X(V_\epsilon)$ the mixed thermal term (Appendix F.17). Large terms cancel in Eq. (F.42), and its individual pieces are not physical coefficients; this is one reason the collapsed and mixed sums need their own convergence checks. The continuum analysis of Sec. 3.8 evaluates this assembly at each (r, λ).

F.8 Exact chiral blocks as spectral generating functions

Replacing state enumeration. The explicit Majorana basis serves for validation but grows rapidly. The production implementation sums the level degeneracies analytically with chiral blocks. A Taylor coefficient of a radially ordered four-point block is the sum, over intermediate states at specified levels, of products of spin-field matrix elements. Dividing these coefficients by the corresponding three energy denominators therefore evaluates the same Rayleigh–Schrödinger sum as Eq. (F.22). This is not a return to the elliptic-integral calculation of the closed-form study: the blocks supply discrete spectral weights, and the operations that follow are coefficient extraction, angular projection and Hamiltonian denominators. The underlying Ising correlators are known independently [13].

Vacuum, thermal and derivative blocks. Place four chiral spin insertions at

$$z_1 = 1, \qquad z_2 = u, \qquad z_3 = uv, \qquad z_4 = uvw, \tag{F.43}$$

and define, for $|u|, |v|, |w| < 1$,

$$\begin{gathered} S = \sqrt{1-uv}\sqrt{1-vw}, \qquad T = \sqrt{1-v}\sqrt{1-uvw}, \qquad \mathcal{D} = \prod_{x\in\{u,v,w,uv,vw,uvw\}} (1-x)^{-1/8}, \\ \mathcal{R} = \sqrt{(S+T)/2}, \qquad \mathcal{U} = \sqrt{(1-u)(1-w)} \end{gathered} \tag{F.44}$$

(the base study writes $\mathcal{Q}$ for $\mathcal{U}$). All roots are continued analytically from the origin, with positive values for small positive real arguments. After removal of simple radial monomials the two

middle-fusion channels ($\eta = 0, 1$) of the vacuum and thermal blocks are

$$\begin{aligned} g_0 &= \mathcal{D}\mathcal{R}, & g_1 &= \frac{\mathcal{D}\mathcal{U}}{2\mathcal{R}}, \\ e_0 &= \frac{\mathcal{D}}{4\mathcal{R}}\big[S(1+uw)+T(u+w)\big], & e_1 &= \frac{\mathcal{D}\mathcal{R}}{2\mathcal{U}}\big[S(1+uw)-T(u+w)\big]. \end{aligned} \tag{F.45}$$

For a partition (A, B) of the four position labels set

$$K_{AB} = uvw\Big[1 - \frac{1}{4}\Big(\sum_{i\in B} z_i - \sum_{i\in A} z_i\Big)\Big(\sum_{i\in B} z_i^{-1} - \sum_{i\in A} z_i^{-1}\Big)\Big], \tag{F.46}$$

and let $K_{13} = K_{(1,3),(2,4)}$ and $K_{14} = K_{(1,4),(2,3)}$. The derivative blocks are

$$d_0 = \frac{\mathcal{D}}{4\mathcal{R}}\big[S(1+uw)K_{13} + T(u+w)K_{14}\big], \qquad d_1 = \frac{\mathcal{D}\mathcal{R}}{2\mathcal{U}}\big[S(1+uw)K_{13} - T(u+w)K_{14}\big]. \tag{F.47}$$

The apparent inverse powers in K_{13} and K_{14} cancel against the displayed uvw and the block factors, so the series are analytic at the origin. Their analytic branches are tested at complex arguments against explicit NS and R form factors (Appendix F.11).

Wick pairings, angular projection and denominators. For each fusion label η, the block is multiplied by the chiral χ factor of one of the three Wick pairings, with the series H of Eq. (F.31):

pairing	chiral factor	middle occupation o
adjacent	$H(u)H(w)$	0
crossed	$H(uv)H(vw)$	1
nested	$H(uvw)H(v)$	1

Here $o = 1$ means a middle state with two χ quanta and $o = 0$ one with none. The real Taylor coefficients of the products are written $G_{nm\ell}$, $E_{nm\ell}$ and $D_{nm\ell}$ for the vacuum, thermal and derivative blocks. The angular projection gives squares for the vacuum and thermal states; for the spin-one derivative state it gives the shifted product $D_{n+1,m+1,\ell+1}E_{nm\ell}$. For a given pairing and fusion label the chain energies are

$$\begin{aligned} E^{(4),\mathrm{chain}}_{0,\eta,o} &= -(2\pi)^4 q^{2\Delta_{\chi,0}} {\sum_{n,m,\ell\geq 0}}' \frac{G^2_{nm\ell}}{(2n+2)(2m+\eta+2\Delta_{\chi,0}o)(2\ell+2)}, \\ E^{(4),\mathrm{chain}}_{\mathcal{E},\eta,o} &= -(2\pi)^4 q^{2\Delta_{\chi,0}} {\sum_{n,m,\ell\geq 0}}' \frac{E^2_{nm\ell}}{(2n+1)(2m-\eta+2\Delta_{\chi,0}o)(2\ell+1)}, \\ E^{(4),\mathrm{chain}}_{d,\eta,o} &= -(2\pi)^4 q^{2\Delta_{\chi,0}} {\sum_{n,m,\ell\geq 0}}' \frac{D_{n+1,m+1,\ell+1}E_{nm\ell}}{(2n+1)(2m-\eta+2\Delta_{\chi,0}o)(2\ell+1)}. \end{aligned} \tag{F.48}$$

The prime excludes only a zero denominator, which corresponds to the external-state projector. A negative denominator, such as that of the lower vacuum intermediate state in a thermal channel, must be retained. Summing over both fusion labels and all three pairings and forming $2E_{\mathcal{E}} - E_d - E_0$ gives $X_{4,\mathrm{chain}}$. The common negative sign reflects three resolvents written with positive excitation-energy denominators. At cutoff N the indices run from 0 to $N-1$; because the derivative block is shifted by one, the coefficient array has shape $(6, N+1, N+1, N+1)$, with the six channels ordered $(g_0, g_1, e_0, e_1, d_0, d_1)$.

Coefficient extraction and convolution. The coefficients of the unregulated Ising blocks are extracted by a discrete Cauchy (Fourier) transform on a three-torus. With M angular points per direction and radius $\varrho = e^{-\varsigma/(N+1)}$,

$$[f]_{nm\ell} \simeq \frac{\varrho^{-n-m-\ell}}{M^3} \sum_{j_1,j_2,j_3=0}^{M-1} f\big(\varrho e^{2\pi \mathrm{i} j_1/M}, \varrho e^{2\pi \mathrm{i} j_2/M}, \varrho e^{2\pi \mathrm{i} j_3/M}\big)\, e^{-2\pi \mathrm{i}(j_1 n + j_2 m + j_3 \ell)/M}. \tag{F.49}$$

M is an even FFT length of about the oversampling factor times $N+1$, and ς is the radius cost. Oversampling controls aliasing; decreasing the radius reduces aliasing but amplifies roundoff through $\varrho^{-n-m-\ell}$. The main grids of the base study use oversampling five and radius cost four at $N=256$, and oversampling four and radius cost 4.5 at $N=384$. These are numerical choices, not physical regulators; the extraction check of Sec. 3.8 compares two grids that differ in the radius. The χ factors are then applied by one-dimensional convolutions along lines parallel to the monomials u, w, uv, vw, uvw or v, and prefix sums of the weighted three-index arrays return every cubical partial sum at once. The prefix test of Sec. 3.8, which repeats the large-N extrapolation on the sums truncated at $N' < N$, uses these partial sums. Independent versions of the convolution and of the coefficient grid are kept for validation.

F.9 Collapsed and mixed contributions

One-dimensional two-point contractions. The second-order energies, the norms of the normalization subtraction and the mixed spin-two term reduce to one-index sums. Let

$$p_0(z)=1, \qquad p_{\mathcal{E}}(z)=\frac{1+z}{2}, \qquad p_d(z)=\frac{(1+z)^3}{8}, \tag{F.50}$$

and $a_f(j)=[z^j](1-z)^{-1/8}p_f(z)$ for $f\in\{0,\mathcal{E},d\}$, with coefficients of negative index zero. With the regulated sequence H_j of Eq. (F.31), set

$$B_f(n)=\sum_{j=0}^{n} a_f(j)H_{n-j}, \qquad \hat{B}_f(n)=\sum_{j=0}^{n} j\,a_f(j)H_{n-j} \tag{F.51}$$

(the base study writes W_f for $\hat{B}_f$). The external states are labelled by

state a	left block l	right block l'	shift s_a	α_a
0	0	0	0	2
$\mathcal{E}$	$\mathcal{E}$	$\mathcal{E}$	0	1
d	d	$\mathcal{E}$	1	1

With $F_2=(2\pi)^2 q^{\Delta_{\chi,0}}$ and $\nu_n=2n+\alpha_a$,

$$E_a^{(2)}=-F_2\sum_{n\geq 0}\frac{B_l(n+s_a)B_{l'}(n)}{\nu_n}, \qquad \mathcal{N}_a:=\langle w_a|w_a\rangle=F_2\sum_{n\geq 0}\frac{B_l(n+s_a)B_{l'}(n)}{\nu_n^2},$$
$$L_a(D_{\mathrm{Ising}})=F_2\sum_{n\geq 0}\frac{1}{\nu_n^2}\Big[(\tfrac{1}{8}-E_a^{(0)})B_l(n+s_a)B_{l'}(n)+\hat{B}_l(n+s_a)B_{l'}(n)+B_l(n+s_a)\hat{B}_{l'}(n)\Big]. \tag{F.52}$$

The finite-regulator second-order observable is $X_2(\lambda)=2E_{\mathcal{E}}^{(2)}-E_d^{(2)}-E_0^{(2)}$ evaluated with these sums, and the normalization contribution is

$$X_{4,\mathrm{norm}}=-2E_{\mathcal{E}}^{(2)}\mathcal{N}_{\mathcal{E}}+E_d^{(2)}\mathcal{N}_d+E_0^{(2)}\mathcal{N}_0. \tag{F.53}$$

These sums are evaluated at much larger cutoffs than the three-index chain: $N_{\mathrm{col}}=200\,000$ in the base study's $N=256$ records and 800 000 in its larger-cutoff records, with the small change between them recorded separately.

Two-dimensional thermal insertions. The mixed thermal term $L_X(V_\epsilon)$ has the insertion V_ϵ either between the two V vertices (central placement) or outside them (side placement). The basic chiral generating functions are

$$\begin{aligned} G_c(u,v)&=(1-u)^{-1/2}(1-v)^{-1/2}(1-uv)^{3/8}H(uv),\\ G_s(u,v)&=(1-u)^{-1/2}(1-v)^{3/8}H(v)(1-uv)^{-1/2}. \end{aligned} \tag{F.54}$$

For either placement the vacuum, energy and derivative factors are $G/\sqrt{2}$, $GP_{\mathcal{E}}/(2\sqrt{2})$ and $GP_d/(2\sqrt{2})$; the polynomials $P_{\mathcal{E},b}$ and $P_{d,b}$, the denominators and the shifted derivative projection are in Appendix F.17. The common factor is $(2\pi)^3 q^{\Delta_{\chi,0}}$, and

$$L_X(V_\epsilon) = X_c + 2X_s, \tag{F.55}$$

where the factor two counts the two side placements. All square partial sums are obtained through N_{mix} (4096 in the base study), and the large-level limit is extrapolated separately (Appendix F.10).

F.10 High-state tails and the regulator limit

The independently determined leading tail. At fixed λ the leading remaining Ising singularity on a face of the coefficient box is $(1-u)^{-1/8}$. Its coefficients $a_0(n) = (\frac{1}{8})_n/n!$ (Appendix F.9) behave as $a_0(n) \sim n^{-7/8}/\Gamma(\frac{1}{8})$, so

$$\sum_{n\geq N} \frac{a_0(n)^2}{2n} \sim \frac{2}{7\Gamma(\frac{1}{8})^2} N^{-7/4}; \tag{F.56}$$

the factor $\frac{2}{7}$ follows by integrating $n^{-11/4}/2$. Exact limits of the generating functions on the faces of the coefficient box determine the coefficient of this tail, with the sign convention

$$X_{4,\mathrm{chain}}(\infty) - X_{4,\mathrm{chain}}(N) = A_{\mathrm{face}}(\lambda, r)\, N^{-7/4} + \text{subleading terms}. \tag{F.57}$$

Appendix F.18 gives the two-index construction of A_{face}. It uses the block singularities and the collapsed norms, not the observed N sequence or any value of c_2. This is the analytically derived coefficient of the face tail that the continuum analysis of Sec. 3.8 subtracts.

What is still fitted at fixed regulator. The subleading powers include the one-face corrections $N^{-11/4}, N^{-15/4}, \ldots$ and overlapping-face contributions such as $N^{-7/2}$. The regulated χ kernel has its next singularities at e^{λ}, which generate terms exponentially small in $N\lambda$; on finite windows they can compete with the power terms. A reasonably large λN is therefore desirable, but this is a planning criterion, not a remainder bound (the analysis of Sec. 3.8 flags points with λN below a minimum). Let $h_n(\lambda)$ be the assembled quantity, Eq. (F.42), with the chain truncated to the cube $[0,n)^3$. With the leading tail held fixed, the corrected sequence is fitted on $n_{\min} \leq n \leq N$ as

$$h_n(\lambda) + A_{\mathrm{face}}\, n^{-7/4} = h(\lambda) + \sum_j \xi_j \Big(\frac{n}{N}\Big)^{-p_j} + \text{exponential terms}, \tag{F.58}$$

with the power sets

$$\{2.75, 3.75\}, \quad \{2.75, 3.75, 4.75\}, \quad \{2.75, 3.5, 3.75, 4.75\}, \quad \{2.75, 3.5, 3.75, 4.75, 5.5\} \tag{F.59}$$

and, for exponential order $n_e = 0, 1, 2, 3$, the optional columns

$$e^{-\lambda(n-n_{\min})}\Big(\frac{n}{N}\Big)^{r-1-j}, \qquad e^{-2\lambda(n-n_{\min})}\Big(\frac{n}{N}\Big)^{2r+3/4-j}, \qquad j = 0, \ldots, n_e - 1. \tag{F.60}$$

The base study's selected ensemble uses at least three power terms (the analysis of Sec. 3.8: three or four), $n_e = 1, 2$, and the two lower endpoints $n_{\min} = 240, 288$ at $N = 384$ and $160, 192$ at $N = 256$, that is $n_{\min} = 5N/8$ and $3N/4$. The fixed-regulator estimate $h(\lambda)$ is the median of the selected limits, and half the difference between the largest and the smallest is the *spectral-model half-spread*, a model diagnostic, neither a standard deviation nor a truncation bound. Small residuals alone do not prove the tail model correct. The continuum analysis of Sec. 3.8 uses these power sets, exponential columns and selection, with the windows $n_{\min} = 5N/8$ and $3N/4$ at every N. That

these windows reproduce the base study's relative windows does not by itself show the same tail model valid at smaller λ; the prefix and cross-N checks of Sec. 3.8 test it.

The mixed thermal sums have their own tail fits in N_{mix}, with powers drawn from $\{1, 2, \frac{11}{4}, 3, 4\}$; the chosen mixed limit is the median of the models with at least three powers and $n_{\min} \geq N_{\mathrm{mix}}/2$ (2048 in the base study). Its spread is recorded separately and not identified with the error of the chain.

Removing the smooth regulator. After the spectral limit, the regulator model is $h(\lambda) = h_0 + k_1\lambda^2 + k_2\lambda^2 \log\lambda + k_3\lambda^3$, the preferred model of Sec. 3.8. The improved low-frequency filter and the removal of the leading cylinder thermal artifact motivate this form; higher terms are not rigorously bounded. The base study varied the regulator order, the λ window, the spectral-tail model, the weighting and the last power (λ^4 instead of λ^3), and it tested omission and scale shifts of the term (F.41). No fit has the physical δ as its variable, but the uncertainty remains an extrapolation uncertainty. A four-parameter model fitted to four regulator points is saturated, and a near-zero residual is then no evidence of accuracy; two of the base study's three regulator families had only four points, which is why more regulator points were needed. The base study's own extrapolations ($N \leq 384$), quoted in Sec. 3.8, were superseded by the analysis given there; its archived records were not: eleven of them are points of that analysis.

For a fixed linear extrapolation $\hat{h}_0 = \sum_i w_i \hat{h}(\lambda_i)$ with input diagnostics ε_i and a regulator-model term $\varepsilon_{\mathrm{model}}$, the error in c_2 is propagated as

$$\frac{1}{b_0^2}\Big(\sum_i |w_i|\varepsilon_i + \varepsilon_{\mathrm{model}}\Big). \tag{F.61}$$

This is the form of the linear sum of Sec. 3.8. It would be a bound only if the ε_i and $\varepsilon_{\mathrm{model}}$ were valid bounds, which the half-spreads and model spreads are not. It also shows why the fixed-λ inputs need better precision than the desired final error. As λ decreases, cancellations between large raw terms and the counterterms make roundoff more important.

F.11 Algebraic and finite-sum checks

The validation suite tests different failure modes:

check	largest reported discrepancy
chiral Virasoro Ward identities (252 identities)	4.22×10^{-16}
exact second-order shell weights (vacuum, energy and derivative correlators)	below 5.4×10^{-16} relative
explicit basis against blocks, fourth-order finite sums	below 2×10^{-13}
blocks at complex arguments against finite-level form factors	about 1.16×10^{-10} (scaled)
alternative finite convolution and sequence implementations	below 2×10^{-11}

The suite also checks the four-spin block sums and the fourth-order formula (F.22) against an independent finite-matrix recursion for the perturbed eigenvector. The discrepancy of the complex-argument blocks includes the truncation of the explicit form factors. The entries are precisions of cross-checks on finite problems. None of these identities bounds the omitted continuum tail; they establish that signs, mode occupations, derivative shifts and normalization subtractions agree between independent implementations on the tested finite problems.

F.12 The protected magnetic dimension as a matching check

At the fixed point the magnetic dimension is protected, $\Delta_\sigma = \frac{1}{8} + \delta$ in this convention [12]. With $g_*^2 = \delta/b_0$, the leading renormalized Hamiltonian gap coefficient must therefore satisfy

$$[g^2](E_\sigma - E_0) = b_0. \tag{F.62}$$

The independently derived local counterterms, including the cylinder improvement of Eq. (F.39), give at $r = 3$ and $\lambda = 1/32$ the value 1.268404004672631 quoted in Sec. 3.8, which differs from b_0 by -3.04×10^{-7}. This value was not used to tune the counterterms. It is a lower-order check of the normalization, not a fourth-order error bound.

F.13 What is exact and what is numerical

The second-order meromorphic continuation (Appendix F.2) and the finite-basis identities are exact analytic or algebraic results. The counterterm classification and the cancellation (F.40) remove the unspecified direct local fourth-order constants within the stated smoothly regulated prescription; this is an argument, not a proof. The fourth-order continuum number is a numerical inference, supported by independent regulator families, the analytically derived leading tail and the lower-order normalization checks; its remaining uncertainty is systematic, not statistical. What is not supplied is a rigorous uniform bound on every nested spectral and regulator remainder, or an independent many-digit enclosure of h_0.

F.14 Checks of the closed form made in the base study

The base study also consolidates a set of formula, elliptic and local checks of the closed-form study (Secs. 3.5 and 3.6), likewise written by GPT-6 astra. Their code is not in the release (Appendix H.5); the value of the closed expression is reproduced, not as an interval enclosure, by `c2_closed_d2`. None of these quantities is an input to the Hamiltonian sums. They certify the value of the closed expression (3.9) and test local steps of the route to Eq. (3.7); they do not prove the identification of the expression with the physical coefficient, which rests on the step of Sec. 3.6.

The certified expression. The expression is Eq. (3.9), with ${}_pF_q(a_1, \dots, a_p; b_1, \dots, b_q; z) = \sum_n \prod_i (a_i)_n / \prod_j (b_j)_n \, z^n/n!$. All three series have excess two and absolutely convergent summands of order n^{-3}. An integer-interval implementation that uses no Gamma or generalized-hypergeometric evaluator and imports no code from the closed-form study encloses the value, with the tail construction below at $n_0 = 4000$, in an interval of width about $2.69823801745 \times 10^{-53}$, and so proves the rounding

$$c_2^{\text{closed}} = -4.06627014813252155274002760857287648752990114118661 8 \tag{F.63}$$

to 51 decimal places. This is a computer-assisted proof about the specified expression. It is not a physical error interval for the Hamiltonian extrapolation, nor a proof of the regulator-identification step of the closed-form study.

Exact tail inequalities and elementary constant bounds. For the absolute value v_n of a term of one of the three series, the ratio is rational,

$$\rho_n = \frac{v_{n+1}}{v_n} = \frac{\prod_{j=1}^{4}(n + a_j)}{(n+1)\prod_{j=1}^{3}(n + b_j)}. \tag{F.64}$$

Rational functions $\phi(n) = \frac{n}{2} + \phi_0 + \phi_1/n + \cdots + \phi_{12}/n^{12}$ are constructed by exact cancellation of the leading powers in $\Xi(n) = \phi(n) - 1 - \rho_n \phi(n+1)$; changing the last coefficient by -1 and $+1$ gives

ϕ_- and ϕ_+. After a positive denominator is cleared and $n = 4000 + m$ substituted, nonnegative polynomial coefficients prove

$$\phi_-(n) - 1 - \rho_n\phi_-(n+1) \le 0, \qquad \phi_+(n) - 1 - \rho_n\phi_+(n+1) \ge 0 \qquad \text{for all } m \ge 0. \tag{F.65}$$

A second exact polynomial inequality bounds $v_n \le v_{n_0}(n_0/n)^2$ and controls the telescoping boundary at infinity. Hence

$$\phi_-(n_0)\, v_{n_0} \le \sum_{n=n_0}^{\infty} v_n \le \phi_+(n_0)\, v_{n_0}. \tag{F.66}$$

The terms of T after index zero are negative, so its enclosed tail is subtracted with the endpoints reversed. For π the implementation uses the alternating rational sums of Machin's identity $\pi = 16\arctan\frac{1}{5} - 4\arctan\frac{1}{239}$. For R_Γ^2 it uses $B(\frac14,\frac12)^2/\pi$ and the convergent split beta integral

$$B(\tfrac14,\tfrac12) = 2^{-1/4}\sum_{n\ge0}\frac{(\frac12)_n}{n!}\frac{2^{-n}}{n+\frac14} + 2^{-1/2}\sum_{n\ge0}\frac{(\frac34)_n}{n!}\frac{2^{-n}}{n+\frac12}. \tag{F.67}$$

Both positive term ratios are below $\frac12$, so after 420 terms each remainder lies between its first omitted term and twice that term. Integer root inequalities enclose the powers of two. All arithmetic uses outward-rounded integer endpoints at scale 10^{110}, and assertions check the exact sign certificates. The closed-form study's own tail check behind the envelope quoted in Sec. 3.5 (Appendix E.15) is of the same type, with $\phi = \frac{n}{2} + \text{const}$ and floating-point endpoints; the higher-order ϕ and the integer arithmetic are what make this enclosure a certified rounding.

Elliptic integration from the period integrands. This check starts from the period integrands, not from the final series. With $K(t) = \int_0^{\pi/2} \mathrm{d}\theta\,(1 - t\sin^2\theta)^{-1/2}$ (parameter convention) and $\mathbf{k}(t) = \big(K(t)^2,\, K(t)K(1-t),\, K(1-t)^2\big)^{\mathsf{T}}$, define

$$\mathbf{P}(a,b) = \mathrm{AC}\int_0^1 t^{a-1}(1-t)^{b-1}\mathbf{k}(t)\,\mathrm{d}t, \qquad \mathbf{p}(a,b) = \frac{4}{\pi^2}\mathbf{P}(a,b), \tag{F.68}$$

where negative exponents mean meromorphic continuation, not a divergent improper integral. Near zero

$$\mathbf{k}(t) \sim \Big(\frac{\pi^2}{4},\ \frac{\pi}{4}\log\frac{16}{t},\ \frac14\log^2\frac{16}{t}\Big)^{\mathsf{T}}. \tag{F.69}$$

The quadrature splits at $\frac12$, subtracts these endpoint terms, restores their elementary continued Mellin integrals and integrates the convergent remainder after the substitution $t = u^{12}/2$; Carlson's R_F supplies K independently of the Frobenius arrays of the closed-form study. One nonsymmetric assembly is

$$\mathbf{a} = 2\mathbf{p}(-\tfrac12,\tfrac74) + 5\mathbf{p}(\tfrac12,\tfrac74), \qquad \mathbf{b} = 5\mathbf{p}(-\tfrac14,\tfrac74) + 2\mathbf{p}(\tfrac34,\tfrac74), \qquad \mathsf{H} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 2 \\ 1 & 2 & -1 \end{pmatrix},$$
$$c_2 = -\frac{R_\Gamma^2}{9} + \frac{\pi R_\Gamma^4}{10368\sqrt{2}}\,\mathbf{a}^{\mathsf{T}}\mathsf{H}\,\mathbf{b}, \tag{F.70}$$

and a separate assembly uses symmetric moments. The observed differences from the enclosed value are

quadrature order	nonsymmetric periods	symmetric moments
96	1.74×10^{-19}	4.78×10^{-29}
192	5.24×10^{-23}	2.28×10^{-34}
384	1.51×10^{-26}	1.04×10^{-39}

These are observed agreements under refinement, not rigorous quadrature-error bounds. The closed-form study's own endpoint-subtracted quadrature with Carlson's integrals (Sec. 3.5, Appendix E.15) is a check of the same kind, implemented separately.

Complex-plane normalization. With the single-valued area pairing $\mathcal{A}(t) = 8\big[K(t)\overline{K(1-t)} + \overline{K(t)}K(1-t)\big]$ (the torus area of Eq. (3.5)) and the continued complex moment

$$\mathcal{M}(a,b) = \mathrm{AC}\int_{\mathbb{C}} \mathrm{d}^2t\,|t|^{2a-2}|1-t|^{2b-2}\mathcal{A}(t)^2, \tag{F.71}$$

the tested period relation is

$$\mathcal{M}(a,b) = 64\,\mathbf{P}(a,b)^{\mathsf{T}}\mathsf{L}(b)\,\mathbf{P}(2-a-b,b), \qquad \mathsf{L}(b) = \begin{pmatrix} 0 & 0 & \sin\pi b \\ 0 & 2\sin\pi b & -2\cos\pi b \\ \sin\pi b & -2\cos\pi b & -\sin\pi b \end{pmatrix}. \tag{F.72}$$

The ordinary convergent moment $\mathcal{M}(\frac{2}{3},\frac{2}{3})$ was evaluated directly on the plane and through the real-period pairing, with the values 24181.76549557689 and 24181.76549557647, a relative difference of 1.72×10^{-14}. Generic convergent moments also agreed. Symbolic checks gave zero residual for the endpoint differential identity, the cut-jump matrix, the reflection reduction and the final symmetric matrix reduction. These tests support the period identity and the normalization in the regimes checked; they do not supply a uniform proof for the physical collisions.

Local contacts. The finite localized-source term B_{31} of Sec. 3 was checked with an elementary-integral residual of about 3.17×10^{-71} at 70 working digits. Direct polar quadrature of the local branch contact, at the source parameter $M=3$, agreed with

$$A_{\mathrm{br}} = 4B_{31} - \frac{\pi^2}{4} = -3.8363599279686606009\ldots \tag{F.73}$$

to about 1.05×10^{-14}; other source parameters gave comparable local agreement. Fixed-shape neutral-charge checks supported the cancellation of the energy-primary term of the shrinking spin pair. In the minimal-subtraction assembly, Eq. (3.7), the local calculation supports the factor $3/4! = \frac{1}{8}$ and the two placements of the coupling counterterm. The two contributions to c_2 there are $-2.12099049015343\ldots$ and $-1.94527965797909\ldots$; the agreement of the second with a_1/b_0 of Appendix F.2 is a comparison made after both calculations, not an input to Eq. (F.42).

The global qualification remains separate. The closed-form study states that its audit does not reconstruct every sectorwise subtraction or prove a regulator-uniform remainder bound across nested collisions (Sec. 3.6). Agreement of local contact terms at fixed nondegenerate modulus does not by itself control the degeneration regions at 0, 1 and ∞, or a simultaneous shrinking of relative scales. The base study's own review of these checks (GPT-6 astra; not the review of Sec. 1.2) reached the same boundary: no error that changes c_2 was found in the tested expressions, but the global regulator comparison was not proved independently. It also noted a point of exposition: the statement that identity projections cancel should distinguish disconnected identity terms from an identity-channel power term with denominator $-1+2\delta$, which is retained and analytically continued; this is not a numerical error. The Hamiltonian result is a different physical consistency check on Eq. (3.7); it does not turn these local checks into a proof of the global comparison.

F.15 Explicit Majorana form factors

Let $\mathbf{m} = (m_1,\ldots,m_{n_{\mathrm{R}}})$ be increasing positive integer R occupations and $\mathbf{s} = (s_1,\ldots,s_{n_{\mathrm{NS}}})$ increasing positive half-integer NS occupations, and

$$f_{\mathrm{R}}(n) = \binom{2n}{n}4^{-n}, \qquad f_{\mathrm{NS}}(s) = \binom{2s-1}{s-\frac{1}{2}}4^{-(s-1/2)}. \tag{F.74}$$

In the real chiral gauge of the implementation (`code/c2_cylinder_d2/code/hamiltonian/ising_basis.py`) the chiral spin-field element between these states is

$$\mathcal{S}(\mathbf{m},\mathbf{s}) = (-1)^{(n_{\mathrm{NS}} \bmod 2)\, n_{\mathrm{R}}}\, 2^{-(n_{\mathrm{R}}+n_{\mathrm{NS}})/2} \prod_{i=1}^{n_{\mathrm{R}}} f_{\mathrm{R}}(m_i) \prod_{j=1}^{n_{\mathrm{NS}}} f_{\mathrm{NS}}(s_j) \times \prod_{i<j} \frac{m_i - m_j}{m_i + m_j} \prod_{i<j} \frac{s_i - s_j}{s_i + s_j} \prod_{i,j} \frac{m_i + s_j}{m_i - s_j}, \tag{F.75}$$

with empty products equal to one. The full spin-field element is the product of the left- and right-chiral factors. This representation, including its phase convention, is checked by chiral Virasoro Ward identities; it gives $\langle\sigma|\sigma|0\rangle = 1$ and $C_{\sigma\sigma\epsilon} = \frac{1}{2}$ in the chosen normalization. Related scaling-limit product formulas are in Ref. [37].

For a fermion mode that creates or removes a positive occupied mode m, the sign is $(-1)^{\#\{k<m\}}$, where $\#\{k < m\}$ counts the occupied modes below m. The diagonal element of the Ramond zero mode is $(-1)^{\#\text{occupied}}/\sqrt{2}$. Products of left and right fermion elements give the thermal field. Angular integration forces equal signed energy changes on the two sides, so the integrated thermal matrix connects pair creation and annihilation and the diagonal of the Ramond zero mode; it is implemented as a sparse matrix, whose angular factor 2π enters only when the Hamiltonian insertion is assembled. A basis state with left and right occupations $\boldsymbol{\ell}$, $\bar{\boldsymbol{\ell}}$ has energy and spin

$$E = \sum_i \ell_i + \sum_j \bar{\ell}_j + \begin{cases} 0 & \text{NS}, \\ \frac{1}{8} & \text{R}, \end{cases} \qquad J = \sum_i \ell_i - \sum_j \bar{\ell}_j, \tag{F.76}$$

and in NS only equal left and right parity is retained. The finite-basis checks built on these elements are those of Appendix F.11; they detect basis-sign and projection errors without assuming the continuum answer.

F.16 Compact quadratures for the local counterterms

The constants of Appendix F.7 are computed from the following finite-domain integrals (`code/c2_cylinder_d2/code/hamiltonian/improved_regulator.py`). Let

$$\mathcal{P}_{\alpha,r}(z) = \sum_{j=0}^{r} \frac{(\alpha)_j}{j!} z^j. \tag{F.77}$$

For $a, b \in (0,1)$ put $Z = (1-a)\big[(1-b^2) + \mathrm{i}b\sqrt{1-b^2}\big]$. The thermal coefficient is

$$c_{t,r} = \int_0^1 \mathrm{d}a \int_0^1 \mathrm{d}b\, (a^2 + b^2 - a^2 b^2)^{3/8}\, |\mathcal{P}_{\Delta_{\chi,0},r}(Z)|^2. \tag{F.78}$$

For $t \in (0,1)$ and $\varphi \in (0, \pi/2)$ put

$$D_t = t^2 + 2t(1-t)\cos\varphi + (1-t)^2, \qquad Z_t = \frac{1-t}{1-t+te^{\mathrm{i}\varphi}}. \tag{F.79}$$

The finite spin-two change is

$$\kappa_r - \kappa_0 = -\pi \int_0^1 \mathrm{d}t \int_0^{\pi/2} \mathrm{d}\varphi\, \frac{\cos(2\varphi)\, t^{11/4}}{(1-t)\, D_t^{\Delta_{\chi,0}}} \Big(|\mathcal{P}_{\Delta_{\chi,0},r}(Z_t)|^2 - 1 \Big). \tag{F.80}$$

For the cylinder term put $y = (1-t)/t$ and

$$F(t,\varphi) = \frac{\mathrm{Re}\big[(e^{\mathrm{i}\varphi} + y)^2\, \mathcal{P}_{\Delta_{\chi,0}-2,r}(Z_t)\, \overline{\mathcal{P}_{\Delta_{\chi,0},r}(Z_t)}\big]}{(1 + 2y\cos\varphi + y^2)^{\Delta_{\chi,0}}}; \tag{F.81}$$

then

$$c_{\mathrm{cyl},r} = -\frac{\Delta_{\chi,0}}{12}\int_0^1 \mathrm{d}t \int_0^{\pi/2} \mathrm{d}\varphi\, \frac{F(t,\varphi)-\cos(2\varphi)}{(1-t)^2}, \qquad r \geq 1. \tag{F.82}$$

The factor two relative to the half-plane expression comes from reflection in x. Tensor Gauss–Legendre rules evaluate these integrals after their domains are mapped to the unit interval; orders 128, 256 and 512 are used for refinement, and the constants are evaluated with 512 points in each direction. The subtraction stays inside the integrand: numerically subtracting separately divergent moments would not define the same calculation.

F.17 Mixed thermal polynomials and denominators

For the central and side kernels of Appendix F.9, the energy polynomials are

$$P_{\mathcal{E},c} = 1 + uv - 2u - 2v, \qquad P_{\mathcal{E},s} = -2 + u + uv - 2u^2 v, \tag{F.83}$$

and the derivative polynomials are

$$\begin{aligned} P_{d,c} &= \frac{uv+1}{4}\big(u^2v^2 + 2u^2v - 4u^2 + 2uv^2 - 6uv + 2u - 4v^2 + 2v + 1\big), \\ P_{d,s} &= -\frac{v+1}{4}\big(4u^4v^2 - 2u^3v^2 - 2u^3v - u^2v^2 + 6u^2v - u^2 - 2uv - 2u + 4\big). \end{aligned} \tag{F.84}$$

For placement $b = c, s$ the coefficient arrays are

$$V^b_{nm} = [u^n v^m]\frac{G_b}{\sqrt{2}}, \qquad E^b_{nm} = [u^n v^m]\frac{G_b P_{\mathcal{E},b}}{2\sqrt{2}}, \qquad D^b_{nm} = [u^n v^m]\frac{G_b P_{d,b}}{2\sqrt{2}}, \tag{F.85}$$

and the derivative angular projection uses $D^b_{n+1,m+1}E^b_{nm}$. The denominators are

placement	vacuum, $d^b_0(n,m)$	energy and derivative, $d^b_{\mathcal{E}}(n,m)$
central	$(2n+2)(2m+2)$	$(2n+1)(2m+1)$
side	$(2n+1)(2m+2)$	$(2n-1)(2m+1)$

so that

$$X_b = (2\pi)^3 q^{\Delta_{\chi,0}} \sum_{n,m\geq 0}\Big[\frac{(2E^b_{nm} - D^b_{n+1,m+1})E^b_{nm}}{d^b_{\mathcal{E}}(n,m)} - \frac{(V^b_{nm})^2}{d^b_0(n,m)}\Big]. \tag{F.86}$$

The side denominator is negative at $n = 0$ and must be kept. The total mixed term is $L_X(V_\epsilon) = X_c + 2X_s$. A function of the form $A(u)B(v)C(uv)$ is generated by diagonal convolutions,

$$[u^n v^m]A(u)B(v)C(uv) = \sum_{j=0}^{\min(n,m)} A_{n-j}B_{m-j}C_j, \tag{F.87}$$

and the finite polynomials are applied by shifts. This reduces the mixed calculation to two-index arrays and permits the much larger N_{mix}.

F.18 The leading face-tail coefficient in full

The coefficient A_{face} of Eq. (F.56) is fixed independently of any fitted cutoff sequence. It is derived from exact face limits of the blocks but evaluated numerically through convergent two-index sums (`code/c2_cylinder_d2/code/hamiltonian/exact_leading_tail.py`; "exact" in the file name describes the source of the asymptotic formula, not an interval enclosure of its floating-point evaluation). With the regulated H of Eq. (F.31) define

$$\Phi_B(u,v) = H(u)(1-v)^{-1/8}H(uv), \qquad \Phi_A(u,v) = H(u)H(v)(1-uv)^{-1/8}, \tag{F.88}$$

with Taylor coefficients B_{ij} and A_{ij}, and set

$$\begin{aligned} E^B_{ij} &= [u^i v^j]\frac{1+v}{2}\Phi_B, \quad & D^B_{ij} &= [u^i v^j]\frac{(1+v)^3}{8}\Phi_B, \\ E^A_{ij} &= [u^i v^j]\frac{1+uv}{2}\Phi_A, \quad & D^A_{ij} &= [u^i v^j]\frac{(1+uv)^3}{8}\Phi_A. \end{aligned} \tag{F.89}$$

The two outer faces are organized by

$$O = \sum_{i,j\ge 0} \frac{1}{2i+2\Delta_{\chi,0}}\left[\frac{(2E^B_{ij} - D^B_{i,j+1})E^B_{ij}}{2j+1} - \frac{B^2_{ij}}{2j+2}\right], \tag{F.90}$$

the adjacent middle face by

$$M_{\mathrm{adj}} = \sum_{i,j\ge 0}\left[\frac{(2E^A_{ij} - D^A_{i+1,j+1})E^A_{ij}}{(2i+1)(2j+1)} - \frac{A^2_{ij}}{(2i+2)(2j+2)}\right], \tag{F.91}$$

and the nested middle face by the collapsed norms of Eq. (F.52),

$$\mathcal{N}_X = 2\mathcal{N}_{\mathcal{E}} - \mathcal{N}_d - \mathcal{N}_0, \qquad M_{\mathrm{nest}} = \frac{H(1)^2\mathcal{N}_X}{(2\pi)^2 q^{\Delta_{\chi,0}}}. \tag{F.92}$$

The leading coefficient is

$$\boxed{A_{\mathrm{face}} = -(2\pi)^4 q^{2\Delta_{\chi,0}}\,\frac{2}{7\Gamma(\frac{1}{8})^2}\,(4O + 2M_{\mathrm{adj}} + M_{\mathrm{nest}}).} \tag{F.93}$$

The base study evaluates it with two-index cutoff 2048 and records O, M_{adj}, M_{nest}, their combination and the sign convention; refinement at smaller and larger auxiliary cutoffs and a comparison with independently fitted leading tails are kept as diagnostics. The continuum analysis of Sec. 3.8 sets this two-index cutoff to a common value for all points and propagates its shift under a change (the "cutoff of the leading-tail calculation" in its error budget). Once Eq. (F.93) is fixed, the residual spectral models of Appendix F.10 account for the subleading powers and exponentials.

F.19 Minimal end-to-end algorithm

For an implementation in another language the logical order is:

1. Fix the unit-field normalization, $\Delta_{\chi,0} = \frac{15}{8}$, $a_1 = -\pi^2/4$ and the common lower-order b_0. Choose r and $\lambda > 0$. No value of c_2 enters the solver.
2. Generate the six Ising block coefficient arrays of Eqs. (F.45) and (F.47) through one index beyond the desired cube; check branches, reality and selected low coefficients against explicit form factors.
3. Construct H_n from Eq. (F.31), apply the three Wick-pairing convolutions, evaluate Eq. (F.48) for each pairing and channel, and return every cubical partial sum.

4. Compute the collapsed energies, norms and spin-two mixed terms of Eq. (F.52) at a much larger cutoff, including the normalization subtraction.
5. Generate the two-index thermal kernels and polynomials of Appendix F.17 and extrapolate their square cutoff separately.
6. Evaluate the compact counterterm integrals of Appendix F.16, form $t_r(\lambda)$ and $-\kappa_r$, and assemble Eq. (F.42), including the exact thermal-squared term.
7. Compute A_{face} from Appendix F.18 and extrapolate the corrected chain sequence, Eq. (F.58), over the predeclared power sets, exponential terms and windows.
8. Repeat at several λ for each regulator order, take the regulator limit only after the spectral limits, and keep model sensitivity and fits with residual degrees of freedom.
9. Convert h_0 to $c_2 = a_1/b_0 + h_0/b_0^2$, and compare with the closed expression only at this final stage.

A faithful reproduction also tests the second-order endpoint anomaly (Appendix F.3), the magnetic normalization (Appendix F.12), the finite-box equivalence of the two implementations (Appendix F.11) and the sensitivity to the coefficient grid; matching a final decimal number alone is not an adequate test.

G c_1 in three dimensions: derivation and conditional control of the omitted spectrum

This appendix gives the derivation and the method of the c_1 package (written by OpenAI GPT-6 astra, in its review revision of 8 October 2026; Sec. 1.2), edited to the notation of this note. Section 4 states the ratio formula, the retained value and its comparison with BRRZ, the implementations, the hypotheses of the estimated range, and the feasibility, box and gap tests; they are not repeated here. Throughout this appendix

$$S \equiv \Delta_\sigma^{\mathrm{SR}} = 0.518148806(24), \qquad E \equiv \Delta_{\mathcal{E}}^{\mathrm{SR}} = 1.41262528(29), \qquad \delta = \frac{s_* - s}{2}, \tag{G.1}$$

as in Table 1, and ϵ is the energy operator of the 3d Ising CFT. Sans-serif letters (S, E, M, V) denote block sums, never these two dimensions. All long decimal strings below are numerical regression targets of the package, not physical significant digits.

The coefficient c_1 is the right derivative of $\Delta_{\mathcal{E}}$ with respect to δ at $\delta = 0$; the long-range side has $\delta > 0$. With the retained value of Sec. 4.2 the expansion reads

$$\Delta_{\mathcal{E}} \simeq 1.41262528 + 0.27315242\,\delta = 1.41262528 + 0.13657621\,(s_* - s), \tag{G.2}$$

with the remainder of the expansion understood. It describes the local slope at s_* only, not a controlled extrapolation over a large range of s.

G.1 Three meanings of precision

Three kinds of uncertainty are kept apart throughout.

- *Numerical.* For a fixed finite list of primary operators with fixed dimensions and operator-product-expansion (OPE) coefficients, the integrals can be evaluated to many digits. Numerical stability then tests the descendant truncation, the quadrature and the subtraction. By itself it says nothing about omitted primaries or about errors of the input correlator.

- *Parametric.* Letting the retained dimensions and couplings vary gives a second uncertainty. The spectrum tables do not provide a calibrated joint probability distribution for their error estimates, so the calculation uses explicit ranges and crossing constraints, not a statistical error model.
- *Physical.* A physical interval must also control the missing conformal families and the possible incompleteness of the selected low-dimensional spectrum. The interval of Sec. 4.3 is conditional on explicit hypotheses about that missing spectrum and on numerical allowances for nonlinear effects. Whenever an interval for c_1 is written, its assumptions are part of the statement.

G.2 Reduction to short-range conformal perturbation theory

The product theory and the perturbing operator. Near the crossover the fixed point is described by the product of the short-range Ising CFT and a generalized free scalar χ [11, 12]. All scalar two-point functions are normalized to unity,

$$\langle\sigma(x)\sigma(0)\rangle = |x|^{-2S}, \qquad \langle\epsilon(x)\epsilon(0)\rangle = |x|^{-2E}, \qquad \langle\chi(x)\chi(0)\rangle = |x|^{-2\Delta_\chi}, \tag{G.3}$$

with

$$\Delta_\chi = 3 - S - \delta, \qquad \mathcal{O} = \sigma\chi, \qquad \Delta_{\mathcal{O}} = 3 - \delta. \tag{G.4}$$

The deformation is $g\mu^\delta \int \mathrm{d}^3x\, \sigma\chi$, with μ the renormalization scale, and the relevant thermal coupling is tuned to the critical trajectory. The generalized free field has Wick-factorized correlators; the Ising factors do not. With a dimensionless renormalized coupling g,

$$\beta(g) = -\delta g + b_0 g^3 + O(g^5), \qquad \gamma_{\mathcal{E}}(g) = a_0 g^2 + O(g^4). \tag{G.5}$$

There is no g^2 term in β, because the undeformed product theory forbids the relevant three-point function ($\langle\mathcal{O}\mathcal{O}\mathcal{O}\rangle = \langle\sigma\sigma\sigma\rangle\langle\chi\chi\chi\rangle = 0$). To first order in δ the infrared fixed point has $g_*^2 = \delta/b_0 + O(\delta^2)$, so $\Delta_{\mathcal{E}} = E + \gamma_{\mathcal{E}}(g_*) = E + (a_0/b_0)\,\delta + O(\delta^2)$, which is $c_1 = a_0/b_0$. Rescaling g by a constant rescales a_0 and b_0 by the same factor and leaves the ratio unchanged.

Why the spin channel supplies the denominator. In the crossover description the long-range spin dimension is protected [12]: $\Delta_\sigma = (3-s)/2 = S + \delta$. If $\gamma_\sigma(g) = a_\sigma g^2 + O(g^4)$, the fixed point gives $\Delta_\sigma = S + a_\sigma\delta/b_0 + O(\delta^2)$, and matching to $S+\delta$ gives $a_\sigma = b_0$. Both coefficients are therefore first-order anomalous-dimension integrals of short-range four-point functions: the energy external state supplies a_0 and the spin external state supplies b_0.

G.3 The integrated four-point functional

Normalization and scale extraction. With e a unit vector, recall from Sec. 4 the four-point function with one external state at infinity, H_O, and write the integral I_O as

$$\begin{aligned} H_O(y) &= \lim_{R\to\infty} R^{2\Delta_O}\langle O(0)\sigma(y)\sigma(e)O(Re)\rangle, \qquad O \in \{\sigma,\epsilon\}, \\ F_O(y) &= |e-y|^{-2(3-S)}\big[H_O(y) - |e-y|^{-2S}\big], \qquad I_O = \mathrm{AC}\int_{\mathbb{R}^3} \mathrm{d}^3y\, F_O(y). \end{aligned} \tag{G.6}$$

The vacuum term removes the identity contraction of the two inserted spins. The symbol AC denotes analytic continuation through the power-divergent short-distance terms, with the relevant deformation tuned critically; it is not a prescription to integrate an unregulated divergent expression numerically. The normalization $a_0 = -2\pi I_\epsilon$, $b_0 = -2\pi I_\sigma$ follows from a second-order amplitude with a Gaussian source. Let

$$D_O(x_1,x_2) = \langle O(\infty)\sigma(x_1)\sigma(x_2)O(0)\rangle - |x_1-x_2|^{-2S}, \tag{G.7}$$

where the state at infinity has unit overlap with the unperturbed state at the origin, and define

$$A_{O,2}(\delta) = \frac{1}{2}\,\mathrm{AC}\int \mathrm{d}^3x_1\,\mathrm{d}^3x_2\,e^{-|x_1|^2-|x_2|^2}\,|x_1-x_2|^{-2(3-S-\delta)}D_O(x_1,x_2). \tag{G.8}$$

Set $x_2 = te$, $x_1 = ty$ and integrate over the direction of e. The measure is $4\pi t^5\,\mathrm{d}t\;\mathrm{d}^3y$, the correlator scales as $D_O(ty,te) = t^{-2S}D_O(y,e)$, and the kernel supplies $t^{-6+2S+2\delta}$. With the factor $\frac{1}{2}$ the remaining scale integral is

$$2\pi\int_0^\infty \mathrm{d}t\,t^{2\delta-1}e^{-t^2(1+|y|^2)} = \pi\Gamma(\delta)(1+|y|^2)^{-\delta}, \tag{G.9}$$

so that $A_{O,2}(\delta) = \pi\Gamma(\delta)\,I_O(\delta;1)$ with the source-dependent integral

$$I_O(\delta;1) = \mathrm{AC}\int \mathrm{d}^3y\,(1+|y|^2)^{-\delta}|e-y|^{-2(3-S-\delta)}\big[H_O(y)-|e-y|^{-2S}\big], \qquad I_O(0;1) = I_O. \tag{G.10}$$

The second argument records the equal Gaussian weights. This amplitude probes the local insertion at the origin against a fixed external state; the source suppresses the second short-distance region, at infinity, so its pole is cancelled by a *single* local operator-renormalization factor, not by twice that factor. With $O_R = Z_O O_{\mathrm{bare}}$ and

$$Z_O = 1 + \frac{a_O}{2\delta}g^2 + O(g^4), \qquad \gamma_O = -\beta(g)\,\partial_g \log Z_O = a_O g^2 + O(g^4), \tag{G.11}$$

finiteness of $Z_O[1 + g^2A_{O,2} + \cdots]$ requires

$$A_{O,2}(\delta) = -\frac{a_O}{2\delta} + O(1), \qquad a_\sigma = b_0, \quad a_\epsilon = a_0. \tag{G.12}$$

Since $\pi\Gamma(\delta)I_O(\delta;1) = \pi I_O/\delta + O(1)$, this gives $a_O = -2\pi I_O$, the normalization of Sec. 4. The common factor -2π cancels from c_1, but keeping it makes the comparison with the published normalization possible. The finite, source-dependent term is not needed at first order. All these identities use the same critical analytic continuation; the Gaussian source does not replace the subtraction of the short-distance power terms.

The folded integration region. Rotational invariance reduces the vector integral to a radius $r = |y|$ and a cosine $\eta = \widehat{y}\cdot e$. Write

$$z = r(\eta + \mathrm{i}\sqrt{1-\eta^2}), \qquad v = |1-z|^2 = 1 - 2r\eta + r^2, \qquad \mathrm{d}\mu = 2\pi r^2\,\mathrm{d}r\;\mathrm{d}\eta. \tag{G.13}$$

Here z only encodes the rotational invariants of a three-dimensional configuration; the integral is not a two-dimensional CFT integral. The conformal maps and the channel identity of Appendix G.4 fold the full integration into

$$\mathcal{R} = \{|z|<1,\ |z|<|1-z|\} = \{|z|<1,\ \mathrm{Re}\,z<\tfrac{1}{2}\}, \tag{G.14}$$

that is, $0<r<1$ and $-1<\eta<\min\{1,(2r)^{-1}\}$. The upper η limit changes at $r=\frac{1}{2}$, which is a breakpoint of the numerical integration.

Block sums. Let $g^{d_a,d_b}_{\Delta,\ell}(z,\bar z)$ be the conformal block with leading OPE behaviour $r^\Delta P_\ell(\eta)$ as $z\to 0$, where P_ℓ is the Legendre polynomial. For the ordered four-point function the external-dimension differences are $d_a = \Delta_1-\Delta_2$ and $d_b = \Delta_3-\Delta_4$; the mixed ordering $(\epsilon,\sigma,\sigma,\epsilon)$ has $(d_a,d_b) = (E-S, S-E)$. In the f normalization of the published spectrum (Appendix G.6), with the rising Pochhammer symbol $(a)_\ell = \Gamma(a+\ell)/\Gamma(a)$, define

$$\begin{aligned} w_\ell &= \frac{\ell!}{(1/2)_\ell}, \qquad \mathsf{S}(z) = \sum_{A\neq\mathbf{1},\ \mathbb{Z}_2\text{-even}} w_{\ell_A} f^2_{\sigma\sigma A}\, g^{0,0}_{\Delta_A,\ell_A}(z), \\ \mathsf{E}(z) &= \sum_{A\neq\mathbf{1},\ \mathbb{Z}_2\text{-even}} w_{\ell_A} f_{\sigma\sigma A} f_{\epsilon\epsilon A}\, g^{0,0}_{\Delta_A,\ell_A}(z), \qquad \mathsf{M}(z) = \sum_{B,\ \mathbb{Z}_2\text{-odd}} w_{\ell_B} f^2_{\sigma\epsilon B}\, g^{E-S,S-E}_{\Delta_B,\ell_B}(z). \end{aligned} \tag{G.15}$$

For example, the direct (odd-channel) representation of the mixed correlator is $H_\epsilon(z) = r^{-S-E}\mathsf{M}(z)$, and its crossed (even-channel) representation is $|1-z|^{-2S}[1+\mathsf{E}(1-z)]$. The two coincide for an exact crossing-symmetric correlator, not necessarily for independently truncated sums.

G.4 Derivation of the folding and its assumptions

Three steps take the integral I_O of Eq. (G.6) over $\mathbb{R}^3$ to the region $\mathcal{R}$ and account for the coefficients and the extra term of the folded expression.

Inversion about the origin. For $y = x/|x|^2$ the Jacobian is $|x|^{-6}$, and

$$H_O\Big(\frac{x}{|x|^2}\Big) = |x|^{2S} H_O(x), \qquad \Big|e - \frac{x}{|x|^2}\Big| = \frac{|e-x|}{|x|}. \tag{G.16}$$

The first identity uses the identical operators O at zero and infinity, which the inversion exchanges. Since $2(3-S)+2S = 6$, the kernel and correlator factors cancel the Jacobian exactly, and

$$I_O = 2\,\mathrm{AC}\int_{|y|<1} \mathrm{d}^3y\, F_O(y). \tag{G.17}$$

The cancellation uses $\Delta_\chi = 3 - S$ at $\delta = 0$; it is not an identity between integrands at finite δ.

The crossed representation on the rest of the ball. Split $|y| < 1$ into $\mathcal{R}$ and $\mathcal{C} = \{|e-y| < |y| < 1\}$, and set $w = e - y$ on $\mathcal{C}$. The exact correlator admits

$$H_O(e-w) = |w|^{-2S}\big[1 + X_O(w)\big], \qquad X_\sigma = \mathsf{S}, \quad X_\epsilon = \mathsf{E}. \tag{G.18}$$

For $O = \epsilon$ this equality selects the even $\sigma\sigma$–$\epsilon\epsilon$ OPE channel; it is *not* a permutation symmetry that interchanges σ and ϵ. The vacuum term cancels, $F_O(e-w) = |w|^{-6}X_O(w)$, and the new domain is $\mathcal{D} = \{|w| < |e-w| < 1\} \subset \mathcal{R}$. Hence

$$I_O = 2\,\mathrm{AC}\int_{\mathcal{R}} \mathrm{d}^3y\, F_O(y) + 2\,\mathrm{AC}\int_{\mathcal{D}} \mathrm{d}^3w\, |w|^{-6} X_O(w). \tag{G.19}$$

Inversion about the second spin insertion. Let $v = |e-w|^2$ and

$$T(w) = e + \frac{w-e}{|w-e|^2}, \qquad |T(w)|^2 = \frac{|w|^2}{v}, \qquad |e - T(w)|^2 = \frac{1}{v}, \qquad |\det \partial T| = v^{-3}. \tag{G.20}$$

The map T sends $\mathcal{R}$ to itself (it exchanges the two conditions $|w| < 1$ and $|w| < |e-w|$) and exchanges its parts with $v < 1$ and $v > 1$, namely $\mathcal{D}$ and its complement, up to boundaries of zero measure. In the complex representative it is $w \mapsto w/(w-1)$ up to conjugation. Both X_O contain pairs of equal external dimensions and only even-spin exchanges, so the pair-exchange factor $(-1)^\ell$ is one and $X_O(Tw) = X_O(w)$. Moreover $|T(w)|^{-6}|\det \partial T| = |w|^{-6}$. The two parts of the last integral of Eq. (G.19) are therefore equal, and

$$\boxed{I_O = \mathrm{AC}\int_{\mathcal{R}} \mathrm{d}^3y\, \big[2F_O(y) + |y|^{-6} X_O(y)\big].} \tag{G.21}$$

The changes of variables are applied with consistent subtractions and analytic continuation, not to bare divergent integrals. They establish Eq. (G.21) for the exact correlator. Replacing its channel sums by independently truncated data introduces a correlator approximation that no change of variables can remove.

Folded integrands and per-family kernels. With $H_\sigma = r^{-2S}(1+\mathsf{S})$ in the direct channel and $H_\epsilon = r^{-S-E}\mathsf{M}$, the implemented linear functionals are

$$I_\sigma = \mathrm{AC}\int_{\mathcal{R}} \mathrm{d}\mu \left\{2v^{-(3-S)}\left[r^{-2S}(1+\mathsf{S}) - v^{-S}\right] + r^{-6}\mathsf{S}\right\}, \tag{G.22}$$

$$I_\epsilon = \mathrm{AC}\int_{\mathcal{R}} \mathrm{d}\mu \left\{2v^{-(3-S)}\left[r^{-S-E}\mathsf{M} - v^{-S}\right] + r^{-6}\mathsf{E}\right\}. \tag{G.23}$$

Both the identity subtraction and the signs of the mixed products must be kept; in particular the energy integrand is not a sum of positive even-channel coefficients. The per-family kernels are

$$\begin{aligned} K_b(\Delta,\ell) &= -2\pi w_\ell\,\mathrm{AC}\int_{\mathcal{R}} \mathrm{d}\mu\,[2v^{-(3-S)}r^{-2S} + r^{-6}]g^{0,0}_{\Delta,\ell}, \qquad K_e(\Delta,\ell) = -2\pi w_\ell\,\mathrm{AC}\int_{\mathcal{R}} \mathrm{d}\mu\, r^{-6} g^{0,0}_{\Delta,\ell}, \\ K_o(\Delta,\ell) &= -4\pi w_\ell \int_{\mathcal{R}} \mathrm{d}\mu\, v^{-(3-S)} r^{-S-E} g^{E-S,S-E}_{\Delta,\ell}, \end{aligned} \tag{G.24}$$

so that, with $f_A = f_{\sigma\sigma A}$, $h_A = f_{\epsilon\epsilon A}$ and $k_B = f_{\sigma\epsilon B}$,

$$b_0 = b_{\mathbf{1}} + \sum_A K_b(\Delta_A,\ell_A) f_A^2, \qquad a_0 = a_{\mathbf{1}} + \sum_A K_e(\Delta_A,\ell_A) f_A h_A + \sum_B K_o(\Delta_B,\ell_B) k_B^2. \tag{G.25}$$

The nonspectral constants come from the remaining terms of Eqs. (G.22) and (G.23),

$$b_{\mathbf{1}} = -4\pi \int_{\mathcal{R}} \mathrm{d}\mu\, v^{-(3-S)}\left[r^{-2S} - v^{-S}\right] \approx -26.0638858592, \qquad a_{\mathbf{1}} = 4\pi \int_{\mathcal{R}} \mathrm{d}\mu\, v^{-3} \approx 44.4132198049 \tag{G.26}$$

(both integrals converge: $v > \frac{1}{4}$ on $\mathcal{R}$, and r^{-2S} is integrable at the origin because $2S < 3$). They are bookkeeping terms, not separate physical observables.

G.5 Conformal blocks and numerical integration

Radial expansion. With the conformal radial coordinate

$$\rho(z) = \frac{z}{(1+\sqrt{1-z})^2}, \qquad \varrho = |\rho|, \qquad \xi = \frac{\mathrm{Re}\,\rho}{|\rho|}, \tag{G.27}$$

where the branch of the square root is continued from $z = 0$, the blocks are evaluated as

$$g^{d_a,d_b}_{\Delta,\ell}(z,\bar z) = (4\varrho)^\Delta \sum_{n=0}^{N} \varrho^n \sum_j B_{n,j} P_j(\xi), \qquad B_{0,j} = \delta_{j\ell}. \tag{G.28}$$

The coefficients solve the quadratic Casimir equation; the matrix recurrence evaluated is given in Appendix G.15. The organization in the radial coordinate follows Ref. [38]; it converges fast in favourable OPE configurations. On the closure of $\mathcal{R}$ the largest radial coordinate is $\max|\rho| = 2-\sqrt{3} \simeq 0.267949$, reached at the corner $z = e^{\mathrm{i}\pi/3}$. This small maximum is why the folded domain is used: equivalent domains with a larger maximum radial coordinate magnify omitted-family and descendant effects, so their truncated values should not be averaged as independent measurements.

A primary is the lowest state of a conformal family, and the terms with $n > 0$ account for its descendants. Increasing N completes a retained family but does not add a primary. Descendant convergence and completeness of the primary spectrum are therefore different error questions.

Analytic continuation at small radius. The r^{-6} terms of Eqs. (G.22) and (G.23) are power divergent at the origin. Choose a subtraction radius $r_0 < \frac{1}{2}$, normally $r_0 = 0.2$, so that the small ball has the complete angular range. On it a separate small-z series is used,

$$g^{0,0}_{\Delta,\ell}(z) = \sum_{n,j} A_{n,j} r^{\Delta+n} P_j(\eta) \tag{G.29}$$

(Appendix G.15). The angular integral keeps only $j = 0$, and gives the explicit small-ball contribution

$$\mathrm{AC}\int_{r<r_0} \mathrm{d}\mu\, r^{-6} g^{0,0}_{\Delta,\ell} = 4\pi \sum_n A_{n,0}\, \frac{r_0^{\Delta+n-3}}{\Delta+n-3}. \tag{G.30}$$

The sum uses 32 levels of the z-series. Terms whose scalar coefficient vanishes are not evaluated as artificial 0/0 expressions. A genuine resonance with a nonzero coefficient would need a separate logarithmic treatment; the implementation rejects an active scalar-series term with $|\Delta+n-3| < 10^{-10}$. This is a diagnostic threshold, not a way to assign a finite part. It is inactive for the retained inputs, where the nearest active power is about 0.41 from resonance. At the conserved stress tensor the null-state consistency checks enforce the required cancellation.

For the rest of the integral, Gauss–Legendre quadrature is used in η, and r is split into $(0, r_0)$, $(r_0, \frac{1}{2})$ and $(\frac{1}{2}, 1)$, with the map $r = r_0 u^4$ on the first interval to resolve the integrable endpoint and linear maps on the others. The principal run uses $N = 24$ and 100×100 nodes in each radial and angular piece. The r^{-6} contribution is left out of the small-ball quadrature, because Eq. (G.30) already includes it. The critical analytic continuation, rather than a cutoff-dependent finite part with unspecified thermal tuning, is part of the definition of the observable. The relevant test is therefore agreement between equivalent analytically continued representations, not agreement between a cutoff integral and the intended critical quantity.

Numerical checks and their scope. The radial block is compared with the independent z-series at complex points and, for scalar exchange with equal external dimensions, with the diagonal hypergeometric expression

$$g_{\Delta,0}(z,z) = \Big(\frac{z^2}{1-z}\Big)^{\Delta/2} {}_3F_2\left(\begin{matrix}\Delta/2,\ \Delta/2,\ \Delta/2-1/2 \\ (\Delta+1)/2,\ \Delta-1/2\end{matrix}; \frac{z^2}{4(z-1)}\right). \tag{G.31}$$

The largest discrepancies are 5.6×10^{-17} against the z-series and 2.3×10^{-16} against Eq. (G.31), floating-point effects at fixed input data. The package's two integrated forms (Sec. 4.2) agree on the retained coefficients; this checks the algebra and the bookkeeping on the *same* folded domain, including the spin-channel identity that reduces one form to the other, and is not an independent validation of the folding.

G.6 Input data and tensor conventions

The selected 60-primary snapshot. The retained selection transcribes the stable extracted operators of dimension below 24 in Tables 3 to 7 of Ref. [14]: 37 internally $\mathbb{Z}_2$-even and 23 internally $\mathbb{Z}_2$-odd primaries (Table 6). It is a selection of published stable entries, not a statement that every exchanged primary below 24 is present. The leading data are replaced coherently by the stress-tensor bootstrap inputs of Ref. [15]:

S	E	$\lambda_{\sigma\sigma\epsilon}$	$\lambda_{\epsilon\epsilon\epsilon}$	$c = c_T/c_B$
0.518148806(24)	1.41262528(29)	1.05185373(11)	1.53244304(58)	0.946538675(42)

Here c_B is the stress-tensor two-point normalization of one free real scalar, and $c = c_T/c_B$ is the dimensionless normalization used in the Ward identities. Shared inputs stay shared under variations: the same $\lambda_{\sigma\sigma\epsilon}$ enters the even ϵ channel and the odd σ channel. The stress-tensor couplings are derived from S, E and c, not treated as independent measured numbers:

$$f_{\sigma\sigma T} = S\sqrt{\frac{3}{8c}}, \qquad f_{\epsilon\epsilon T} = E\sqrt{\frac{3}{8c}}, \tag{G.32}$$

so the Ward ratio is $f_{\epsilon\epsilon T}/f_{\sigma\sigma T} = E/S$. A simultaneous sign change of both stress couplings is an operator convention and does not change the products that enter.

What the printed errors mean. In Table 1 of Ref. [15], the rigorous entries among these five inputs are S and E, under that bootstrap setup; the printed errors of the two OPE coefficients and of c are not established there as rigorous bounds. The errors of the higher families in Ref. [14] describe spreads of extracted data over 60 spectra at the fixed derivative cutoff $\Lambda = 43$; they contain no certified $\Lambda \to \infty$ uncertainty. For example, Ref. [18] reports an extraction near 7.297 at $\Lambda = 30$ for the even scalar listed at 7.2535(51) in the retained table. This does not determine the true dimension, but it shows why the printed spreads cannot be read as complete errors. The strong sensitivity of c_1 to the ϵ' family and to the odd spin-three coupling near $\Delta = 4.638$ makes this limitation relevant to the final interval; the package supplies no calibrated independent cross-check of the quoted error of that coupling.

Normalization conversion and signs. The λ convention meant here is the tensor convention of Kos, Poland and Simmons-Duffin [16] and of Reehorst [17], related to the table convention by $\lambda = 2^{\ell/2} f$; the symbol is not a convention-independent OPE normalization. The block contraction factor in the λ convention is $\ell!/[2^\ell (1/2)_\ell]$; multiplied by the conversion 2^ℓ of the two couplings, it gives $w_\ell = \ell!/(1/2)_\ell$ in Eq. (G.15). For scalar exchange the conventions agree.

The signs of $f_A h_A$ are physical relative signs. Nine rows, all in the selected $[\epsilon\epsilon]_0$ trajectory (family `ee0`), have a negative product; replacing all couplings by absolute values would change the mixed correlator and the answer. The overall sign of one exchanged primary cancels from its product and is immaterial.

Odd spins in the reflected mixed ordering. The lightcone-normalized blocks of Ref. [14] behave as $g^{\mathrm{SD}}_{\Delta,\ell} \sim (-1)^\ell w_\ell (4\varrho)^\Delta P_\ell(\xi)$, whereas the blocks of Eq. (G.28) have leading coefficient $+1$. In the ordering $(\epsilon, \sigma, \sigma, \epsilon)$ the two OPE pairs are reflected, and reversing the ordered scalar pair contributes a second factor $(-1)^\ell$ in the OPE product, which cancels the sign of the lightcone block. The coefficient of the mixed block is therefore $+w_\ell f^2_{\sigma\epsilon B}$, as in M, with no further $(-1)^\ell$. For $0 < z < 1$ the reflected radial-state decomposition must have positive leading squared amplitudes; its all-descendant positivity is formulated with the prefactor Q of Eq. (G.42), not asserted for every coefficient of the raw mixed block. The generalized-free prefactor and positive-state tests of Appendix G.14 check this convention. As an independent calibration of the stress family, a free real scalar has $S = \frac{1}{2}$, $c = 1$, and Eq. (G.32) gives $w_2 f^2_{\sigma\sigma T} = \frac{8}{3} \cdot \frac{3}{32} = \frac{1}{4}$.

G.7 Retained-spectrum evaluation

The package stores the per-family kernels and contributions, not only the final ratio, which exposes normalization changes and large cancellations. For example, the even ϵ family contributes about 108.0654 to a_0 and the odd σ family about -147.3028, before the other families and the nonspectral constant $a_{\mathbf{1}}$ are added; the comparatively small a_0 of Sec. 4.2 should not be reconstructed from rounded individual terms.

Numerical convergence at fixed inputs. Raising the descendant order from 24 to 28 and the quadrature from 100^2 to 128^2 nodes changes the ratio by about 1.7×10^{-12}; moving r_0 from 0.2 to 0.15 changes it by about 3×10^{-14}. Quadrature refinement is therefore not the route to a substantially smaller physical error.

Comparison with the published normalization. The component-by-component agreement with BRRZ in Sec. 4.2 uses the same normalization $a_0 = -2\pi I_\epsilon$, $b_0 = -2\pi I_\sigma$. These comparisons are compatible at the published accuracy; they are not independent measurements to be combined into a smaller error. The precision of the retained integral and the uncertainty of the full CFT remain separate.

G.8 Retained-input variations

In this and the following subsections a_0, b_0 denote the retained reference coefficients of Sec. 4.2, $c_1^{(0)} = a_0/b_0$, and Δa, Δb denote shifts away from them.

Linearized integrated objective. The first variation is

$$\mathrm{d}c_1 = \frac{\mathrm{d}a_0 - c_1^{(0)}\,\mathrm{d}b_0}{b_0}. \tag{G.33}$$

By Eq. (G.25) the coupling derivatives are elementary: for an even primary the variations with f_A and h_A are proportional to $K_e h_A - 2c_1^{(0)} K_b f_A$ and $K_e f_A$, and for an odd primary the derivative is $2K_o k_B$. Dimension derivatives are finite differences of the family kernels. The five shared low inputs S, E, $\lambda_{\sigma\sigma\epsilon}$, $\lambda_{\epsilon\epsilon\epsilon}$ and c are varied coherently through Eq. (G.32): varying an external dimension changes the powers in the integrands and all affected blocks, as well as the associated exchanged operator, so these inputs are varied globally and not counted again as independent entries of the σ, ϵ or T rows.

For every other input u_k with quoted error e_k, the scaled displacement is $x_k = (u_k - u_k^{(0)})/(2e_k)$. The range $|x_k| \le 1$ is a twice-quoted-error hypothesis, not a two-standard-deviation statement. The coupling derivatives are analytic. For an exchanged dimension, the coefficient of the integrated objective is the centred kernel difference at $\Delta \pm e_k$ multiplied by the full scale $2e_k$. For a shared low input it is the centred difference of the full ratio at $u_k^{(0)} \pm 2e_k$. Nonlinear corrections to this affine objective are allowed separately (Appendix G.10).

Retained-input bound for the denominator. The conversion of a numerator range into a ratio interval also needs a bound on the denominator. Every retained independent input is placed within twice its quoted error, keeping shared quantities coherent. For each nonleading family the dimension is sampled at $\Delta + n e_\Delta$, $n = -2, \dots, 2$, together with the coupling corners $f \pm 2e_f$ and, for an even family, $h \pm 2e_h$; if the interval of f contained zero, $f = 0$ would be added to the candidates, because b_0 depends on f^2 (no retained row triggers this). The smallest and largest sampled shifts of each family's contribution to b_0 are kept and summed over families; the shared low inputs are evaluated at their coherent $\pm 2e_k$ endpoints and added. This calculation uses descendant order 20 and 80×80 nodes. The sampled envelope is

$$-0.00142738811843 \lesssim \Delta b_{\mathrm{in}} \lesssim 0.00142355554030, \tag{G.34}$$

rounded outward to $|\Delta b_{\mathrm{in}}| \le 0.001428$. It does not impose crossing or the $h_{\epsilon'}$ restriction of Appendix G.9, so it remains applicable after those constraints shrink the input region. It is a sampled numerical allowance, not an interval-arithmetic extremum over every continuous parameter value, and its adequacy is one of the assumptions of the estimated range.

G.9 Conditional control of the omitted spectrum

The hypotheses and their limitations. The main scenario assumes the gaps

$$G_{\mathrm{even}} = 8, \qquad G_{\mathrm{odd}} = 7 \tag{G.35}$$

of Sec. 4.3: every unlisted primary exchanged in the relevant scalar OPEs has dimension at least the corresponding G. This allows arbitrary omitted spectral weight above the thresholds, subject to crossing and positivity; it does not assert that the retained list is complete up to its largest dimension, 24. Neither threshold is proved, and additional unlisted low-dimensional operators remain a possible cause of failure of the narrow interval. The input ranges are hypotheses as well: the actual retained CFT data are assumed to lie inside the twice-quoted-error box. Exact crossing and reflection positivity are imposed on the full theory, not on the finite retained approximation alone.

Evidence motivating the thresholds, not proving them. The gap hypotheses refer to spatial-parity-even primaries in the indicated internal $\mathbb{Z}_2$ sectors. Spatial and internal parity are different labels: a spatial-parity-odd state does not contribute to these two-scalar OPEs merely because its internal parity is appropriate. The selected list has 11 even primaries below dimension 8. Published spectral extractions and fuzzy-sphere studies [14, 18, 19, 20] give the following nearby unlisted families; the entries are estimates, not gap bounds.

internal parity	spin	estimated dimension	evidence and status
odd	3	7.53(6)	fuzzy sphere [20]; an additional trajectory is also discussed in extremal spectra
odd	6	7.73(5); about 7.3	Ref. [20]; lower extremal-functional estimate in Ref. [18]
even	0	about 8.55 to 8.92	differing extremal extractions compared in Ref. [18]; not a certified interval
even	2	about 8.6	higher trajectory in the spectral plots of Ref. [18]; approximate reading
even	6	about 8.78	described as an unstable extraction in Ref. [18]; tentative

The fuzzy-sphere results of Ref. [20] explain why an odd threshold of 8 cannot simply be kept: allowed unlisted odd states are estimated below 8. A threshold of 7 leaves only a margin of about 0.3 to 0.7 to the nearby spin-six estimates, and the associated OPE data do not justify assuming that missing states just above the threshold have negligible weight. None of these approximate couplings is used as a precision input; the remainder bounds must accommodate them. The comparisons certify neither gap. In particular, Figures 1 and 2 of Ref. [14] contain unstable extracted entries, and the main scenario assumes that no such unretained entry below its threshold is an additional genuine exchanged primary; calling an entry an artifact does not establish that assumption. The evidence motivates testing $(8, 7)$, and the feasibility and sensitivity tests (Sec. 4.4 and Appendix G.13) expose its tension and its consequences.

The published constraint on $\lambda_{\epsilon\epsilon\epsilon'}$. Let ϵ' be the next retained internally even scalar after ϵ, with reference dimension 3.82968 and $h_{\epsilon'} = f_{\epsilon\epsilon\epsilon'} = 1.5360$; for a scalar the f and λ conventions coincide. Reehorst's conditional bootstrap analysis gives $\lambda_{\epsilon\epsilon\epsilon'} = 1.5362(12)$ [17]. Its assumptions include an isolated nondegenerate ϵ', the next even scalar above dimension 6 and the next odd scalar above dimension 3, which are compatible with the spectral assumptions used here. To allow for the rounding of the printed bound, the retained-input box is intersected with

$$1.5349 \le h_{\epsilon'} \le 1.5375. \tag{G.36}$$

This applies a published conditional constraint; it is not a new semidefinite-programming computation, nor a weighted average of independent experiments. If its sparsity or nondegeneracy hypotheses are not assumed, the restriction must be removed and the envelope recomputed.

The identical-channel positive remainder. For an identical external scalar of dimension $D \in \{S, E\}$, write its reduced correlator as $P + R$, where P is the retained block sum including the identity and R the missing sum. In radial variables, reflection positivity gives nonnegative expansion coefficients on the collinear axis, $R(r, 1) = \sum_p c_p r^p$ with $c_p \ge 0$; each descendant adds a nonnegative integer level to the primary dimension, so with a missing-primary gap G every $p \ge G$. For $0 < r_1 < r_2$ this gives

$$0 \le R(r_1, 1) \le \left(\frac{r_1}{r_2}\right)^G R(r_2, 1), \tag{G.37}$$

and the same gap controls the full omitted family. Choose a real $0 < x < \frac{1}{2}$ and let $A = [x/(1-x)]^{2D}$, $r_1 = \rho(x)$, $r_2 = \rho(1-x)$. Crossing gives $P(x) + R(x) = A[P(1-x) + R(1-x)]$, with $R(x)$ evaluated

at radius r_1 and $R(1-x)$ at r_2. Solving for the remainder at the larger radius and using Eq. (G.37),

$$R(r_2,1) = A^{-1}R(r_1,1) + A^{-1}P(x) - P(1-x) \le A^{-1}\Big(\frac{r_1}{r_2}\Big)^G R(r_2,1) + A^{-1}P(x) - P(1-x), \quad \text{(G.38)}$$

and therefore

$$R(r_2,1) \le \frac{A^{-1}P(x) - P(1-x)}{1 - A^{-1}(r_1/r_2)^G}, \quad \text{(G.39)}$$

provided the denominator is positive. The numerator must be maximized over the allowed retained inputs and include the nonlinear and numerical allowances. A negative upper bound is a consistency failure, not permission to replace a sizeable negative number by zero. Different reference points give valid bounds at different radii: if B_i bounds the reference norm at r_i, then

$$R(r,1) \le \min_{i:\, r \le r_i} B_i \Big(\frac{r}{r_i}\Big)^G. \quad \text{(G.40)}$$

The minimum is justified because each reference bound is separately necessary; it is not a statistical combination. The radial-expansion framework is that of Ref. [38].

Even mixed and odd remainders. The even mixed remainder is not positive, but the Cauchy–Schwarz inequality bounds it by the two identical-channel norms,

$$\big|R_{\sigma\sigma;\epsilon\epsilon}(r,\xi)\big| \le \sqrt{R_{\sigma^4}(r,1)\, R_{\epsilon^4}(r,1)}. \quad \text{(G.41)}$$

This is why the analysis uses all three scalar correlators, including $\langle\epsilon\epsilon\epsilon\epsilon\rangle$, although that correlator does not enter the slope functional directly. In the odd mixed channel positivity applies to

$$Q(\rho,\bar\rho)\, R_{\rm odd}, \qquad Q(\rho,\bar\rho) = \Big|\frac{1+\rho}{1-\rho}\Big|^{2(E-S)}, \quad \text{(G.42)}$$

not to the raw mixed-block coefficients. At real odd-channel reference points $x > \frac{1}{2}$, mixed crossing and Eq. (G.41) bound this normalized odd norm; radial suppression is applied to the normalized quantity, and the bound at a complex integration point is divided by Q again. Applying positivity directly to the raw series, without this prefactor, would be invalid.

Let $U_\sigma(u)$ and $U_\epsilon(u)$ be the identical-channel envelopes of Eq. (G.40), and $d_m^{\rm min}(x)$ the smallest allowed retained mixed residual, with the direct-minus-crossed normalization of Eq. (G.46) and including its nonlinear allowance. At a real odd reference point $x > \frac{1}{2}$ mixed crossing reads

$$0 = d_m(x) + x^{-S-E} R_{\rm odd}(x) - (1-x)^{-2S} R_{\sigma\sigma;\epsilon\epsilon}(1-x). \quad \text{(G.43)}$$

Before the common numerical padding (Appendix G.10), the normalized odd reference norm at $u_x = \rho(x)$ is therefore bounded by

$$B_{\rm odd}(x) = x^{S+E}\, Q(u_x,u_x)\Big[-d_m^{\rm min}(x) + (1-x)^{-2S}\sqrt{U_\sigma(\rho(1-x))\, U_\epsilon(\rho(1-x))}\Big]. \quad \text{(G.44)}$$

Each valid odd reference supplies a radial bound, and the envelope used at radius u is

$$U_o(u) = \min_{x:\, u \le \rho(x)} B_{\rm odd}(x) \Big(\frac{u}{\rho(x)}\Big)^{G_{\rm odd}}. \quad \text{(G.45)}$$

The raw odd remainder at a complex point is bounded by $U_o(|\rho|)/Q(\rho,\bar\rho)$. As in the identical channel, a materially negative upper reference norm is a consistency failure. The identical-channel inequality and the mixed normalization were tested on exact generalized-free correlators; these tests check the algebra of the envelope and its implementation, not the completeness assumptions about the Ising spectrum.

G.10 Joint crossing and input optimization

The crossing residuals. The residuals are unnormalized differences of the direct and crossed correlators, not divided by an average correlator magnitude. With the further sum $\mathsf{V}(z) = \sum_{A\neq\mathbf{1},\ \mathbb{Z}_2\text{-even}} w_{\ell_A} h_A^2 g^{0,0}_{\Delta_A,\ell_A}(z)$, they are, for each retained input configuration,

$$\begin{aligned} d_\sigma(z) &= |z|^{-2S}\big[1+\mathsf{S}(z)\big] - |1-z|^{-2S}\big[1+\mathsf{S}(1-z)\big], \\ d_\epsilon(z) &= |z|^{-2E}\big[1+\mathsf{V}(z)\big] - |1-z|^{-2E}\big[1+\mathsf{V}(1-z)\big], \\ d_m(z) &= |z|^{-S-E}\mathsf{M}(z) - |1-z|^{-2S}\big[1+\mathsf{E}(1-z)\big]. \end{aligned} \tag{G.46}$$

The identity contributions are explicit, and the external dimensions and all occurrences of shared couplings vary coherently. These definitions fix the signs and normalization of the reference bounds, the Jacobian and the affine constraints below.

The affine necessary-constraint problem. Crossing constrains which retained-input shifts and omitted contributions can coexist. The retained inputs enter through the scaled variables x_k of Appendix G.8; except for the intersected $h_{\epsilon'}$ range, $-1 \le x_k \le 1$ is a displacement of at most twice a quoted error. Shared low inputs are parametrized once, and there are 154 variables. For residual type $q \in \{\sigma^4, \epsilon^4, \text{mixed}\}$ at point z_j,

$$d_{qj}(x) = d^{(0)}_{qj} + \sum_k J_{qjk} x_k + e_{qj}(x), \qquad |e_{qj}(x)| \le p_{qj}. \tag{G.47}$$

The columns of J are centred secants of the retained residual at $x_k = \pm 1$, with the other inputs at their reference values. The bound on e_{qj} is a sampled numerical allowance, not a proved uniform bound. For each nonleading family the affine remainder is evaluated at scaled dimension displacements $-1, -\frac{1}{2}, 0, \frac{1}{2}, 1$ and scaled coupling displacements $-1, 0, 1$; the largest absolute remainder at each crossing point is multiplied by 1.5 and summed over families. For each shared low input, twice the absolute difference between the mean of its two endpoint residuals and the reference residual is added. Finally the floor $2\times10^{-10}(1+|d^{(0)}|)$ is added. The factors 1.5 and 2 and the floor are chosen allowances, not consequences of a proved uniform Hessian bound. (The "margins" of Sec. 4.3 are these paddings, the objective allowance 5×10^{-5} and the sampled denominator, reference and quadrature allowances.) Adding separate per-family and low-input samples also does not enclose cross terms between a shared low input and a family input uniformly; the nonlinear tests of Appendix G.14 sample such combined shifts but do not close this logical gap.

The array has 92 point entries: 35 equally spaced real identical-channel references from 0.15 to 0.49, 15 equally spaced odd references from 0.64 to 0.85, and the grid $z = 0.2 + 0.1 i_1 + 0.12 i_2\,\mathrm{i}$ with $i_1 = 0,\dots,6$ and $i_2 = 0,\dots,5$. Some configurations serve more than one role, so 92 counts entries, not independent constraints. Blocks are evaluated at descendant level 40 for the principal optimization.

Exact crossing requires a nonzero omitted contribution t_{qj} to cancel the retained residual. With a conditional bound $t_{qj} \in [-B^-_{qj}, B^+_{qj}]$, Eq. (G.47) becomes two affine inequalities once the nonlinear padding is included. Real-axis positivity gives one-sided improvements; complex-point bounds use magnitudes. The truncated correlator is never forced to satisfy crossing with its tail set to zero.

The allowed residual ranges follow from the reference envelopes. Put $u = |\rho(z)|$, $u' = |\rho(1-z)|$ and $Q_z = Q(\rho(z), \overline{\rho(z)})$. For an identical external scalar of dimension $D \in \{S, E\}$, with $U_S = U_\sigma$ and $U_E = U_\epsilon$, define

$$T^d_D = |z|^{-2D} U_D(u), \qquad T^c_D = |1-z|^{-2D} U_D(u'). \tag{G.48}$$

In general the retained residual lies in $[L_D, U^{\text{res}}_D] = [-(T^d_D + T^c_D),\ T^d_D + T^c_D]$. On the real axis a one-sided improvement needs a sign condition: for $x = \min\{z, 1-z\} < \frac{1}{2}$ put

$$q_D(x; G) = \Big[\frac{\rho(x)}{\rho(1-x)}\Big]^G \Big(\frac{1-x}{x}\Big)^{2D} \le 1. \tag{G.49}$$

Then the direct positive tail on the side of the smaller radius is at most q_D times the crossed one, and the exact retained residual there, $T^c - T^d$ in terms of the actual tails, is nonnegative. At the nominal points the largest values are about 0.6629 for $D = S$ and 0.7121 for $D = E$, excluding the symmetric point $z = \frac{1}{2}$, where the residual vanishes identically. The implementation checks Eq. (G.49) and stops if it fails, so a different gap or point selection cannot silently keep the sharpening. Under this condition the ranges are

$$[L_D, U_D^{\mathrm{res}}] = \begin{cases} [0,\, T_D^c], & 0 < z < \frac{1}{2}, \\ [-T_D^d,\, 0], & \frac{1}{2} < z < 1. \end{cases} \tag{G.50}$$

For the mixed residual put

$$T_m^o = |z|^{-S-E} \frac{U_o(u)}{Q_z}, \qquad T_m^e = |1-z|^{-2S} \sqrt{U_\sigma(u')\, U_\epsilon(u')}. \tag{G.51}$$

The general range is $[-T_m^o - T_m^e,\, T_m^o + T_m^e]$; on the real axis its upper end is reduced to T_m^e, because the direct odd remainder is nonnegative. These bounds constrain the retained residuals, not the missing residual of opposite sign. Writing any allowed range as $[L_{qj}, U_{qj}]$, the necessary inequalities on the affine part are

$$J_{qj} \cdot x \le U_{qj} - d_{qj}^{(0)} + p_{qj}, \qquad -J_{qj} \cdot x \le -L_{qj} + d_{qj}^{(0)} + p_{qj}. \tag{G.52}$$

Each pair is divided by $\max\{|d_{qj}^{(0)}|, \max_k |J_{qjk}|, p_{qj}, 10^{-10}\}$ before it is passed to the solver, which does not change the feasible region. The result is a linear program, solved with the HiGHS backend of SciPy. It is not a full semidefinite bootstrap calculation, and feasibility of these finitely many necessary inequalities is not an existence proof of a CFT.

Iterating input constraints and reference bounds. Starting from the input box and Eq. (G.36), each reference numerator of Eq. (G.39) is maximized over the current feasible input region. Equations (G.39) to (G.45) then give omitted-tail bounds, which tighten the allowed crossing residuals and hence the input region; the references are recomputed in the smaller region and the step is repeated. New inequalities are intersected with the existing ones, not averaged. The nominal calculation makes ten constraint updates; the archived doubled-padding run makes eight (Appendix G.13 compares matched ten-update runs). A fixed number of updates is not a convergence criterion. The intermediate intervals of the iteration use symmetric magnitude envelopes for the tails; they are stages of the construction, not the reported result, which uses the sign-aware odd integral of Appendix G.11.

The reference bounds include a chosen multiplicative factor 1.001, intended to allow for tiny external-dimension effects; this too is a numerical allowance, not an interval-certified treatment of all continuous dependence on the dimensions. The final problem has 552 scaled inequalities in 154 variables, plus the variable bounds. For the contribution of the retained inputs the objective is

$$F_{\mathrm{in}} = \frac{\Delta a_{\mathrm{retained}} - c_1^{(0)} \Delta b_{\mathrm{retained}}}{b_0}, \tag{G.53}$$

and its final linear extrema are

$$-0.001584618275 \lesssim F_{\mathrm{in,lin}} \lesssim 0.001256144319. \tag{G.54}$$

A further 5×10^{-5} is allowed on each side for the nonlinearity of the integrated objective; the separately sampled sum of objective remainders is about 2.3952×10^{-5}, below this allowance. No covariance matrix is inferred by fitting crossing residuals, and no independent-Gaussian interpretation is given to the box.

The central input vector is excluded. Inserting $x = 0$ into the *final* nominal constraints $Ax \le B$ violates the 68 rows of Sec. 4.4 (those with $B_i < -10^{-9}$, all in the $\sigma\sigma\sigma\sigma$ channel); the smallest scaled right-hand side is about -0.727632. The central tabulated data are therefore not a feasible correlator input for the nominal relaxation. This concerns the vector of input data; its integrated value $c_1^{(0)}$ can, and does, lie inside the final interval. The displacement needed for this fixed problem is measured by

$$t_* = \min_{x,t} t \quad \text{subject to} \quad Ax \le B, \quad -t \le x_k \le t, \quad t \ge 0, \tag{G.55}$$

with the $h_{\epsilon'}$ restriction kept. The saved nominal matrices give $t_* = 0.5735097802$, that is $2t_* = 1.1470195604$ quoted errors. This is the smallest *maximum* displacement over the inputs; it does not say that every input moves by that amount and has no Gaussian interpretation. Equation (G.55) freezes the nominal reference bounds, secants and padding, so it does not give the feasibility threshold of a rebuilt input box, whose arrays and iteration change with the box (Sec. 4.4). The exclusion shows a tension among the central extracted spectrum, the missing-spectrum hypotheses and the chosen numerical relaxation. Under a valid remainder construction, underestimated tabulated errors or missing low-dimensional spectral weight are natural explanations; the calculation does not decide between them, does not establish a new operator, and does not prove that these possibilities exhaust the limitations of the numerical envelopes. Nor does it turn an empirical error spread into a rigorous physical range by shifting central values until crossing is feasible.

G.11 Sign-aware integration of the odd missing contribution

Why absolute values lose information. Integrating an absolute pointwise tail bound gives a valid conditional magnitude estimate, but it can be loose. In the normalized odd channel

$$Q\, R_{\mathrm{odd}}(\varrho, \xi) = \sum_{p,j} c_{p,j}\, \varrho^p P_j(\xi), \qquad c_{p,j} \ge 0, \quad p \ge G_{\mathrm{odd}}, \tag{G.56}$$

where the radial-state energy p is a primary dimension plus an integer descendant level, so it is not restricted to integers. A reference norm bounds $\sum_{p,j} c_{p,j} r_{\mathrm{ref}}^p$. For one such state, the Legendre polynomial is integrated against the actual odd contribution to a_0, whose kernel carries an overall minus sign. Separating a positive integration weight W, which includes $4\pi\, \mathrm{d}\mu\, v^{-(3-S)} r^{-S-E}/Q$, define

$$M_j(p; r_{\mathrm{ref}}) = -\int_{\mathcal{R}} W(z) \Big(\frac{|\rho(z)|}{r_{\mathrm{ref}}}\Big)^p P_j\Big(\frac{\mathrm{Re}\,\rho}{|\rho|}\Big). \tag{G.57}$$

A reference norm B then bounds the integrated omitted contribution by B times the extrema of $M_j(p; r_{\mathrm{ref}})$ over the allowed states. The angular integration is done before the worst case is taken, and the positive and negative extrema are in general unequal.

Coverage of spins and continuous exponents. The main calculation evaluates $0 \le j \le 64$, radial energies from G_{odd} to $G_{\mathrm{odd}} + 60$ in steps of 0.1, and 100 quadrature nodes in each variable. Three analytic bounds cover the rest. For higher spins, the Legendre bound implied by Eq. 18.14.7 of Ref. [34],

$$|P_j(\xi)| \le \min\Big\{1,\ \sqrt{\frac{2}{\pi(j+1/2)\sqrt{1-\xi^2}}}\Big\}, \tag{G.58}$$

is used with $j + \frac{1}{2}$ replaced by $J + \frac{3}{2}$, a common upper bound for every $j \ge J + 1$; the cap at 1 prevents an artificial divergence of the angular estimate at the endpoints. Between sampled exponents with step h, the second derivative with respect to p brings down $\log^2(|\rho|/r_{\mathrm{ref}})$, and since $|\rho|/r_{\mathrm{ref}} < 1$ a uniform interpolation allowance is

$$\frac{h^2}{8} \int_{\mathcal{R}} W(z) \Big(\frac{|\rho|}{r_{\mathrm{ref}}}\Big)^{G_{\mathrm{odd}}} \log^2\Big(\frac{|\rho|}{r_{\mathrm{ref}}}\Big). \tag{G.59}$$

Above the last sampled exponent, the absolute kernel is bounded by its magnitude at the gap times $[(2-\sqrt{3})/r_{\mathrm{ref}}]^{60}$; only reference radii larger than $2-\sqrt{3}$ are used for this bound. These are analytic envelopes evaluated by floating-point quadrature. They do not certify the input or nonlinear assumptions, but they control the tails in the state indices instead of stopping a scan at a convenient value.

Numerical effect. In the nominal scenario the symmetric magnitude estimate of the odd contribution to a_0 is 0.01652521405, while the sign-aware upper end is only 0.004677262775:

$$\Delta a_{\mathrm{odd}} \in [-0.01652521405,\ +0.004677262775] \tag{G.60}$$

under the hypotheses, about $[-0.0013485,\ +0.0003817]$ after division by b_0. The best upper reference in this run is $x = 0.76$, with $r_{\mathrm{ref}} \simeq 0.34237382$, and the sampled maximum occurs at $p = 7$, $j = 2$. Raising the quadrature to 160 nodes and the explicit spin cutoff to 96 and halving the exponent step changes the positive allowance to about 0.004672754193; the sampled maximum itself is stable to about 5×10^{-15}, and most of the change is the intentionally reduced interpolation allowance. The main interval uses the larger, original allowance. Because the odd remainder enters with this definite sign, lowering the odd gap mainly lowers the lower end of the range (Sec. 4.4).

G.12 The conditional result and its error budget

Combining numerator and denominator corrections. The nominal tail estimates are

$$|\Delta b_{\mathrm{tail}}| \le 7.02687010 \times 10^{-5}, \qquad |\Delta a_{\mathrm{even}}| \le 0.004552089294, \tag{G.61}$$

together with the asymmetric odd interval of Eq. (G.60). With Eq. (G.54), the allowances in units of c_1 (numerator contributions divided by b_0) are

contribution	lower	upper
crossing-constrained linear input variation	−0.00158462	+0.00125614
nonlinear integrated-input allowance	−0.00005000	+0.00005000
missing even contribution to a_0	−0.00037145	+0.00037145
missing odd contribution to a_0	−0.00134846	+0.00038166
missing contribution to b_0, times $-c_1^{(0)}$	−0.00000157	+0.00000157

These are allowed ranges, not independent standard deviations, and they are added linearly. The denominator is not frozen in the conversion to the ratio; exactly,

$$c_1 - c_1^{(0)} = \frac{\Delta a - c_1^{(0)} \Delta b}{b_0 + \Delta b}. \tag{G.62}$$

The retained-input bound $|\Delta b_{\mathrm{in}}| \le 0.001428$ of Eq. (G.34) plus the tail bound keeps the denominator away from zero. With the total scaled numerator range $[F_-, F_+]$ and $d_b = 0.001428 + |\Delta b_{\mathrm{tail}}|$, the conversion is

$$\Big[c_1^{(0)} + \frac{F_-}{1 - d_b/b_0},\ c_1^{(0)} + \frac{F_+}{1 - d_b/b_0}\Big], \qquad \text{provided} \quad F_- \le 0 \le F_+, \quad 0 \le d_b < b_0. \tag{G.63}$$

The nominal range is $[F_-, F_+] = [-0.003356090758,\ 0.002060824936]$, so both conditions hold. Taking the smaller allowed denominator for both ends is conservative only because the two ends have opposite signs; it is not a general rule for converting ratios. The code checks both conditions with explicit exceptions and does not use a first-order expansion of the denominator for the quoted range. The result is the nominal interval [0.269795922707, 0.275213500746]; the archived doubled-padding run with eight updates gives [0.269273823430, 0.275410362633] (Sec. 4.3 quotes both to seven digits).

Nominal and enlarged padding. The outward-rounded estimated range $[0.2690, 0.2760]$ of Sec. 4.3 covers these two runs. The retained value and the estimated range are reported separately. The range extends about -0.0042 and $+0.0028$ from the rounded retained value 0.2732, not symmetrically; its rounded midpoint, 0.2725, is not a new best fit or a feasible optimized correlator. The range has no claimed confidence level, and its rounding covers the specified doubled-padding test, not arbitrary changes of the input box, the gaps or the numerical allowances. All endpoints remain conditional on the omitted-primary gaps, on the validity of the retained-input ranges and of the nondegenerate-ϵ' constraint, and on the adequacy of the sampled nonlinear, denominator and quadrature allowances. The finite crossing relaxation supplies necessary conditions only; its feasible points and extrema need not be realized by an exact CFT. No assumption-independent physical interval is established.

G.13 Sensitivity scans: method and further rows

The scans of Sec. 4.4 rerun the same final method with ten constraint updates each; they are not alternative estimators to average. In the box scan each quoted error e_k is replaced by me_k, and the integrated sensitivities, the sampled denominator box, the crossing secants and padding, and the reference-bound iteration are all recomputed. The physical $h_{\epsilon'}$ window of Eq. (G.36) is held fixed, not scaled with the table errors, so its limits in the scaled variable are recomputed from the changed parameter scale. In the gap scan the central spectrum and the twice-quoted-error box are unchanged; the same level-40 basis is reused, while the radial suppression, the reference bounds, the crossing constraints and the integrated tails are recomputed at each threshold. A third scan multiplies every crossing padding by a constant with ten updates, which isolates the effect of the padding from that of the iteration count. The rows not shown in Sec. 4.4 are:

scan	setting	ten-update interval
box, m (box radius $\pm 2m$ quoted errors)	0.5	infeasible
	0.7	[0.27050311, 0.27455152]
	0.9	[0.27003354, 0.27506550]
	1.5	[0.26848108, 0.27600079]
	3	[0.26380302, 0.27869935]
gaps $(G_{\text{even}}, G_{\text{odd}})$	$(6.5, 7)$	[0.26850350, 0.27679517]
	$(7, 6)$	[0.26573717, 0.27697080]
	$(6, 6)$	[0.26419915, 0.27833854]
	$(8, 5.5)$	[0.26156563, 0.27688176]
crossing-padding multiplier	1 (nominal)	[0.26979592, 0.27521350]
	2	[0.26927464, 0.27540968]
	3	[0.26887007, 0.27557227]

Every feasible range listed contains the retained value $c_1^{(0)}$, but that does not remove the infeasibility of the central input vector. The scans keep the chosen 5×10^{-5} allowance for the integrated objective while recomputing the other quantities; enlarging the box does not certify this fixed allowance on the enlarged domain, so the wider-box outputs are numerical stress diagnostics, not validated confidence intervals. The sampled nonlinear maxima, the incomplete control of mixed input cross terms and the absence of interval arithmetic apply throughout.

The iterations are finite and not claimed to be converged. At the end of the original histories the nominal lower end still changed by order 10^{-6} per update, and the eight-update $(6, 6)$ history by order 1.5×10^{-4}, so the ten-update $(6, 6)$ row above differs correspondingly from that archived history. Assuming valid outer allowances, further intersections can tighten the feasible region; stopping earlier is no reason to claim a sharper result. Improving the precision requires resolving the tension between the spectrum and the inputs and certifying the numerical envelopes, not rounding the nominal endpoints more finely.

G.14 Validation

Nonlinear crossing and solver checks. The nonlinear test uses a fixed pseudorandom seed, 6087, and 34 configurations: sampled interior points, parameter corners and extrema of the linear optimization. It recomputes the full nonlinear retained residual and compares the affine error with its allocated padding. The largest ratio is 0.666645: below one, but about two thirds of the allowance, so the chosen factor 1.5 provides essentially the sampled margin rather than a large safety reserve. This is evidence that the padding is adequate on the tested configurations, not a proof that it covers every point of the continuous box. The test of the integrated objective samples nine dimension values and the coupling corners family by family, with a separate treatment of the global low inputs; the summed sampled remainder is 2.3952×10^{-5}, against the allowance 5×10^{-5}. Summing such samples cannot by itself give a certified statement. The two final objective linear programs have primal–dual gaps of order 10^{-18} and largest scaled constraint violations of about 8.87×10^{-10}. These tests establish solver consistency for the floating-point affine problem; they do not certify that the problem contains every physical correlator consistent with the stated gaps.

Descendants, integration and positive radial states. Raising the descendant order of the crossing basis from 40 to 48 changes the largest central residual by 5.75×10^{-11}, a Jacobian entry by 3.30×10^{-14} and a padding entry by 7.10×10^{-14}; the bases at orders 32, 40 and 48 are all kept. Changing the quadrature of the integrated envelopes from 80 to 220 nodes changes the even contribution to a_0 by about 1.01×10^{-7}, or 8.3×10^{-9} after division by b_0; the spin and odd magnitudes are much more stable. Separate low-level checks of the normalized odd radial expansion support the use of Eq. (G.42); they supplement the positivity argument, they do not replace it.

Executable guards and the remaining logical limits. The final runner reads $c_1^{(0)}$ and b_0 from the current retained-spectrum output, derives the outward-rounded denominator allowance from the input-box output, and derives the scaled $h_{\epsilon'}$ limits from the actual parameter scale; the chosen objective allowance and the physical $h_{\epsilon'}$ window are explicit settings in one configuration file. The numerical guards raise exceptions that stay active when Python assertions are disabled, among them the sign condition of Eq. (G.49) and the conditions of Eq. (G.63), and the table of the estimated range is generated only after checking that it contains the computed nominal and doubled-padding intervals. The verification script tests the level-40 and level-48 arrays (residual, Jacobian and padding), the normalized radial-state output, the generalized-free remainder tests, the integrated-tail quadrature and the angular and exponent envelopes, as well as the central values and the optimality of the linear programs. These are numerical regression checks: no passing test establishes completeness of the spectrum, a uniform bound on the nonlinearity in the inputs, or the existence of a CFT at an extremum of the linear program.

One limitation of the even mixed-tail construction is its dependence, through Eq. (G.41), on the positive ϵ^4 remainder. The retained $\langle\epsilon\epsilon\epsilon\epsilon\rangle$ correlator can have a substantially larger crossing mismatch than $\langle\sigma\sigma\sigma\sigma\rangle$ or the mixed correlator. It does not enter the slope functional directly, but it weakens the auxiliary norm bound, so a small mixed residual alone cannot replace the full remainder analysis.

G.15 The implemented conformal-block recurrences

The small-z series. For external differences d_a, d_b write

$$g^{d_a,d_b}_{\Delta,\ell}(z) = \sum_{n\geq 0}\sum_{j\geq 0} A_{n,j} r^{\Delta+n} P_j(\eta), \qquad A_{0,j} = \delta_{j\ell}. \tag{G.64}$$

Let $C_{\Delta,\ell} = \Delta(\Delta-3)+\ell(\ell+1)$ and $D_{n,j} = (\Delta+n)(\Delta+n-3)+j(j+1)-C_{\Delta,\ell}$. The coefficients obey

$$\begin{aligned} D_{n,j}A_{n,j} = {}& \frac{j}{2j-1}(\Delta+n+j-2-d_a)(\Delta+n+j-2+d_b)A_{n-1,j-1} \\ & + \frac{j+1}{2j+3}(\Delta+n-j-3-d_a)(\Delta+n-j-3+d_b)A_{n-1,j+1}, \end{aligned} \tag{G.65}$$

where the first term is absent for $j=0$ and out-of-range coefficients vanish. At a null-state denominator the numerator must vanish consistently; otherwise the routine raises an error. Only the scalar angular coefficient $A_{n,0}$ enters the analytic continuation of Eq. (G.30); for block validation the series is also evaluated at complex points with all angular components.

The radial matrix recurrence. The coefficients $B_{n,j}$ of Eq. (G.28) are computed by matrices acting on vectors of Legendre coefficients (matrices carry hats). Let $\hat{M}$ represent multiplication by ξ and $\hat{D}$ differentiation with respect to ξ:

$$\hat{M}_{j+1,j} = \frac{j+1}{2j+1}, \qquad \hat{M}_{j-1,j} = \frac{j}{2j+1}, \qquad \hat{D}_{k,j} = \begin{cases} 2k+1, & k = j-1,\, j-3,\, \cdots \geq 0, \\ 0, & \text{otherwise.} \end{cases} \tag{G.66}$$

With the Chebyshev matrix polynomials $T_m(\hat{M})$ and $U_m(\hat{M})$, built by the usual second-order recurrence, define

$$\hat{A} = 2\hat{M}^2 - \hat{I}, \qquad \hat{B} = 2\hat{M}(\hat{I}-\hat{M}^2)\hat{D}, \qquad a = -\frac{d_a}{2}, \quad b = \frac{d_b}{2}, \quad c = a+b, \quad p = ab. \tag{G.67}$$

(These a, b, c, p and the u below are local symbols: here c is not c_T/c_B, and p and u are not the radial energy and radius of the preceding subsections.) Let $\boldsymbol{B}_n$ be the vector of $B_{n,j}$; $\boldsymbol{B}_0$ has only its entry ℓ equal to one. For each $1 \leq m \leq n$ put $u = \Delta+n-m$ and accumulate a right-hand side acting on $\boldsymbol{B}_{n-m}$:

- for even $m = 2(k+1)$, add $-4\big[u\big(-\hat{A}U_k(\hat{A}) + U_{k-1}(\hat{A}) - \tfrac{1}{2}\hat{I}\big) + U_k(\hat{A})\hat{B}\big]\boldsymbol{B}_{n-m}$, with $U_{-1}=0$;
- for odd m, add $8c\big[u\,T_m(\hat{M}) + (\hat{M}^2-\hat{I})U_{m-1}(\hat{M})\hat{D}\big]\boldsymbol{B}_{n-m}$;
- for every m, add $16p\,(-1)^{m-1} m\,T_m(\hat{M})\boldsymbol{B}_{n-m}$.

The jth component of the total right-hand side, divided by $D_{n,j}$, is $B_{n,j}$. The same quadratic Casimir eigenvalue governs both series, but the recurrences and the evaluation coordinates differ. The implementation uses matrices of size $\ell+N+8$ and returns angular spins up to $\ell+N$. Denominators smaller than 10^{-9} are treated as null candidates, whose numerator residual must be below 10^{-8}; these are floating-point consistency thresholds, not regulators of a genuine nonzero null-state obstruction, and the code fails rather than keep an inconsistent coefficient.

G.16 The retained input spectrum

Table 6 lists the central values actually evaluated, grouped by internal $\mathbb{Z}_2$ parity and by the trajectory labels `ss0`, `ee0`, `ss1`, `se0` and `other`. These labels are bookkeeping inherited from the published selection; they do not prove a classification of the CFT spectrum. All rows come from the stable-spectrum tables of Ref. [14], except that the leading scalars and the stress tensor (marked †) are replaced by the inputs of Ref. [15] and by Eq. (G.32). For even rows the columns are $(f, h) = (f_{\sigma\sigma A}, f_{\epsilon\epsilon A})$, and for odd rows the single coefficient is $f_{\sigma\epsilon B}$, all in the f normalization of Appendix G.6. The overall sign of a primary is conventional, but the signs of the even products are kept. The decimal places specify the numerical inputs, not a guaranteed physical accuracy. The package's input files also hold every quoted dimension and coupling error, the values before replacement and the source table of each row; the stress-tensor row has zero independent errors because its variations are inherited from the shared low inputs, which is not a claim that the stress couplings are exact.

Table 6: The retained input spectrum of the c_1 package: 37 internally $\mathbb{Z}_2$-even and 23 internally $\mathbb{Z}_2$-odd primaries, with central values. †: replaced by Ref. [15] (ϵ, σ) or by the Ward identity, Eq. (G.32) (T, at $\Delta = 3$).

family	ℓ	Δ	f	h
internally $\mathbb{Z}_2$*-even*: $f = f_{\sigma\sigma A}$, $h = f_{\epsilon\epsilon A}$				
`ss0`†	2	3	0.3261376622	0.8891467104
`ss0`	4	5.022665	0.069076	0.24792
`ss0`	6	7.028488	0.0157416	0.066136
`ss0`	8	9.031023	0.003685	0.017318
`ss0`	10	11.0324141	0.00087562	0.0044811
`ss0`	12	13.033286	0.00020992	0.00115174
`ss0`	14	15.033838	5.065×10^{-5}	0.00029484
`ss0`	16	17.034258	1.228×10^{-5}	7.517×10^{-5}
`ss0`	18	19.034564	2.98935×10^{-6}	1.91408×10^{-5}
`ss0`	20	21.0347884	7.2954×10^{-7}	4.8632×10^{-6}
`ss0`	22	23.034983	1.78412×10^{-7}	1.23201×10^{-6}
`ee0`	4	6.42065	0.0019552	−0.110247
`ee0`	6	8.4957	0.000472	−0.0431
`ee0`	8	10.562	0.0001084	−0.0139
`ee0`	10	12.5659	2.598×10^{-5}	−0.004437
`ee0`	12	14.633	6.1×10^{-6}	−0.001224
`ee0`	14	16.6174	1.417×10^{-6}	−0.0003791
`ee0`	16	18.678	3.547×10^{-7}	-9.72×10^{-5}
`ee0`	18	20.654	7.99×10^{-8}	-2.84×10^{-5}
`ee0`	20	22.651	1.83×10^{-8}	-7.58×10^{-6}
`ss1`	0	3.82968	0.053012	1.536
`ss1`	2	5.50915	0.0105745	0.69023
`ss1`	4	7.38568	0.00237745	0.22975
`ss1`	6	9.32032	0.00055657	0.06949
`ss1`	8	11.2751	0.00013251	0.0198
`ss1`	10	13.241	3.234×10^{-5}	0.005459
`ss1`	12	15.2301	7.64×10^{-6}	0.001538
`ss1`	14	17.1944	1.93×10^{-6}	0.000386
`ss1`	16	19.195	4.568×10^{-7}	0.0001107
`ss1`	18	21.172	1.153×10^{-7}	2.798×10^{-5}
`ss1`	20	23.167	2.74×10^{-8}	7.45×10^{-6}
`other`†	0	1.41262528	1.05185373	1.53244304
`other`	2	7.0758	0.0004773	0.21882
`other`	4	8.941	0.0001173	0.08635

family	ℓ	Δ	f	h
other	6	10.975	2.437×10^{-5}	0.02775
other	0	6.8956	0.0007338	0.1279
other	0	7.2535	0.000162	0.1874
internally $\mathbb{Z}_2$*-odd*: $f = f_{\sigma\epsilon B}$				
se0	2	4.180305	0.38915941	
se0	3	4.63804	0.1385	
se0	4	6.112674	0.1077052	
se0	5	6.709778	0.04191549	
se0	6	8.08097	0.0286902	
se0	7	8.747293	0.01161255	
se0	8	10.0623	0.00745	
se0	9	10.77075	0.003115	
se0	10	12.0492	0.00194	
se0	11	12.787668	0.000823634	
se0	12	14.0383	0.0004983	
se0	13	14.80006	0.000215	
se0	14	16.0305	0.0001291	
se0	15	16.81009	5.587×10^{-5}	
se0	16	18.025	3.13×10^{-5}	
se0	17	18.81794	1.44219×10^{-5}	
se0	18	20.01947	8.442×10^{-6}	
se0	19	20.8246	3.69×10^{-6}	
se0	20	22.0152	2.131×10^{-6}	
se0	21	22.83035	9.512×10^{-7}	
other†	0	0.518148806	1.05185373	
other	0	5.2906	0.057235	
other	2	6.9873	0.017413	

H Reproduction

Everything in this note is reproduced from the folder `SRI/` of the public repository; paths below are relative to it unless they are said to be in a package.

H.1 Layout

- `code/<pkg>/`: the code and records of each package. Production code is included verbatim, at its git commit or as a dated snapshot with a sha256 list (`public/SNAPSHOT.sha256`); the tools added for the public replay are in `public/` and `public.mk`.
  - `dual_d1`: the compendium of the dual of the chain (C_3^{SR} to C_8^{SR}, the certificates, the record of C_8^{SR}), pruned to what the replay and the cited checks need;
  - `indep_d1`: the independent re-derivation of the inputs of C_5^{SR} to C_8^{SR};
  - `integ_d1`: the magnetization-scheme calculation of C_8^{SR} (Appendix I): code, frozen records and the engine outputs they are rebuilt from;
  - `c2_closed_d2`: the final-method study of c_2 and the three Monte Carlo configurations of the review;
  - `c2_cylinder_d2`: the cylinder-Hamiltonian solver, its cluster records and the continuum analysis;
  - `c1_d3`: the c_1 package (review revision of 8 October 2026) with the review's two re-implementations in `review_indep/`;

  - `closure_d1`: the checks of the two arguments of Appendix D (no quoted number): an independent check of the closure argument's finite algebra (matrix-word charges, the radial-power identity, the $\mathcal{O}_g\mathcal{O}_g$ product), and in `prescription/` the prescription argument's own checker (rerun by `make test`) with an independent re-derivation of the β recurrence.

- `docs/<pkg>/`: the documentation of each package (README, REPRODUCE, PROVENANCE, DATA, REVIEWS, requirements, `results_map.md` with every quoted number, its record, rule and tolerance, and TESTS with the export tests on the symmetry cluster).
- `results/quoted_numbers.json`: every quoted number of the paper with its macro and its text as printed, the TeX of the two closed-form equations of the dual, and the cells of the table of short-range coefficients; `results/sr_values.json`: machine-readable values for other folders (Sec. H.4).
- `verify/`: `sri_lib.py` (the replay helper: PASS/FAIL/INFO/SKIP table, rounding rule) and `check_sr_values.py`; `note/`: this note and `tools/check_note.py`.

H.2 Commands and cost

From `SRI/`:

```
make test          # fast tests of every package              about 12 min
make replay        # every quoted number from shipped records about 17 min
make fetch-data    # the hosted inputs (Zenodo data record; see below)
make replay-full   # the heavier reruns                       about 35 min
make check-values  # results/sr_values.json against the tree
make note          # tools/check_note.py of this note
make -C code/<pkg> -f public.mk replay   # one package (also test, replay-full)
```

Measured on the symmetry cluster (seconds, Intel Xeon Gold 6148 / AMD EPYC 7532 nodes, Python 3.12, unpacked in node-local storage; `docs/<pkg>/TESTS_<pkg>.md`):

package	`test`	`replay`	`replay-full`
`dual_d1`	23 / 26	65 / 70	544 / 556 (a)
`indep_d1`	258 / 308 (b)	229 / 264	(c)
`integ_d1`	249 / 249 (b)	78 / 85	(c)
`c2_closed_d2`	4 / 4	39 / 41	–
`c2_cylinder_d2`	170 / 152	3 / 3	10 / 9
`c1_d3`	16 / 14	469 / 484 (d)	1474 / 1486

(a) After `fetch-data`. (b) On 8 cores. (c) Prints the production rerun. (d) Plus 66 s (Intel) for the two re-implementations in `review_indep/`. Further targets: `indep_d1 replay-raw RAW=<dir>` (349 s; the assembly from the raw per-sector engine outputs), `c2_closed_d2 rerun-mc-smoke` (24 s) and `mc-batch-check` (169 s; reruns the first batch of one Monte Carlo chunk with the production flags; bitwise identical on AMD, within 1.6×10^{-11} of the largest entry on Intel).

Rule. A replay recomputes each quoted number of its package from shipped records, rounds it half-up to the precision at which the paper prints it, and compares it with the printed text in `results/quoted_numbers.json` (for a value with its uncertainty in parentheses, the uncertainty is in units of the last digit). It prints a PASS/FAIL table and exits non-zero on any FAIL. Numbers that the note prints but the paper does not, and literature values, give INFO rows. The regenerated records are compared with the shipped ones within the tolerances of Table 7.

H.3 Map of the quoted numbers

Table 7 gives, for each result, the package, the record (path in the package), the target that checks it and the tolerance beyond the rounding rule; `results/quoted_numbers.json` gives the paper's macro for each. In the table and below, G stands for `results/continuum_2026-10/generations/20261015T000632Z_99019_1e66b9`, the current generation of the cylinder analysis (named by `results/continuum_2026-10/CURRENT`).

Table 7: Reproduction map. Target: `make -C code/<package> -f public.mk <target>`, or `make <target>` from `SRI/`.

result	package, target	record (in the package)	tolerance
C_1^{SR}, C_2^{SR}	`dual_d1`, replay	Refs. [3, 4]; registry `data/coefficients.json`	INFO (literature)
C_3^{SR} to C_7^{SR}	`dual_d1`, test, replay	Eq. (2.12) with mpmath at 50 digits; `data/coefficients.json`	rounding
Eq. (2.12)	`dual_d1`, test, replay	`data/coefficients.json`, the closed forms as the compendium's TeX states them (verbatim blocks in `public/compendium_extract.json`), the candidates proved by the C_6^{SR}, C_7^{SR} input maps	exact equality of every rational coefficient
certificates of C_5^{SR} to C_7^{SR}	`dual_d1`, replay; replay-full	C_5^{SR} (D_3) and C_6^{SR} certificates, C_6^{SR} and C_7^{SR} input maps, C_7^{SR} certificate at weights 2 to 5 (replay); the complete certificate task, with the hosted weight-6 part (replay-full)	exact zero residual (exit 0); input-map reports equal the recorded ones
$\pi_0 C_8^{\mathrm{SR}} = 88$	`dual_d1`, test, replay	`data/archived/C8_rational_completion/result.json /C8_weight_zero`	exact: the record, the full-Jacobian script on the 35 rational inputs and two separate rational sums agree
C_8^{SR}	`dual_d1`, test, replay	`data/archived/C8_zeta_candidate/result.json /C8_assembled_without_candidate`	rounding; recomputed from its 36 inputs within 10^{-95}
"about 85 digits"	`dual_d1`, test	the record and its inputs	printing floor about 2×10^{-85} (INFO)
Eq. (2.15)	`dual_d1`, test, replay	`data/coefficients.json` ($n = 8$), `result.json`	exact coefficients; residual 3.085×10^{-86} below the printing floor $10^{-(\mathrm{digits}-15)}$
fresh P_6 evaluations	`dual_d1`, replay-full	the 66 404-term P_6 expression at 70 and 100 digits	
largest input pull through C_7^{SR}	`indep_d1`, replay	`RESULTS_independent.json` against `spec/comparison/archive_comparison.json`; `RESULTS_archive_comparison.json`	INFO (not printed by the paper; $0.915 \to 0.9$); regenerated records below
C_8^{SR} independent and its pull	`indep_d1`, replay (replay-raw)	`RESULTS_C8_independent.json /step1/outputs/C8`; `RESULTS_C8_archive_comparison.json`; `public/pooled_sector_record.json`	INFO (not printed by the paper); central values 10^{-24} relative, linear and replicate errors 10^{-12}, sector-pooled errors 10^{-6} of the quoted error

result	package, target	record (in the package)	tolerance
C_8^{SR}, magnetization scheme (Eq. (2.17))	`integ_d1`, replay	`results/C8_record_v2.json` `/C8`, `/C8_error`; every record from two bubbles to C_8^{SR} rebuilt from `results/m*_runs/` by the verbatim gates	rounding; records byte-identical (sha256)
numbers of Appendix I (inputs, gates, comparison)	`integ_d1`, replay	the records and the outputs of the verbatim tools	25-digit inputs equal to the record; "to 10^{-n}" means in $[10^{-n-1/2}, 10^{-n+1})$; counts equal
engine builds and production sectors	`integ_d1`, test	`results/builds.json`; sectors of four production chunks rerun	build ids equal; $\lvert\mathrm{rerun} - \mathrm{shipped}\rvert \le 4\varepsilon B$ per coefficient (B the engine's bound)
c_2 closed form (Eq. (3.9))	`c2_closed_d2`, test, replay	`outputs/summary.json /C2`; the study's eight-stage rerun	rounding; 50-digit evaluation within 10^{-45}
c_2 Monte Carlo	`c2_closed_d2`, test, replay	`mc/run_{A,B,C}/out_*.txt` (142 chunks); `public/mc_combined.json` `/combined/C2`, `/combined/C2_se`	rounding; recombination 10^{-12} relative; `summ.py` output equal to the production logs
c_2 cylinder, its relative difference from the closed form and $N_{\max}$ (also the c_2 row of the paper's table of perturbative Hamiltonian truncation)	`c2_cylinder_d2`, replay	`G/C2_estimate.json` `/final/C2`, `/final/dC2`, from `G/continuum_points.json`	rounding ($N_{\max}$: INFO, not printed by the paper); fits recomputed twice (verbatim and independent) within 10^{-9} relative
same, from the raw records	`c2_cylinder_d2`, replay-full	16 production, 25 extraction-check and 17 archived records, auxiliary inputs	discrete choices equal; chain 10^{-7} absolute; fit diagnostics $10^{-6} + 10^{-3}\lvert x\rvert$
c_1 ($d = 3$)	`c1_d3`, replay	`outputs/tables/current_summary.json /retained/C1`; the 20-stage workflow rerun from scratch	rounding; rerun within 5×10^{-10}
c_1 estimated range	`c1_d3`, replay	`refinement/final_8_7_pad1_summary.json` and `refinement/final_8_7_pad2_summary.json` (`/interval`) of the rerun; `current_summary.json` `/recommended_working_summary/interval`	rounding; envelopes within 2×10^{-8}
two independent implementations of c_1 (Table 2)	`c1_d3`, replay	`review_indep/` (two scripts) on `data/inputs/retained_spectrum.json`	each within 10^{-12} of the package; their agreement, a bound, INFO (not printed by the paper)
feasibility and sensitivity (Sec. 4)	`c1_d3`, replay-full	the review-revision diagnostics and the 11 gap and margin and 8 box scenarios	status and endpoints within 2×10^{-8}
c_1 BRRZ	`c1_d3`, replay	Ref. [12]	INFO (literature)

result	package, target	record (in the package)	tolerance
numbers of orders (eight computed, seven in closed form, two published)	`dual_d1`, test	`data/coefficients.json`	equal; INFO for those the paper does not print

H.4 Values for other folders

`results/sr_values.json` publishes each value with all the digits of its record, its status, and the file and JSON pointer where it lives in the tree; `make check-values` resolves every entry. The stable pointers are

pointer	quantity	file (in `SRI/code/`)
`/d1/C/1` to `/d1/C/7`	C_1^{SR} to C_7^{SR}	`dual_d1/data/coefficients.json`
`/d1/C/8`	C_8^{SR} (record)	`dual_d1/data/archived/C8_zeta_candidate/result.json`
`/d1/magnetization/C8`	C_8^{SR}, magnetization scheme	`integ_d1/results/C8_record_v2.json`
`/d2/c2/closed`	c_2, closed form	`c2_closed_d2/outputs/summary.json`
`/d2/c2/mc`	c_2, Monte Carlo	`c2_closed_d2/public/mc_combined.json`
`/d2/c2/cylinder`	c_2, cylinder	`c2_cylinder_d2/G/C2_estimate.json`
`/d3/c1/value`	c_1	`c1_d3/outputs/tables/current_summary.json`
`/d3/c1/range`	estimated range of c_1	`c1_d3/outputs/tables/current_summary.json`
`/d3/S`, `/d3/E_SR`	$\Delta_\sigma^{\mathrm{SR}}$, $\Delta_\mathcal{E}^{\mathrm{SR}}$ ($d = 3$)	`c1_d3/data/inputs/low_inputs.json`

The resummation folder reads C_1^{SR} to C_8^{SR}, c_2 (closed form), c_1 with its estimated range, and the two $d = 3$ inputs at these pointers.

H.5 What cannot be reproduced

- *The compendium's raw integrals cannot be regenerated*, from this folder or from the compendium's original archive: 21 ordered-integration stage programs are missing from the archive, so the documented raw regenerations stop at the first missing stage. What is offered instead is the replay of every quoted number from the shipped records, the exact certificate checks of C_5^{SR} to C_7^{SR}, and the independent re-derivation of every input of C_5^{SR} to C_8^{SR} (`indep_d1`).
- Six inputs of C_8^{SR} ($W, S, L_3, M_{21}, M_{12}, N_2$) are single numerical records with no exact expression; no C_8^{SR} certificate exists.
- Hosted inputs: the weight-6 C_7^{SR} certificate (57 MB, needed only by `dual_d1 replay-full`) and the raw per-sector outputs of `indep_d1` (131 MB, as a 31 MB archive; needed only by `replay-raw`) are in the project's Zenodo data record, DOI 10.5281/zenodo.23466766; `make fetch-data` downloads and checks them (the certificate can also be taken from the compendium's original archive).
- `dual_d1` needs a case-sensitive work directory: the compendium has two input files whose names differ only in case.
- The base study's checks of the closed expression of c_2 (the integer-interval evaluation to 51 decimal places, its tail inequalities, elliptic integration and local checks; Appendix F.14) are described but their code is not in the release; the value of the closed expression is reproduced

by `c2_closed_d2` (50 digits, not as an interval enclosure). The closure argument's own script of finite checks was not supplied and is not part of the release (Appendix D.15).

- The release has no git history, so it cannot show the order in which the magnetization-scheme calculation committed its designs and its data (Appendix I.7): `integ_d1` checks that the files its tools ask git about equal the committed blobs, and the code inspections checked the order on the history.
- Production reruns (not needed for any replay): `indep_d1` about 2.2×10^4 core-hours on the symmetry cluster; the engine runs of `integ_d1` about 1.6×10^4 core-hours on the symmetry cluster; the Monte Carlo of `c2_closed_d2` about 135 core-hours; the cylinder records of `c2_cylinder_d2` about 1.5 node-hours on the symmetry cluster, about 470 service units on SDSC Expanse, and, again on the symmetry cluster, the two rebuilt $N = 1024$ grids (2.6 to 2.7 hours each) and twelve extraction-check jobs (0.3 to 1.6 hours each). A Monte Carlo rerun is statistically equivalent, not identical, unless the same binary, weights and seeds are used.

H.6 Numbers of this note that the paper does not quote

The decimal strings of Table 2 and Sec. 3.5 are evaluations of the closed forms (mpmath, 60 digits; the same digits as `results/sr_values.json`). The inputs of Table 3, the pulls, the estimators and the checks of Sec. 2.6 are in the result records of `code/indep_d1/` (and its `RESULTS.md`, `RESULTS_C8.md`); the eighth-order sub-results of Sec. 2.5 in those of `code/dual_d1/`; the constants of Sec. 3.5 in `code/c2_closed_d2/outputs/summary.json`; the per-order table and the error budget of Sec. 3.8, and the superseded first analysis, in `G/C2_estimate.json`, `G/REPORT.md` and `results/continuum_2026-10/evidence/rev4/` of `code/c2_cylinder_d2/`; and the intervals and sensitivity tables of Sec. 4 in `code/c1_d3/outputs/tables/` (the box scan of Sec. 4 at $m = 2$ is the package's; the review's own scan gave $[0.26698, 0.27693]$). The NRG residuals of Sec. 2.9 are checked in the NRG folder of the repository [2]. The magnetization-scheme values of Sec. 2.7 and Appendix I other than the one the paper quotes are in the records of `code/integ_d1/results/` (Appendix I.8), and the replay of `integ_d1` checks each of them.

I The eighth order from the integrable point: the magnetization scheme

This appendix gives the third calculation of C_8^{SR} summarized in Sec. 2.7: its derivation, method, checks and records. It starts from the wall gas of Appendix A.3, with the wall weight W_n of Eq. (A.31), but it uses nothing of the matched two-coupling prescription of Sec. 2.1 (no hard core, no finite interval, no endpoint source) and no input of the compendium or of the independent re-derivation. Its code and records are the package `code/integ_d1/` (Appendix H). Its result is numerical in the sense of Sec. 1.3.

Notation. As elsewhere $\bar\delta = 1 - s$; v is the charge of the wall weight W_n (at $\bar\delta = 0$ a pair of opposite walls at distance r has weight r^{-2v}), and $w = 1 - v$. In this appendix h_{m} is a uniform magnetic field (not the current coupling h of the dual), $\bar m$ is the magnetization, the mean of $m(X)$, ξ is the expansion variable of the magnetization, $\hat u$, $\hat z$ and $\hat F$ are the couplings and the beta function of the magnetization scheme (Appendix I.4), and f_{ijk} is the coefficient of $\hat u^i \hat z^j \bar\delta^k$ in $\hat F$. A "bubble" is a pair of consecutive opposite walls, which bounds one reversed domain.

I.1 The integrable point

At $s = 1$ the wall gas, with the weights $\prod_{i<j}(X_j - X_i)^{2vq_iq_j}$, is the Coulomb gas of Anderson, Yuval and Hamann for the anisotropic Kondo model [5, 6], and equivalently that of the ohmic spin–boson model with dissipation strength $\alpha = v$ [7]. These models are solvable by Bethe ansatz [8, 9]. For $\bar\delta > 0$ the interaction $f_{\bar\delta}$ of Eq. (A.5) breaks the integrability, and the long-range fixed point lies at $\bar\delta > 0$. The integrable point can therefore supply the $\bar\delta = 0$ part of the flow exactly, to all orders in the wall fugacity, while the dependence on $\bar\delta$ needs perturbation theory in kink clusters.

The observable is the magnetization of the gas on the infinite line in a uniform field $h_{\mathrm m}$ coupled to $m(X)$. Relative to the all-up configuration, a configuration of walls carries the extra weight $e^{-2h_{\mathrm m}\Xi}$, where Ξ is the total length of the reversed domains (as in Ξ_{2m} of Eq. (A.31)). No hard core is imposed: the short-distance divergences near $v = 1$ are regulated by analytic continuation in v (Appendix I.5).

I.2 The magnetization at $s = 1$ from the Bethe ansatz

Bethe-ansatz equations. The spin sector of the anisotropic Kondo model is an inhomogeneous six-vertex (XXZ) chain in its gapless regime, with anisotropy γ, N bulk sites at rapidity 0 and the impurity at rapidity Θ; $\Theta \to \infty$ is the weak-coupling scaling limit [8, 9]. With the kernels

$$a_n(\theta) = \frac{1}{2\pi}\,\frac{\gamma\sin(n\gamma)}{\cosh(\gamma\theta) - \cos(n\gamma)}, \qquad \hat a_n(\omega) = \int a_n(\theta)\,e^{\mathrm{i}\omega\theta}\,\mathrm{d}\theta = \frac{\sinh\big((\pi - n\gamma)\omega/\gamma\big)}{\sinh(\pi\omega/\gamma)}, \tag{I.1}$$

the roots fill $\theta < \theta_F$ in a field, and the impurity root density σ_{imp} solves

$$\sigma_{\mathrm{imp}}(\theta) + \int_{-\infty}^{\theta_F} a_2(\theta - \theta')\,\sigma_{\mathrm{imp}}(\theta')\,\mathrm{d}\theta' = a_1(\theta - \Theta) \quad (\theta < \theta_F), \qquad \bar m = 1 - 2\int_{-\infty}^{\theta_F}\sigma_{\mathrm{imp}}(\theta)\,\mathrm{d}\theta. \tag{I.2}$$

The impurity magnetization is $\frac12 - \int\sigma_{\mathrm{imp}}$, and $\bar m \in [0,1]$ is twice it. The zero-field bulk density is $1/(4\cosh(\pi\theta/2))$, so spinon energies scale as $e^{-\pi\theta/2}$, and $h_{\mathrm m}/T_K \propto e^{-\pi L/2}$ with $L = \theta_F - \Theta$ and T_K the Kondo scale.

Wiener–Hopf solution. With $x = \theta_F - \theta$ and $\varsigma(x) = \sigma_{\mathrm{imp}}(\theta_F - x)$, Eq. (I.2) becomes $\varsigma(x) + \int_0^\infty a_2(x-y)\varsigma(y)\,\mathrm{d}y = a_1(L - x)$ for $x > 0$. With $p = \pi/\gamma$ the kernel factorizes,

$$\begin{aligned} 1 + \hat a_2(\omega) &= \frac{2\sinh((p-1)\omega)\cosh\omega}{\sinh(p\omega)} = \Phi_+(\omega)\Phi_-(\omega), \\ \Phi_-(\omega) &= \sqrt{\frac{2\pi(p-1)}{p}}\,\frac{\Gamma(1 + \mathrm{i}p\omega/\pi)\,e^{-\mathrm{i}\omega c/\pi}}{\Gamma(1 + \mathrm{i}(p-1)\omega/\pi)\,\Gamma(\frac12 + \mathrm{i}\omega/\pi)}, \end{aligned} \tag{I.3}$$

with $c = p\log p - (p-1)\log(p-1)$ and $\Phi_+(\omega) = \Phi_-(-\omega)$, which tends to 1 at infinity. The standard argument gives $\int_0^\infty\varsigma = [e^{\mathrm{i}\omega L}\hat a_1/\Phi_-]_+(0)/\Phi_+(0)$, where $[\cdot]_+$ is the part analytic in the upper half plane. With $\omega = \mathrm{i}\pi t/p$ the Gamma functions combine into one Mellin–Barnes integral,

$$\bar m = 1 + \frac{1}{\sqrt\pi}\,\frac{1}{2\pi\mathrm{i}}\int_{c_0-\mathrm{i}\infty}^{c_0+\mathrm{i}\infty}\frac{\Gamma(t)\,\Gamma\big(\frac12 - (1-v)t\big)}{\Gamma(1+vt)}\,\xi^{-t}\,\mathrm{d}t, \qquad -1 < c_0 < 0, \tag{I.4}$$

with $v = 1 - \gamma/\pi$ and $\xi = e^{\gamma(L + c/\pi)} \propto (h_{\mathrm m}/T_K)^{-2(1-v)}$. The identification $v = 1 - \gamma/\pi$ is forced by the exponent: perturbation theory in the wall fugacity y proceeds in powers of $y^2 h_{\mathrm m}^{-2(1-v)}$, the Bethe ansatz in powers of $h_{\mathrm m}^{-2\gamma/\pi}$. The free-fermion point $\gamma = \pi/2$ is then the Toulouse point $v = \frac12$, and the isotropic limit $\gamma \to 0$ is $v \to 1$.

The two expansions. Closing the contour to the left picks up the poles of $\Gamma(t)$ at $t = -n$ and gives the large-field series

$$\bar{m} = 1 - \frac{1}{\sqrt{\pi}} \sum_{n\geq 1} \frac{(-1)^{n+1}\,\Gamma\big(\frac{1}{2} + n(1-v)\big)}{n!\,\Gamma(1-nv)}\,\xi^n. \tag{I.5}$$

Closing to the right picks up the pole of $\Gamma(t)$ at $t = 0$, which cancels the 1, and the poles at $t = s_m = (m + \frac{1}{2})/(1-v)$, and gives a series odd in h_{m},

$$\bar{m} = \frac{1}{\sqrt{\pi}} \sum_{m\geq 0} \frac{(-1)^m\,\Gamma(1+s_m)}{m!\,(m+\frac{1}{2})\,\Gamma(1+vs_m)}\,\xi^{-s_m}. \tag{I.6}$$

At $v = 0$ Eq. (I.5) sums to $\bar{m} = (1+\xi)^{-1/2}$, the free spin. At $v = \frac{1}{2}$ only odd powers survive, since $1/\Gamma(1-\frac{n}{2}) = 0$ for even n, and $\bar{m} = \frac{2}{\pi}\arctan(h_{\mathrm{m}}/\Gamma_0)$, the resonant level of width Γ_0.

Checks.

- *The Wiener–Hopf algebra.* The integral equation (I.2) was solved numerically by Nyström discretization and compared with Eq. (I.4), including the shift c. At $\gamma = 0.4$, 0.7 and 1.2 ($v = 0.873$, 0.777, 0.618) and $L = -6$, -2, 0, 2 and 6, where $\bar{m}$ runs from 0.9996 to 4×10^{-5}, the differences in $\bar{m}$ are at most 8×10^{-13}.
- *Against the wall gas, with no Bethe ansatz.* The ratio $R = c_2/c_1^2$ of the coefficients of ξ^2 and ξ in Eq. (I.5) does not depend on the normalization of ξ:

$$R_{\mathrm{BA}} = \frac{\sqrt{\pi}\,\Gamma(\frac{5}{2}-2v)\,\Gamma(1-v)^2}{2\,\Gamma(1-2v)\,\Gamma(\frac{3}{2}-v)^2}. \tag{I.7}$$

  On the gas side R is the connected two-bubble cluster with the field weight,

$$R_{\mathrm{gas}} = \frac{(4v-3)\,A_2}{2\,\Gamma(2-2v)^2}, \qquad A_2 = \Gamma(3-4v)\int_0^1 \big(t(1-t)\big)^{-2v}\Big[I(t) - \frac{1}{2}\Big]\,\mathrm{d}t,$$
$$I(t) = \int_0^\infty \Big[\frac{(t+g)^{2v}(g+1-t)^{2v}}{g^{2v}(1+g)^{2v}} - 1\Big]\,\mathrm{d}g, \tag{I.8}$$

  which converges for $v < \frac{1}{2}$ (where R_{BA} is the analytic continuation of the formula). The two agree exactly at $v = 0$ ($R = \frac{3}{2}$) and to 2×10^{-13}, 3×10^{-13}, 4×10^{-12}, 4×10^{-10} and 1×10^{-9} at $v = 0.1$, 0.2, 0.3, 0.4 and 0.45, each within the quadrature error of the gas integral.
- *Higher clusters.* The later stages reproduce the exact two-, three- and four-bubble clusters at $\bar{\delta} = 0$ (Appendix I.6).

The derivation covers $\frac{1}{2} < v < 1$ directly (real roots, $0 < \gamma < \frac{\pi}{2}$); for $v < \frac{1}{2}$ the formula is the analytic continuation in v, which the second check tests. The Bethe-ansatz equations and the identification of the gas with the anisotropic Kondo model are taken from the literature; everything from the impurity integral equation on is derived here.

I.3 What the integrable point can and cannot test

At $\bar{\delta} = 0$ the change $u \to H/4$ brings the flow of Sec. 2.2 to coordinates in which $H = 4u$ exactly. The freedom that remains is the shifts $z \to z + u\,\chi(u,z)$, with u readjusted so that $H = 4u$ still holds. Write F_{ij} for the coefficient of $u^i z^j$ of F in these coordinates; $F(0,z) = z$. Through total degree four in (u,z), the order that C_8^{SR} needs, F has ten coefficients with $i \geq 1$. The linearized action of the shifts on them has rank eight (in exact arithmetic, at random flows), so exactly two combinations are invariant:

$$F_{10} = 1, \qquad I_2 = F_{20} - \tfrac{1}{2}F_{11} - F_{12}. \tag{I.9}$$

In the inputs of the hard-core scheme (Sec. 2.3), $I_2 = b + \frac{f}{4} - \frac{a}{2} - \frac{c}{8} - A$. With the closed forms of a, b, c, f (Sec. 2.3) and $A = -8 - 4\zeta(2) - 2\zeta(3) + 16\log 2$ (Appendix B.3) it vanishes identically: the rational, $\zeta(2)$, $\zeta(3)$ and $\log 2$ parts cancel separately, and $\log 3$, $\log^2 2$ and $\mathrm{Li}_2(-\frac{1}{2})$ cancel between b and $f/4$. The Bethe-ansatz flow of Appendix I.4, whose $\bar\delta = 0$ part is already in these coordinates, gives $F_{10} = f_{100} = 1$ and $I_2 = f_{200} - \frac{1}{2}f_{110} - f_{120} = 0$ (to 10^{-54} in the first fit and 10^{-63} in the frozen matching). The agreement is a check of the compendium's $s = 1$ sector by integrability.

All other $s = 1$ coefficients through degree four can be changed by the shifts: among them J_9, and with it $A_9[\varepsilon^0\ell^1]$, and the current-type inputs L_3, N_2 and K_v of C_8^{SR} (the direction of J_9 lies in the span of the linearized action). They are coordinate choices of the hard-core scheme, and only C_8^{SR} as a whole is physical. The $s = 1$ solution alone therefore cannot test A_9 or J_9; the calculation below instead computes C_8^{SR} in its own scheme.

I.4 The magnetization scheme

Couplings. For the gas of bare charge v in the field h_{m}, at $0 < \bar\delta < 1$, define

$$K_h = v\,(2h_{\mathrm{m}})^{-\bar\delta}, \qquad 1 + \hat z = K_h\,\bar m^{\,2-\bar\delta}, \qquad \hat u = \frac{2-\bar\delta}{4}\,\frac{\mathrm{d}\log\bar m}{\mathrm{d}\log h_{\mathrm{m}}}. \tag{I.10}$$

Then, identically by these definitions,

$$h_{\mathrm{m}}\frac{\mathrm{d}\hat z}{\mathrm{d}h_{\mathrm{m}}} = (1+\hat z)\,\hat H, \qquad \hat H = 4\hat u - \bar\delta, \tag{I.11}$$

and the only function to be computed is $\hat F$ in

$$h_{\mathrm{m}}\frac{\mathrm{d}\hat u}{\mathrm{d}h_{\mathrm{m}}} = 2\hat u\,\hat F(\hat u,\hat z,\bar\delta), \qquad \hat F = \hat z + \sum_{i+j+k\ge 1} f_{ijk}\,\hat u^i\hat z^j\bar\delta^k, \qquad f_{0j0} = 0, \quad f_{100} = 1. \tag{I.12}$$

The flow is in $\log h_{\mathrm{m}}$, and Eqs. (I.11) and (I.12) have the form of Eq. (A.23) with $(\hat u,\hat z,\hat F,\hat H)$ in place of (u,z,F,H).

Fixed point and invariants. $\hat H = 0$ gives $\hat u_* = \bar\delta/4$ exactly, and $\hat F(\bar\delta/4,\hat z_*,\bar\delta) = 0$ fixes $\hat z_*$. Since $\hat H_{\hat z} = 0$ and $\hat H_{\hat u} = 4$, the trace and determinant of Eq. (A.48) per unit $\log h_{\mathrm{m}}$ are $T_h = \frac{\bar\delta}{2}\hat F_{\hat u}$ and $D_h = -2\bar\delta(1+\hat z_*)\hat F_{\hat z}$ at the fixed point. The field has dimension $1 - \Delta_\sigma = 1 - \bar\delta/2$, so per unit logarithm of length

$$T = \Big(1 - \frac{\bar\delta}{2}\Big)\frac{\bar\delta}{2}\,\hat F_{\hat u}, \qquad D = -\Big(1 - \frac{\bar\delta}{2}\Big)^2 2\bar\delta\,(1+\hat z_*)\,\hat F_{\hat z}, \tag{I.13}$$

and Eqs. (2.6) and (2.8) give the C_n^{SR}. The leading $\hat F = \hat z + \hat u$ gives $T = \bar\delta/2 + O(\bar\delta^2)$ and $D = -2\bar\delta + O(\bar\delta^2)$, hence $C_1^{\mathrm{SR}} = -2$, $C_2^{\mathrm{SR}} = \frac{1}{2}$. The next orders are, for example,

$$C_3^{\mathrm{SR}} = \frac{39}{16} + 2f_{001} - 2f_{011} - \frac{1}{2}f_{110}, \qquad C_4^{\mathrm{SR}} = f_{101} + \frac{1}{2}f_{200} - f_{001}f_{110} - \frac{1}{4}f_{110} - \frac{1}{2}, \tag{I.14}$$

and C_8^{SR} is affine in its two new coefficients, $\partial C_8^{\mathrm{SR}}/\partial f_{202} = 2$ and $\partial C_8^{\mathrm{SR}}/\partial f_{301} = \frac{3}{4}$; the $\bar\delta = 0$ coefficient f_{400} enters as $f_{400}/4$, the counterpart of $J_9/4$ in Eq. (2.10).

Which coefficients, from how many kinks. $\hat u \propto y^2$ counts bubbles, so f_{ijk} comes from connected clusters of $i+1$ bubbles, that is $2i+2$ kinks; k is the number of logarithm insertions from expanding $f_{\bar\delta}$, and the dependence on $\hat z$ comes from the expansion in w about $v = 1$. C_8^{SR} needs the f_{ijk} with $i+j+k \le 4$, 29 of them:

coefficients	content	kinks (integration dimension)	first needed for
f_{ij0}	$\bar\delta = 0$, from Eq. (I.5)	none	C_3^{SR}
f_{0jk}, $k \geq 1$	one bubble, all orders in $\bar\delta$	2 (closed form in derivatives of Γ)	C_3^{SR}
f_{101}; f_{111}; $f_{121}, f_{102}, f_{112}, f_{103}$	two bubbles, one to three logarithms	4 (2D)	C_4^{SR}; C_5^{SR}; C_6^{SR} to C_8^{SR}
f_{201}, f_{211}	three bubbles, one logarithm	6 (4D)	C_6^{SR}, C_7^{SR}
f_{202}	three bubbles, two logarithms	6 (4D)	C_8^{SR}
f_{301}	four bubbles, one logarithm	8 (6D)	C_8^{SR}

The $\bar\delta = 0$ coefficients follow from Eq. (I.5): with $K_h = v$ at $\bar\delta = 0$, Eq. (I.10) expresses $\hat u$ and $\hat z$ through ξ at $v = 1 - w$, and eliminating ξ gives $\hat F(\hat u, \hat z, 0)$ to all orders; through degree four the overdetermined matching has residual 10^{-63}. They include f_{400}: the $s = 1$ part of C_8^{SR} is exact in this scheme, and no nine-kink integral is needed. No current-type input is needed either, since $\hat H$ is exact.

I.5 Kink clusters and their continuation to $v = 1$

Clusters. The $\bar\delta$-dependent coefficients come from the connected Mayer clusters of n bubbles per unit length (field scaled to $2h_{\mathrm{m}} = 1$),

$$J_n = \frac{1}{n!} \int \prod_{a=1}^{n} \mathcal{W}_a \sum_{\text{connected graphs}} \prod_{(ab)} \varphi_{ab}, \tag{I.15}$$

where $\mathcal{W}_a$ is the weight of bubble a (its pair factor and field factor), and the bond φ_{ab} is $C_{ab} - 1$ for disjoint bubbles, with C_{ab} the cross-pair factor of W_n between them, and -1 for overlapping ones, which alternation excludes. The expansion of $f_{\bar\delta}$ to order k gives k logarithm insertions. A symbolic map, derived once per order, turns the Laurent coefficients of the J_n at $v = 1$ into the f_{ijk}.

Radial factor and sectors. Every monomial of the integrand is homogeneous of degree $-2nv$ in the gaps (the cross-pair factors are scale invariant), so after the overall size is scaled out the radial integral is exact: $J_n^{(k)} = \sum_j \Gamma^{(j)}(\alpha_n) A_j^{(k)}$ with $\alpha_n = 2n - 1 - 2nv$, the logarithm insertions giving the derivatives $\Gamma^{(j)}$, and $A_j^{(k)}$ angular integrals over a simplex of dimension $2n - 2$. The angular domain is split into primary Cheng–Wu sectors (the largest gap set to 1), refined by iterated Binoth–Heinrich splitting until every sum of gaps is a monomial times a positive function: 187 sectors for three bubbles and 4777 for four, of which 3445 are computed, using a mirror symmetry of the bubble patterns.

Exponents. In a sector every term is $\prod_k t_k^{a_k + b_k w}$ times an analytic function. A set G of gaps that shrinks together at rate ρ, possibly as several disjoint contacts, contributes $\rho^{a_G + b_G w}$ with $a_G = (|G| - 1) + 2S_G$ at $v = 1$, where S_G is the sum of $q_i q_k$ over all pairs whose separation shrinks with ρ. A single kink–antikink contact gives ρ^{-2+2w}; two simultaneous contacts give ρ^{-3+4w} and three ρ^{-4+6w}. Every Taylor term whose integrated exponent is at most -1 at $w = 0$ is continued analytically, which produces poles in w. No term has $a_k \leq -1$ with $b_k = 0$ (a divergence independent of v) for $n \leq 4$.

Continuation. Each sector variable has $[0, 1] = [0, \tau] \cup [\tau, 1]$. On $[\tau, 1]$ the integral is done by Gauss–Legendre quadrature in $\log t$. On $[0, \tau]$ the Taylor coefficients of the analytic part are

obtained by the trapezoid rule on the circle $|t| = \tau$ and integrated analytically against $t^{c+n} \log^p t$, which continues every non-integrable power. The radius τ is set by the analyticity radius of the regular part: sampled for three bubbles, and for four bubbles a rigorous radius from Rouché's theorem with a joint polydisc check. All quantities are truncated Laurent series in w, so the Laurent coefficients of the clusters at $v = 1$ come out directly. The engine is written in C++, in double-double arithmetic (about 32 digits) for three bubbles; for four bubbles it runs in double precision with double-double accumulators, the largest-bound sectors entirely in double-double. It gives a first-order roundoff bound for every sector.

What the poles test. The raw Laurent coefficients carry information in their pole parts as well as their finite parts, and all of them are computed. The checks are that the f_{ijk} have no negative powers of w (the analyticity relations, listed symbolically for each order), the exact $\bar{\delta} = 0$ clusters from Eq. (I.5), and the closed forms of C_3^{SR} to C_7^{SR}.

I.6 Gates and results

The plan fixed the stages, their checks and their stopping rules before any multi-bubble cluster with logarithm insertions was computed; each stage gated the next.

- *One bubble.* The poles cancel in $\hat{F}(0, \hat{z}, \bar{\delta})$ through $\bar{\delta}^3$, and $f_{001} = -\gamma_E$.
- C_3^{SR}. With $f_{001} = -\gamma_E$, $f_{011} = -\gamma_E - 2\zeta(2)$ and $f_{110} = 4\zeta(2)$, $C_3^{\mathrm{SR}} = \frac{39}{16} + 2\zeta(2)$, to 2×10^{-38} (absolute) numerically.
- *Two bubbles (C_4^{SR}, C_5^{SR}).* The cluster at $\bar{\delta} = 0$, continued to $v = 1$, matches the Bethe ansatz to 4×10^{-32}, and all 36 analyticity relations hold to 10^{-28}. The value of f_{101} implied by C_4^{SR} was printed in advance as a blind check; the computation gave $f_{101} = -36.49853744243762\ldots$. C_4^{SR} agrees to 3×10^{-28} and C_5^{SR} to 10^{-26} (absolute).
- *Three bubbles, one logarithm (C_6^{SR}, C_7^{SR}).* The engine matches the exact three-bubble cluster to 10^{-23} at real v and 10^{-28} at complex v, and the analyticity relations hold to 10^{-27}. With the results hashed before the comparison, C_6^{SR} agrees to 4×10^{-24} and C_7^{SR} to 2×10^{-24} (absolute).
- *Three bubbles, two logarithms (f_{202}).* The $\bar{\delta}^0$ part matches the Bethe ansatz to 4×10^{-23}, the $\bar{\delta}^1$ part the frozen one-logarithm record to 9×10^{-24}, and all twelve $\bar{\delta}^2$ analyticity relations hold to 4×10^{-24}. $f_{202} = 172415.83018563 \pm 1.7 \times 10^{-7}$, the difference between two quadrature grids.
- *Four bubbles, one logarithm (f_{301}).* Three complete runs: a fine and a coarse grid at contour-radius ratio 0.2, and a radius-check run at 0.3; in each, the sectors with the largest roundoff bounds were recomputed entirely in double-double arithmetic (8 at ratio 0.2, 14 at 0.3). The $\bar{\delta}^0$ part matches the exact four-bubble cluster to 4×10^{-11}, and all twelve $\bar{\delta}^1$ analyticity relations hold to 7×10^{-12}. A confirmatory check of the $\bar{\delta}^0$ part at two complex values of v gives 1.7×10^{-14} and 1.6×10^{-12} (the second under amendment A of Appendix I.7).
- *Roundoff, measured.* 56 further sectors were recomputed in double-double arithmetic at the fine grid: the 16 with the next-largest bounds, 24 drawn by bound decile, and 16 from the two top deciles as a held-out sample fixed in advance. The difference between double and double-double is at most 2.5×10^{-3} of the first-order bound in the top 16 sectors and the top decile, which together carry about 98% of the bound, at most 1.3×10^{-2} in the next decile, and reaches 0.41 only in sectors whose bound is negligible.
- *Result.* $f_{301} = -514275.75 \pm 0.17$; the uncertainty, 0.174, is the linear sum of the fine–coarse difference 0.016, the fine–radius difference 0.022 and the validated roundoff estimate 0.137.

The inputs were hashed and frozen before C_8^{SR} was assembled. The same assembly reproduces the closed forms of Eq. (2.12):

	magnetization scheme	closed form	relative difference
C_3^{SR}	5.72736813369645287294483	5.72736813369645287294483	3×10^{-39}
C_4^{SR}	−12.19619149266966002908757	−12.19619149266966002908757	2×10^{-29}
C_5^{SR}	29.23781158099003956882694	29.23781158099003956882694	3×10^{-28}
C_6^{SR}	−58.66569747422530894971715	−58.66569747422530894971715	7×10^{-26}
C_7^{SR}	51.06898861292810883118393	51.06898861292810883118394	4×10^{-26}

and gives

$$C_8^{\mathrm{SR}} = 247.381179745789 \pm 0.131, \tag{I.16}$$

quoted as 247.38118 ± 0.13 in Eq. (2.17). The uncertainty is the linear sum of the propagated input uncertainties: $\frac{3}{4}\times 0.174 = 0.13$ from f_{301}, 3.3×10^{-7} from f_{202} and 3.6×10^{-15} from all other inputs. Table 8 lists the 29 inputs, from which Eq. (I.13) reassembles C_3^{SR} to C_8^{SR}: from the printed 25-digit values C_8^{SR} equals the record to all its printed digits, and C_3^{SR} to C_7^{SR} come out to 10^{-25} to 2×10^{-22} relative (the table above used the unrounded inputs). The computation used about 2.7×10^4 core-hours on the symmetry cluster, including pilots and failed jobs; the engine runs behind the records, which a rerun repeats, took about 1.6×10^4.

Table 8: The 29 inputs of C_8^{SR} in the magnetization scheme, as frozen in the record of C_8^{SR} (Appendix I.8). Index ijk: $\hat{u}^i\hat{z}^j\bar{\delta}^k$, with $i+1$ bubbles and k logarithms. The $\bar{\delta}^0$ entries come from the exact Bethe-ansatz formula (matching residual 10^{-63}); the one-bubble entries are exact or high-precision numerics; the two-bubble entries are good to about 10^{-26} relative and the three-bubble $\bar{\delta}^1$ entries to about 10^{-22}. (a) f_{202} and f_{301} are printed in full only for exact reassembly; they are accurate only to $\pm1.7\times10^{-7}$ and ±0.174.

	monomial	value	source	closed form
f_{100}	$\hat{u}$	1	Bethe ansatz (exact)	1
f_{110}	$\hat{u}\hat{z}$	6.57973626739290574588966 1	Bethe ansatz (exact)	$4\zeta(2)$
f_{120}	$\hat{u}\hat{z}^2$	42.00883821061607434137304	Bethe ansatz (exact)	
f_{130}	$\hat{u}\hat{z}^3$	174.5896405924634503829158	Bethe ansatz (exact)	
f_{200}	$\hat{u}^2$	45.29870634431252721431787	Bethe ansatz (exact)	
f_{210}	$\hat{u}^2\hat{z}$	777.0075210550932975552262	Bethe ansatz (exact)	
f_{220}	$\hat{u}^2\hat{z}^2$	8099.313348527777075847784	Bethe ansatz (exact)	
f_{300}	$\hat{u}^3$	3848.445467772377268238629	Bethe ansatz (exact)	
f_{310}	$\hat{u}^3\hat{z}$	115632.5319783609240553502	Bethe ansatz (exact)	
f_{400}	$\hat{u}^4$	545193.0244029872263588049	Bethe ansatz (exact)	
f_{001}	$\bar{\delta}$	−0.57721566490153286060651 21	one bubble	$-\gamma_E$
f_{011}	$\hat{z}\bar{\delta}$	−3.86708379859798573355134 2	one bubble	$-\gamma_E - 2\zeta(2)$
f_{021}	$\hat{z}^2\bar{\delta}$	−8.09809574633483001454378 3	one bubble	
f_{031}	$\hat{z}^3\bar{\delta}$	−13.46681348232748267372698	one bubble	
f_{002}	$\bar{\delta}^2$	5.292133223877124175286	one bubble	
f_{012}	$\hat{z}\bar{\delta}^2$	43.26016741947440932389841	one bubble	
f_{022}	$\hat{z}^2\bar{\delta}^2$	194.2323046570014639460987	one bubble	
f_{003}	$\bar{\delta}^3$	−61.14566451935350750983253	one bubble	
f_{013}	$\hat{z}\bar{\delta}^3$	−767.6867345496617072942533	one bubble	
f_{101}	$\hat{u}\bar{\delta}$	−36.4985374424376232979 5634	two bubbles	
f_{111}	$\hat{u}\hat{z}\bar{\delta}$	−404.1590010240865956167455	two bubbles	
f_{121}	$\hat{u}\hat{z}^2\bar{\delta}$	−2982.066378636498938272215	two bubbles	
f_{102}	$\hat{u}\bar{\delta}^2$	810.9918099349166764628352	two bubbles	
f_{112}	$\hat{u}\hat{z}\bar{\delta}^2$	14885.335418594594510562 8608	two bubbles	
f_{103}	$\hat{u}\bar{\delta}^3$	−23498.571644954480587913 84	two bubbles	
f_{201}	$\hat{u}^2\bar{\delta}$	−3157.008605154717628460989	three bubbles	

	monomial	value	source	closed form
f_{211}	$\hat{u}^2\hat{z}\bar{\delta}$	$-75994.36349692199322461628$	three bubbles	
f_{202}	$\hat{u}^2\bar{\delta}^2$	$172415.8301856328384692596$	three bubbles	(a)
f_{301}	$\hat{u}^3\bar{\delta}$	$-514275.7548978331544915535$	four bubbles	(a)

I.7 Comparison, disclosures and review

Comparison. The comparison stage was run after the last code inspection had approved, and it refused to run unless the record of C_8^{SR} was the current frozen one, bound to the current records of its inputs and to a passing confirmatory check. With Δ the magnetization-scheme value minus the reference:

reference	value	Δ	$\Delta/0.131$	Δ/own error
record, Eq. (2.14)	$247.3812048049935\ldots$	-2.506×10^{-5}	-0.0002	–
independent, Sec. 2.6	242.825 ± 2.371	$+4.556$	$+34.8$	$+1.92$

(The comparison stage of the calculation used the rounded 242.83 ± 2.37, which gives $\Delta=+4.551$.)

Disclosures. The review of the code approved quoting the value on condition that the following are stated.

- The central value was frozen, by hash, before any comparison. A first version, 247.38118 ± 0.077 with the same central value, is superseded because its roundoff estimate had not been validated.
- The session coordinating the calculation saw the comparison when that first version was frozen. The session running the computation was exposed to it later: after the central values and the design of the additional runs for the error bar were frozen, but before the first record of the confirmatory check was computed (from finished runs, against its fixed criterion, which it failed) and before the error bar was evaluated.
- Two stopping rules fixed in advance fired during that work. The author decided both amendments after seeing the data; the failing first records stay frozen.

  A. *Confirmatory check.* The second complex point missed its criterion 10^{-12} by roundoff (1.6×10^{-12}, against that run's own first-order roundoff bound 1.7×10^{-9}). A double-double rerun (about 1.7×10^{4} core-hours) was not made; the criterion was amended to at most 10^{-11} and within the run's own roundoff bound.

  B. *Roundoff validation.* The global rule, a stop if in any sampled sector the difference between the double and double-double results exceeds 0.1 of the first-order bound, fired on sectors of negligible bound; applied as written it would have made the uncertainty of C_8^{SR} about 16. The rule was amended to apply the same factor-ten margin decile by decile, with a stop only for a ratio above 0.1 among the sectors ranked 9 to 24 by bound (the top 8, recomputed entirely in double-double, are checked separately) or in the two top deciles, and it was tested on the fresh held-out sample, fixed before its data, which stayed within the margin.

Review. OpenAI Codex reviewed the plan in three rounds, approving it before any multi-bubble cluster with logarithm insertions was computed, and inspected the code six times; the sixth inspection approved quoting the value with the disclosures above. After each of the last two rounds of fixes, every current record was rebuilt byte for byte with the revised checks.

What the calculation establishes and what it does not.

- It gives $C_8^{\mathrm{SR}} = 247.38118 \pm 0.13$ in a scheme that shares no input with the compendium or with the independent re-derivation, and that never uses A_9. It agrees with the record to 2.5×10^{-5}. Since $C_8^{\mathrm{SR}} - \frac{1}{4}J_9$ of the independent re-derivation agrees with the compendium (Sec. 2.6), the low independent value lies in its J_9, and within J_9 in $A_9[\varepsilon^0\ell^1]$, the one piece that deviates; whether that deviation is a fluctuation with an underestimated error or a systematic error is not known.
- With a different regulator (analytic continuation in v instead of a hard core) and different responses (the magnetization on the infinite line instead of finite-interval responses with an endpoint source), it reproduces the closed forms of C_3^{SR} to C_7^{SR} to 7×10^{-26} and the record of C_8^{SR} within its uncertainty. This is numerical support for the prescription of Sec. 2.1 and for the compendium's integrations, not a proof of either.
- The uncertainty is an empirical numerical estimate from three runs and measured roundoff: not a standard deviation and not a bound, and it cannot exclude an error common to all three runs. At 5×10^{-4} relative it can neither identify nor certify the rational coefficients of Eq. (2.15).
- It assumes, in its own coordinates, the same two-coupling description of the gas as the prescription (the analyticity relations test it order by order), the identification $\Delta_{\mathcal{E}} = \Delta_-$ of Eq. (2.6), the Bethe-ansatz equations of the anisotropic Kondo model and its identification with the gas at $s = 1$. It does not test whether the gas, or the dual, describes the critical chain.

I.8 Records

The records of the calculation are frozen and identified by their sha256 hashes; they are in `code/integ_d1/results/`, `make test` checks the hashes, and `make replay` rebuilds each current record byte for byte from the shipped engine outputs. The record of C_8^{SR} contains Table 8, the values of C_3^{SR} to C_7^{SR} and the amendment texts, and it is bound to the records of its inputs:

record	sha256
C_8^{SR} (`C8_record_v2.json`)	`0d463b29d9ec474cacf2ab96afff62ae56b0ec262d766684ebbf0b7b8b20ec0c`
two bubbles (`M2_flow_coefficients.json`)	`33cc2e1e9a6e9a57510b0b3c6a7ef419258317be892774c3054d04852161e833`
three bubbles, one logarithm (`M3_flow_coefficients.json`)	`4328cab931ea96e953e9e6ba3789dfbfc0859c94f26fc8d38c1b1161eb271bce`
f_{202} (`M4A_flow_coefficients.json`)	`584562ca72485e0f57c2f1e8ac6a05dcc5cc07b2acc73659a39587bd61641048`
f_{301} (`M4B_flow_coefficients_v2.json`)	`90542a62db08d875995f2944e84ec7d6ff8987acc1b50892cd459d7e632d9899`
confirmatory check (`M4B_cx_check_v2.json`)	`66287b2af2f4c5352a1c15dc07edf6b381c2215f54b5b979fe726f4febf0f1fb`

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